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Cs601 Reading Material:

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Updates Will Be Provided On Weekly Basis In Group's Description.

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Communication: Sharing of Information (Local or remote).

Telecommunications: Communication at a Distance (includes telephony, telegraph, and television etc.)

Data communications: Exchange of data between two devices via some form of transmission media.

Effectiveness of a Data Communication System:

The effectiveness of a Data Communication System depends on the following factors:

Delivery:

When data is sent from one place to another correctly and successfully, this is called delivery.

Accuracy:

When data is sent, it must be accurate, which means that there must be no errors.

Timeliness:

For data to be considered timely, it must be sent within a reasonable amount of time.

Jitter

The difference in packet arrival times in a network, which affects how consistently data is sent at regular intervals, is called jitter.

A data communications system has five components:

~ Sender

~ Receiver

~ Transmission medium

~ Protocol

~ Message

Following are the different forms of Information

~ Text

~ Numbers

~ Images

~ Audio

~ Video

Data Flow between two devices in the following modes:

Simplex: In simple terms, simplex is like talking on a one-way street. While in simplex mode, data can only go from one device to another. The device that receives the data cannot send it back. Like a TV remote control, you can tell it what to do (like changing stations), but it doesn't

get any information from the TV.

Half Duplex: It is like talking on a walkie-talkie. Data can be sent and received in this mode, but not at the same time. Like switching places in a chat. People listen to each other when they talk and talk to each other when they listen. Changing between sending and getting is possible, but not both at the same time.

Full Duplex: Communication that works both ways is like a normal phone call. There is time for both people to talk and listen. Like having a street that goes both ways, data can flow in both directions at the same time. This is like how we talk to each other in real life—we can both talk and listen at the same time, which makes dialogue smooth and natural.

Network: Interconnection of a set of devices capable of communication

Host: In computer networks, a host is any device that is connected to the network, like a computer or printer. Each host has its own unique IP address, which lets other devices on the network talk to each other and share data.

Connecting Device: In networks, a connecting device is either hardware or software that lets devices talk to each other by controlling network traffic and managing data transfers easier.

A network must be able to meet a certain number of criteria such as:

~ Performance

~ Reliability

~ Security

Physical Network Attributes:

Link: It refers to a way for two or more devices to talk to each other or a connection that lets them share info. Different types of media, like wired (like Ethernet cables and fiber-optic lines) or wireless (like Wi-Fi and Bluetooth) technologies, can be used to make links.

Type of Connection

Point-to-Point

In networking, point-to-point connections let two devices talk to each other directly over a fixed link, making it easy for them to share data.

Multipoint

Multipoint links, on the other hand, let more than two devices share the same communication link, though they do have to compete for the available bandwidth. This lets more than two
devices talk to each other over a shared network.

Physical Topologies

~ Links + Nodes = Topology

~ Topology: Physical Layout of Network

~ Physical Topologies:

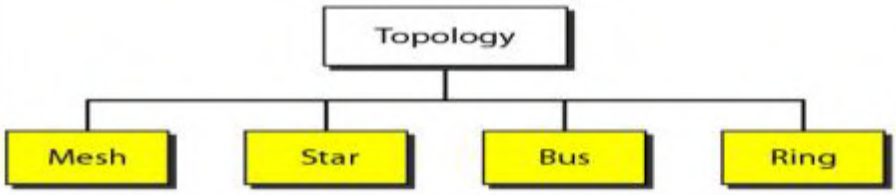
Mesh: A network setup where each device is connected to every other device through individual point-to-point links.

Star: In a star network topology, every device is connected through a dedicated point-to-point link solely to a central controller, often referred to as a hub.

Bus: A single extended cable serves as a backbone to connect all devices within a network in a multipoint configuration.

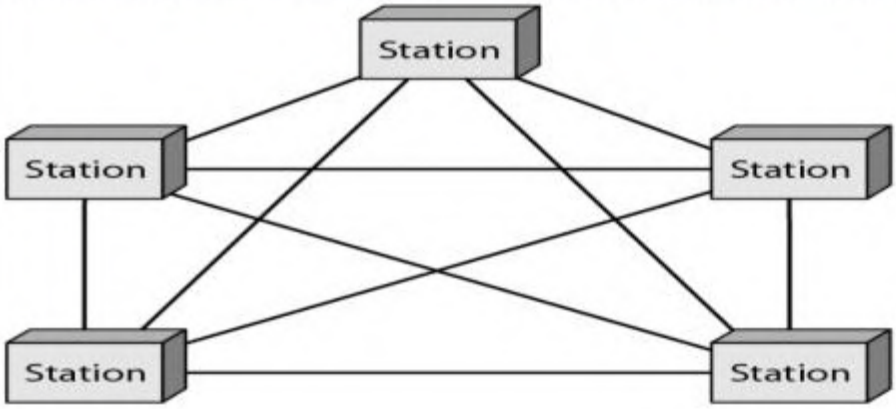
Ring: Every device maintains an exclusive point-to-point link with only the two neighboring devices on either side of it.

Physical Topologies



Mesh Topology:

A network setup where each device is connected to every other device through individual point-to-point links.

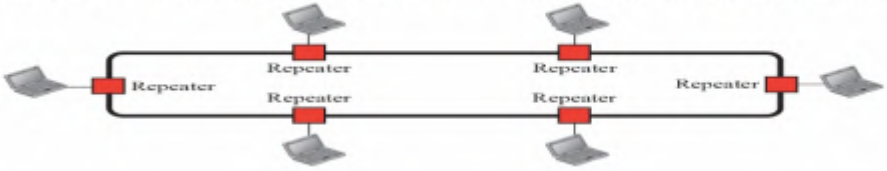


Bus Topology:

A single extended cable serves as a backbone to connect all devices within a network in a multipoint configuration.

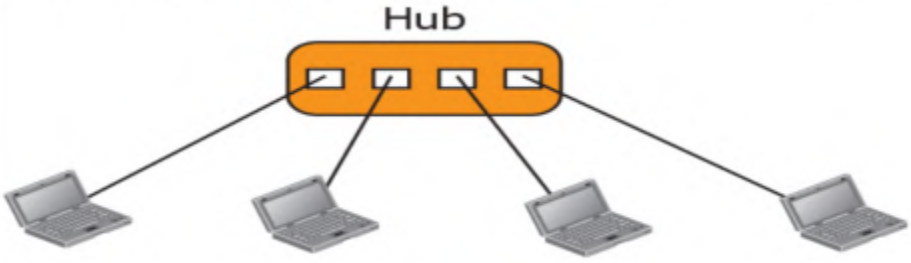
Ring Topology:

Every device maintains an exclusive point-to-point link with only the two neighboring devices on either side of it.



Star Topology:

In a star network topology, every device is connected through a dedicated point-to-point link solely to a central controller, often referred to as a hub.



Networks Types

Network classification:

Networks can be classified in various ways based on different criteria.

~ Size

~ Geographical Coverage

~ Ownership

Local Area Networks

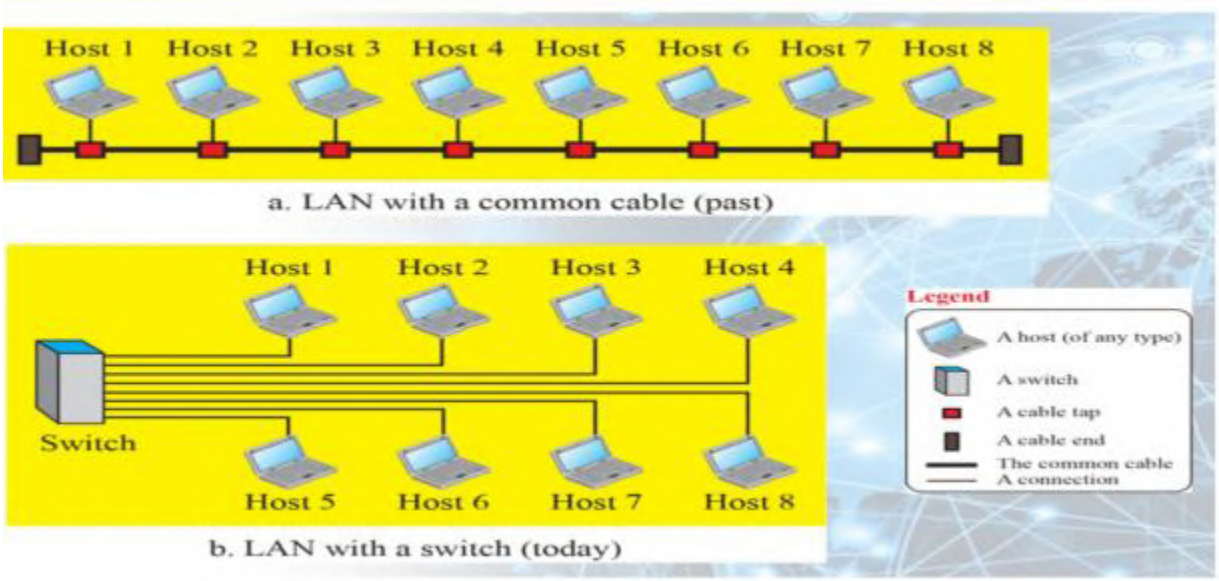
~ Usually Privately owned

~ Connects some hosts in a single office, building, or campus

~ Can be as simple as two PCs and a printer in someone's home office

~ Can extend throughout a company

~ Host Address



Wide Area Network

~ Wider geographical span than a LAN

~ Spans a town, a state, a country, or even the world

~ Interconnects connecting devices such as switches, routers, or modems

~ Normally created and run by communication companies

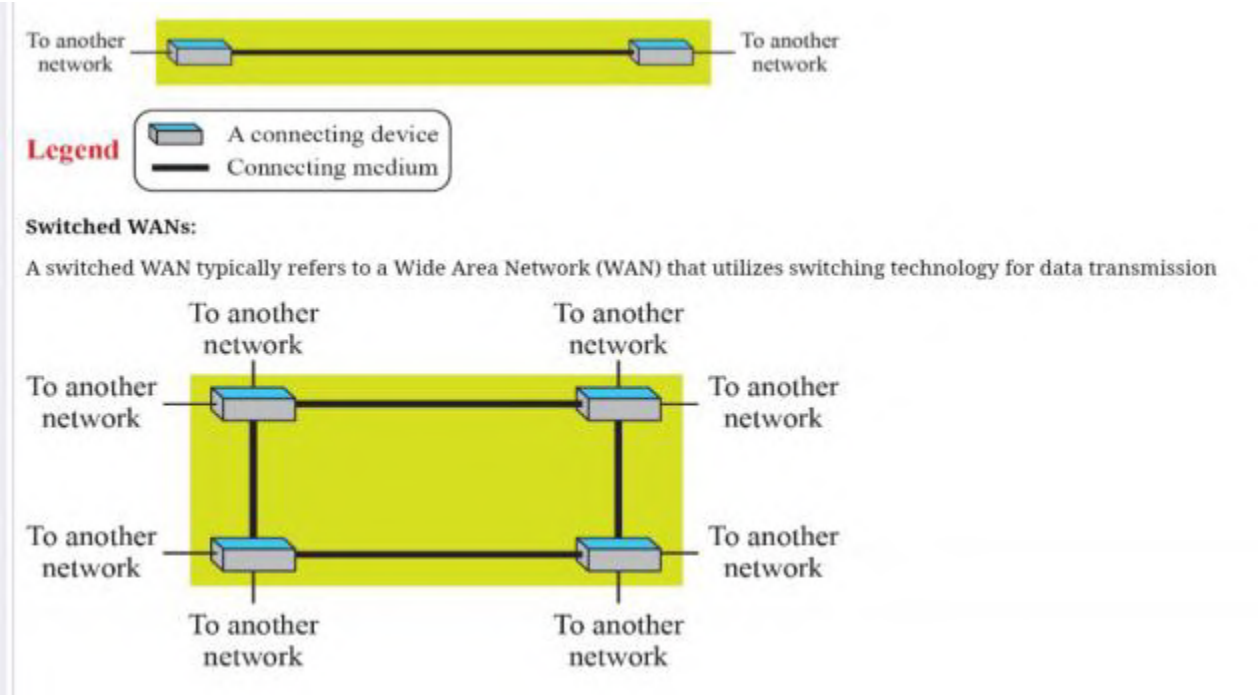
~ Point-to-Point WAN

~ Switched WAN

~ Internetwork

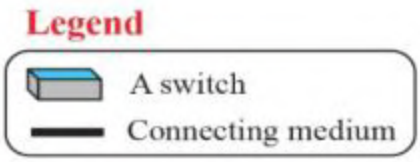
Point-to-Point WANs:

A Point-to-Point Wide Area Network (WAN) is a type of network connection that establishes a direct link or communication path between two specific locations or endpoints.



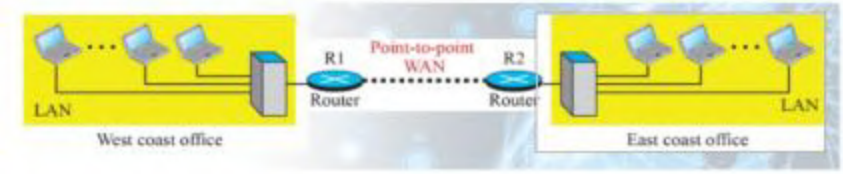
Switched WANs:

A switched WAN typically refers to a Wide Area Network (WAN) that utilizes switching technology for data transmission



Internetwork:

is a network that connects multiple individual networks or subnetworks together, allowing them to communicate and share information across a larger, interconnected system.



Switching:

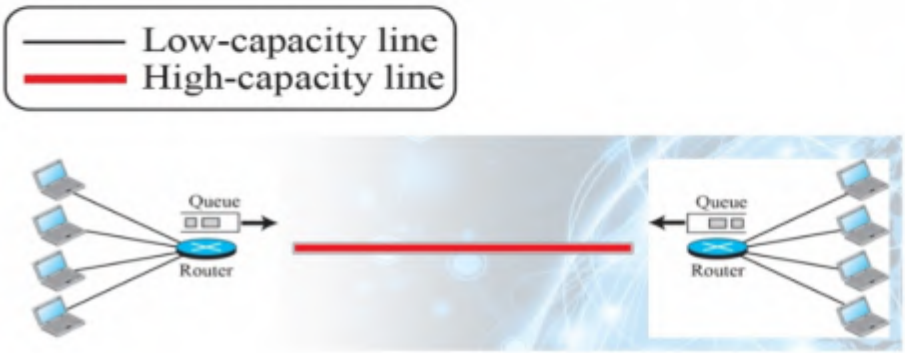
- Circuit-Switched Network: A dedicated link, referred to as a circuit, remains constantly established between the two end systems, while the switch can solely control its activation or deactivation.
- Packet- Switched Network: A packet-switched network is a type of data communication network in which digital data is divided into small packets for transmission.

Circuit Switched Network:

A circuit-switched network is a type of telecommunication network in which a dedicated communication path or circuit is established between two devices for the duration of their conversation.

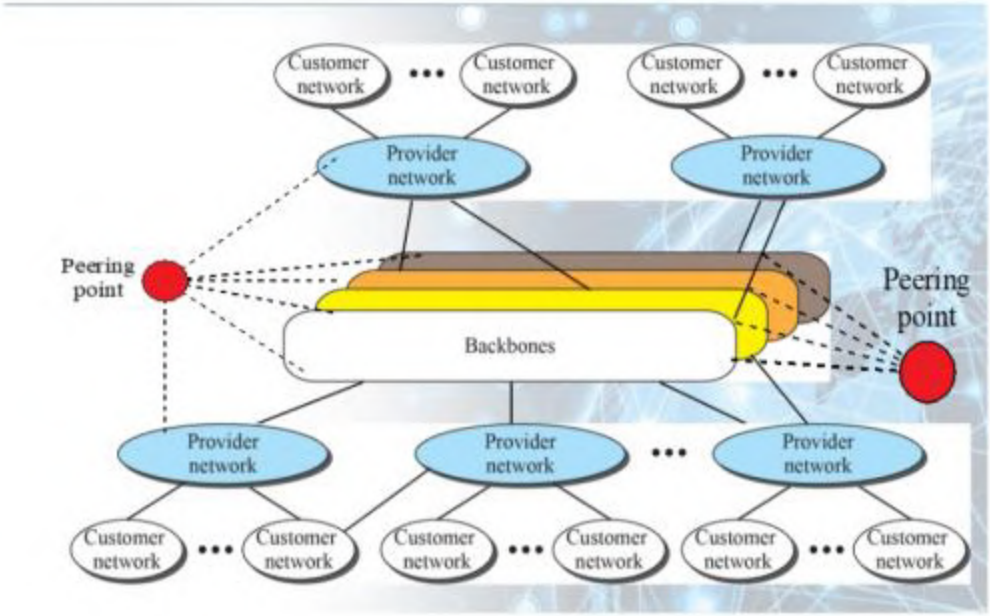
Packet Switched Network:

A packet-switched network is a type of data communication network that transmits digital information in discrete packets of data.



The Internet

- An internet (note the lowercase i) is two or more networks that can communicate with each other
- The Internet (uppercase I), and is composed of thousands of interconnected networks.
- Accessing the Internet



Internet History

Telegraph and Telephone networks, before 1960: Constant-rate communication only

ARPANET: short for Advanced Research Projects Agency Network, was one of the earliest and most significant computer networks that laid the foundation for the development of the modern internet.

MILNET: short for Military Network, was a computer network used by the United States Department of Defense (DoD) in the 1980s.

CSNET: short for Computer Science Network, was a computer network that played a significant role in the early development of the internet.

NSFNET: The National Science Foundation Network, commonly known as NSFNET, was a pivotal computer network that played a crucial role in the development and expansion of the internet in the United States.

Internet Standards and Administration

Internet draft: An Internet draft is a provisional document, still under development, and lacking official status, typically having a lifespan of approximately six months.

Request for Comments (RFC) : Based on guidance from internet authorities, a draft could be released as a Request for Comment (RFC).

~ Proposed Standard

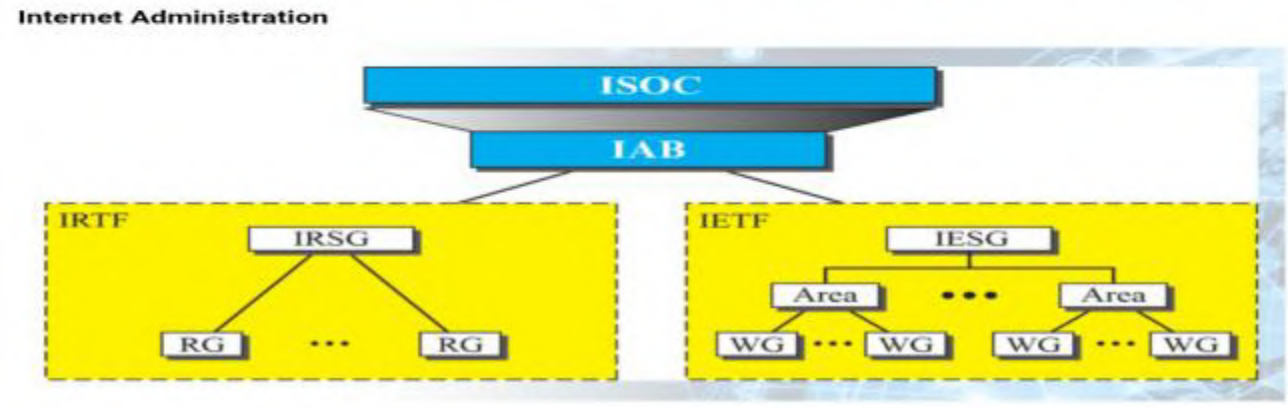
~ Draft Standard

~ Internet Standard

~ Historic

~ Experimental

~ Informational



Second Week Reading Material

Protocol Layering - Introduction

Protocol

Rules that both the sender and receiver and all intermediate devices need to follow to be able to communicate effectively.

Protocol Layering

Simple Communication: Exchange of information between two or more individuals or entities using straightforward and easily understood language or methods. We may need only one simple protocol.

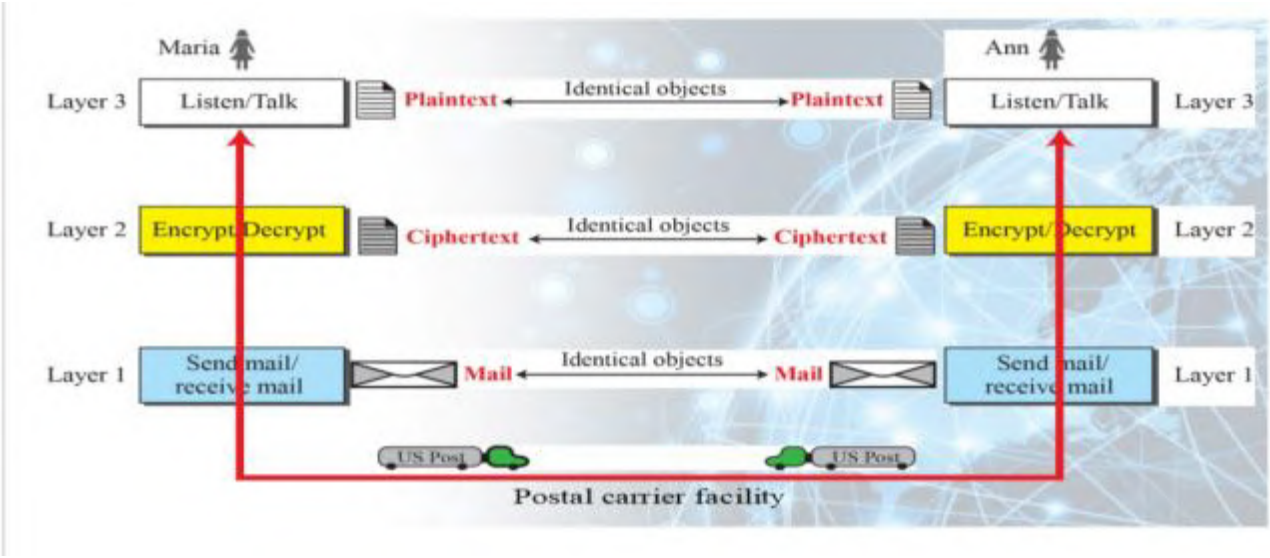
Complex Communication: The task is divided between different layers, in that case we need a protocol at each layer, or protocol layering.

Protocol Layering - Example Scenario 1

In the picture below, communication between Maria and Ann takes place in one layer, face to face, in the same language.

Protocol Layering - Example Scenario 2

In the below scenario, Maria and Ann are using encryption/decryption technique so their ideas do not disclose to other people



Protocol Layering - Advantages and Disadvantages

Advantages

~ Modularity

~ Separation of Service & Implementation

~ Reduced Complexity & Cost

Disadvantages

Complexity: Layering introduces multiple protocols, each responsible for a specific function or task.

Overhead: Each layer adds overhead in the form of headers, trailers, and additional processing.

Protocol Layering - Principles

~ Bidirectional Communication à Each Layer performs two opposite tasks in each direction.

~ Two objects under each layer at both sites should be identical.

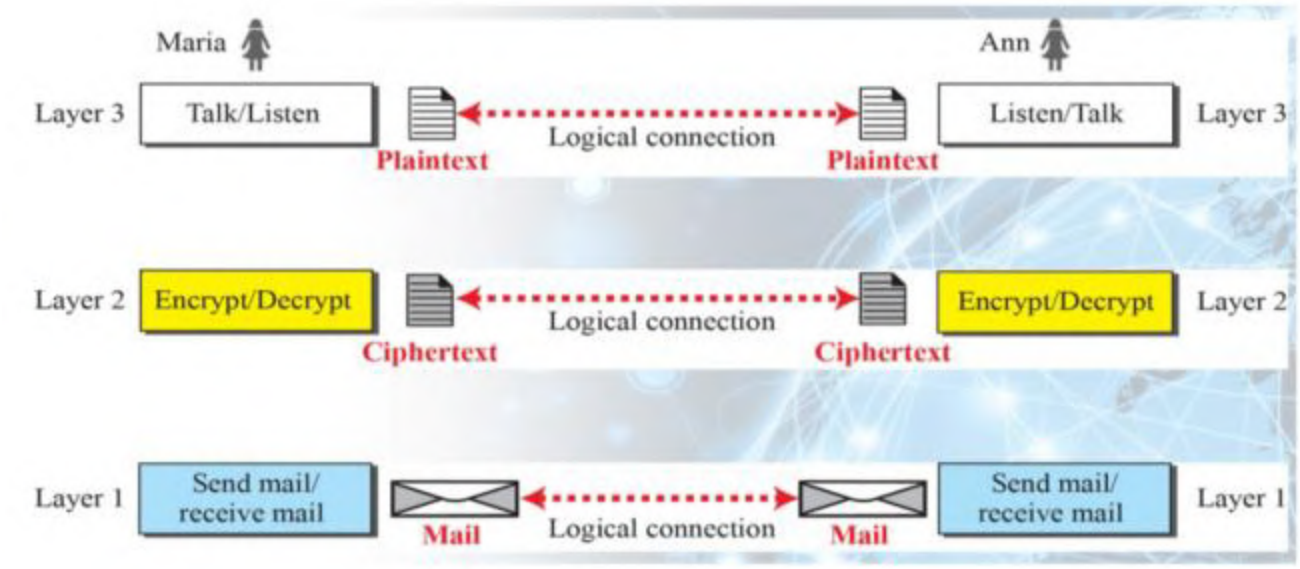
Protocol Layering - Logical Connections

Logical Connections

~ Imaginary connection between each layer

Protocol Layering

Now Maria and Ann can imagine that there is a logical connection at each layer through which they can send their data created from that layer.



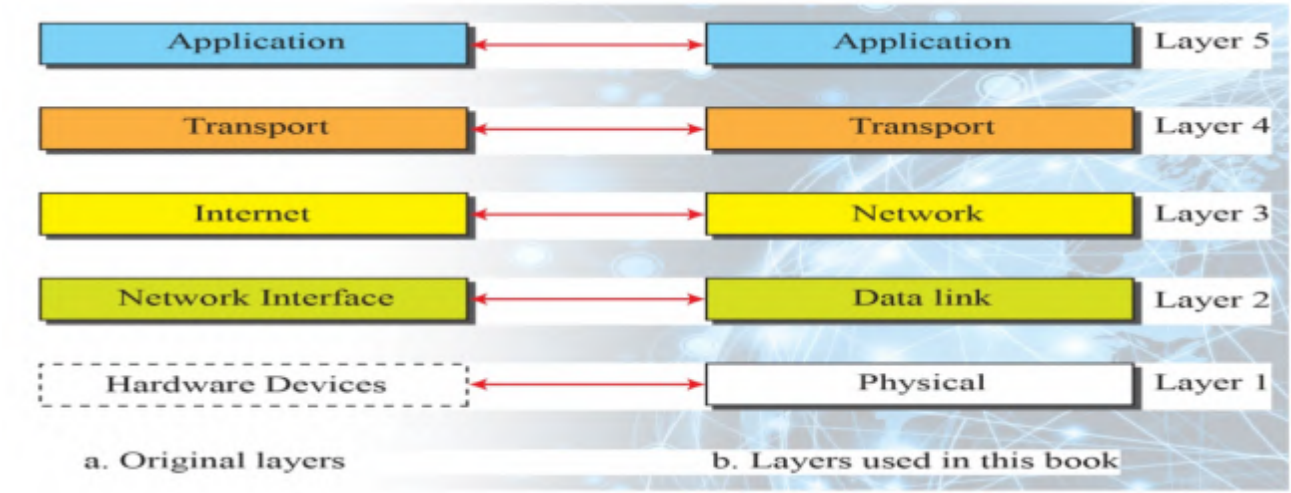
TCP/IP Protocol Suite

~ Protocol suite used on the Internet today.

~ Each Layer provides specific functionality.

~ Transmission Control Protocol/Internet Protocol (TCP/IP) is a hierarchical protocol suite used for data communication in computer networks.

~ Presented in 1973 and chosen to be the official protocol of Internet in 1983



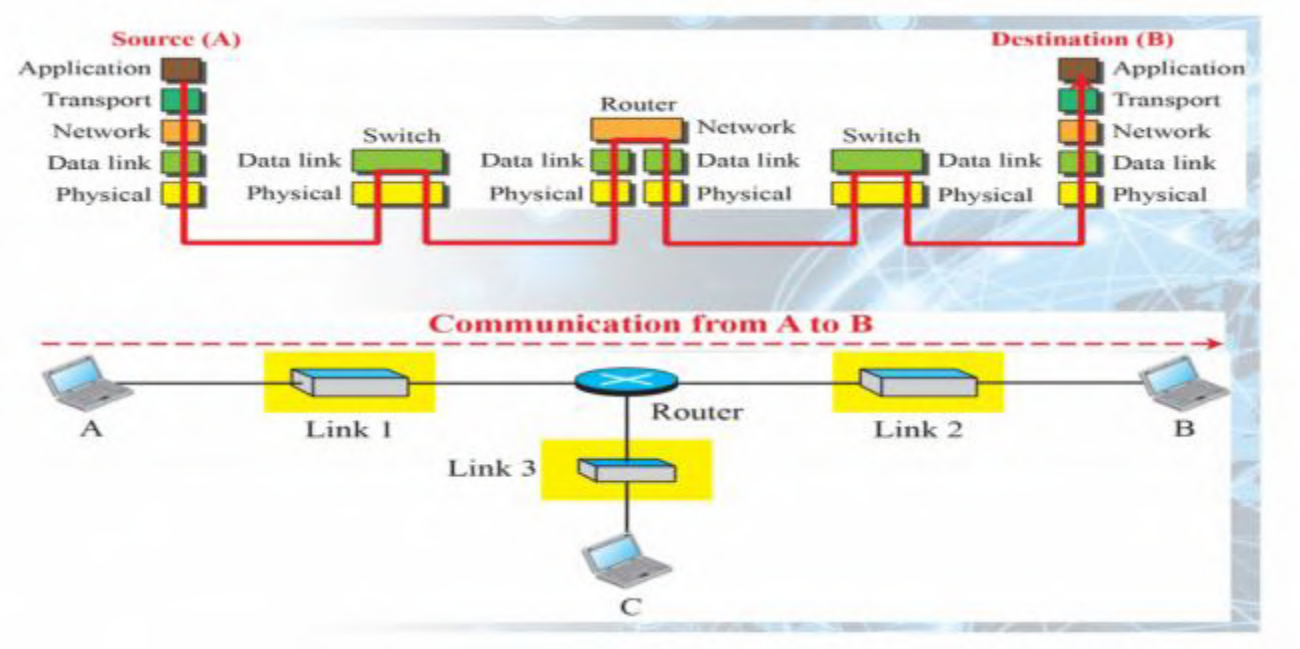
TCP/IP Protocol Suite - Layered Architecture

~ In the below figure, let us assume that computer A communicates with computer B.

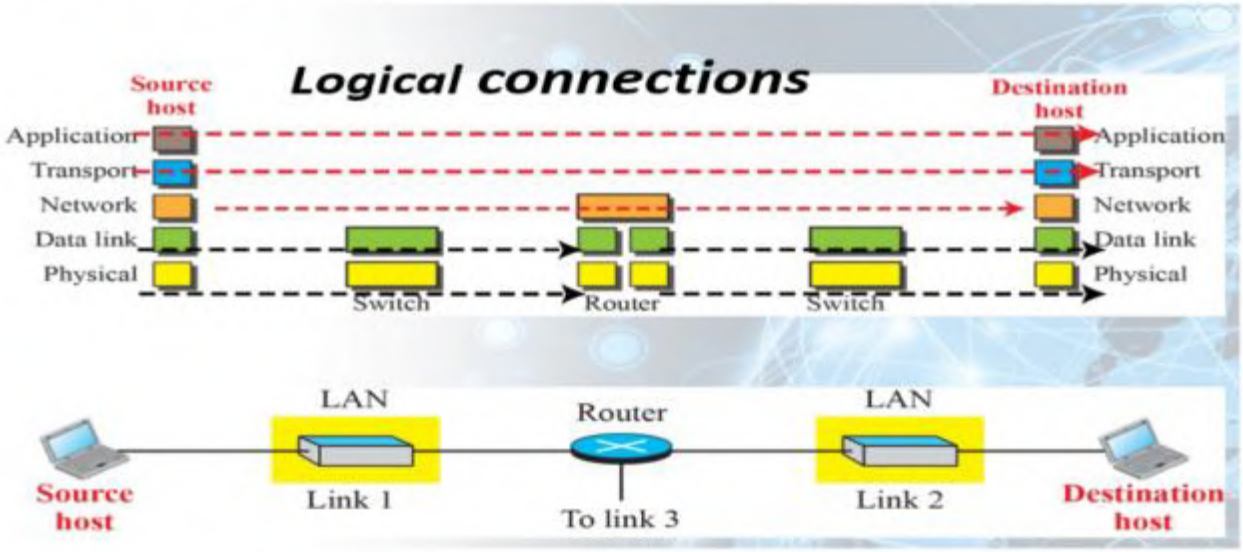
~ Both hosts participate in all five layers of the communication process.

~ The router is involved in only three layers, there is no transport or application layer in a router as the router is used only for routing.

~ A link-layer switch is involved only in two layers, datalink and physical.



The following diagram show the **Logical Connections between TCP/IP Layers.**

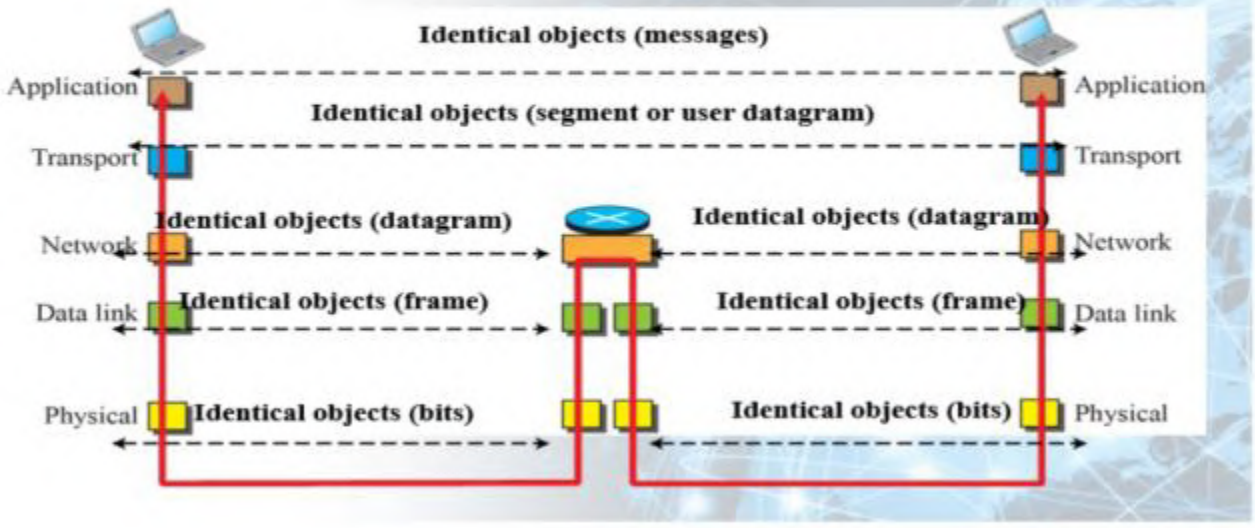


Now the other figure shows the identical objects of each layer:

For example,

Application layers exchanges the identical objects as "messages" and Physical layers as "bits".

Notes: We have not shown switches because they don't change objects.



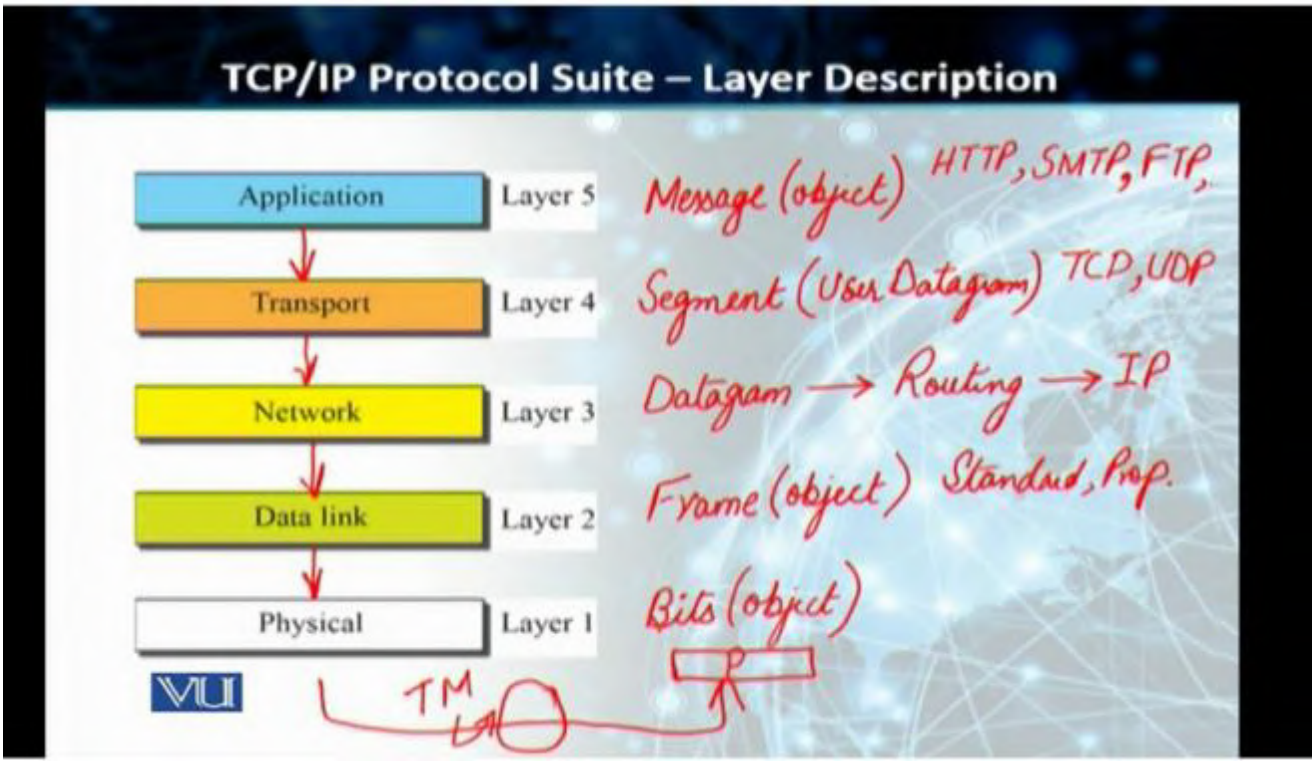
The following figure shows the protocols and the objects used in Layer 3-5.

~ HTTP, SMTP and FTP protocols are used in Application layer

~ TCP and UDP in Transport layer

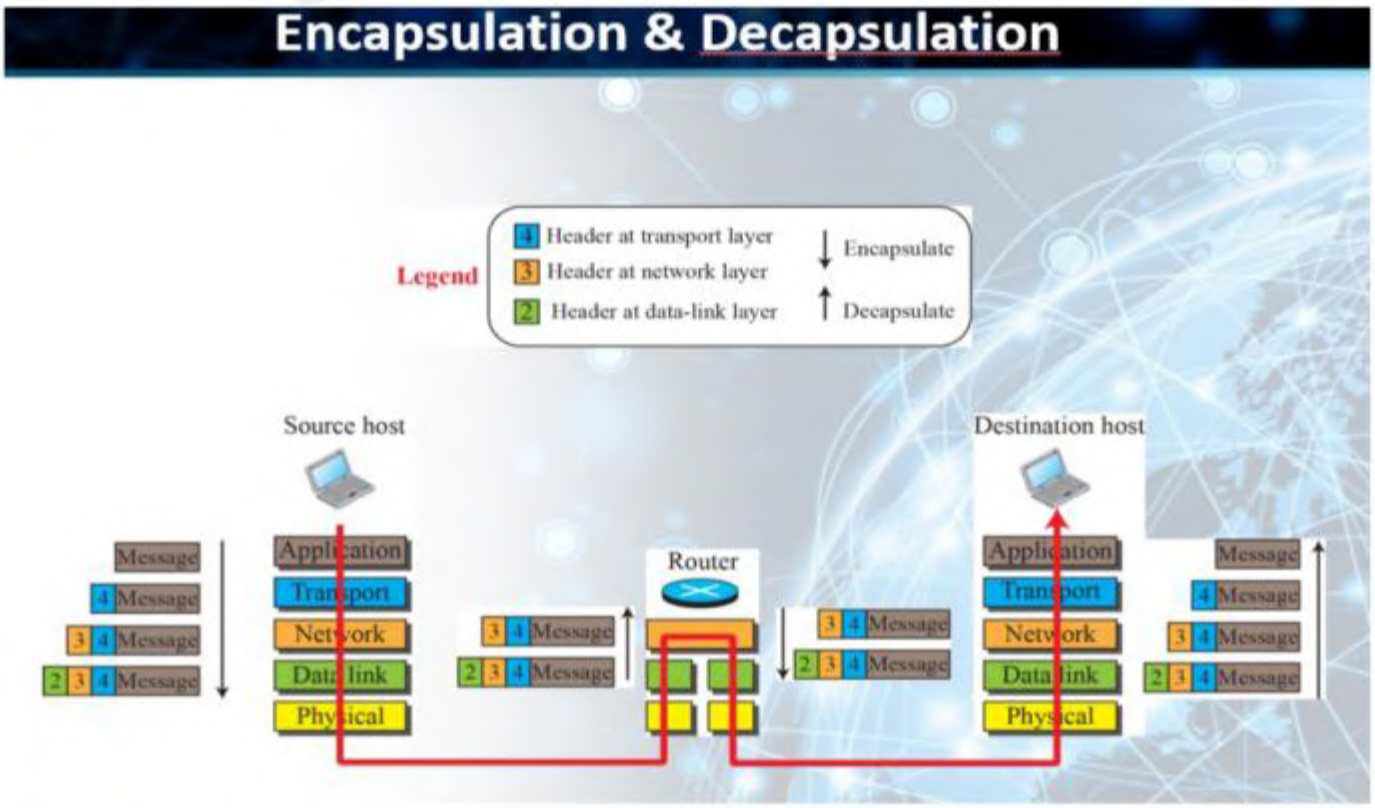
~ IP protocol in Network layer

PDNTA

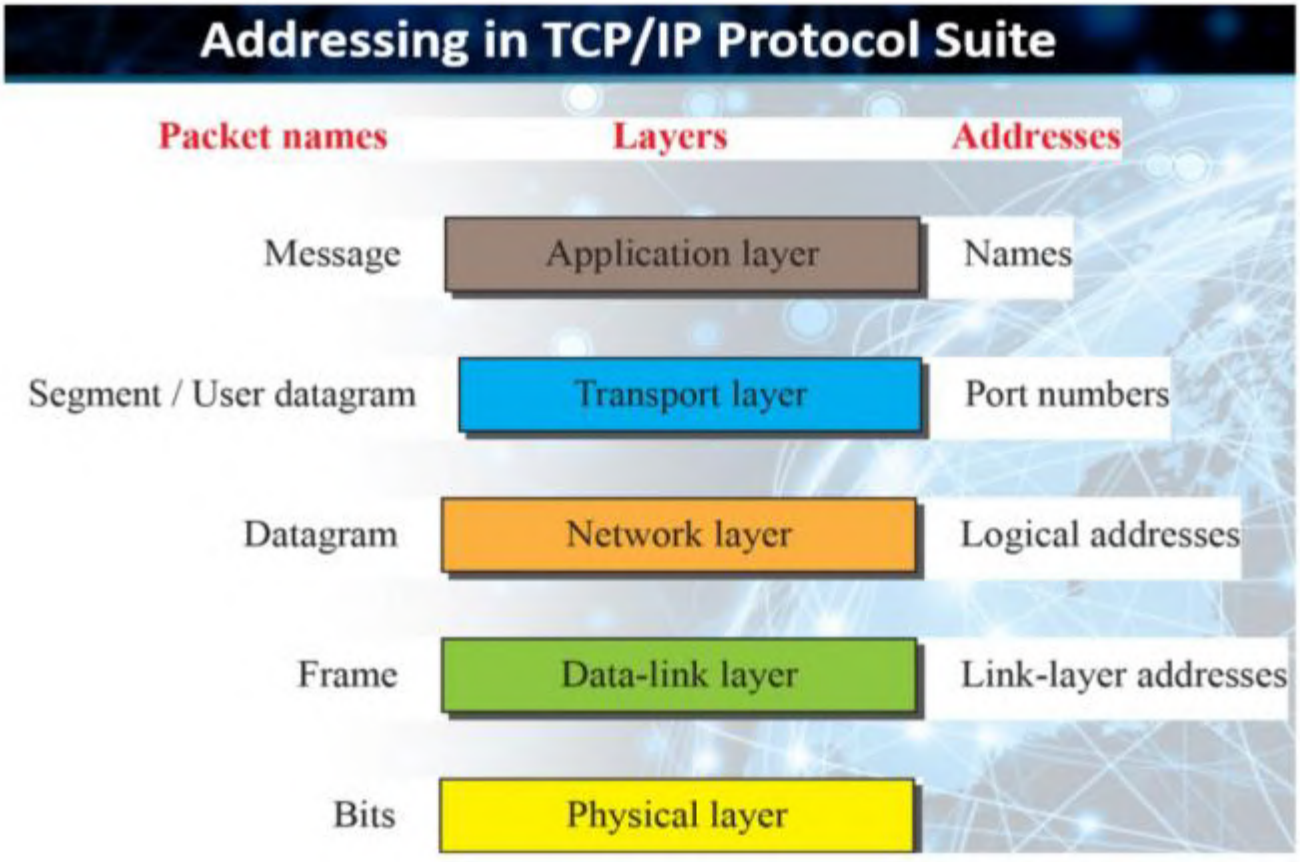


Encapsulation is the process of enclosing data from higher network levels with headers from lower layers in order to facilitate transmission, similar to placing a letter inside an envelope.

Decapsulation refers to the procedure of unsealing the envelope, eliminating the headers, and gaining access to the initial data upon reception. These operations guarantee the appropriate encapsulation and decapsulation of data as it traverses various network layers.

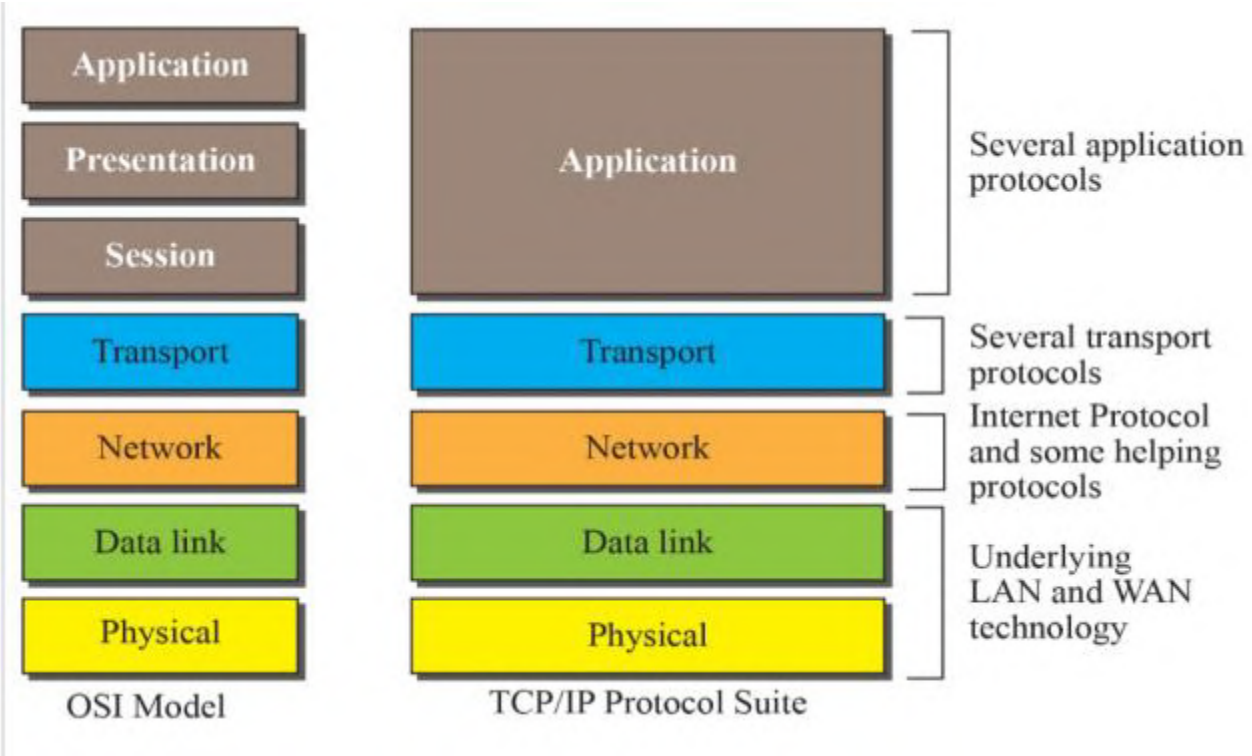


The following shows the addressing done at each layer:



The number and layers are both different in TCP/IP protocol suite as compared with OSI model.

The following figure clearly depicts the difference:



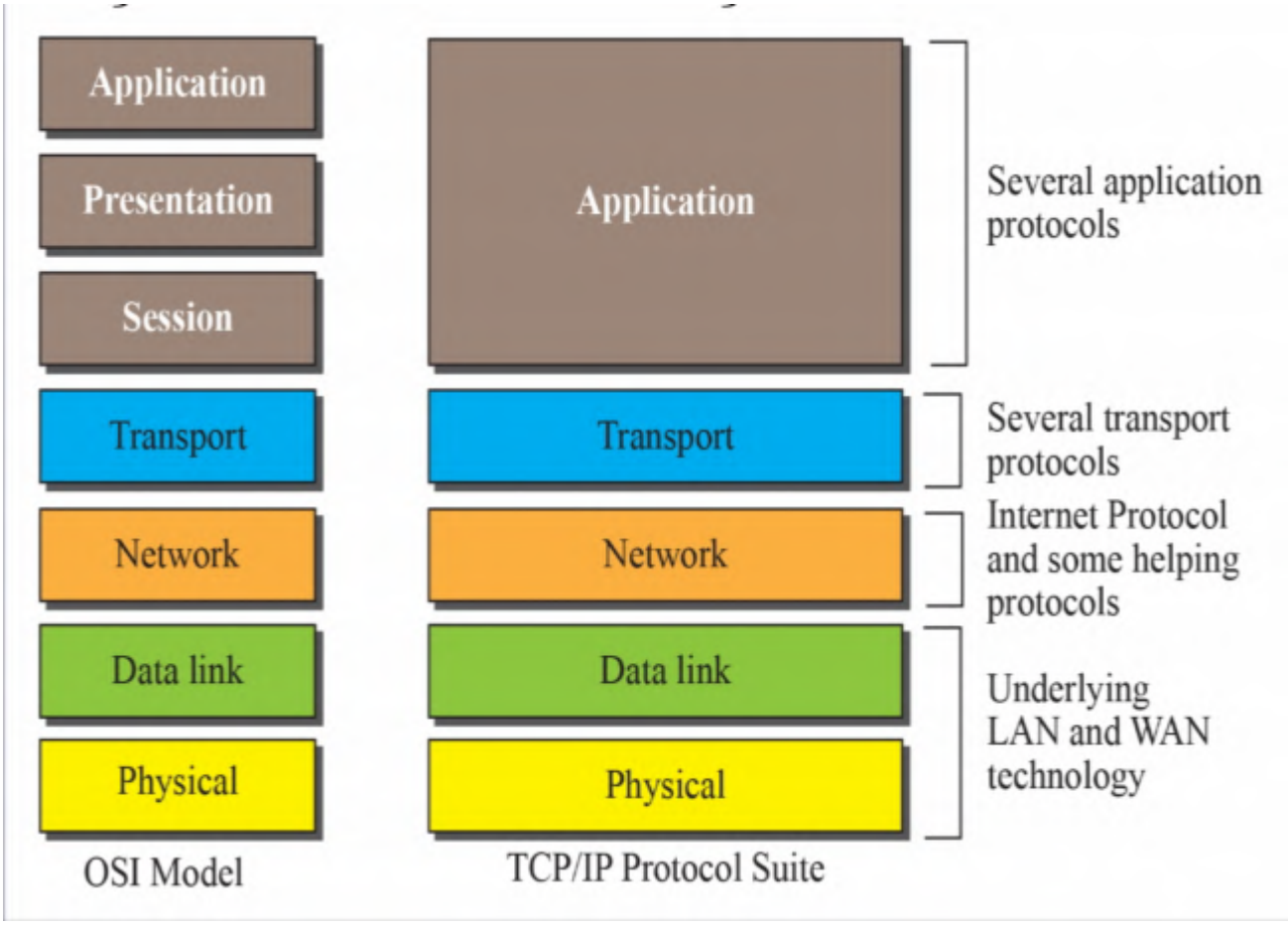
There are several reasons OSI did not replace TCP/IP, three of them are mentioned below:

~ OSI was completed when TCP/IP was fully in place

~ Some layers in OSI not fully defined

~ Performance of TCP/IP better than that of OSI

We will discuss the bottom two layers i.e. the Physical and Datalink layers in our course.



The following topics in these two layers will be covered:

~ Analog & Digital Transmission

~ Transmission Media

~ Switching

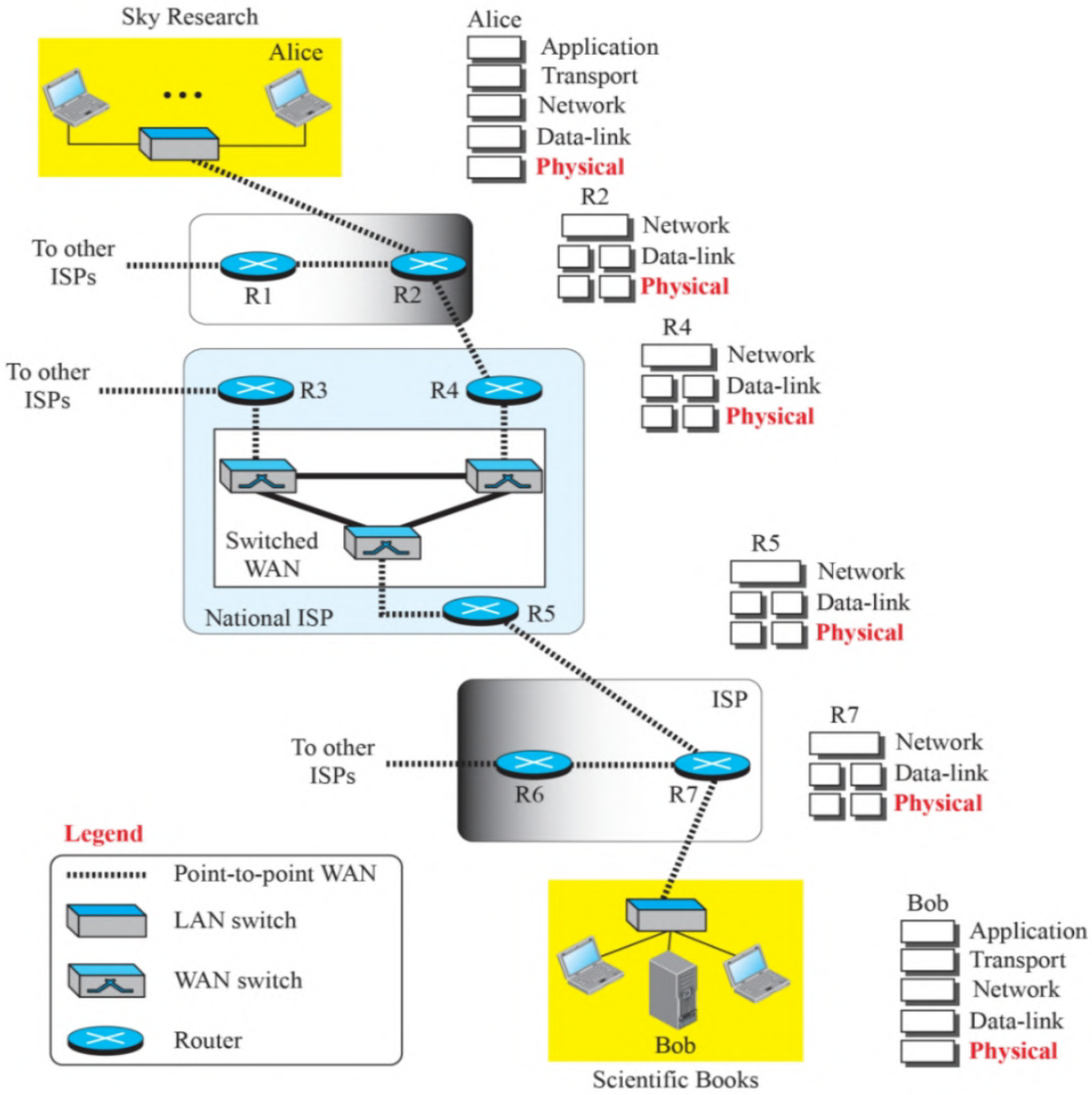
~ Error Detection and Correction

~ Media Access and Data Link Control

~ Wired and Wireless LANs

[CS601 Week - 3]

The following figure shows the layers involved in the communication between two users i.e. Alice and Bob.



The data signals can take two forms: Digital and Analog

Digital Data:

Digital data is information that is stored in a precise numerical form, most often in the binary system, where data is stored as groups of 0s and 1s. In a digital representation, each number (bit) stands for a different piece of data. Digital data is accurate, simple to change, and can be sent, saved, and processed quickly by computers and other electronic systems. Noise and degradation are less likely to happen during transfer, so it can be used for long-term storage and accurate communication.

Analogue Data:

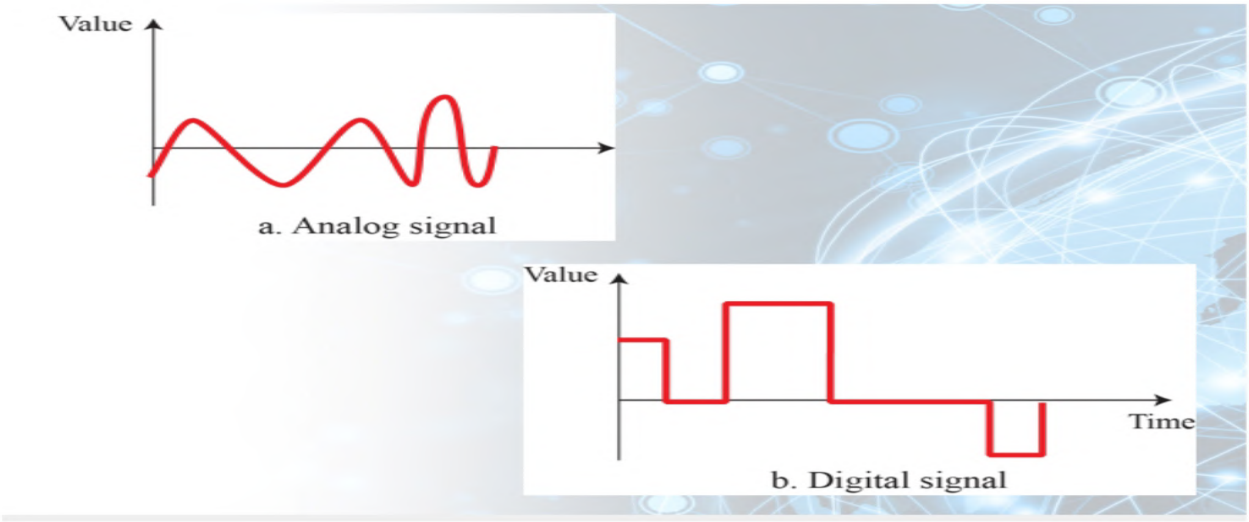
On the other hand, analogue data uses constant signals to show information. Within a certain time frame, these signals can have a lot of different meanings. Natural sounds, human voices, and temperatures in the surroundings are all examples of analogue data. An analogue signal is different from digital signals because it is constant and could have any value within a certain range. In audio and video signals, for example, analogue signals are used when the information they carry changes smoothly and constantly. During transfer and conversion, noise and degradation can happen to analogue data.

Analog Signal:

A continuous, smooth waveform that uses an infinite range of values to describe information is called an analogue signal. Like nature sounds or changes in temperature, it changes all the time.

Digital Signal:

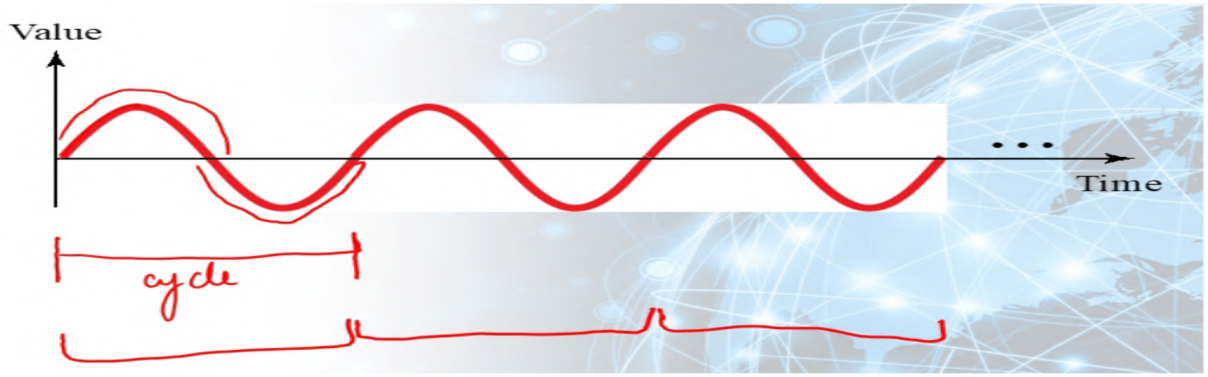
A digital output only has a limited number of values, which are usually shown as 0s and 1s. In computers and other modern electronics, digital signals are used to reliably and accurately send and store data.



Periodic analogue signals:

Periodic analogue signals are signals that exhibit a recurring pattern within a defined time frame. They demonstrate consistent and repetitive behaviour, which allows for easy analysis within the field of signal processing.

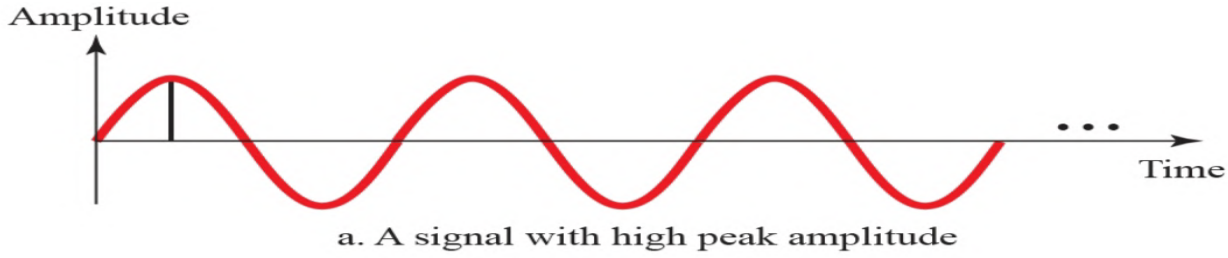
Let's take a look at a simple periodic analogue signal, specifically a sine wave.



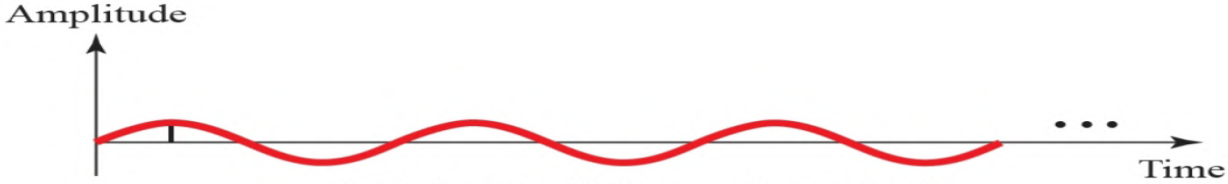
A simple periodic analog signal, visualized as a sine wave, displays a consistent rhythmic motion at regular intervals. Sine waves are fundamental in explaining natural phenomena like sound waves and electrical currents. They have a singular frequency and amplitude, making them vital in signal processing and mathematics.

Composite Periodic Analog Signal:

A composite periodic analog signal is complex, composed of multiple sine waves with different frequencies and amplitudes. These waves combine to form intricate waveforms, representing complex real-world signals like music or speech. Professionals use this breakdown method in fields like telecommunications and audio engineering for effective analysis and manipulation.



a. A signal with high peak amplitude



b. A signal with low peak amplitude

Frequency refers to the number of cycles that a wave completes within a given time period, usually measured in Hertz (Hz). It determines various characteristics, such as pitch in sound waves and colour in light waves.

The formula for frequency (f) is:

f=1/t. Where:

~ f represents the frequency in Hertz (Hz).

~ t represents the period of the wave, which is the time it takes to complete one full cycle, measured in seconds (s).

Period		*Frequency*	
Unit	*Equivalent*	*Unit*	*Equivalent*
Seconds (s)	1 s	Hertz (Hz)	1 Hz
Milliseconds (ms)	10^{-3} s	Kilohertz (kHz)	10^{3} Hz
Microseconds (μs)	10^{-6} s	Megahertz (MHz)	10^{6} Hz
Nanoseconds (ns)	10^{-9} s	Gigahertz (GHz)	10^{9} Hz
Picoseconds (ps)	10^{-12} s	Terahertz (THz)	10^{12} Hz

Examples of frequency calculation:

The power we use at home has a frequency of 60 Hz. The period of this sine wave can be determined as follows:

The period (TT) of a wave is the reciprocal of its frequency (ff). In this case, where the frequency is 60 Hz, the period can be calculated using the formula:

✓ T=1/f

✓ Substituting the given frequency:

✓ T=1/60 Hz

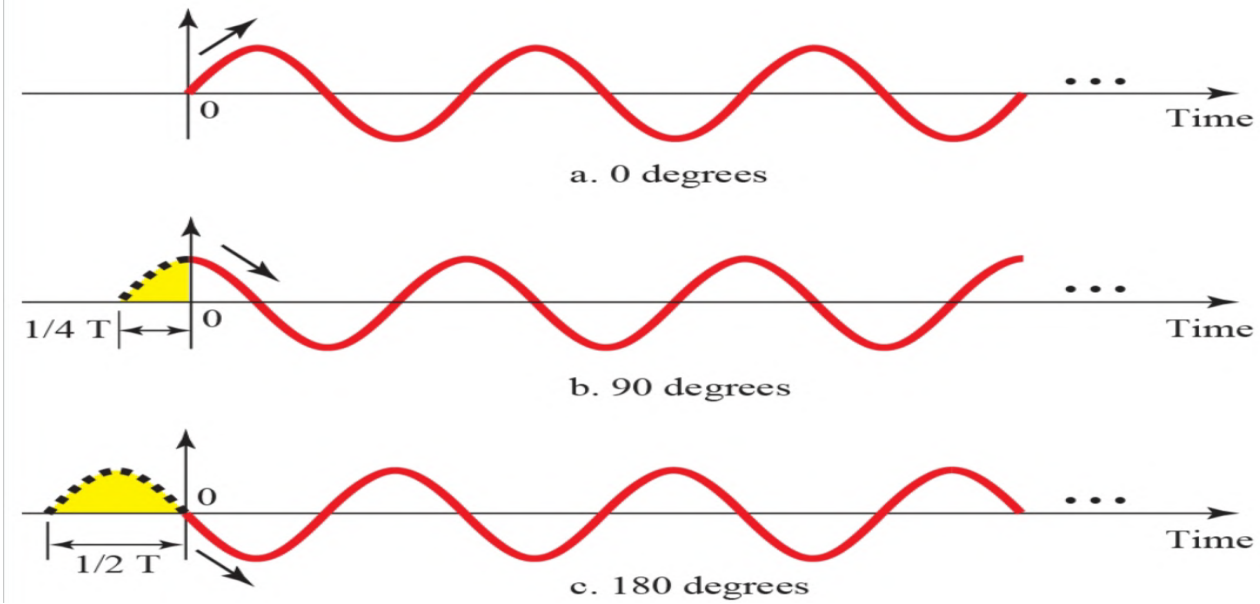
✓ T=0.0166 seconds or 16.66 msec

Phase or Phase Shift refers to the position of a point on a waveform in relation to a reference point, usually measured in degrees or radians. This is an important concept in signal processing and telecommunications, as it helps with wave synchronization and alignment.

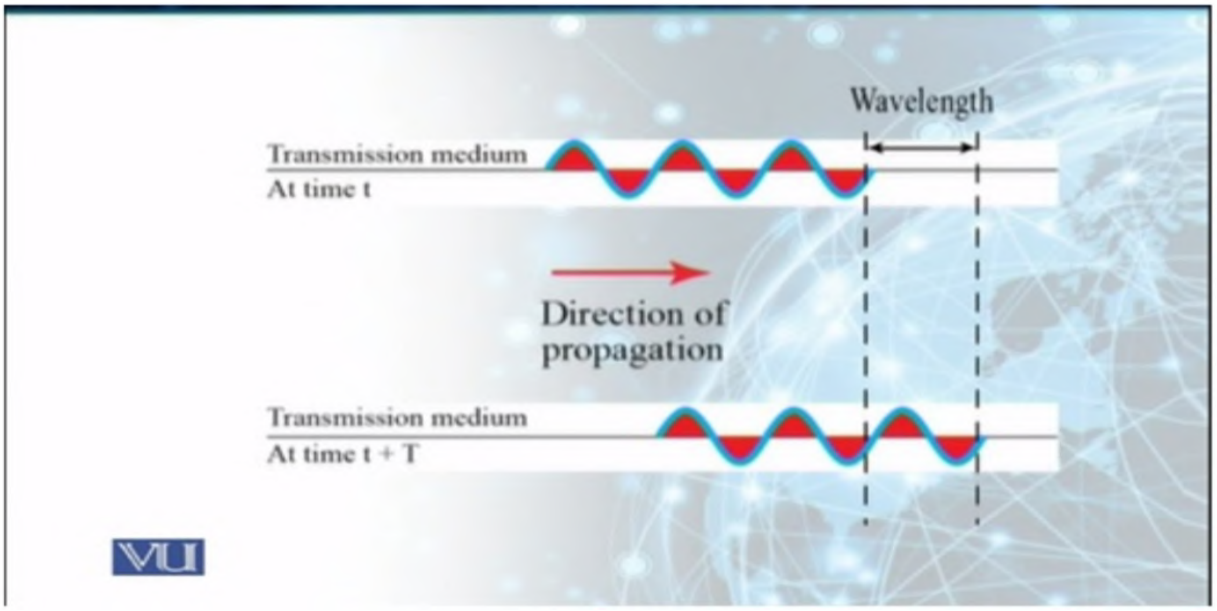
Formula:

$$360\ degrees = 2\pi\ radians$$

$$1\ degree = 2\pi/360\ radians$$



Wavelength is another characteristic of a signal traveling through a transmission medium. Wavelength binds the period or the frequency of a simple sine wave to the propagation speed of the medium.



The formula for wavelength (λ) is:

[redacted] Corrected : $v = f * \lambda$

Where:

✓ λ is the wavelength in meters (m).

✓ v is the velocity of the wave in meters per second (m/s).

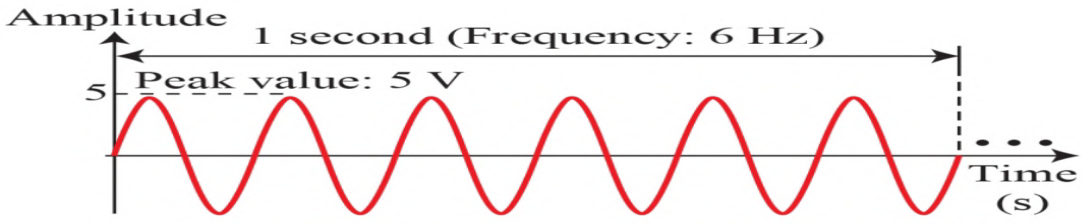
✓ f is the frequency of the wave in Hertz (Hz).

✓ A sine wave is comprehensively defined by its amplitude, frequency, and phase.

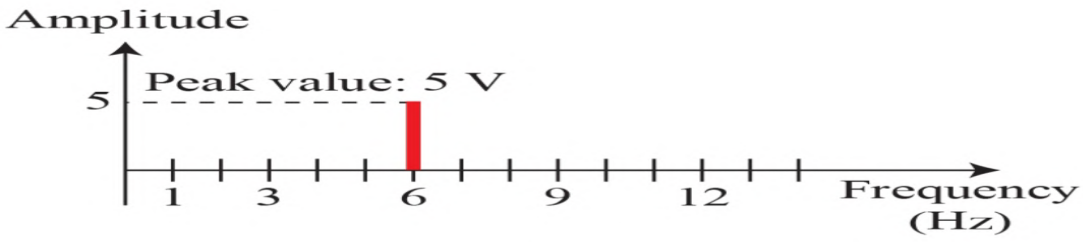
✓ We have been showing a sine wave by using what is called a time domain plot.

✓ The time-domain plot shows changes in signal amplitude with respect to time (it is an amplitude-versus-time plot).

✓ Phase is not explicitly shown on a time-domain plot.

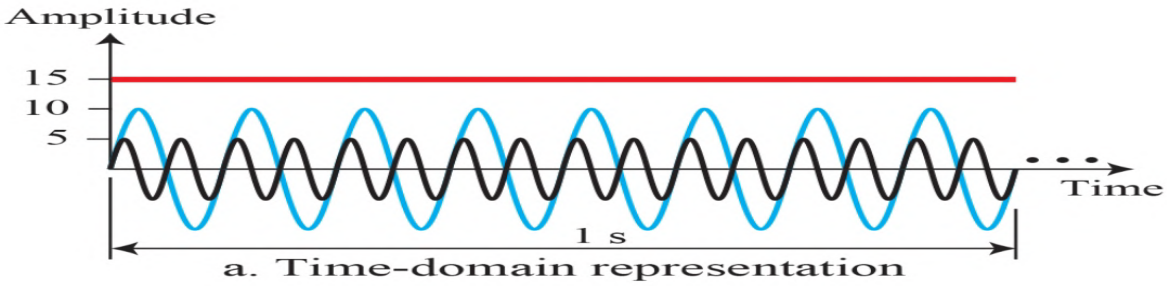


a. A sine wave in the time domain

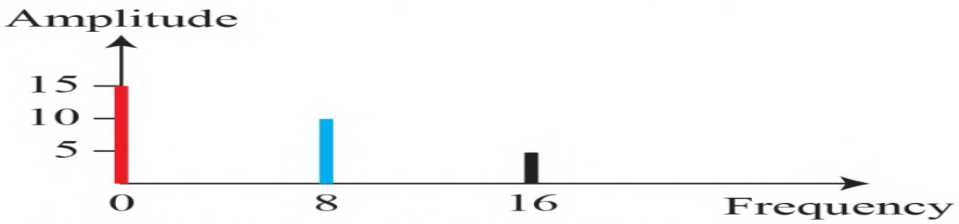


b. The same sine wave in the frequency domain

The frequency domain is more compact and useful when we are dealing with more than one sine wave. For example, Figure 3.9 shows three sine waves, each with different amplitude and frequency. All can be represented by three spikes in the frequency domain.



a. Time-domain representation

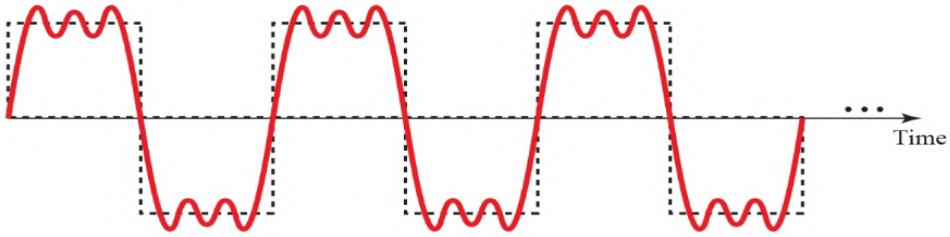


b. Frequency-domain representation

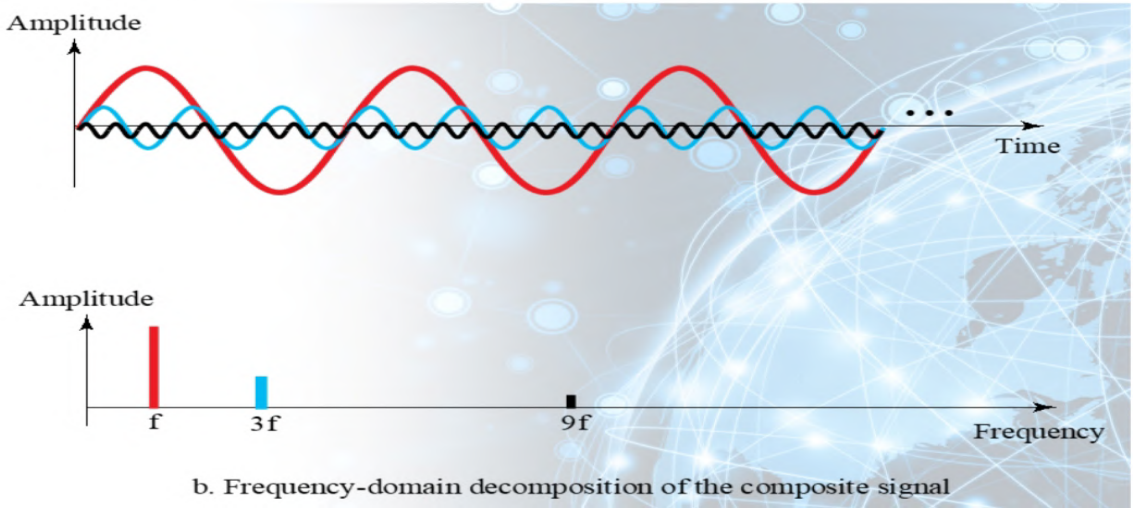
✓ Single Sine Wave can only carry limited information

✓ Composite Signal is made up of multiple simple sine waves

✓ Can be periodic or non-periodic



Following figure shows the decomposition of composite periodic signal:



b. Frequency-domain decomposition of the composite signal

A communication channel or network's bandwidth is its frequency range. It is the difference between the higher and lower frequencies in a continuous band, measured in Hertz.

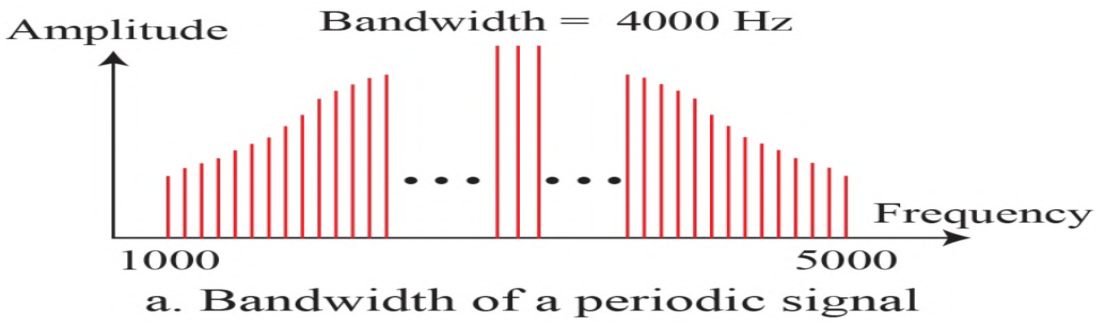
In radio transmission, a channel's bandwidth determines its frequency range. In signal processing, a wide bandwidth transmits more data but uses more resources. Thus, understanding bandwidth is crucial to optimising communication systems to ensure the channel or network can handle data transmission frequencies.

Bandwidth can be used in two different contexts with two different measuring values:

✓ Bandwidth in Hertz

✓ Bandwidth in bits per second

Bandwidth of a composite signal:

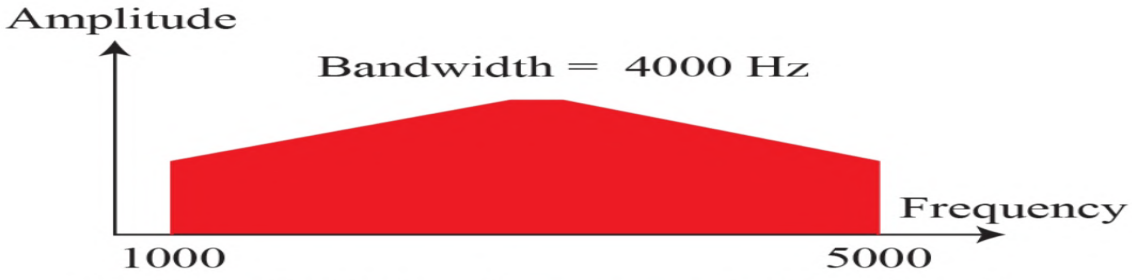


a. Bandwidth of a periodic signal

$B = f_h - f_l$ bandwidth = higher frequency - lower frequency

$B = 5000 - 1000$

$B = 4000 \text{ Hz}$



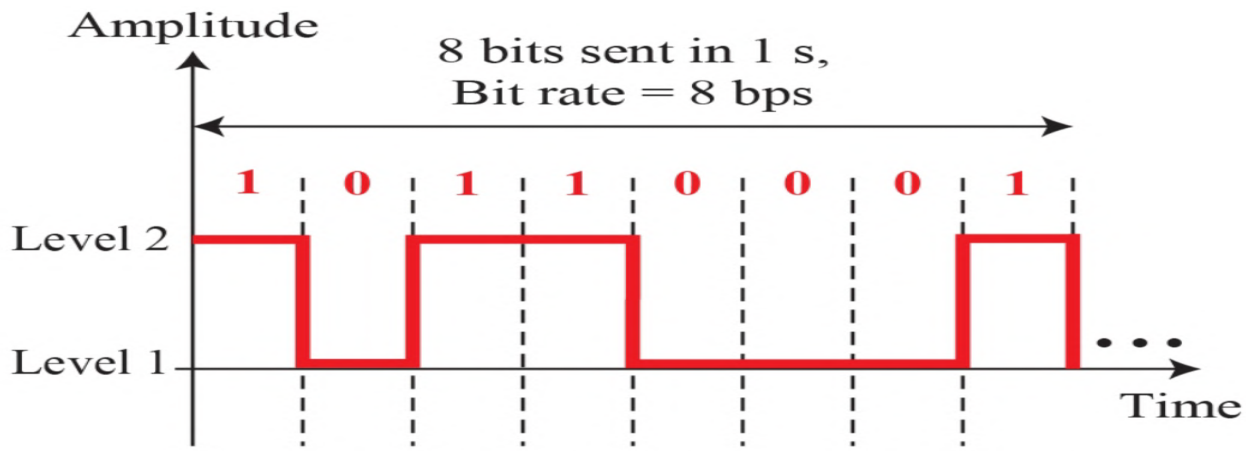
b. Bandwidth of a nonperiodic signal

In addition to being represented by an analog signal, information can also be represented by a digital signal. For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels.

Information can also be represented by a digital signal

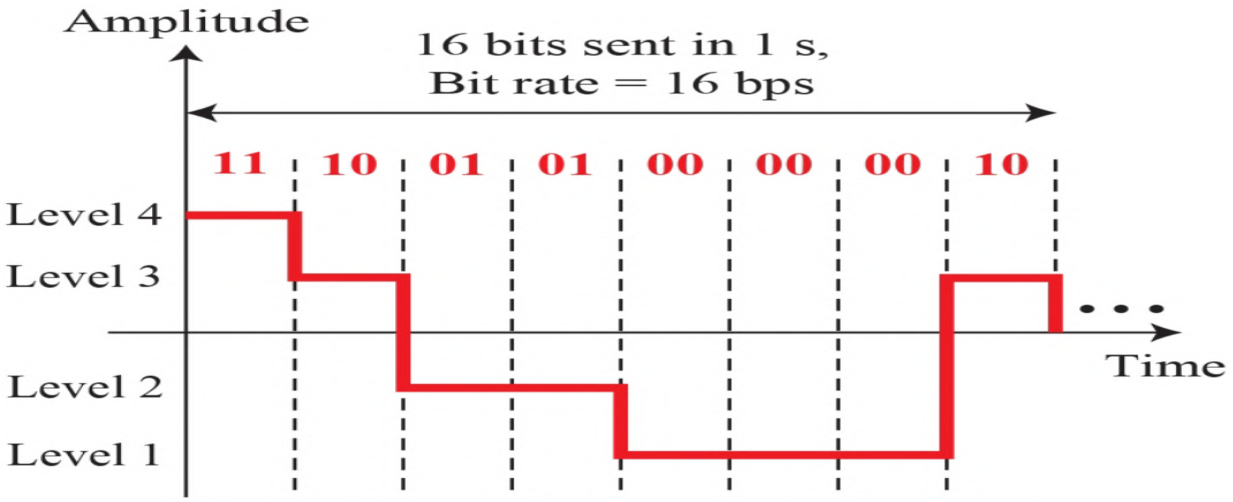
✓ For example, a 1 can be encoded as a positive voltage and a 0 as zero voltage

✓ A digital signal can have more than two levels so that we can send more than one bit for each level



a. A digital signal with two levels

In general, if a signal has L levels, each level needs log2 L bits. So, we can send log2 2 = 1 bit per level



b. A digital signal with four levels

$\log_2 4 = 2$ bits

Example:

A digital signal has eight levels. How many bits are needed per level?

Number of bits per level:

✓ Log2 L

log base 2 (8) . means you are asking 2 to what powers 8 .

✓ Log2 (8)

=3

Hence 3 bits are needed per level.

A digital signal has nine levels. How many bits are needed per level? We calculate the number of bits by using the formula.

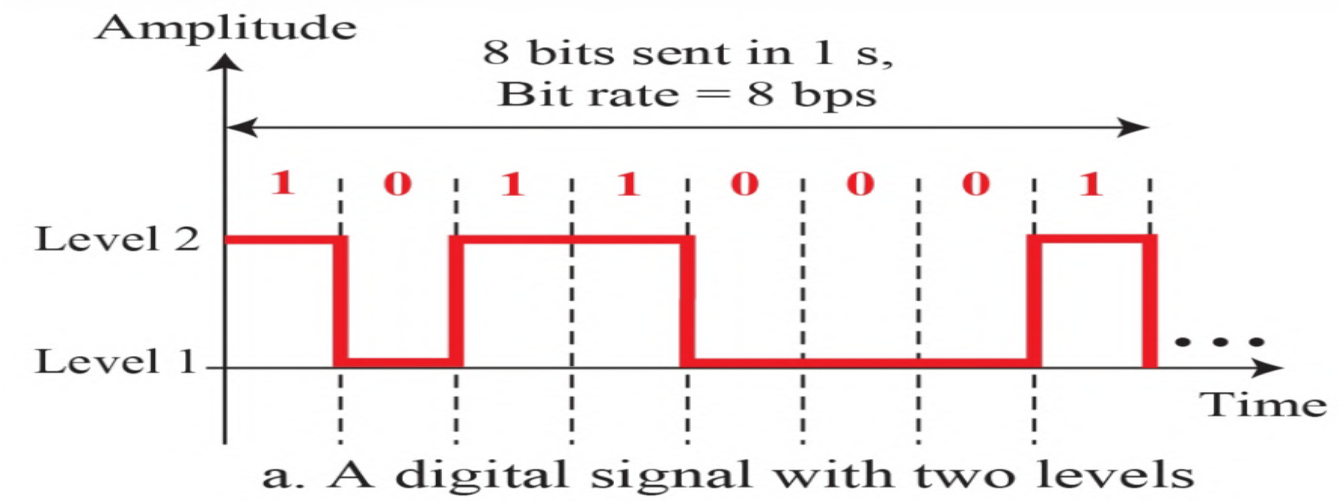
We calculate the number of bits by using the formula. Each signal level is represented by 3.17 bits. However, this answer is not realistic. The number of bits sent per level needs to be an integer as well as a power of 2. For this example, 4 bits can represent one level.

Bit Rate

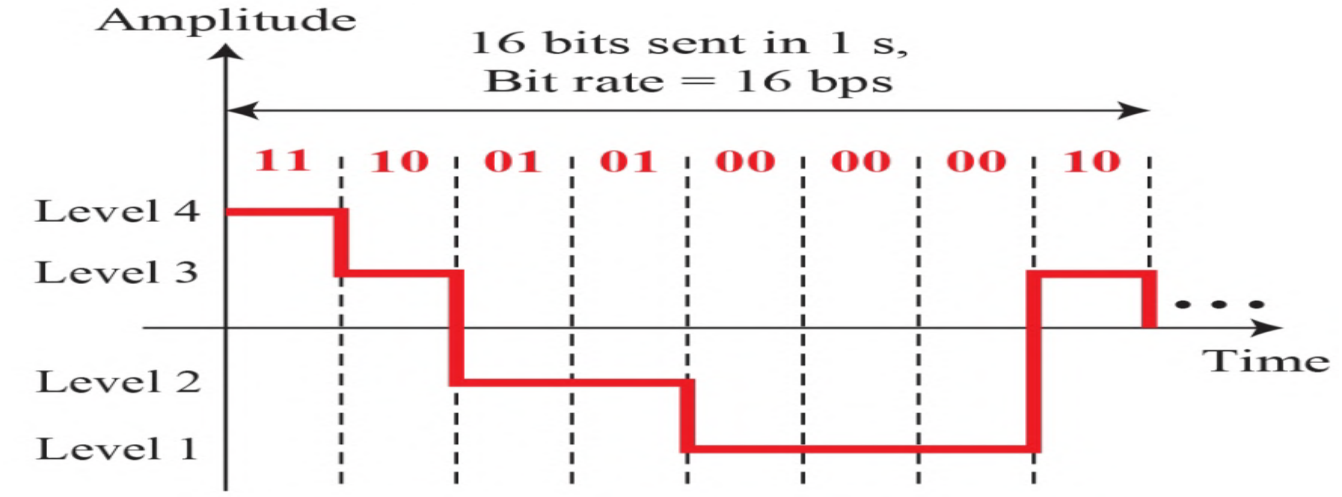
✓ Number of bits sent in 1 second.

✓ Bit Rate is expressed in bits per second (bps)

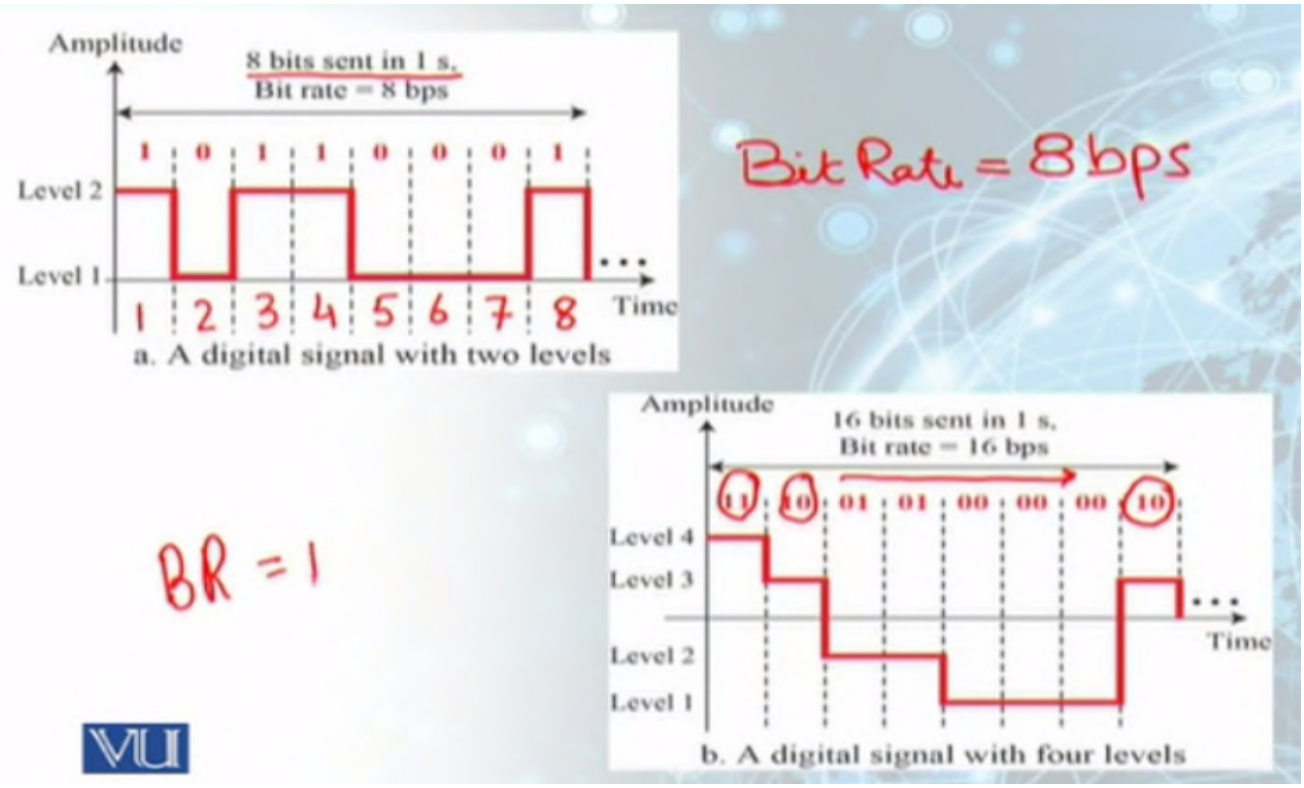
✓ Most digital signals are non-periodic, and thus period and frequency are not appropriate characteristics.



a. A digital signal with two levels



b. A digital signal with four levels

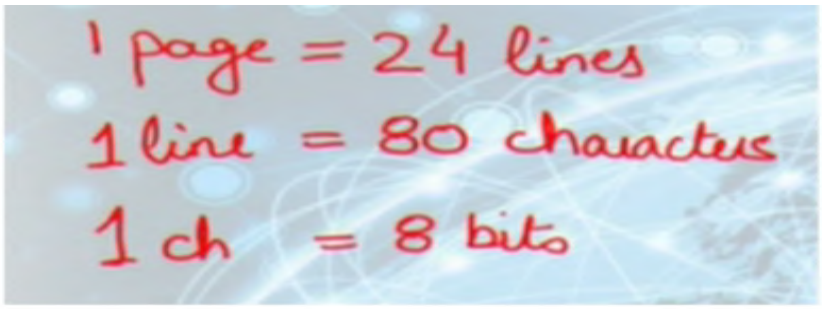


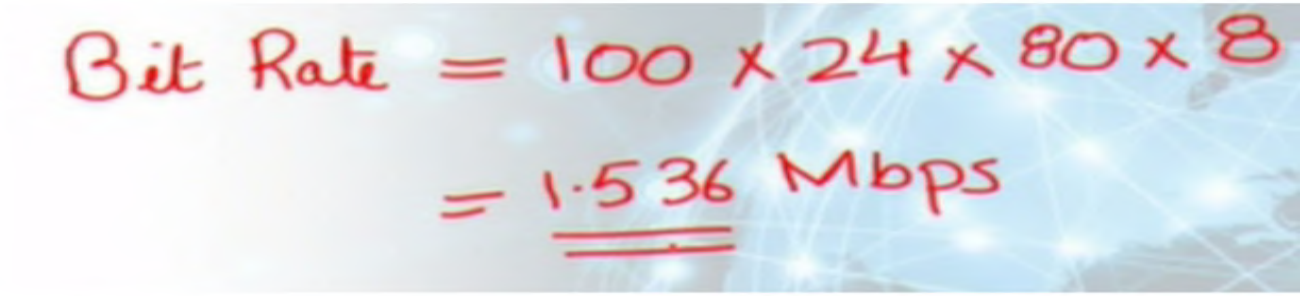
Bit Rate

Most digital signals are nonperiodic, and thus period and frequency are not appropriate characteristics. Another term—bit rate (instead of frequency)—is used to describe digital signals. The bit rate is the number of bits sent in 1s, expressed in bits per second (bps). Figure 3.17 shows the bit rate for two signals.

Example

Assume we need to download text documents at the rate of 100 pages per second. What is the required bit rate of the channel?

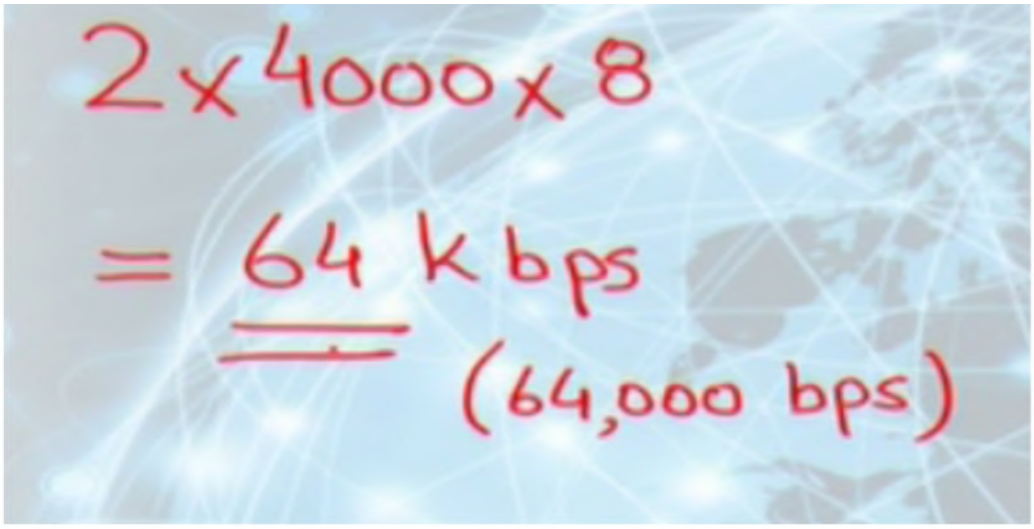




$100 \times 24 \times 80 \times 8 = 1{,}536{,}000 \text{ bps} = 1.536 \text{ Mbps}$

Example

A digitized voice channel is made by digitizing a 4-kHz bandwidth analog voice signal. We need to sample the signal at twice the highest frequency (two samples per hertz). We assume that each sample requires 8 bits. What is the required bit rate?



A page is an average of 24 lines with 80 characters in each line. If we assume that one character requires 8 bits, the bit rate

$2 \times 4000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$

Bit Length

We discussed the concept of the wavelength for an analog signal: the distance one cycle occupies on the transmission medium. We can define something similar for a digital signal: the bit length. The bit length is the distance one bit occupies on the transmission medium.

Bit Length = Propagation Speed × Bit Duration

Digital Signal as Composite Analog Signal

✓ Based on Fourier analysis, a digital signal is a composite analog signal

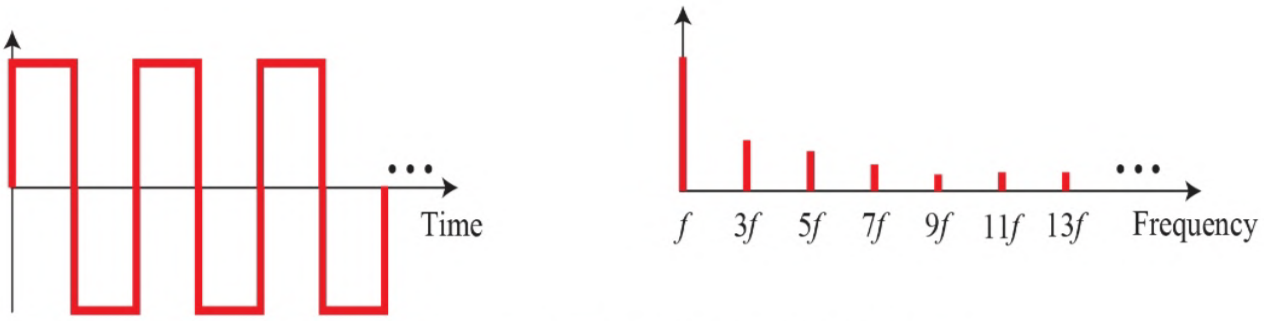
✓ A digital signal, in the time domain, comprises connected vertical and horizontal line segments.

✓ A vertical line in the time domain means a frequency of infinity.

✓ A horizontal line in the time domain means a frequency of zero.

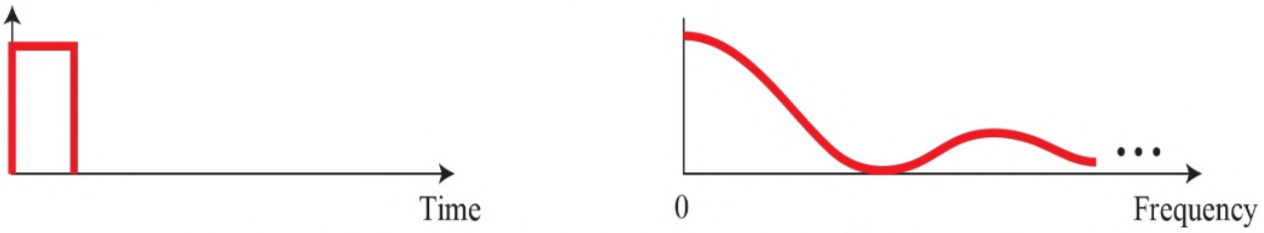
✓ Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

Digital Signal as Composite Analog Signal



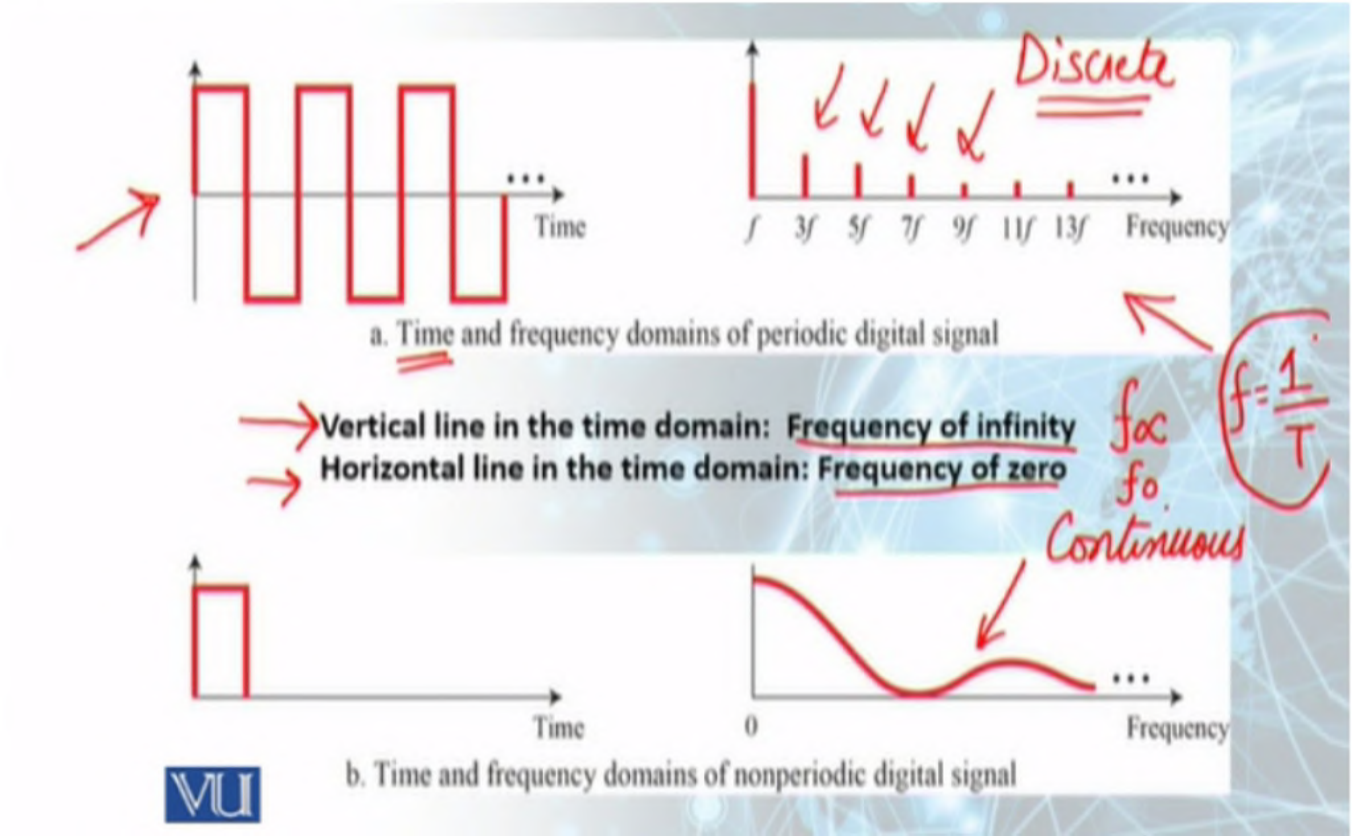
a. Time and frequency domains of periodic digital signal

Vertical line in the time domain: Frequency of infinity Horizontal line in the time domain: Frequency of zero



b. Time and frequency domains of nonperiodic digital signal

Digital Signal as Composite Analog Signal



Transmission of Digital Signals

✓ Digital signal, periodic or non-periodic, is a composite analog signal with frequencies between zero and infinity (Infinite Bandwidth)

~ Two approaches for transmission:

✓ Baseband Transmission: Baseband transmission is a method of transmitting digital signals over a communication medium, such as a cable or a fiber optic line, in which the entire bandwidth of the medium is used to send a single stream of data.

✓ Broadband Transmission: Broadband transmission is a method of sending multiple signals simultaneously over a communication medium, such as a cable, fiber optic line, or wireless channel, by dividing the available bandwidth into multiple frequency channels.

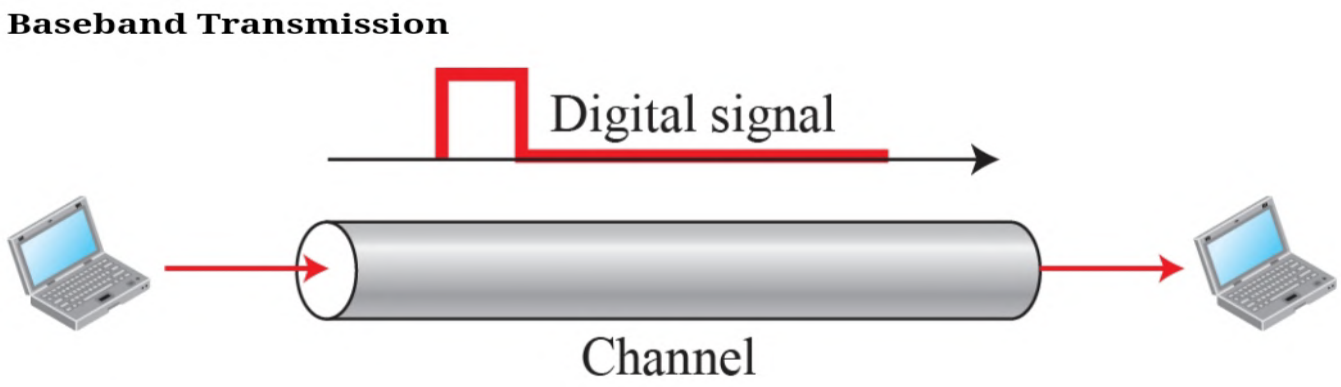
Transmission of Digital Signals

✓ A vertical line in the time domain means a frequency of infinity

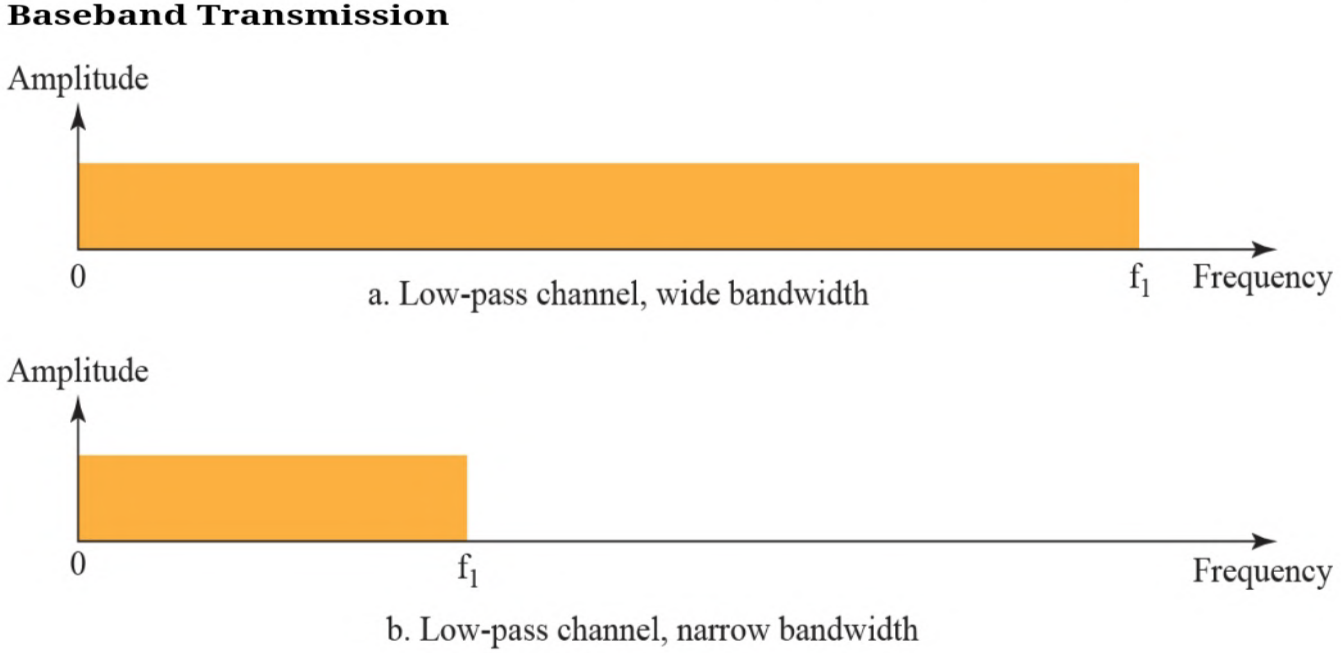
✓ A horizontal line in the time domain means a frequency of zero.

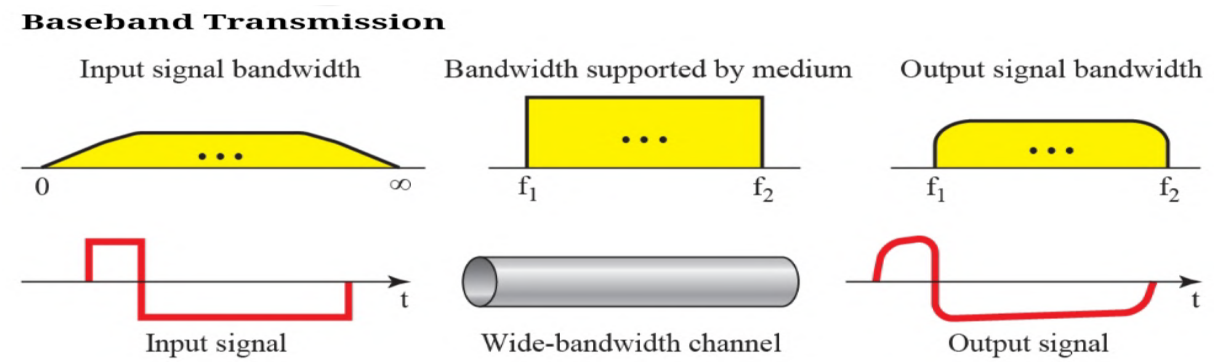
✓ Going from a frequency of zero to a frequency of infinity implies all frequencies in between are part of the domain.

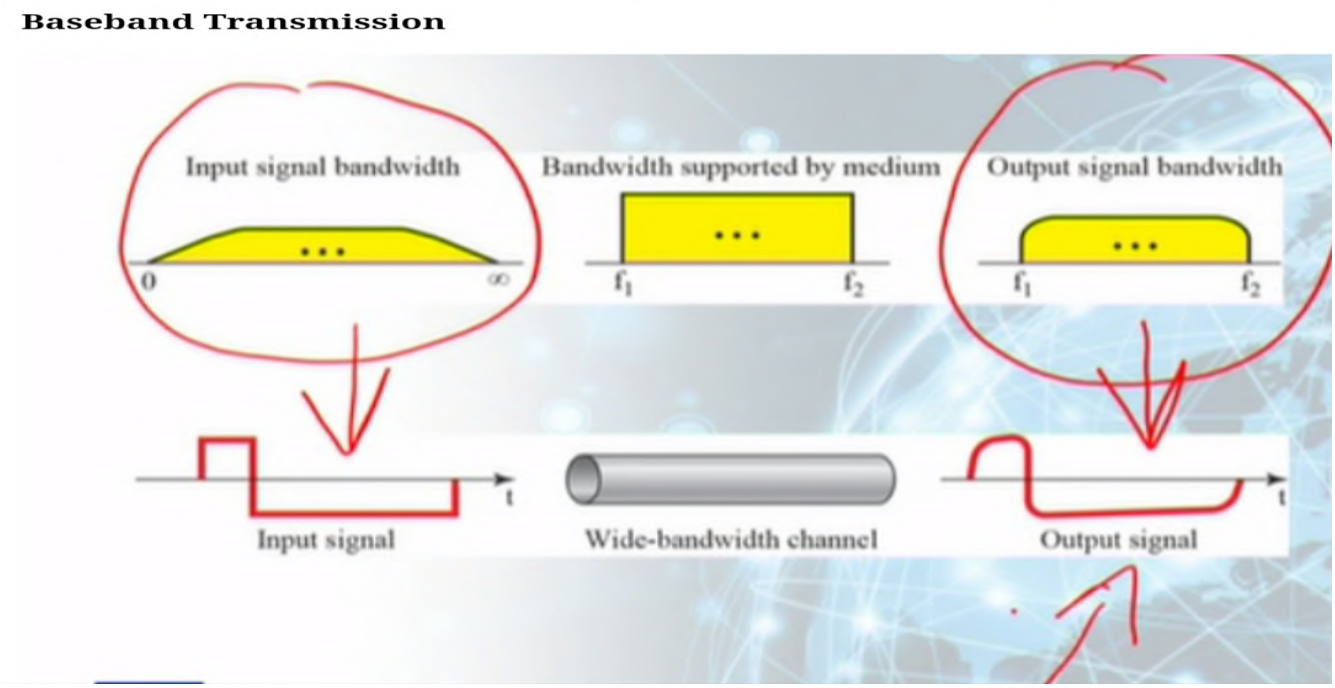
Baseband Transmission



Sending a Digital Signal without changing it to an Analog Signal







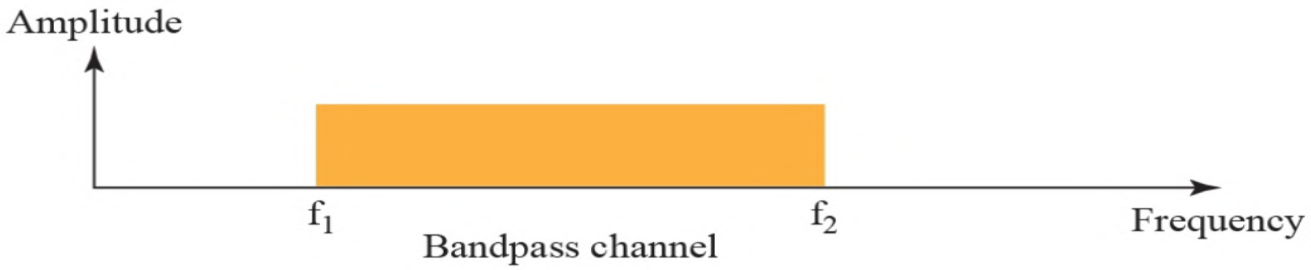
Broadband Transmission (Modulation)

✓ Changing the Digital signal to an Analog signal for transmission

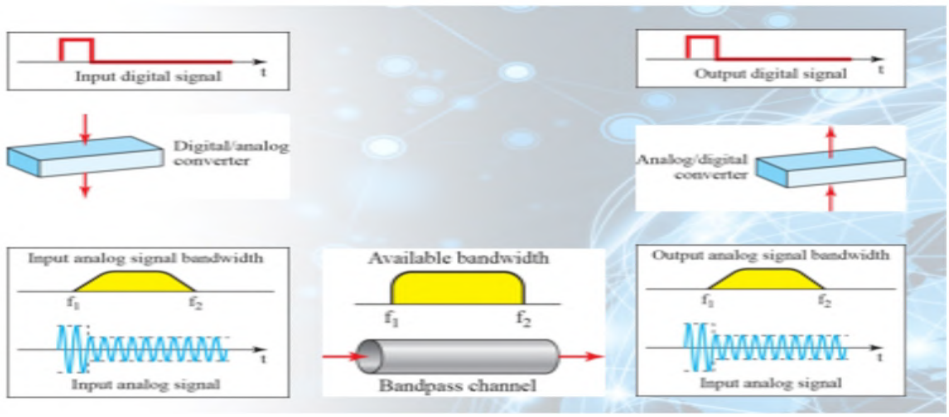
✓ Modulation allows us to use a bandpass channel—a channel with a bandwidth that does not start from zero

✓ More available than a low-pass channel

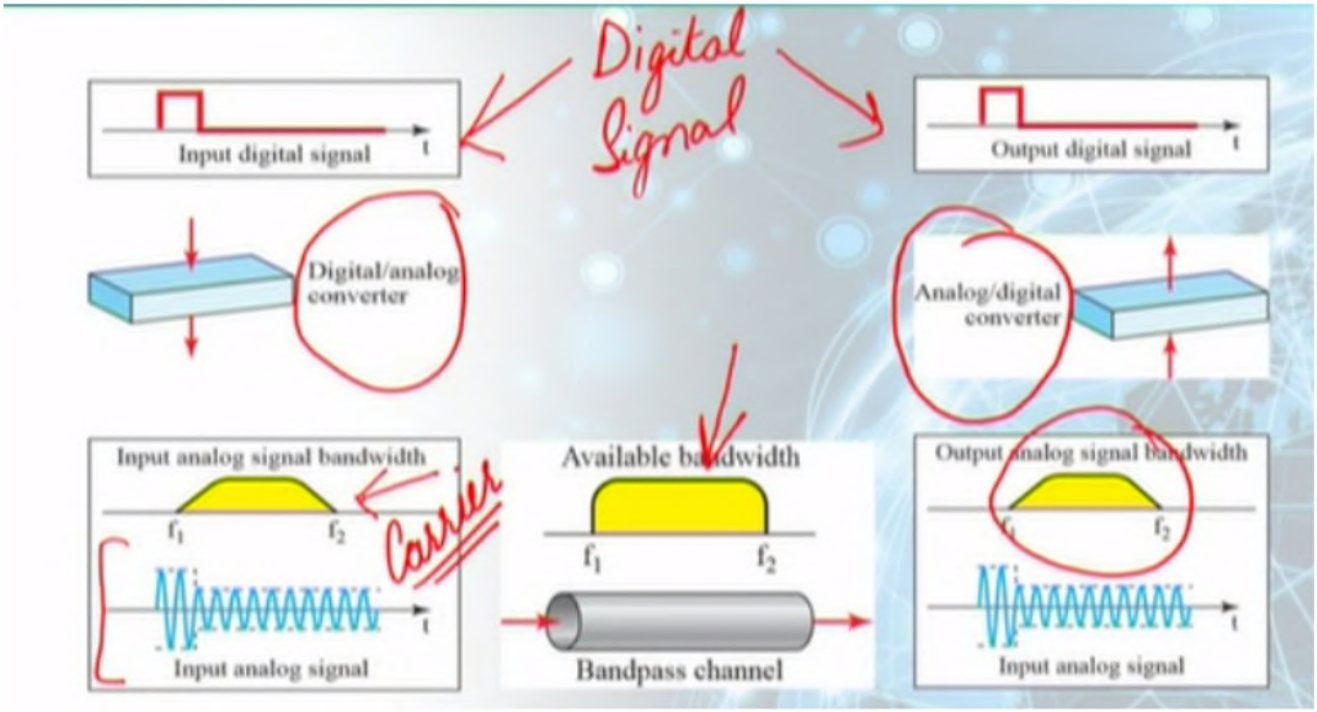
Broadband Transmission (Modulation)



Broadband Transmission (Modulation)



Broadband Transmission (Modulation)



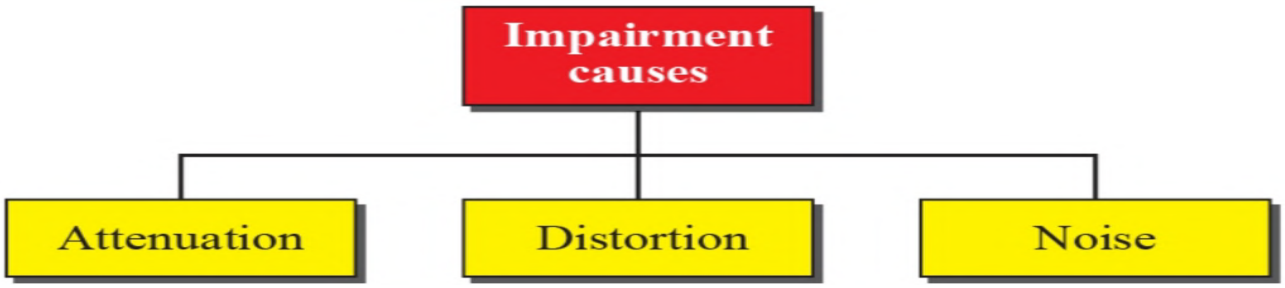
Transmission Impairments

✓ Transmission media are not perfect

✓ Cause Signal impairments

✓ Signal sent is not the same as the signal received

Causes of Transmission Impairment



Transmission Impairment

What is sent is not what is received. Three causes of impairment are attenuation, distortion, and noise

Transmission Impairment

Signals travel through transmission media, which are not perfect. The imperfection causes signal impairment. This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium.

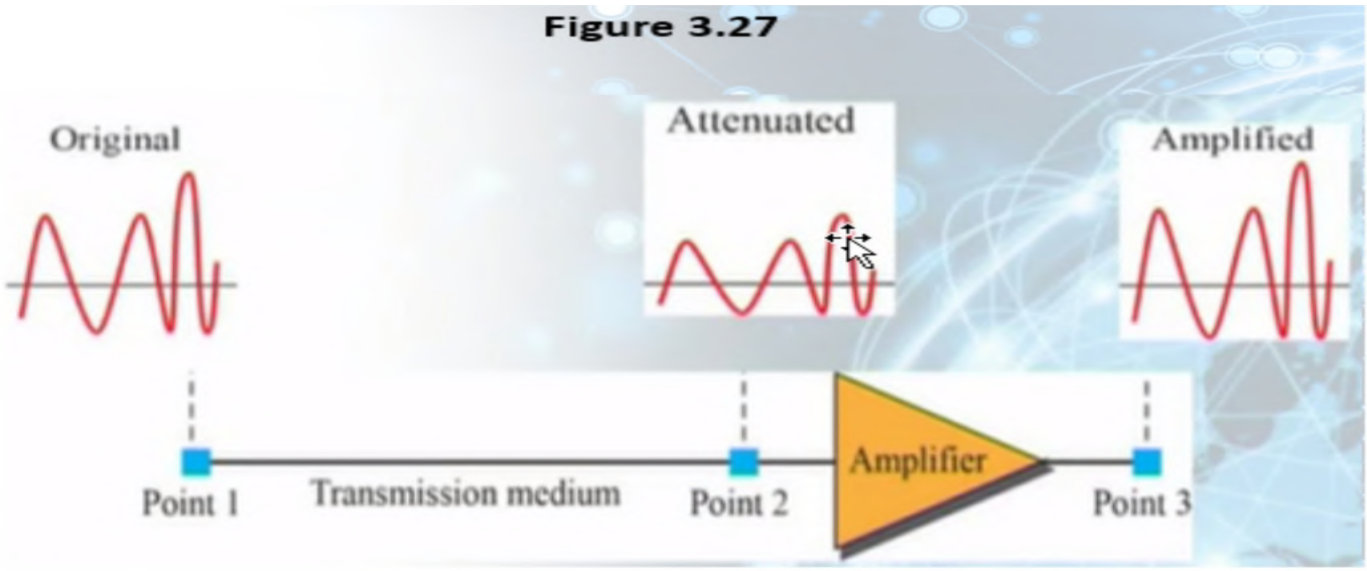
Attenuation

Attenuation means a loss of energy. When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium. That is why a wire carrying electric signals gets warm, if not hot, after a while. Some of the electrical energy in the signal is converted to heat.

To compensate for this loss, amplifiers are used to amplify the signal. Figure 3.27 shows the effect of attenuation and amplification.

Attenuation and amplification

Figure 3.27



Example 3.26

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that P2 = 0.5 P1. In this case, the attenuation (loss of power) can be calculated as

$$10 \log_{10} P_2/P_1 = 10 \log_{10} (0.5\ P_1)\ /P_1$$

$$= 10 \log_{10} 0.5 = 10 \times (-0.3) = -3 \text{ dB}.$$

A loss of 3 dB (−3 dB) is equivalent to losing one-half the power.

Example 3.27

A signal travels through an amplifier, and its power is increased 10 times. This means that P2 = 10P1. In this case, the amplification (gain of power) can be calculated as

$$10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{10P_1}{P_1} =$$

$$10 \log_{10} 10 = 10(1) = 10 \text{ dB}$$

Example 3.29

Sometimes the decibel is used to measure signal power in milliwatts. In this case, it is referred to as dBm and is calculated as dBm = 10 log10 Pm, where Pm is the power in milliwatts. Calculate the power of a signal if its dBm = −30.

Solution

We can calculate the power in the signal as

$$\text{dB}_\text{m} = 10 \log_{10} \longrightarrow dB_m = -30 \longrightarrow$$

$$\log_{10} P_m = -3 \longrightarrow P_m = 10^{-3} \text{ mW}$$

WAttenuation and Amplification - Decibel

✓ Unit of Signal strength is Decibel or dB

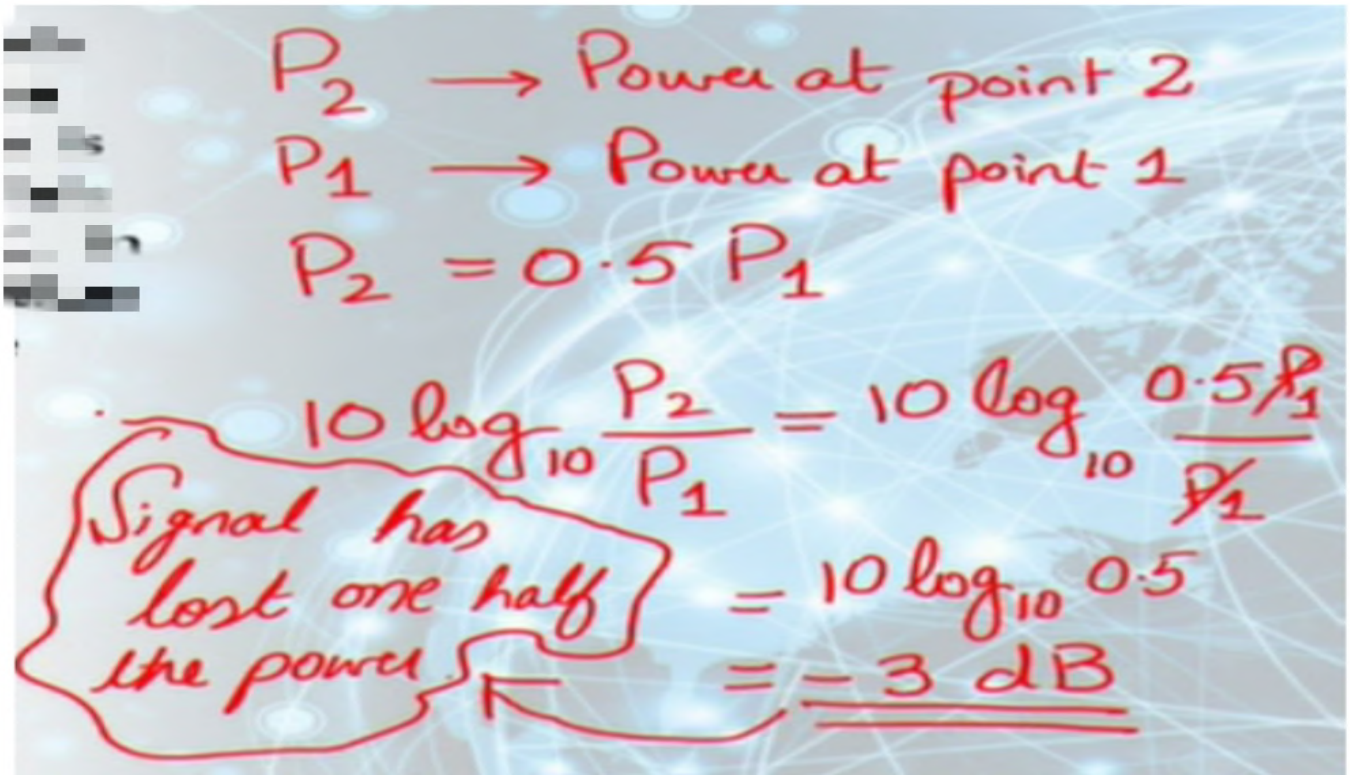
✓ Decibel (dB) measures the relative strengths of two signals or one signal at two different points

$$10 \log_{10} P_2/P_1$$

Decibel is negative if a signal is attenuated and positive if signal is amplified

Example

Suppose a signal travels through a transmission medium and its power is reduced to one half. This means that P2 = 0.5 P1. In this case, the attenuation (loss of power) can be calculated as



Example

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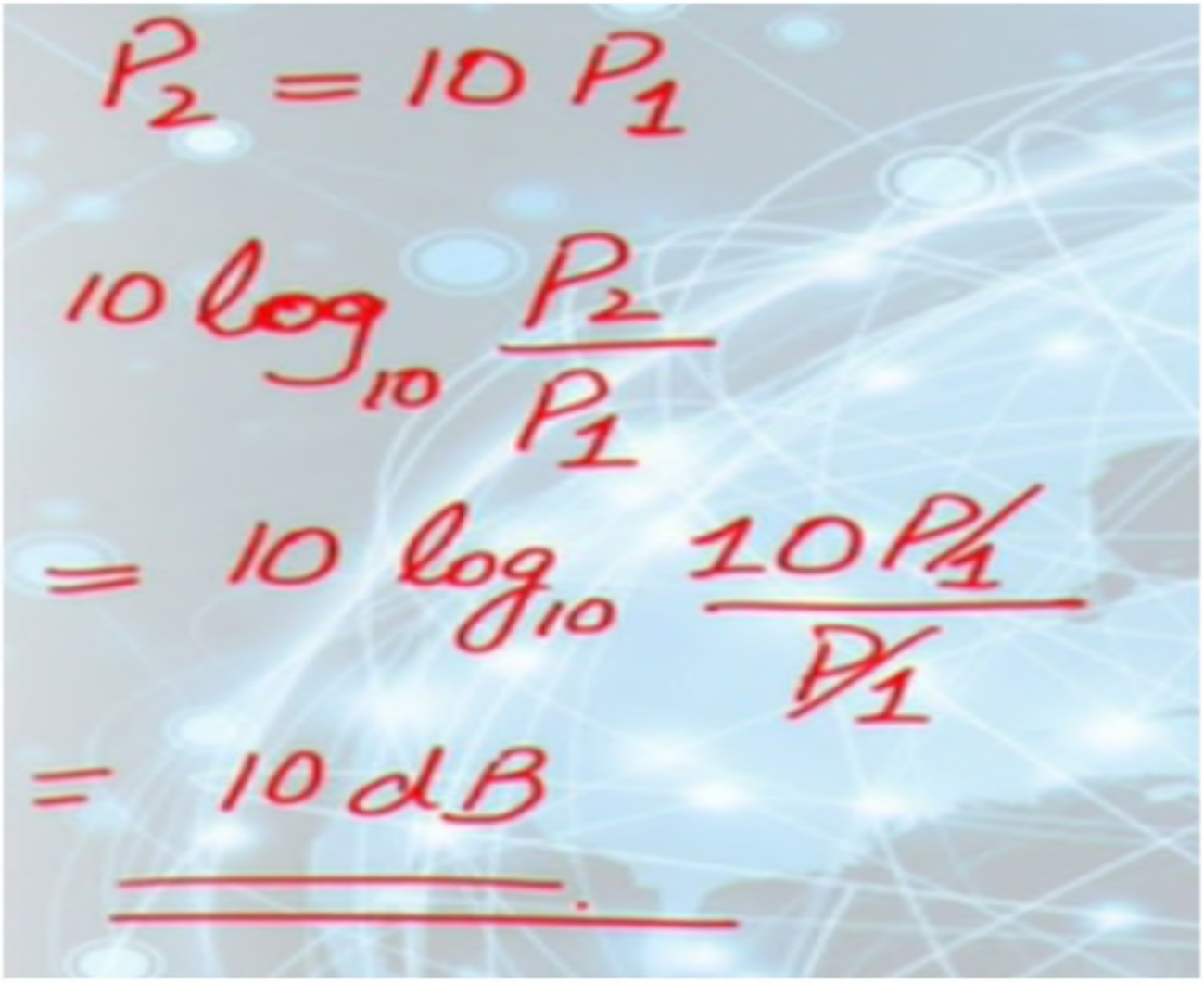


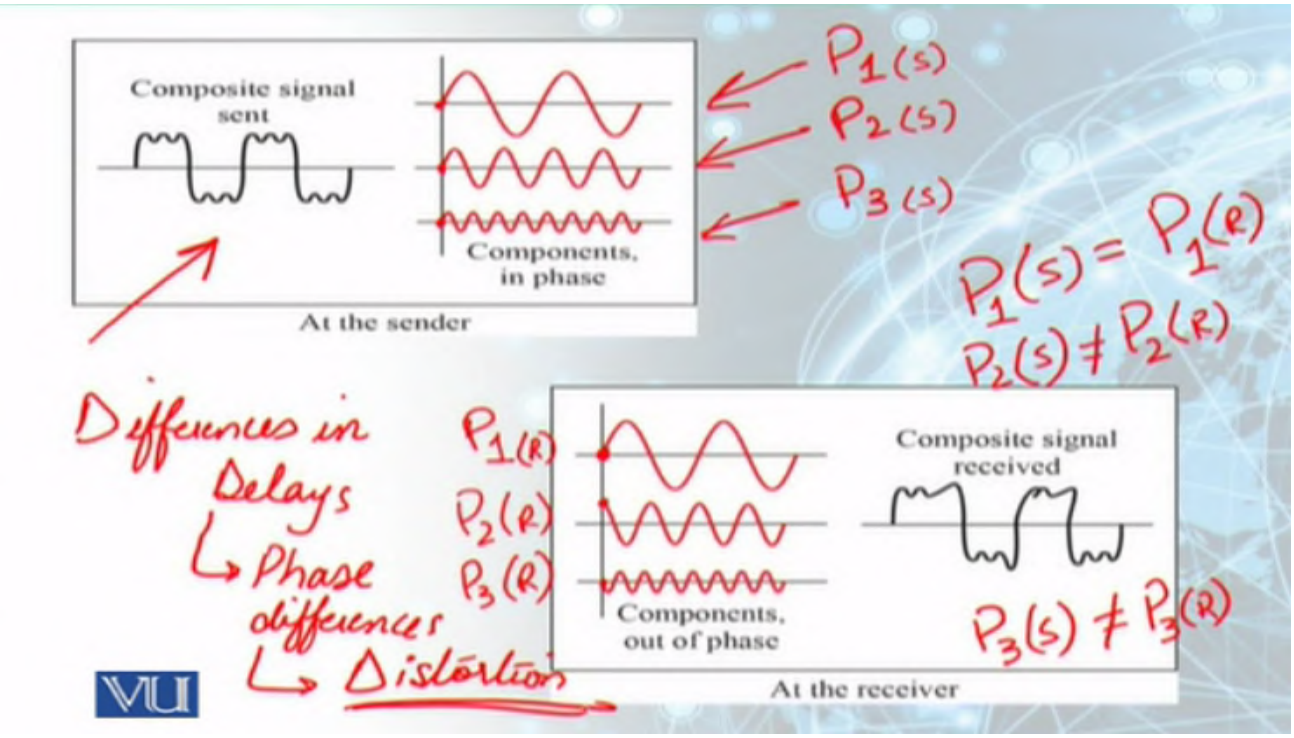
Distortion

✓ Distortion means that the signal changes its form or shape.

✓ Distortion can occur in a composite signal made of different frequencies.

✓ Each signal component has its own propagation speed (see the next section) through a medium and, therefore, its own delay in arriving at the destination.

✓ Differences in delay may create a difference in phase if the delay is not the same as the period duration.



Noise

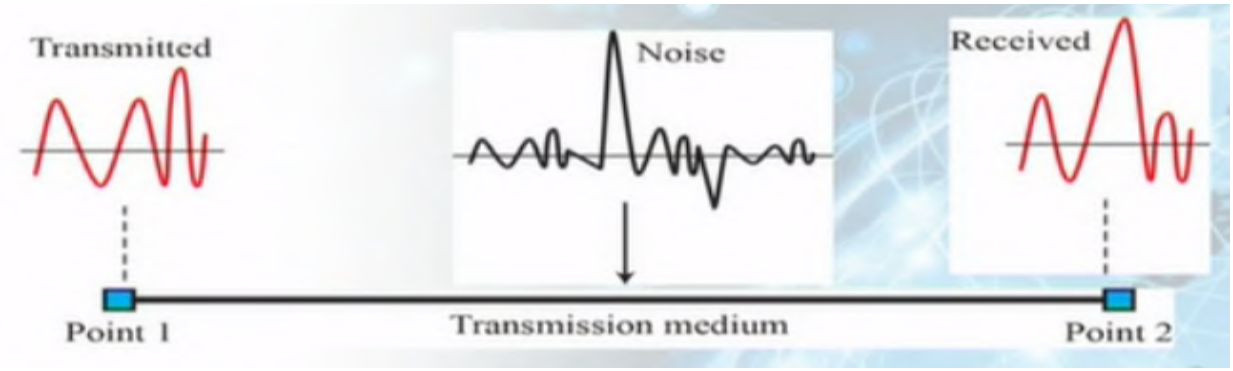
✓ Noise is another cause of impairment.

✓ Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.

✓ Thermal noise is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.

✓ Induced noise comes from sources such as motors.

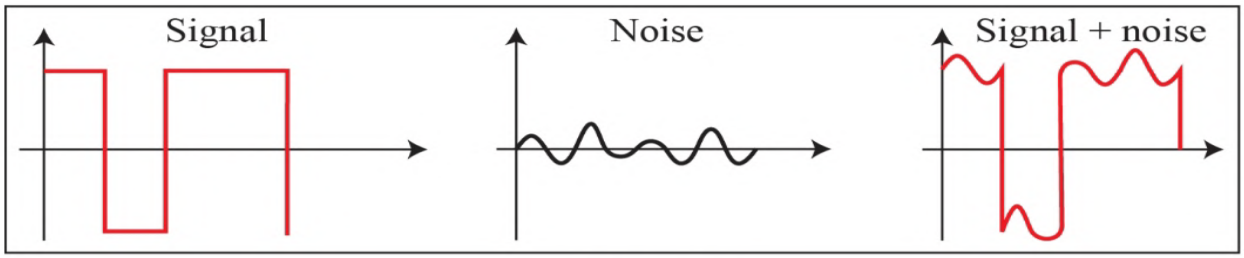
✓ Crosstalk is the effect of one wire on the other.



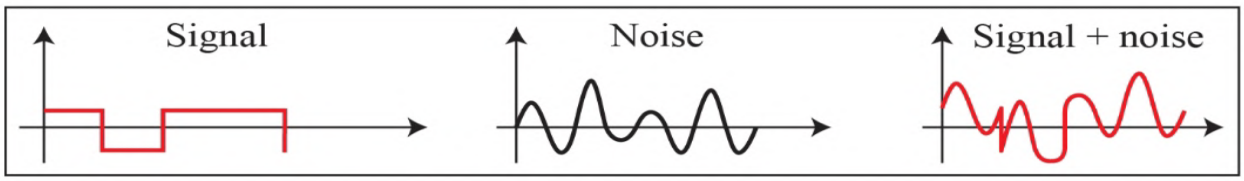
Noise – Signal to Noise Ratio (SNR)

Signal to Noise Ratio (SNR) is used to find the theoretical bit rate limit of a signal

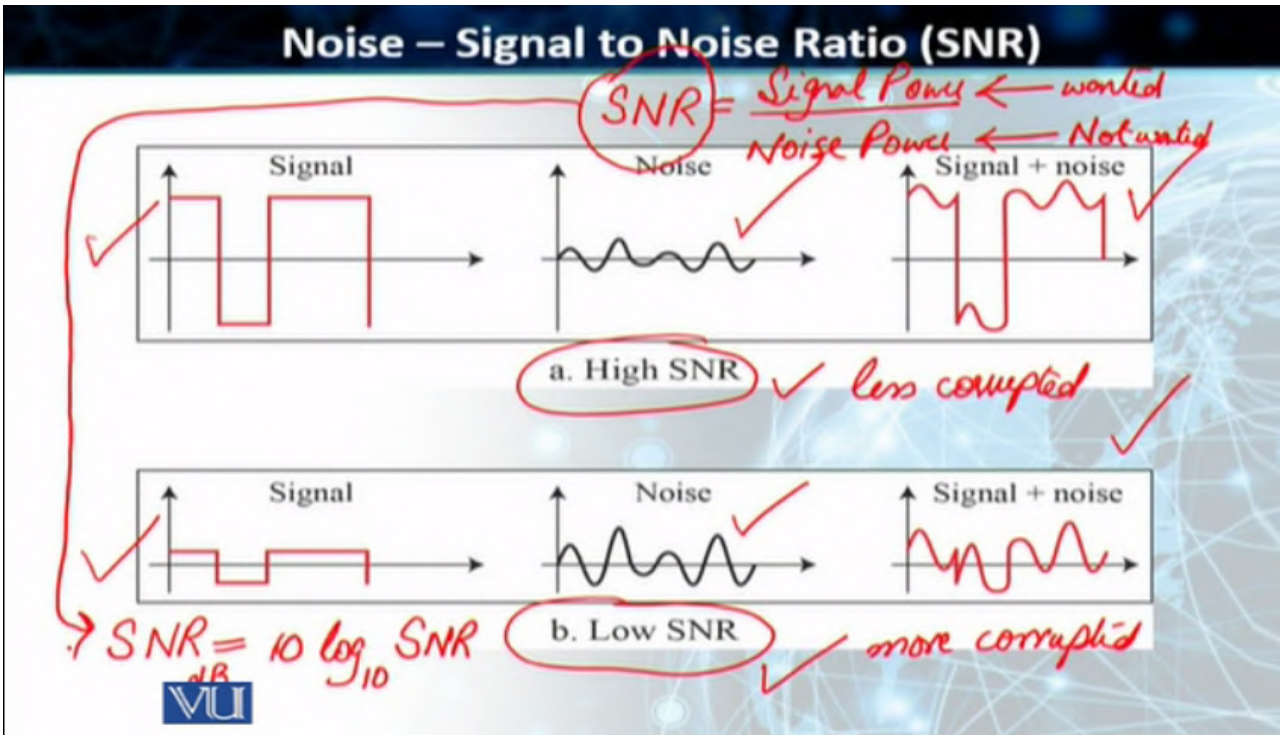
$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$



a. High SNR

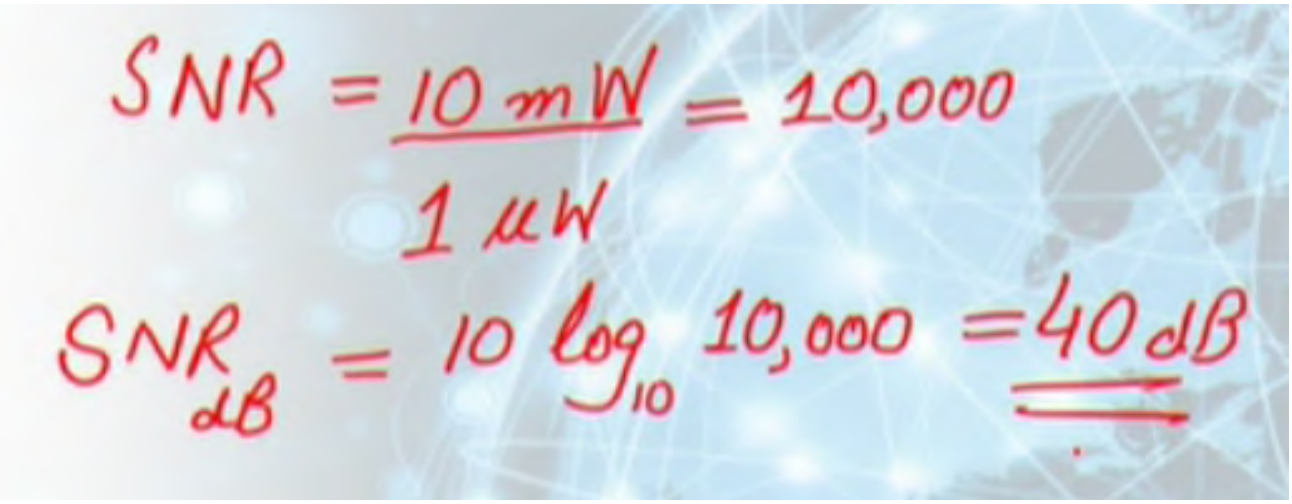


b. Low SNR



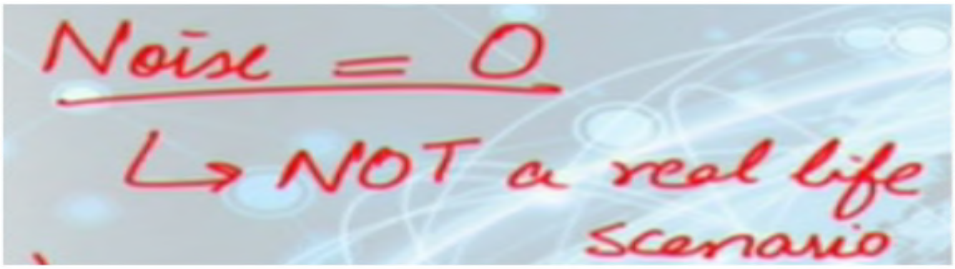
Example

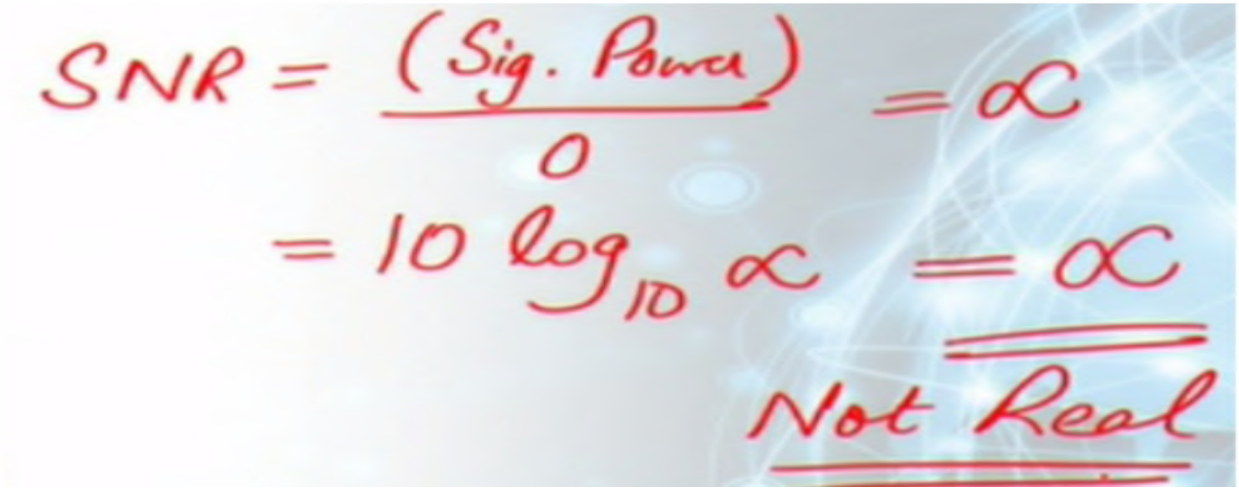
The power of a signal is 10 mW and the power of the noise is 1 μW; what are the values of SNR and SNRdB?



Example

The values of SNR and SNRdB for a noiseless channel are calculated as





Data Rate Limits

✓ How fast we can send data, in bits per second, over a channel?

Data Rate depends on 3 factors:

✓ The Bandwidth available

✓ The level of the signals we use

✓ The level of noise

Two theoretical formulas developed to calculate the data rate:

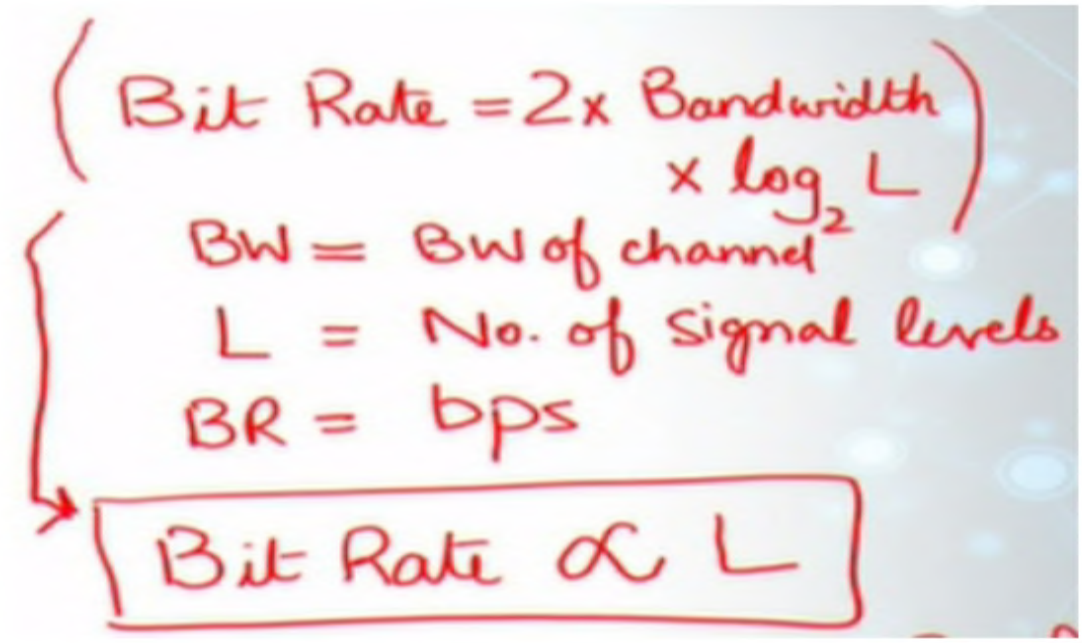
✓ one by Nyquist for a noiseless channel

✓ another by Shannon for a noisy channel

Noiseless Channel: Nyquist Rate

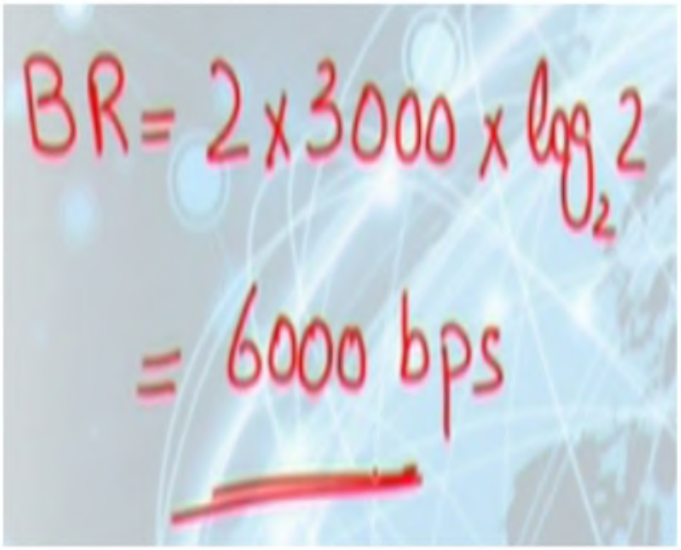
✓ For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

✓ Finding balance between Bit rate and System Reliability



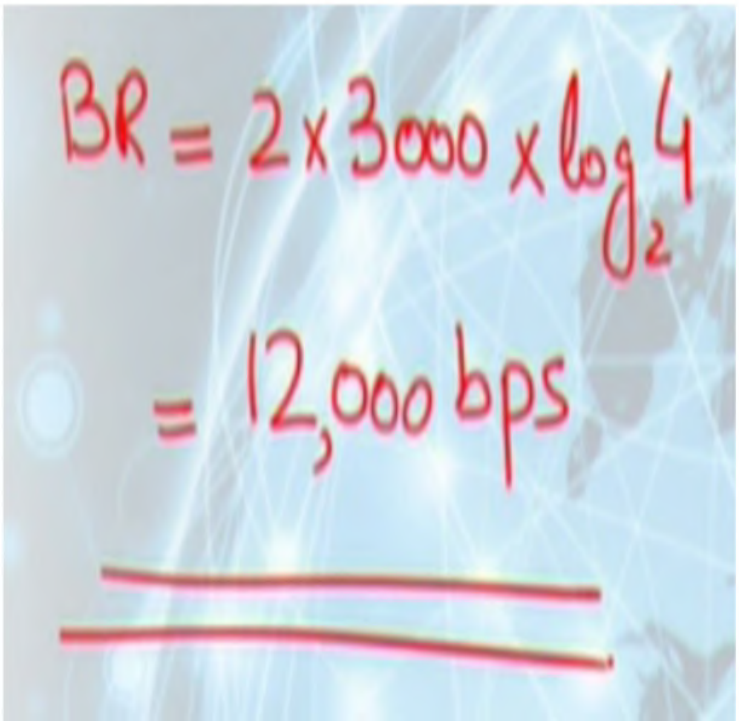
Example

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as



Example

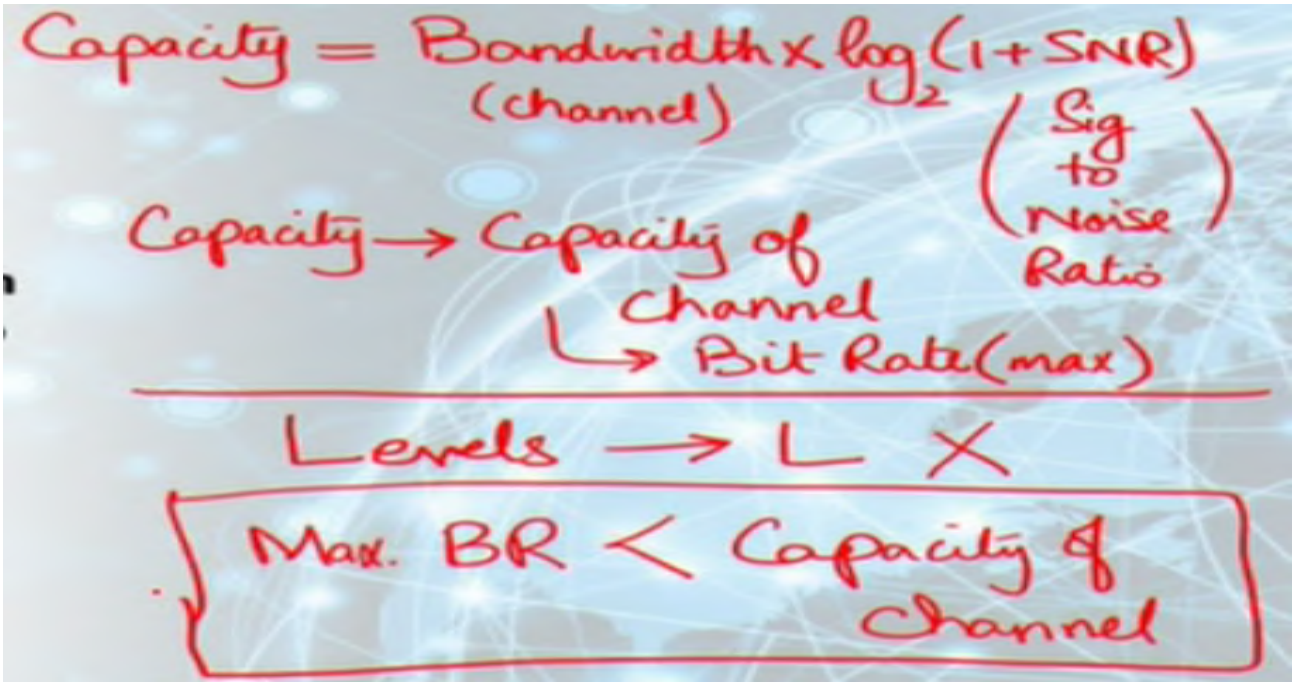
Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as



Noisy Channel : Shannon Capacity

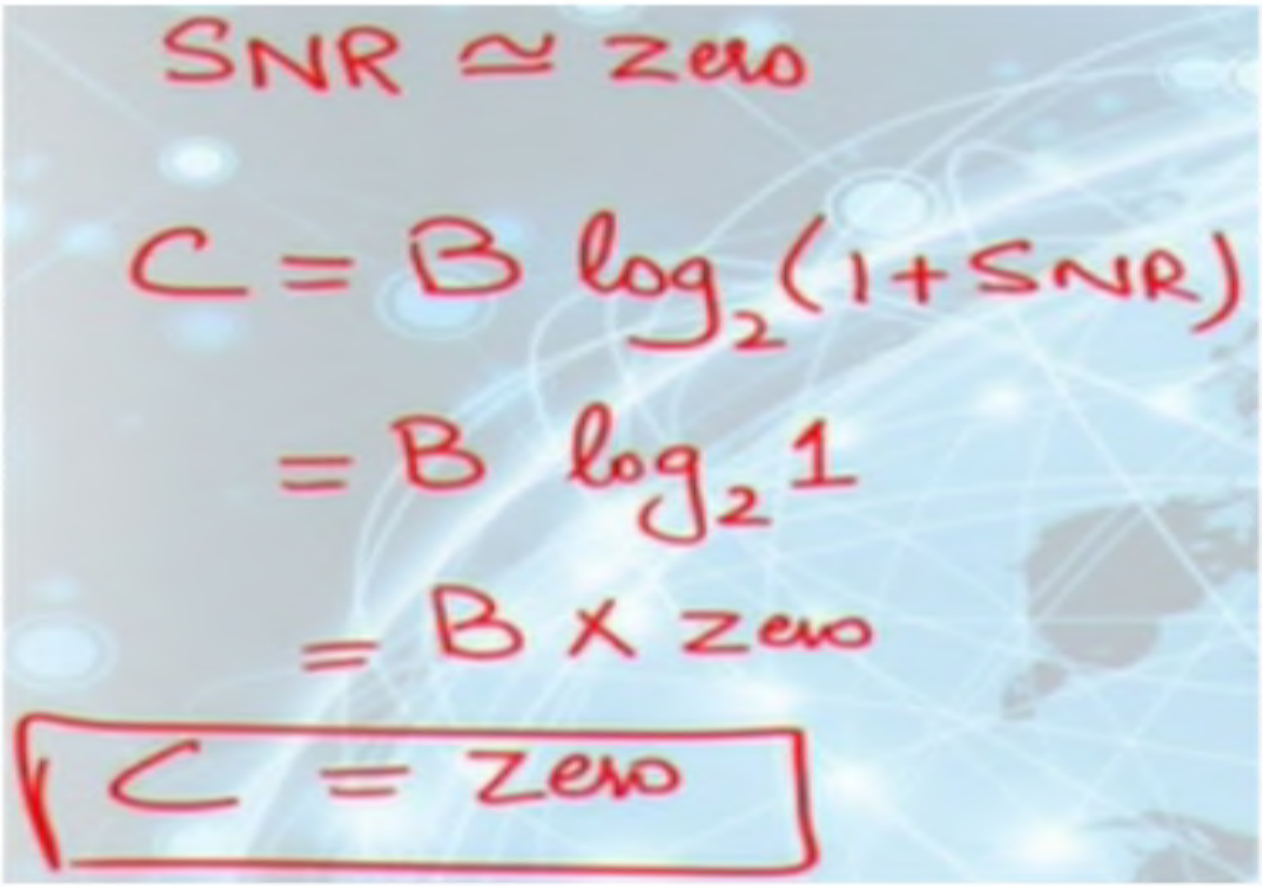
✓ We cannot have a noiseless channel; the channel is always noisy

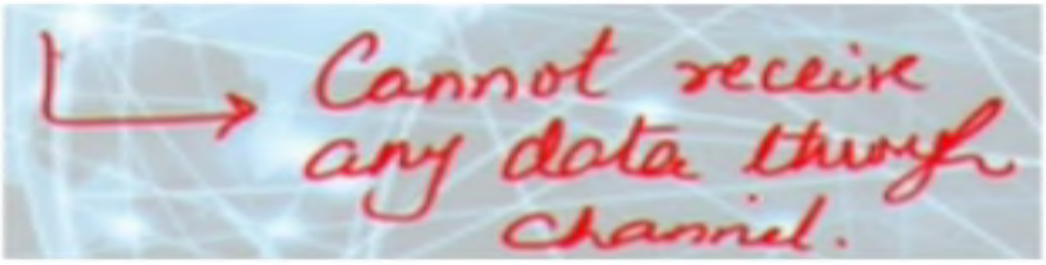
✓ In 1944, Claude Shannon introduced a formula, to determine the theoretical highest data rate for a noisy channel:



Example

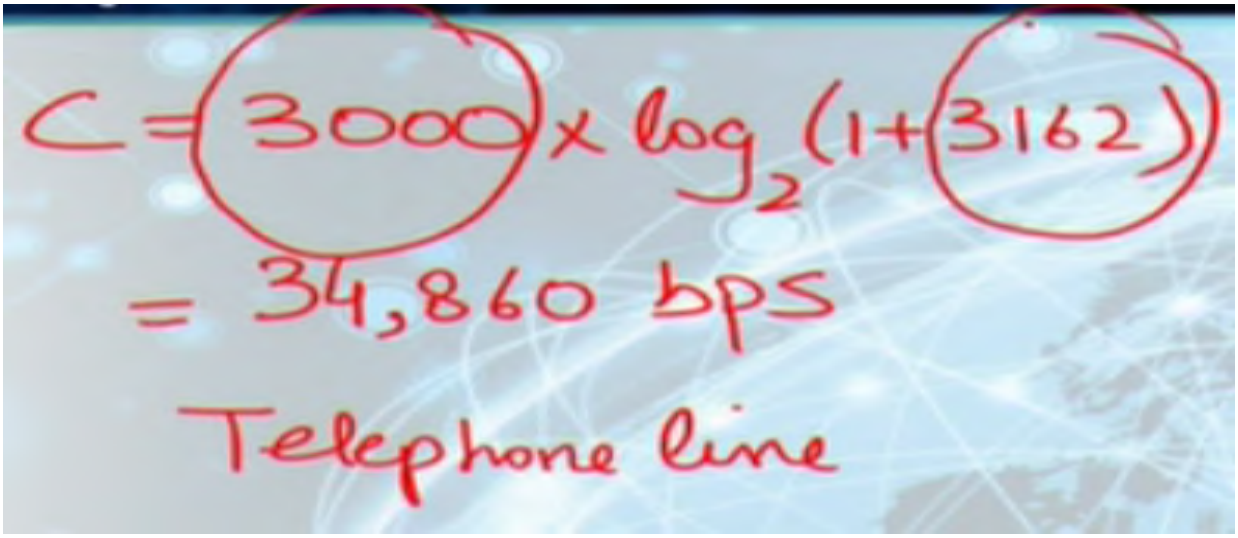
Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as





Example

Theoretical highest bit rate of a Telephone line with a Bandwidth of 3000 Hz assigned for data communication. SNR is usually 3162. The capacity is calculated as:



Using Both Limits

In practice, we need to use both methods to find the limits and signal levels

Shannon's formula gives us the upper limit while the Nyquist formula gives us the signal levels

Example

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$C = B \log_2 (1 + \text{SNR}) = B \log_2(1 + 0)$$

$$= B \log_2 1 = B \times 0 = 0$$

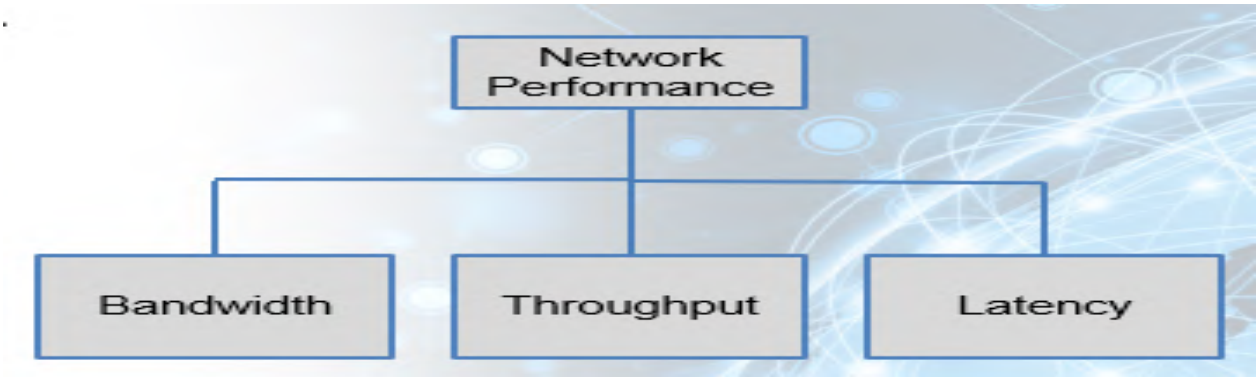
This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Network Performance

Data transmission (in form of Signal) over a network and how network behaves is important.

More important is the performance of the network.

Network Performance



There are 3 characteristics of network performance.

Bandwidth

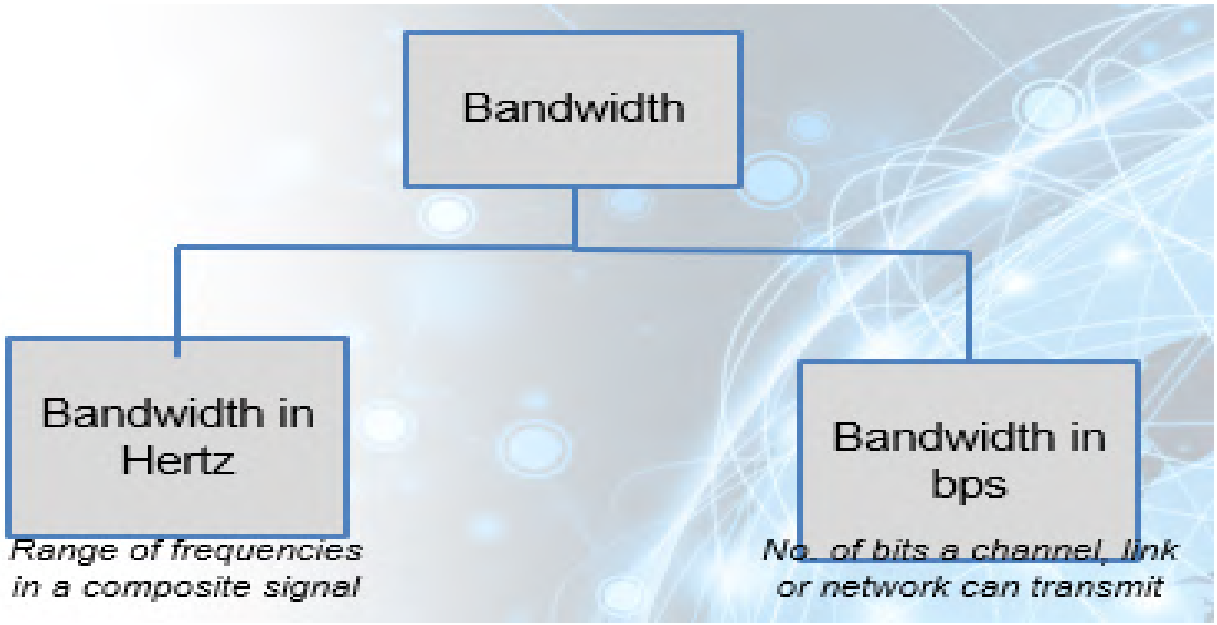
✓ An important characteristic that measures Network Performance

✓ Bandwidth can be used in two different contexts with two different measuring values:

✓ Bandwidth in Hertz

✓ Bandwidth in bits per second

Bandwidth



Throughput

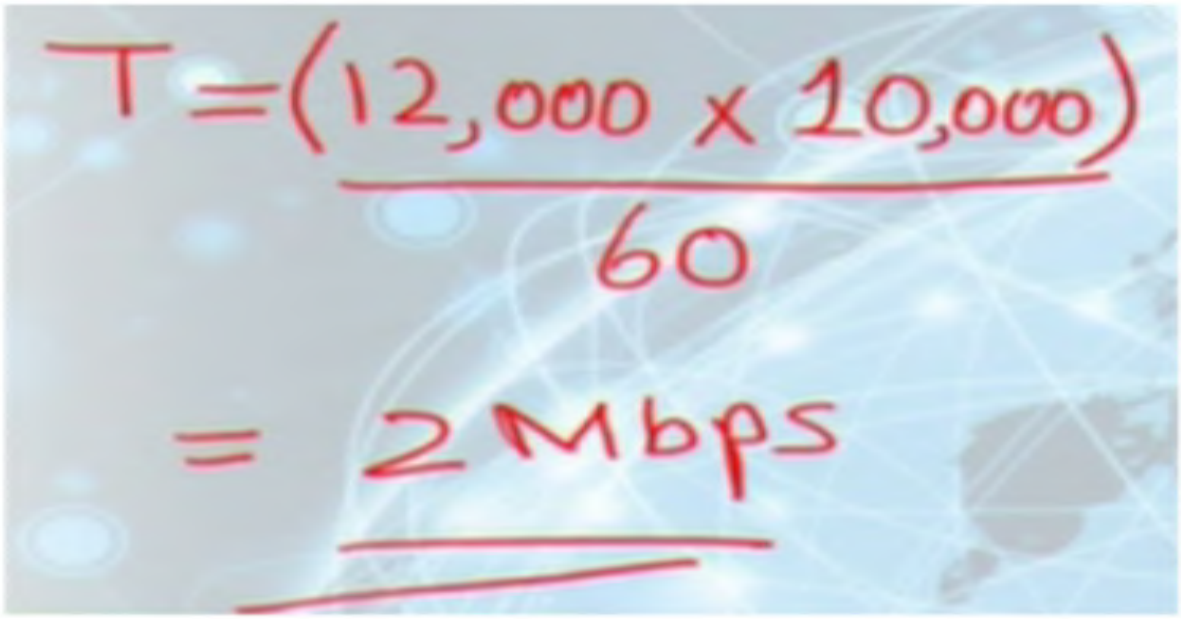
✓ Measure how fast we can send data through a network.

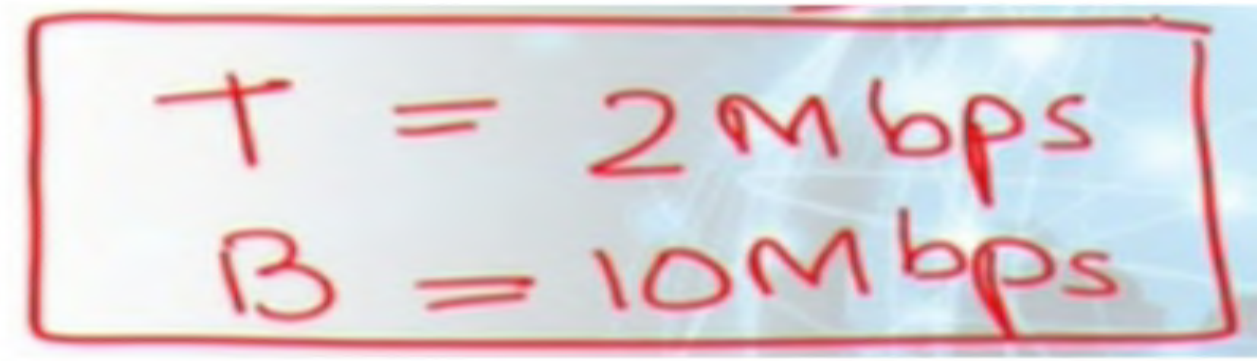
✓ Bandwidth is not the same as Throughput

✓ A link may have a bandwidth of B bps, but we can only send T bps through this link with T always less than B

Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?





Throughput

The throughput is a measure of how fast we can send data through a network. Although, at first glance, bandwidth in bits per second and throughput seem the same, they are different. A link may have a bandwidth of B bps, but we can only send T bps through this link with T always less than B.

Latency

The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source.

We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency = propagation time + transmission time + queuing time + processing delay

Example

A network with bandwidth of 10 Mbps can pass only an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

Solution

We can calculate the throughput as

Throughput = (12,000X10,000)/60 = 2Mbps

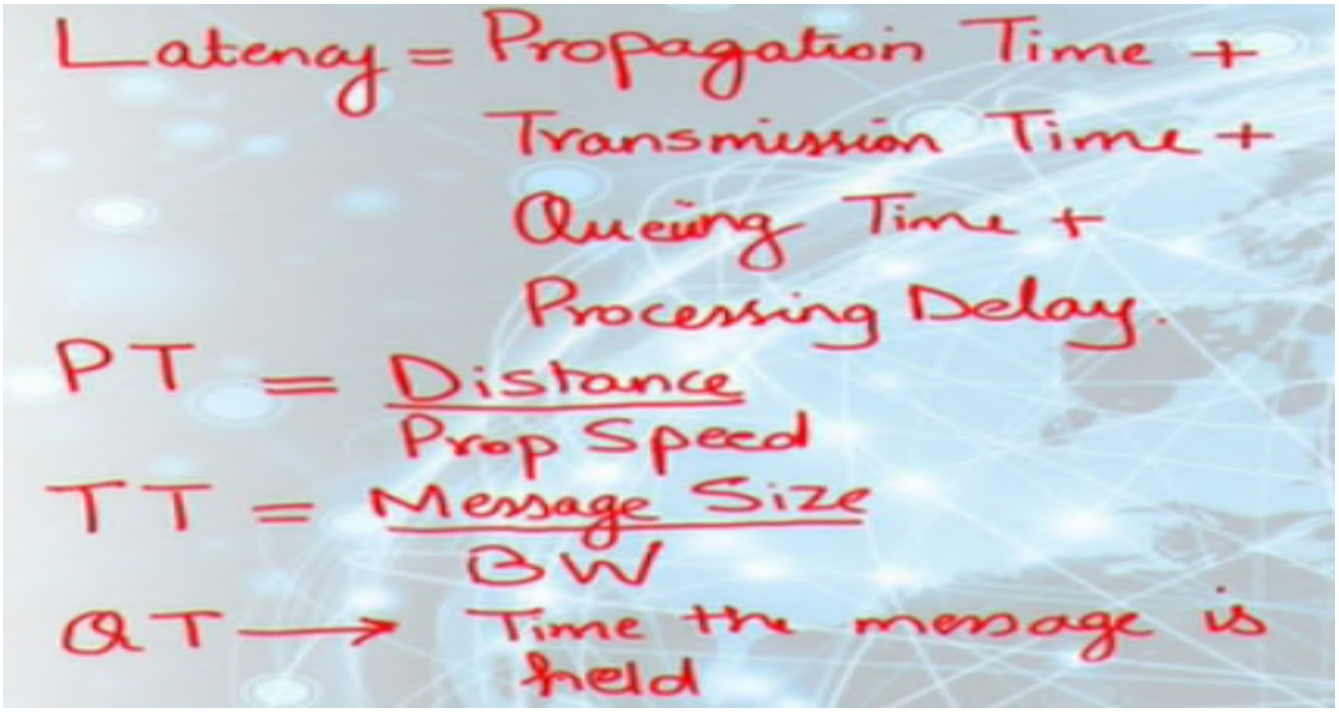
Throughput=Amount of Data Transferred/Total Time

In this case, the "Amount of Data Transferred" is the product of the average frames per minute and the average bits per frame

The throughput is almost one-fifth of the bandwidth in this case.

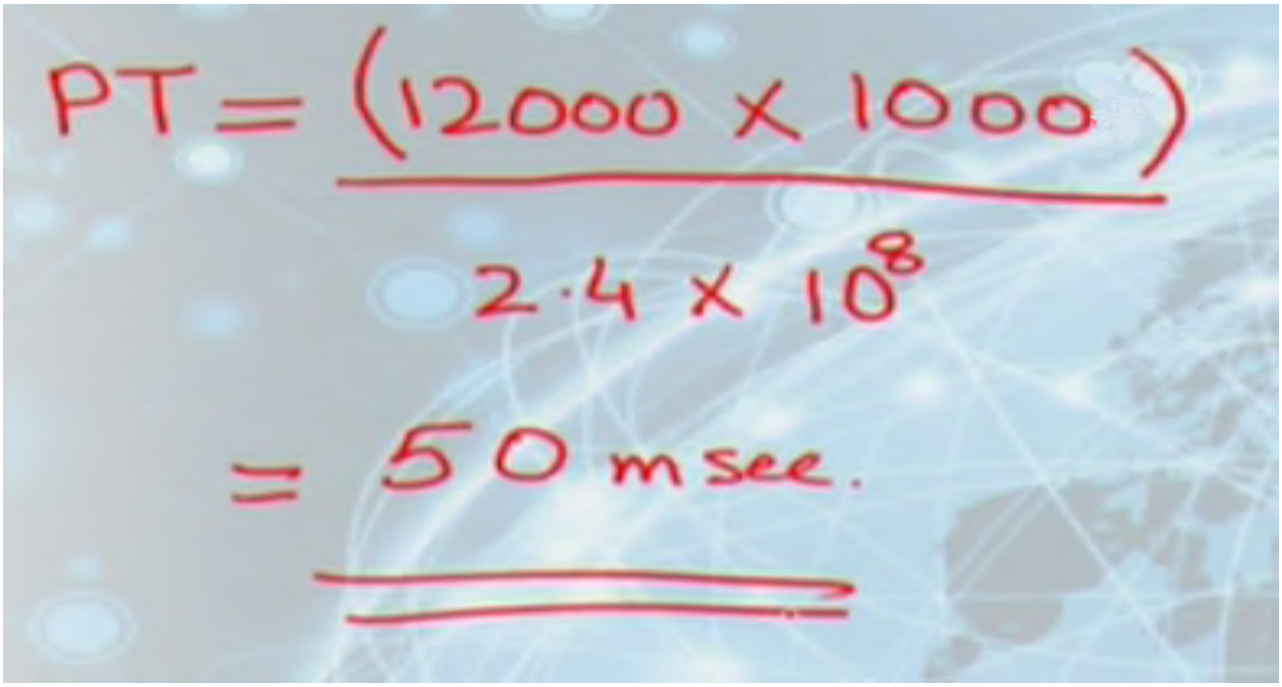
Latency or Delay

Latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source



Example

What is the propagation time if the distance between the two points is 12,000 km? Assume the propagation speed to be 2.4 × 108 m/s in cable.



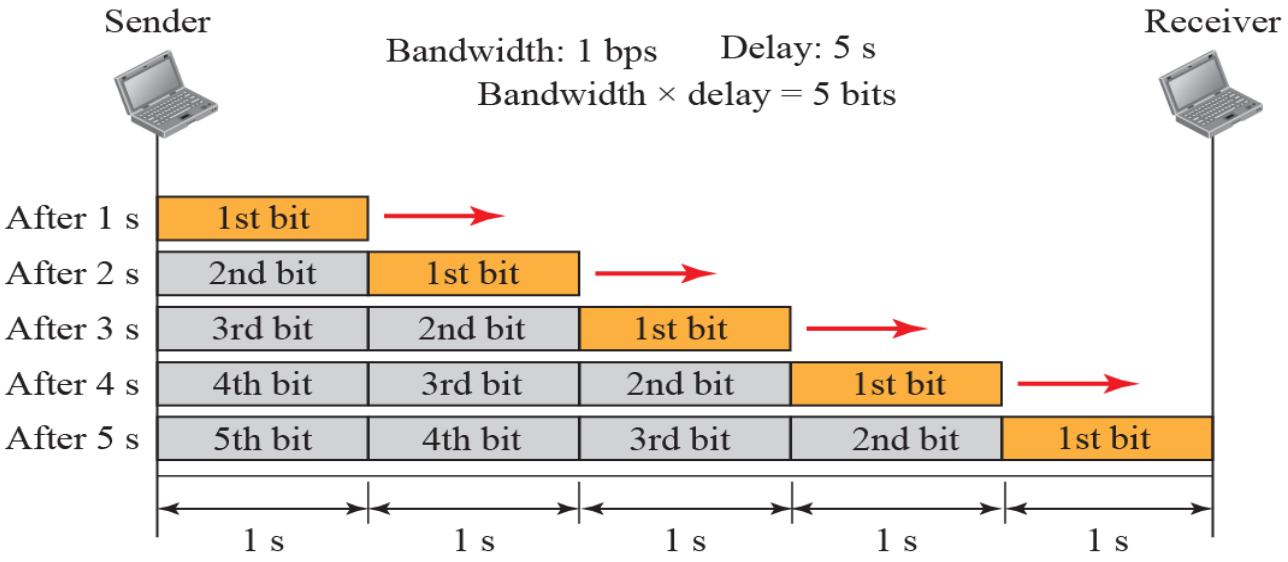
Delay – Bandwidth Delay Product

✓ Bandwidth and delay are two performance metrics of a link

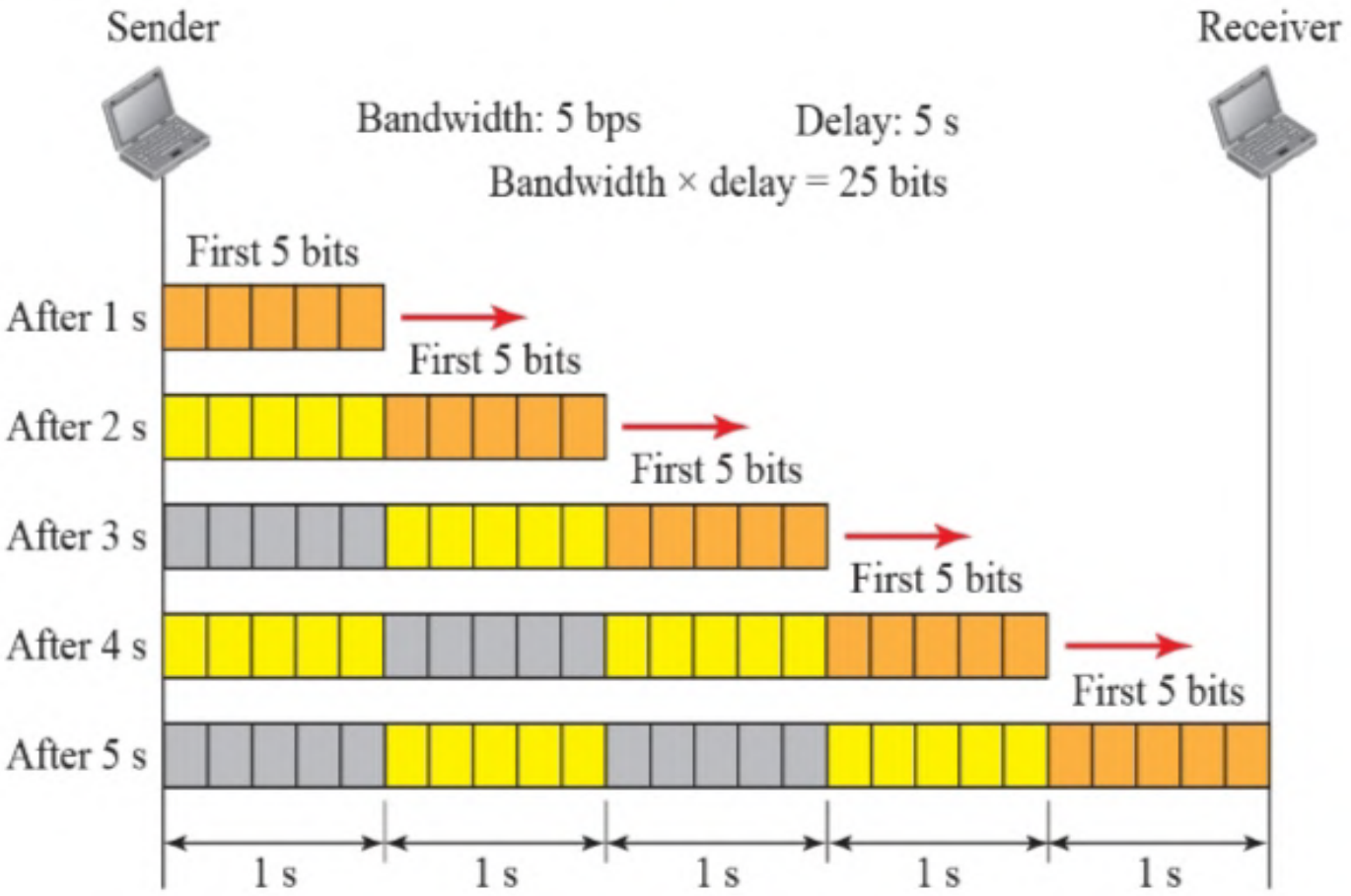
✓ Product of the two, The Bandwidth-Delay Product defines the number of bits that can fill a link

Bandwidth-Delay Product

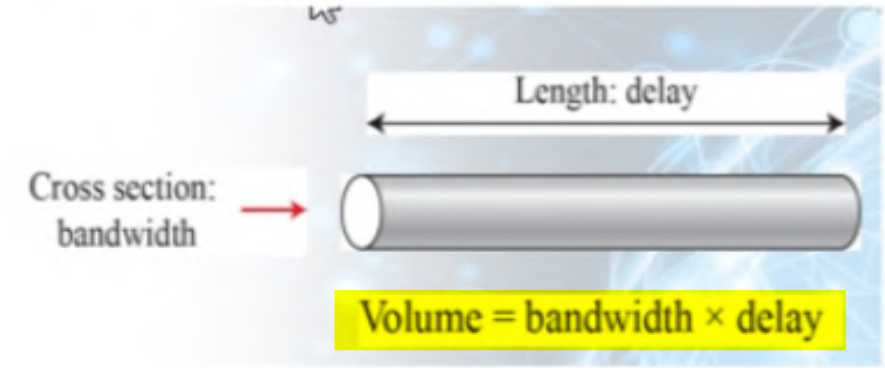
Case 1



Case 2



Bandwidth-Delay Product



We can think about the link between two points as a pipe. The cross section of the pipe represents the bandwidth, and the length of the pipe represents the delay.

We can say the volume of the pipe defines the bandwidth-delay product, as shown in above.

Delay - Jitter

Jitter is a problem if different packets of data encounter different delays and the application using the data at the receiver site is time-sensitive (audio and video data, for example)

CS601 ~ Week•4

When it comes to digital communication, one important process is digital-to-digital conversion. This process involves converting data from one digital format to another. Understanding this conversion process is crucial for effectively transmitting digital information across different communication channels.

There are various methods for converting digital signals to digital format:

Line coding is a crucial process in which binary data is transformed into digital signals, enabling smooth transmission over a communication channel. It guarantees a seamless coordination between the sender and receiver.

Block coding involves the grouping of a set of bits and encoding them into a specific code. This method improves the integrity of the data and allows for the detection and correction of errors.

Scrambling is a valuable technique that helps to enhance the efficiency of data transmission by

randomising the data. By doing so, it minimises the chances of having long sequences of identical bits and optimises the use of the communication channel.

Let's discuss the data and signal elements:

The smallest unit of data representation is a bit. The value can be either 0 or 1, serving as the foundation for digital information processing and storage.

In digital communication, a signal element refers to the smallest unit of a digital signal. It represents a particular voltage level or phase change and is utilised to transmit data over a communication channel.

Understanding data and signal rates is essential for optimising efficiency and speed in digital communication.

The data rate, which is measured in bits per second (bps), represents the quantity of data elements that are transmitted within a one -second interval. Understanding digital information interchange requires a grasp of the fundamental building blocks: bits represented by 0s and 1s. A higher data rate enables faster communication and data transfer by transmitting a larger amount of data in a shorter period of time.

The signal rate, measured in baud, refers to the quantity of signal elements that are transmitted within a single second. Signal elements consist of pulses, phases, and modulated signals. Signal rate is primarily concerned with the physical alterations in the signal that transmit the data, rather than the actual content being transmitted. The signal rate is also known as the pulse or modulation rate.

Line coding converts digital data into digital signals, a key digital communication procedure. Computer memory stores text, numbers, graphics, music, and video as sequences of bits (0s and 1s). Digital information is built from these bits.

Line coding ensures efficient and reliable data transfer over communication lines.

Line coding schemes:

Following are the 5-line coding schemes:

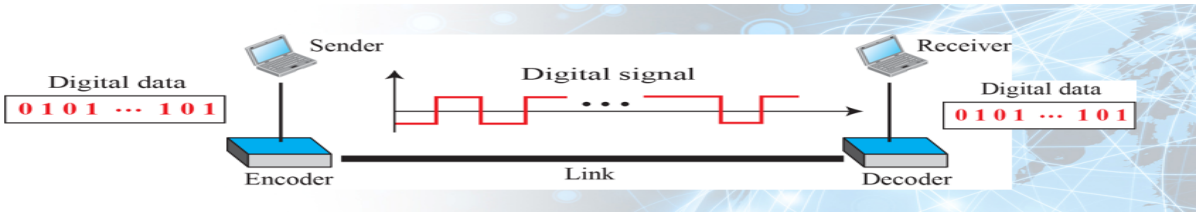
✓ Unipolar

✓ Polar

✓ Bipolar

✓ Multilevel

✓ aMultitransition



Line coding schemes:

Following are the 5-line coding schemes:

✓ Unipolar

✓ Polar

✓ Bipolar

✓ Multilevel

✓ Multitransition

✓ Unipolar Scheme:

Unipolar line coding involves placing all digital signal levels on one side of the time axis, above or below. In binary, 1s are represented by a positive or non-zero voltage level, while 0s are zero. Avoiding signal time axis crossing simplifies signal detection and synchronisation.

For unipolar line coding:

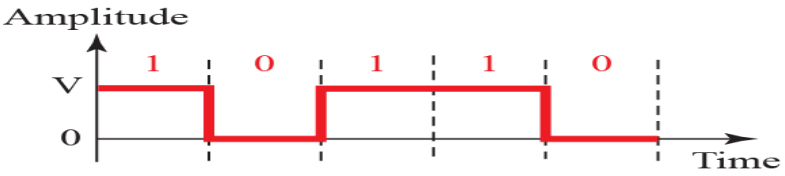
✓ The positive voltage (+V) is represented by binary 1.

✓ Binary 0 is zero voltage.

Simple unipolar line coding is easy to implement and decode. Even with several zeros in the data, the constant positive signal reduces signal power efficiency and transmits power consistently.

Unipolar line coding is employed in short-distance communication systems since it is easy to build and decode and power efficiency is less important.

In Unipolar NRZ, a binary 1 has a constant voltage level throughout the bit period, while a binary 0 has no voltage. Unipolar NRZ retains voltage signal polarity for all binary values, unlike other NRZ methods. However, it only uses positive voltage for binary 1 and zero voltage for binary 0.

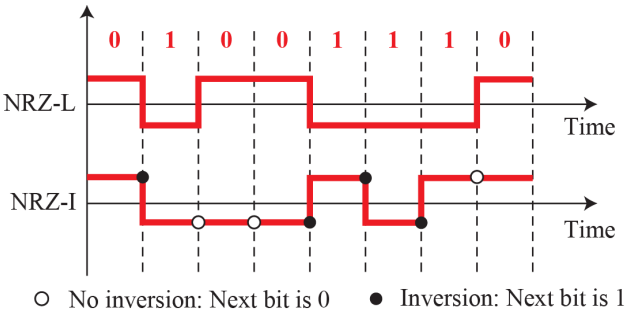


Polar-NRZ Line Coding:

Binary 1 has a constant voltage level, positive or negative, during the bit period.

The voltage level of 0 in binary is opposite (negative or positive) throughout the bit period.

In Polar NRZ, the voltage level for a binary 1 remains constant during the bit period, while the voltage level for a binary 0 remains constant but reversed. The signal is not reset to zero each bit period in this technique. Instead, it keeps the voltage constant throughout each bit.

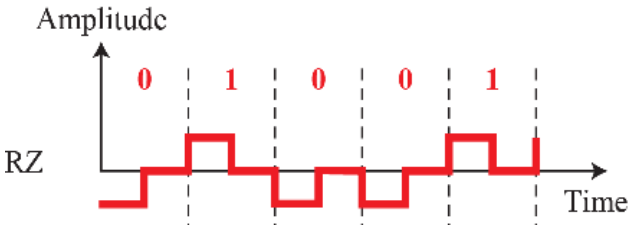


The Polar RZ Line Coding Scheme:

In binary, 1 is represented by a positive voltage level in the first half of the bit period and zero value in the second half.

The binary number 0 is represented by a negative voltage level for half the bit period and zero for the other half.

Polar RZ signals always revert to zero voltage (or baseline) midway through each bit period, independent of bit value. This improves synchronisation and reduces issues caused by long sequences of identical bits, a common concern in non-return-to-zero methods.



Polar NRZ and Polar RZ were separate polar line coding methods. In Polar NRZ, the voltage level remains constant during the bit period, whether it's 1 or 0. However, Polar RZ voltage returns to zero halfway through each bit cycle. Consider communication requirements like bandwidth efficiency and synchronisation while choosing between these systems.

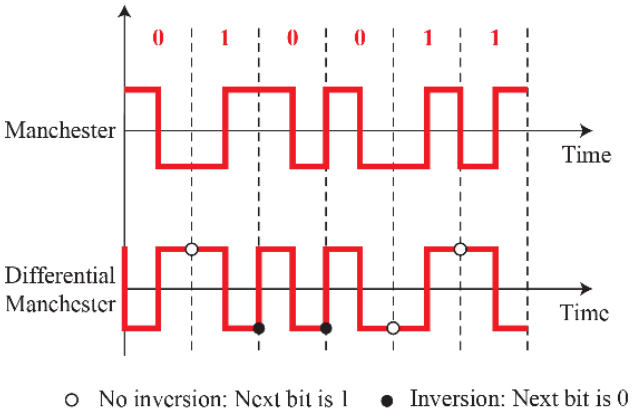
Polar Biphase, also known as Manchester encoding, is a line coding scheme commonly employed in digital communication. Within this scheme, every bit period is split into two equal intervals. In each bit period, the polarity of the voltage signal switches back and forth, allowing for regular transitions that help the sender and receiver stay in sync.

When considering Polar Biphase:

Binary 1 is indicated by a change from a positive voltage level to a negative voltage level (or vice versa) during the bit period. This transition is commonly known as a 'Manchester transition' or 'Manchester violation.'

In binary, a 0 is represented when there is no transition within the bit period, which can happen at the beginning or end of the period.

Polar Biphase encoding possesses the convenient feature of being self-clocking. This means that the receiver is able to extract the clock signal directly from the received data stream, thanks to the guaranteed transition in the middle of each bit period. This property is highly advantageous in preventing synchronisation errors, making it a popular choice for Ethernet LANs and other digital communication systems that require precise clock recovery.



AMI/Pseudoternary Bipolar Schemes:

AMI uses zero voltage for binary 0 and positive and negative voltage for binary 1. AMI's 1s alternate polarity is unique. Code for "1101" is "+0-+". AMI is used in T1 and E1 lines for data transmission and synchronisation.

Digital communication systems encode data using pseudoternary line coding. Balanced line codes ensure equal positive and negative voltage levels. Telecoms and networks utilise this coding strategy to send binary data over a channel. It reduces errors and ensures data transmission and is dependable and efficient.

Binary 1 has a continuous zero voltage in pseudoternary, while binary 0 alternates between positive and negative. AMI has alternating polarities, but Pseudoternary has no voltage change for binary 1s. This encoding method is used in older telecommunication systems to maintain a balanced distribution of positive and negative voltage levels for long-distance transmission.

Communication applications often use bipolar systems like AMI and Pseudoternary. These systems encode binary data with positive, negative, and zero voltage.

Block coding converts one amount of bits into more. Often used in digital signal encoding. In mB/nB encoding,'m' represents the number of bits in the original data block and 'n' represents the encoded block, which is usually larger.

Reason for Block Coding:

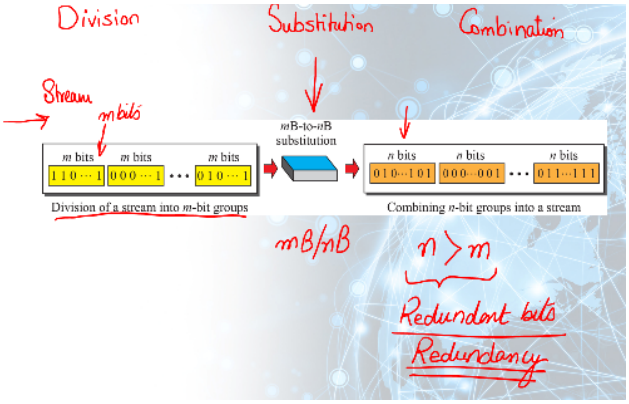
Block coding adds signal redundancy, improving signal reliability. Redundancy adds data. Additional components aid receiver synchronisation, error identification, and correction.

Synchronisation requires transmitter and receiver alignment at the start and end of each data block. This minimises errors and streamlines data flow. Synchronisation is needed to decode the signal and recover the data.

Benefits of Block Coding:

✓ Block coding includes redundancy to improve data transmission reliability. Redundancy detects and fixes transmission problems to maintain data integrity.

✓ Efficient Synchronisation: Redundant bits help sender-receiver synchronisation. Accurate signal decoding requires correct synchronisation.



Using block coding 4B/5B with NRZ-I line coding

In digital communication systems, the utilisation of 4B/5B block coding alongside NRZ-I line coding is a widely employed technique. This approach is particularly prevalent in high-speed networks and interfaces like Gigabit Ethernet and Fibre Channel. Now, let's analyse the meaning behind this combination:

In 4B/5B block coding, groups of 4 bits are assigned to distinct 5-bit code words. There are additional benefits to using this encoding method compared to regular NRZ encoding. It helps maintain a balance between 1s and 0s and also offers error detection capabilities. This scheme enables a wider variety of patterns compared to simple NRZ encoding, which enhances the reliability of data transmission.

NRZ-I Line Coding: NRZ-I, also known as Non-Return-to-Zero Inverted, is a line coding technique that utilises transitions to represent binary 1s and the absence of transitions to represent binary 0s. NRZ-I operates in a self-clocking manner, eliminating the need for an external clock signal to ensure synchronisation. On the contrary, it utilises the transitions within the data itself to ensure synchronisation between the sender and receiver.

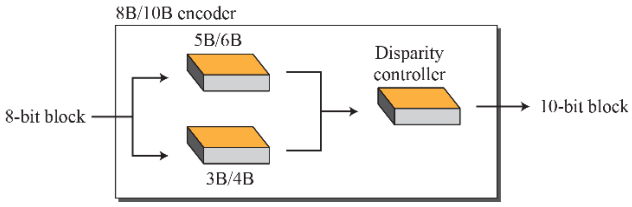
4B/5B mapping codes

Data Sequence	*Encoded Sequence*	*Control Sequence*	*Encoded Sequence*
0000	11110	Q (Quiet)	00000
0001	01001	I (Idle)	11111
0010	10100	H (Halt)	00100
0011	10101	J (Start delimiter)	11000
0100	01010	K (Start delimiter)	10001
0101	01011	T (End delimiter)	01101
0110	01110	S (Set)	11001
0111	01111	R (Reset)	00111
1000	10010		
1001	10011		
1010	10110		
1011	10111		
1100	11010		
1101	11011		
1110	11100		
1111	11101		

Block coding converts one amount of bits into more. Often used in digital signal encoding. In mB/nB encoding,'m' represents the number of bits in the original data block and 'n' represents the encoded block, which is usually larger.

8B/10B block encoding is a technique used in data transmission to ensure reliable and efficient communication.

8B/10B is a block encoding scheme commonly utilised in high-speed digital communication to optimise data transmission and maintain synchronisation and error detection. In this scheme, 8 bits of data are assigned to distinct 10-bit code words. This mapping ensures a balanced distribution of 1s and 0s, which facilitates synchronisation and enhances error-detection capabilities.



Scrambling

Scrambling is a technique used in data communication to improve the reliability of data transmission and to ensure that the data can be accurately recovered at the receiving end.

✓ B8ZS and HDB3 are methods of scrambling data to make sure it's sent in a way that prevents synchronization and data integrity issues, especially over long-distance communication lines.

✓ Biphase schemes, such as Manchester encoding, are often used for LAN (Local Area Network) communication due to their advantages in terms of synchronization and low error rates.

✓ Biphase schemes suitable for LAN but not for Long Distance

✓ Block Coding + NRZ-I solves synch issue but has DC component.

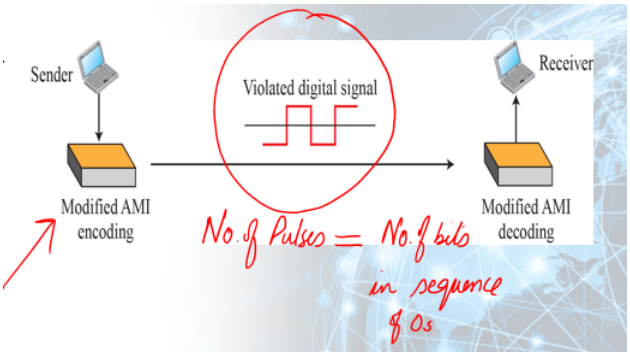
✓ Bipolar AMI has a narrow bandwidth (no DC Component) but synch issue (long series of 0s)

✓ The system needs to insert the required pulses based on the defined scrambling rules.

AMI is used with scrambling.

✓ In AMI encoding, a 0-bit is represented alternately by a positive voltage (mark) and a negative voltage (space), while a 1-bit is represented by no voltage change (zero).

✓ This alternate marking of 0s ensures that there are frequent voltage transitions, which is important for clock recovery and synchronization.



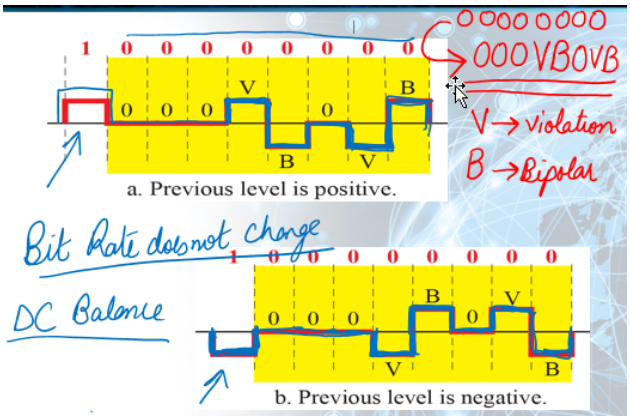
Types of Scrambling Techniques

Two common scrambling techniques are B8ZS and HDB3

Bipolar with 8-Zero Substitution (B8ZS): B8ZS is a specific method of scrambling. When you have a long string of consecutive zeros in your data, B8ZS replaces these with a special code to keep the data balanced and maintain timing. This substitution helps ensure that there are enough transitions in the data to keep everything synchronized during transmission.

High-density bipolar 3-zero (HDB3): also focuses on keeping the data balanced and ensuring that there are enough transitions. When there are four consecutive zeros in the data, HDB3 replaces them with a unique pattern to maintain the balance of ones and zeros in the data, which is crucial for reliable communication over long distances.

Two cases of B8ZS scrambling technique



✓ In this technique, eight consecutive zero-level voltages are replaced by the sequence 000VB0VB.

✓ The V in the sequence denotes violation: this is a nonzero voltage that breaks an AMI rule of encoding (opposite polarity from the previous).

✓ The B in the sequence denotes bipolar, which means a nonzero level voltage in accordance with the AMI rule.

Types of Scrambling Techniques

Two common scrambling techniques are B8ZS and HDB3

Bipolar with 8-Zero Substitution (B8ZS): B8ZS is a specific method of scrambling. When you have a long string of consecutive zeros in your data, B8ZS replaces these with a special code to keep the data balanced and maintain timing. This substitution helps ensure that there are enough transitions in the data to keep everything synchronized during transmission.

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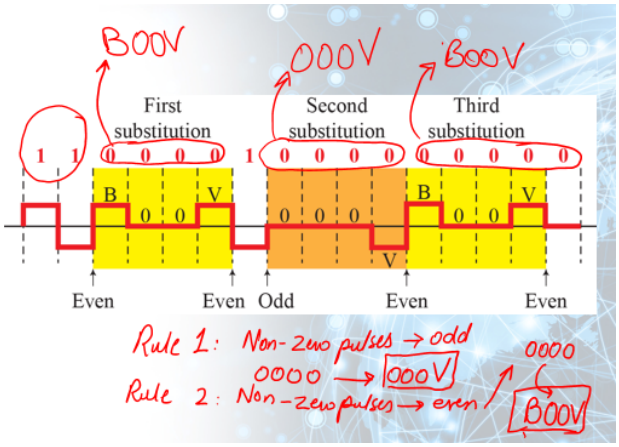
Different situations in HDB3 scrambling technique.

In HDB3 four consecutive zero-level voltages are replaced with a sequence of 000V or B00V.



✓ If the number of nonzero pulses after the last substitution is odd, the substitution pattern will be 000V, which makes the total number of nonzero pulses even.

✓ If the number of nonzero pulses after the last substitution is even, the substitution pattern will be B00V, which makes the total number of nonzero pulses even.

✓ The V in the sequence denotes violation; this is a nonzero voltage that breaks an AMI rule of encoding.

✓ The B in the sequence denotes bipolar, which means a nonzero level voltage in accordance with the AMI rule.

Analog-to-digital Conversion

Analog Data to Digital Data: Analog data to digital data conversion involves sampling the analog signal, quantizing the sampled values into discrete levels, encoding these values into binary form, and then using digital systems to process, store, or transmit the digital representation of the original analog information.

Process of Digitization: The process of digitization involves converting analog information into digital data, making it easier to store, process, and transmit.

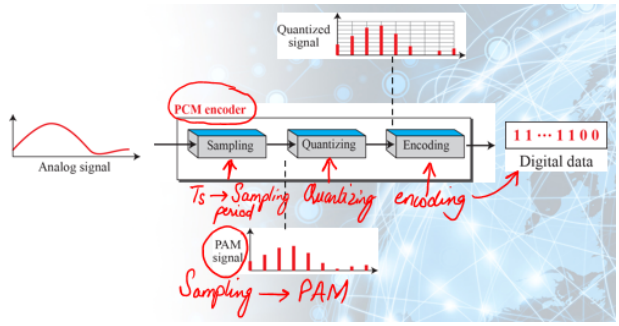
Pulse Code Modulation (PCM)

Pulse Code Modulation (PCM) is a widely used method for digitally representing analog signals, such as audio or video, in a format that can be easily processed and transmitted by digital systems.

Delta Modulation (DM)

Delta Modulation (DM) is a simple method used to digitally encode analog signals, typically for voice or low-quality audio transmission.

Components of PCM encoder



Pulse Code Modulation (PCM)

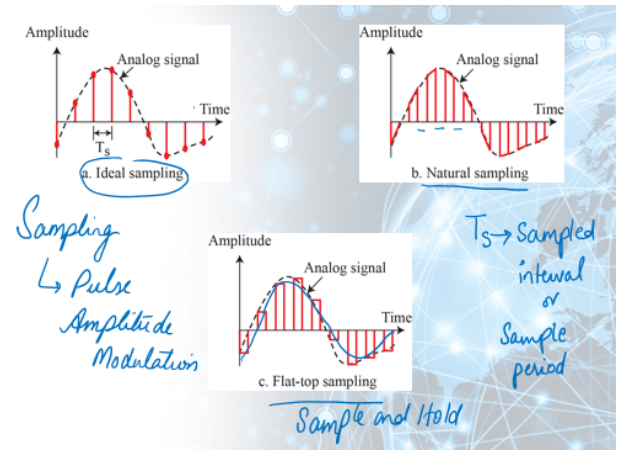
Pulse Code Modulation (PCM) is a widely used method for digitally representing analog signals, such as audio or video, in a format that can be easily processed and transmitted by digital systems.

✓ Sampling: The first step in PCM is to take samples of the analog signal at regular intervals.

✓ Quantization: Each of the sampled values is then quantized, which means assigning a numerical value to represent the amplitude of the sample.

✓ Encoding: The quantized values are encoded into a digital format, typically using binary code (0s and 1s).

Three different sampling methods for PCM



Ideal Sampling:

✓ Often referred to as Nyquist sampling, is a theoretical concept in signal processing.

✓ According to the Nyquist-Shannon theorem, to accurately represent an analog signal in digital form, you should sample it at a rate that is at least twice the signal's highest frequency (the Nyquist rate).

Natural Sampling:

✓ Also known as impulse or instantaneous sampling, is a practical approach to capturing analog signals.

✓ In natural sampling, the analog signal is sampled at specific points in time when an impulse or a brief sampling pulse occurs.

Flat-Top Sampling:

✓ Flat-top sampling is a modification of natural sampling that helps reduce the effects of quantization noise in analog-to-digital conversion.

✓ In flat-top sampling, the sampling pulse has a finite duration (flat top) rather than being instantaneous.

Nyquist Sampling Rate

Nyquist's theorem states that to accurately capture an analog signal, the sampling rate should be at least twice the signal's maximum frequency (Nyquist rate).

Nyquist à fs = 2fh

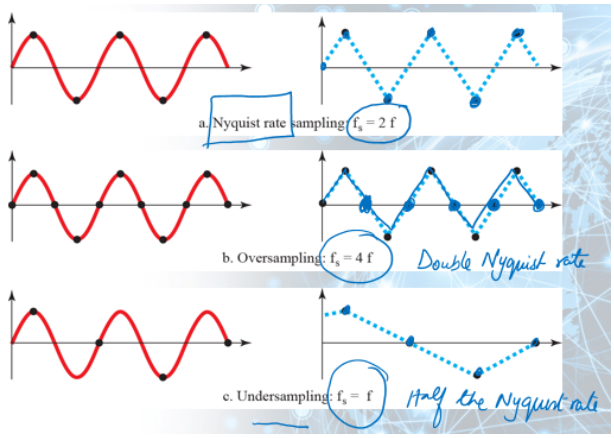
Sampling Period (Ts):

The analog signal is sampled every Ts s, where Ts is the sample interval or period.

Sampling rate or sampling frequency:

The inverse of the sampling interval is called the sampling rate or sampling frequency and denoted by fs, where fs = 1/Ts.

Nyquist Sampling Rate



Sampling sine wave at three sampling rates:

fs = 4f (2 times the Nyquist rate)

✓ Oversampling involves taking more samples of a signal than the minimum required by the Nyquist-Shannon theorem.

✓ By oversampling, you collect more data points per unit of time, which can improve the accuracy and reliability of the captured information.

fs = 2f (Nyquist rate)

Nyquist's theorem states that to accurately capture an analog signal, the sampling rate should be at least twice the signal's maximum frequency (Nyquist rate).

fs = f (one-half the Nyquist rate)

✓ Under sampling, on the other hand, involves sampling a signal at a rate lower than the Nyquist rate.

√Undersampling intentionally captures fewer data points to reduce data volume or handle multiple signals within a limited bandwidth.

Quantization & encoding of a sampled signal.

✓ Sampling à Series of pulses with amplitude values between min and max signal amplitude

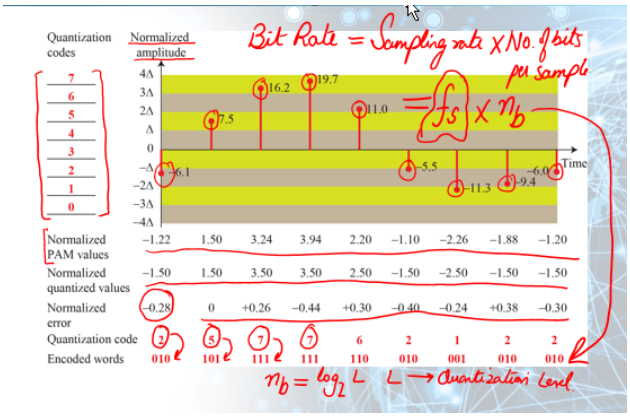
✓ Infinite set with non-integral values not suitable for encoding

✓ We quantize the sampling output into certain levels based on range of amplitudes and how much accuracy is needed.

Quantization

After sampling, each of the sampled values is then quantized, which means assigning a numerical value to represent the amplitude of the sample.

Quantization & encoding of a sampled signal.



✓ After each sample is quantized and the number of bits per sample is decided, each sample can be changed to an nb-bit code word. In the above Figure, A quantization code of 2 is encoded as 010; 5 is encoded as 101 and so on.

✓ The number of bits for each sample is determined from the number of quantization levels.

✓ If the number of quantization levels is L, the number of bits is nb = log2 L.

✓ In the above example L is 8 and nb is therefore 3.

✓ Bit rate = sampling rate x number of bits per sample = fs x nb

Example:

The human voice normally contains frequencies from 0 to 4000 Hz. So the sampling rate and bit rate are calculated as follows

$$\text{Sampling rate} = 4000 \times 2 = 8000 \text{ samples/s}$$

$$\text{Bit rate} = 8000 \times 8 = 64{,}000 \text{ bps} = 64 \text{ kbps}$$

Delta Modulation (DM)

✓ PCM is a very complex technique.

✓ Delta modulation is a simpler technique.

✓ PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample.

✓ No code words in delta modulation; bits are sent one after another.

Modulator:

The modulator is used at the sender site to create a stream of bits from an analog signal.

Delta δ:

✓ The process records the small positive or negative changes, called delta δ.

✓ If the delta is positive, the process records a 1

✓ If it is negative, the process records a 0.

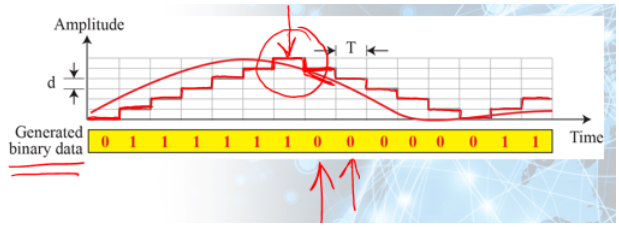
Staircase:

✓ The process also needs a base against which the analog signal is compared.

✓ The modulator builds a second signal that resembles a staircase.

✓ Finding the change is then reduced to comparing the input signal with the gradually made staircase signal.

The process of delta modulation

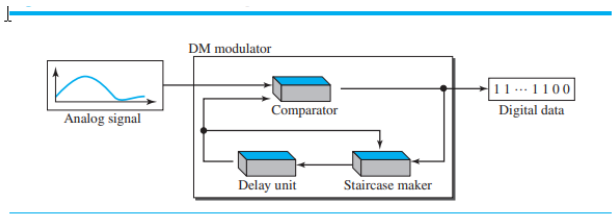


The process of delta modulation

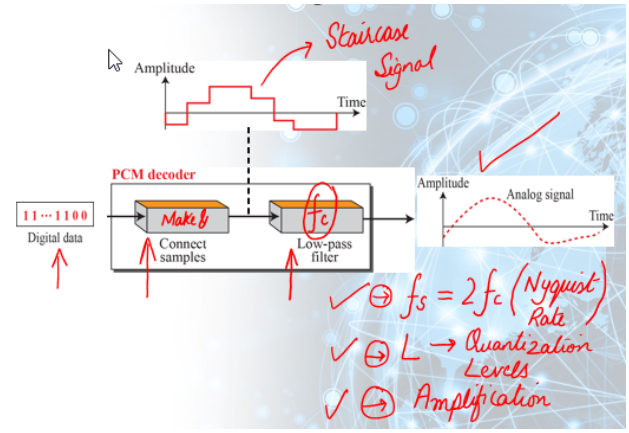
The modulator, at each sampling interval, compares the value of the analog signal with the last value of the staircase signal. If the amplitude of the analog signal is larger, the next bit in the digital data is 1; otherwise, it is 0.

The output of the comparator, however, also makes the staircase itself. If the next bit is 1, the staircase maker moves the last point of the staircase signal δ up; if the next bit is 0, it moves it δ down.

Note that we need a delay unit to hold the staircase function for a period between two comparisons.



Original Signal Recovery- PCM Decoder



✓ The recovery of the original signal requires the PCM decoder.

✓ The decoder first uses circuitry to convert the code words into a pulse that holds the amplitude until the next pulse.

✓ After the staircase signal is completed, it is passed through a low-pass filter to smooth the staircase signal into an analog signal.

✓ The filter has the same cutoff frequency as the original signal at the sender.

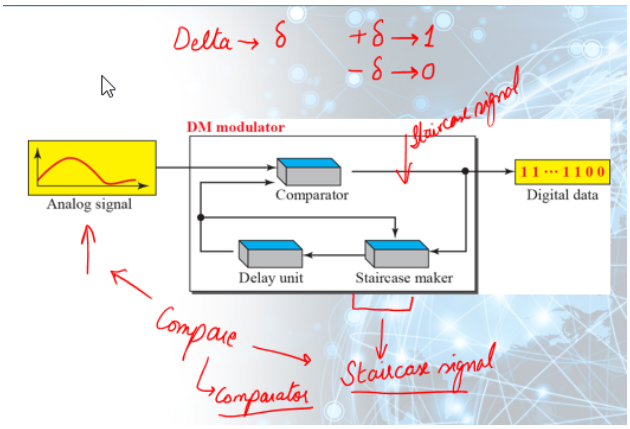
✓ If the signal has been sampled at (or greater than) the Nyquist sampling rate and if there are enough quantization levels, the original signal will be recreated.

✓ The maximum and minimum values of the original signal can be achieved by using amplification.

Delta Modulation (DM)

- Delta modulation is a simpler technique.
- DM finds the change from the previous sample.
- No code words in delta modulation; bits are sent one after another.

Delta Modulation Components



Delta Modulation Components

✓ The modulator, at each sampling interval, compares the value of the analog signal with the last value of the staircase signal.

✓ If the amplitude of the analog signal is larger, the next bit in the digital data is 1; otherwise, it is 0.

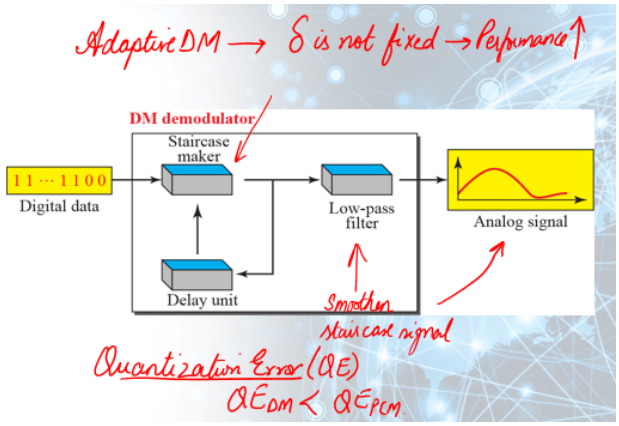
✓ The output of the comparator, however, also makes the staircase itself.

✓ If the next bit is 1, the staircase maker moves the last point of the staircase signal δ up.

✓ If the next bit is 0, it moves it δ down.

✓ We need a delay unit to hold the staircase function for a period between two comparisons.

Delta De modulation Components



Delta De modulation Components

✓ The demodulator takes the digital data and, using the staircase maker and the delay unit, creates the analog signal.

✓ The created analog signal, however, needs to pass through a low-pass filter for smoothing.

Adaptive DM

✓ A better performance can be achieved if the value of δ is not fixed.

✓ In adaptive delta modulation, the value of δ changes according to the amplitude of the analog signal.

Wiring: when we are considering the wiring is the data stream.

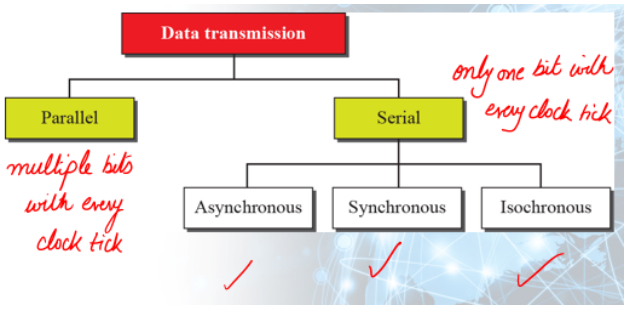
Data Stream: Do we send 1 bit at a time; or do we group bits into larger groups and, if so, how?

Transmission Modes: Parallel or Serial Transmission

In parallel mode, multiple bits are sent with each clock tick.

In serial mode, 1 bit is sent with each clock tick.

Data transmission modes



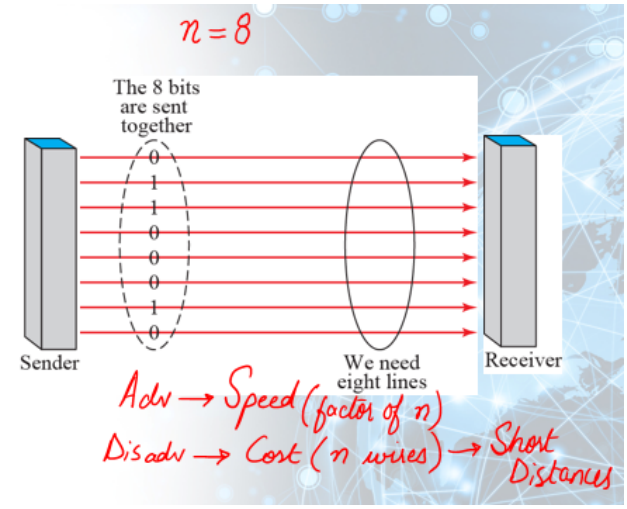
Parallel Transmission

✓ Binary data (1s ad 0s) organized in groups of 'n' bits.

✓ We send 'n' bits at a time instead of just one.

✓ 'n' wires required to send 'n' bits at one time.

Parallel Transmission



✓ Computers produce and consume data in groups of bits. By grouping, we can send data n bits at a time instead of 1. This is called parallel transmission.

✓ In parallel transmission, we use n wires to send n bits at one time. That way each bit has its own wire, and all n bits of one group can be transmitted with each clock tick from one device to another.

✓ In the above figure, we can see that how parallel transmission works for n = 8.

✓ Typically, the eight wires are bundled in a cable with a connector at each end.

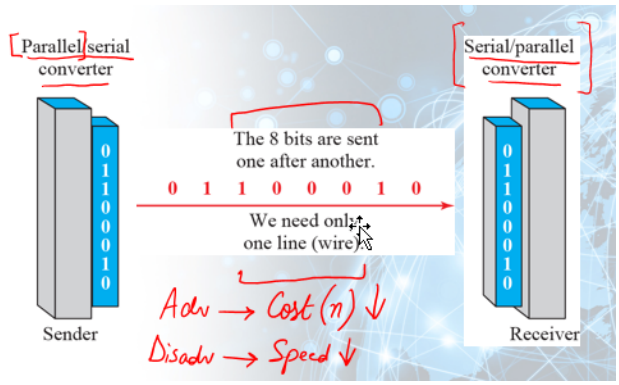
Advantage:

Speed: Parallel transmission can increase the transfer speed by a factor of n over serial transmission.

Disadvantage:

Cost: Parallel transmission requires n communication lines (wires in the example) just to transmit the data stream. It is expensive so parallel transmission is usually limited to short distances.

Serial Transmission



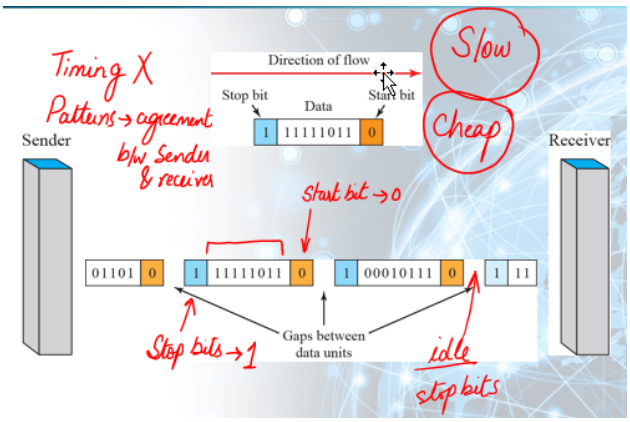
In serial transmission one bit follows another, so we need only one communication channel rather than n to transmit data between two communicating devices. It also reduces the cost of transmission over parallel by roughly a factor of n.

Since communication within devices is parallel, conversion devices are needed at the interface between the sender and the line that is performing parallel-to-serial conversion and between the line and the receiver which is conducting serial-to-parallel conversion.

Serial transmission occurs in one of three ways: asynchronous, synchronous, and isochronous.

Asynchronous Transmission

In asynchronous transmission, we send 1 start bit (0) at the beginning and 1 or more stop bits (1s) at the end of each byte.



Asynchronous transmission is so named because the timing of a signal is unimportant. Information is received and translated by agreed upon patterns. If those patterns are followed, the receiving device can retrieve the information without regard to the rhythm in which it is sent.

Start Bit: An extra bit is added to the beginning of each byte to alert the receiver to the arrival of a new group. This bit, usually a 0, is called the start bit.

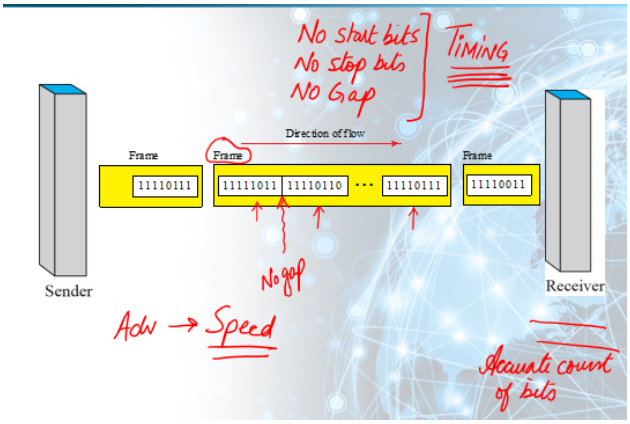
Stop Bit: To let the receiver know that the byte is finished, 1 or more additional bits are appended to the end of the byte. These bits, usually 1s, are called stop bits.

By this method, each byte is increased in size to at least 10 bits, of which 8 bits is information and 2 bits or more are signals to the receiver.

Idle channel: In addition, the transmission of each byte may then be followed by a gap of varying duration. This gap can be represented either by an idle channel or by a stream of additional stop bits.

Synchronous Transmission

In synchronous transmission, we send bits one after another without start or stop bits or gaps.



✓ In the above figure it is showing divisions between bytes, in reality, those divisions do not exist.

✓ The sender puts its data onto the line as one long string.

✓ When a sender wants to transmit data in separate bursts, they need to insert a specific sequence of 0s and 1s, signifying idle periods, in between these segments.

✓ The receiver counts the bits as they arrive and groups them into 8-bit units.

✓ In the absence of these idle intervals, as well as start and stop bits, there is no inherent mechanism to assist the receiving device in adjusting its bit synchronization during the data transmission process.

✓ As a result, precise timing becomes essential, as the accuracy of the received information relies entirely on the receiving device's ability to maintain an accurate bit count as the data is received.

Isochronous Transmission

✓ Used to send data in a predictable and timely manner.

✓ Data is sent at a fixed and constant rate, maintaining a precise timing schedule.

✓ This method is commonly used for real-time applications, such as audio and video streaming, where it's crucial to deliver data at a consistent pace to ensure smooth playback or communication.

CS601 – Week 5

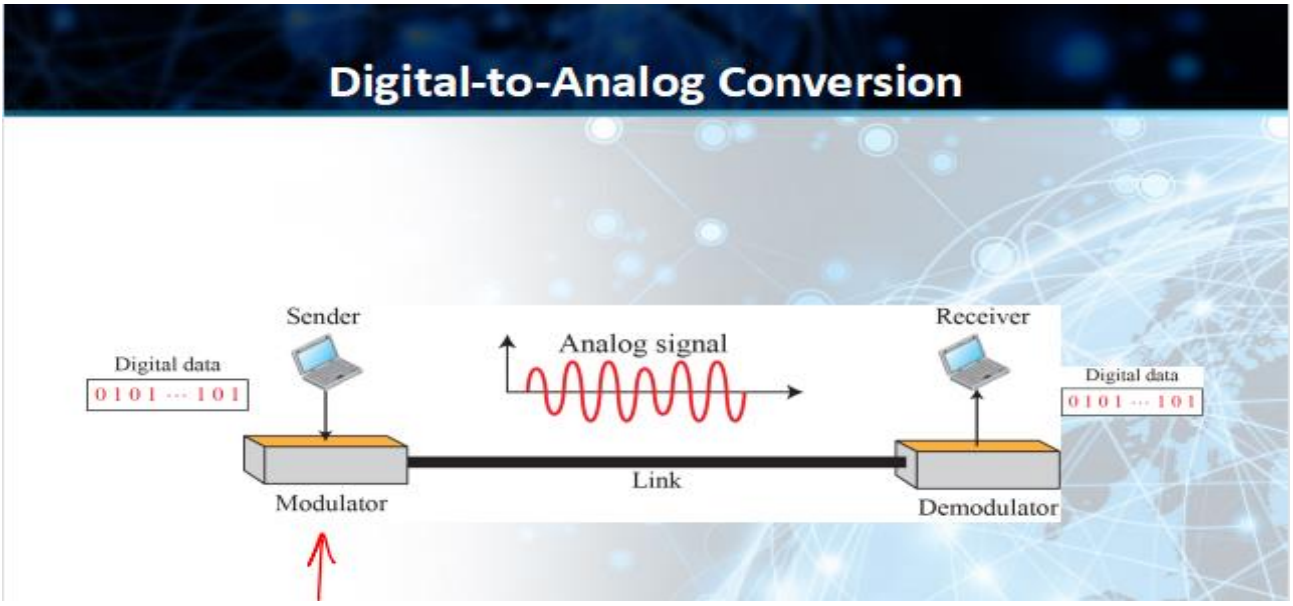
Digital-to-Analog Conversion :

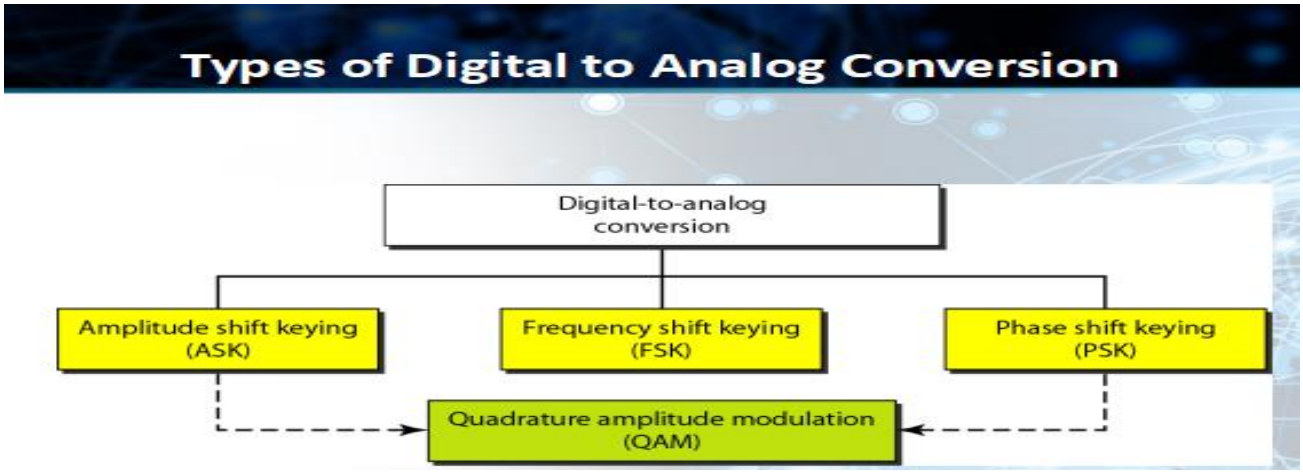
- ✓ Process of changing one of the characteristics of analog signal based on the information in digital data

  A sine wave is defined by 3 characteristics:

- Amplitude
- Frequency
- Phase

- ✓ By changing one of these characteristics, we can use it to represent a digital signal





Before we discuss specific methods of digital-to-analog modulation, two basic issues must be reviewed:

Bit and Baud rates and The Carrier Signal:

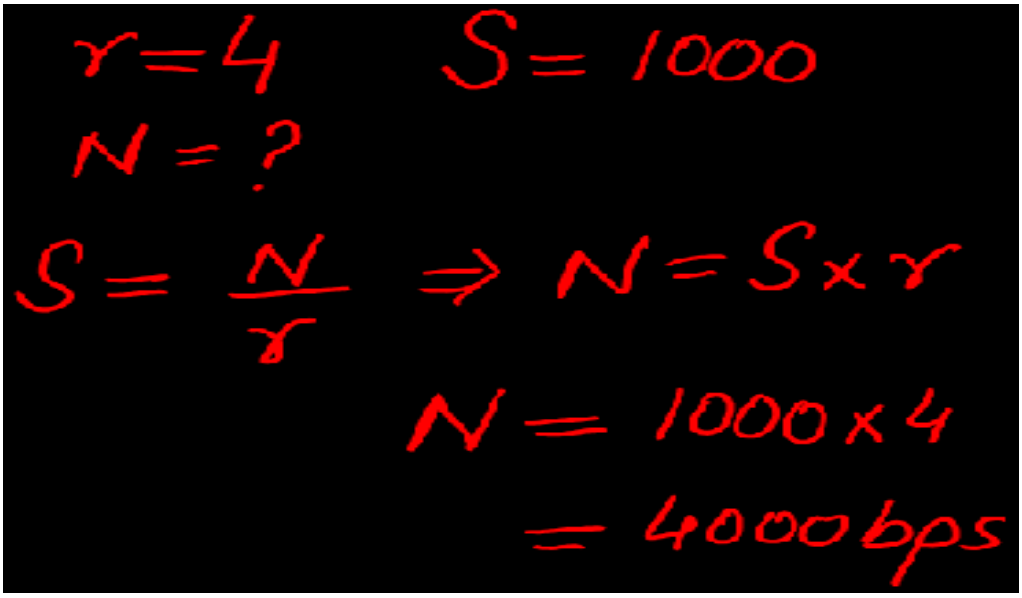
- The Baud rate refers to the total number of signal units transmitted in one second.
- The Bit rate refers to the total Bits transmitted in one unit time.

In telecommunications, a carrier wave, carrier signal, or just carrier, is a waveform (usually sinusoidal) that is modulated (modified) with an information-bearing signal (called the message signal or modulation signal) for the purpose of conveying information.

- In Analog Transmission of Digital Data, Baud Rate is less than or equal to the Bit Rate
- ✓ Data Element vs. Signal Element
- ✓ Data Rate vs. Signal Rate
- Bandwidth Required ∝ Signal Rate (except FSK)
- Carrier Signal

Example:

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate?



Example:

An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

$N = 8000$ bps
$S = 1000$ baud
$L = ?$
$r = ?$

$$S = \frac{N}{r} \Rightarrow r = \frac{N\ (\text{bits})}{S\ (\text{bauds})}$$

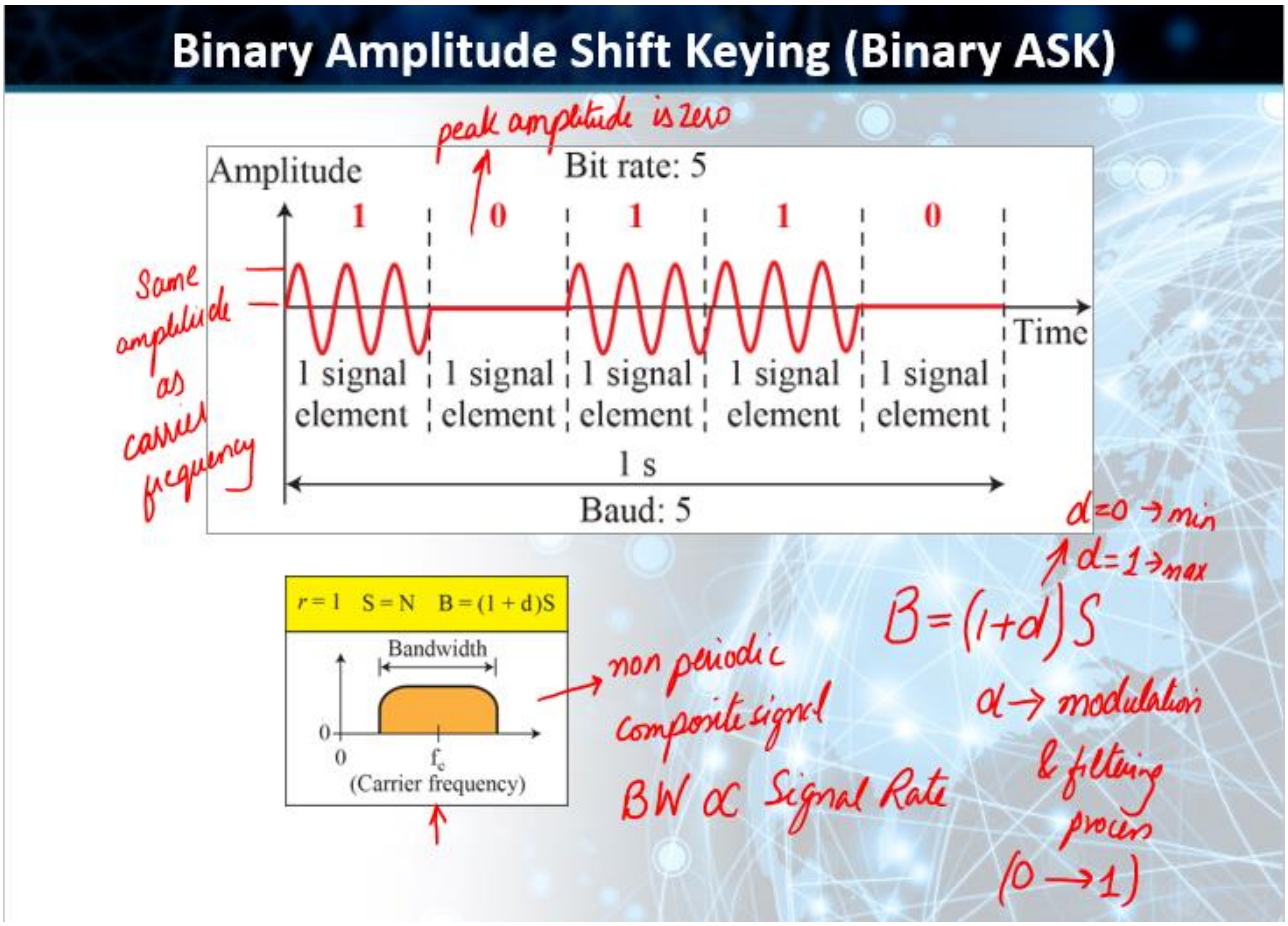
$$\rightarrow r = 8 \text{ bits/baud}$$

$$r = \log_2 L$$

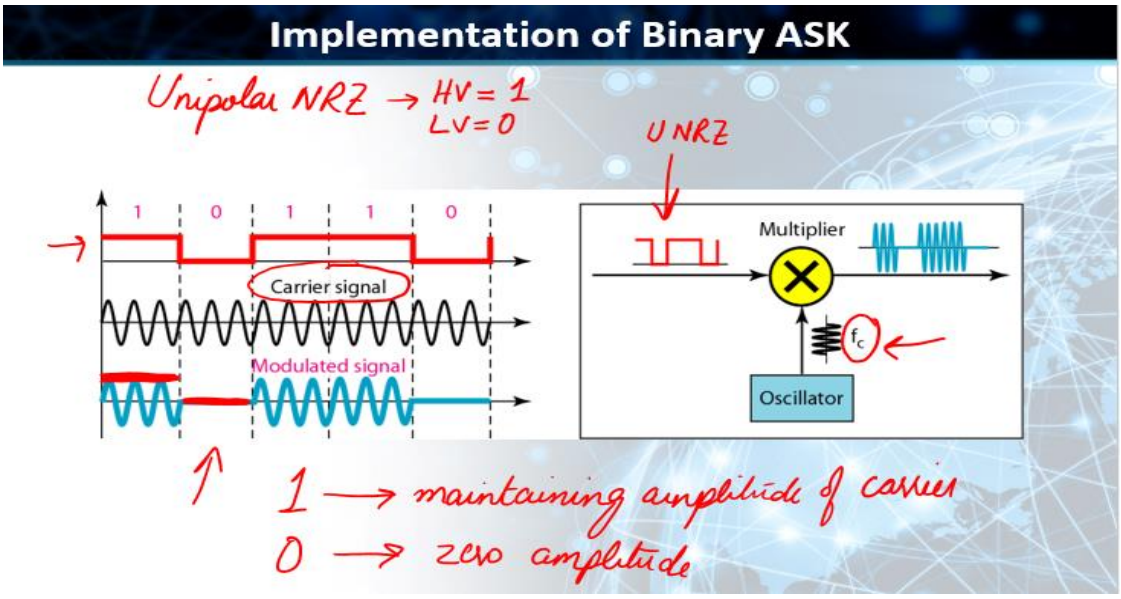
$$\rightarrow L = 2^r = 2^8 = 256$$

Amplitude Shift Keying (ASK)

- The amplitude of the carrier signal is varied to create signal elements
- Both frequency and phase remain constant while the amplitude changes
- Binary ASK or On-Off Keying (OOK)



Implementation of Binary ASK or On-Off Keying (OOK)



Example

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with d = 1?

$f_c = 250\,kHz$

$B = (1+d)S$

$B = (1+1)S$

$B = 2S$

$r = 1$

$B = 2\left(\frac{N}{r}\right)$

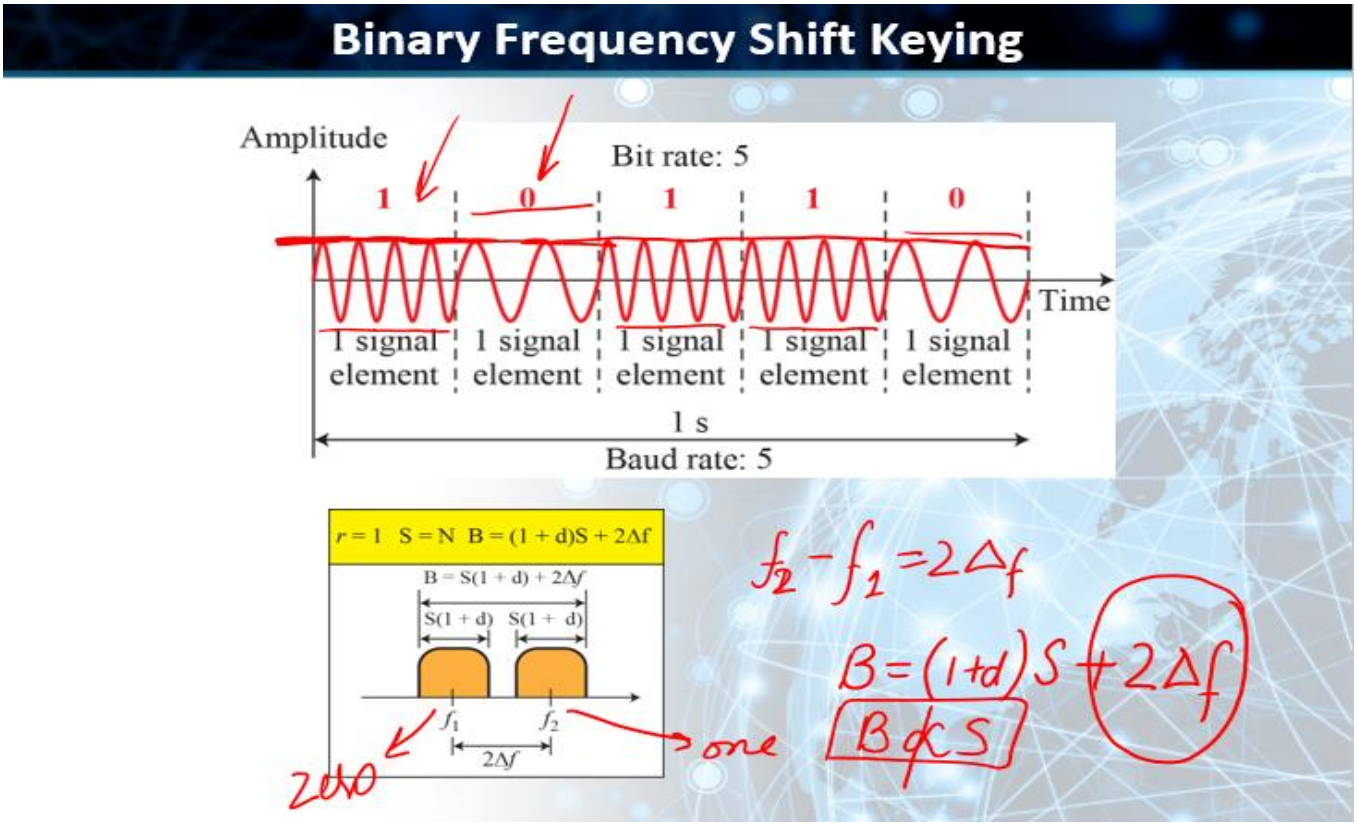
$B = 2N \quad (r=1)$

$N = \frac{B}{2} = \frac{100\,kHz}{2} = 50\,kbps$

Frequency Shift Keying (FSK)

- The frequency of the carrier signal is varied to represent data
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes

- Both peak amplitude and phase remain constant



Frequency Shift Keying (FSK)

- The frequency of the carrier signal is varied to represent data
- Both peak amplitude and phase remain constant

Example:

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

$B = (1+d) \times S + 2\Delta f$

$\Delta f = 50\,kHz$

$B = 2S + 50$

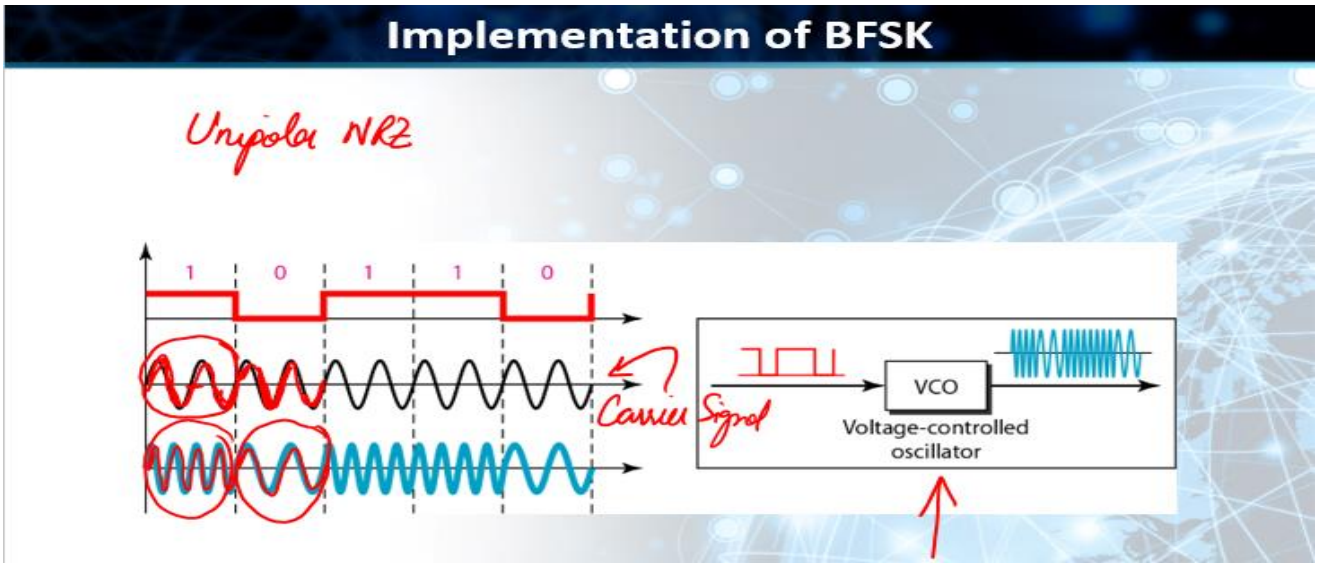
$2S + 50 = 150$

$2S = 50\,kHz$

$S = 25\,kbaud$

$N = S$

$N = 25\,kbps$

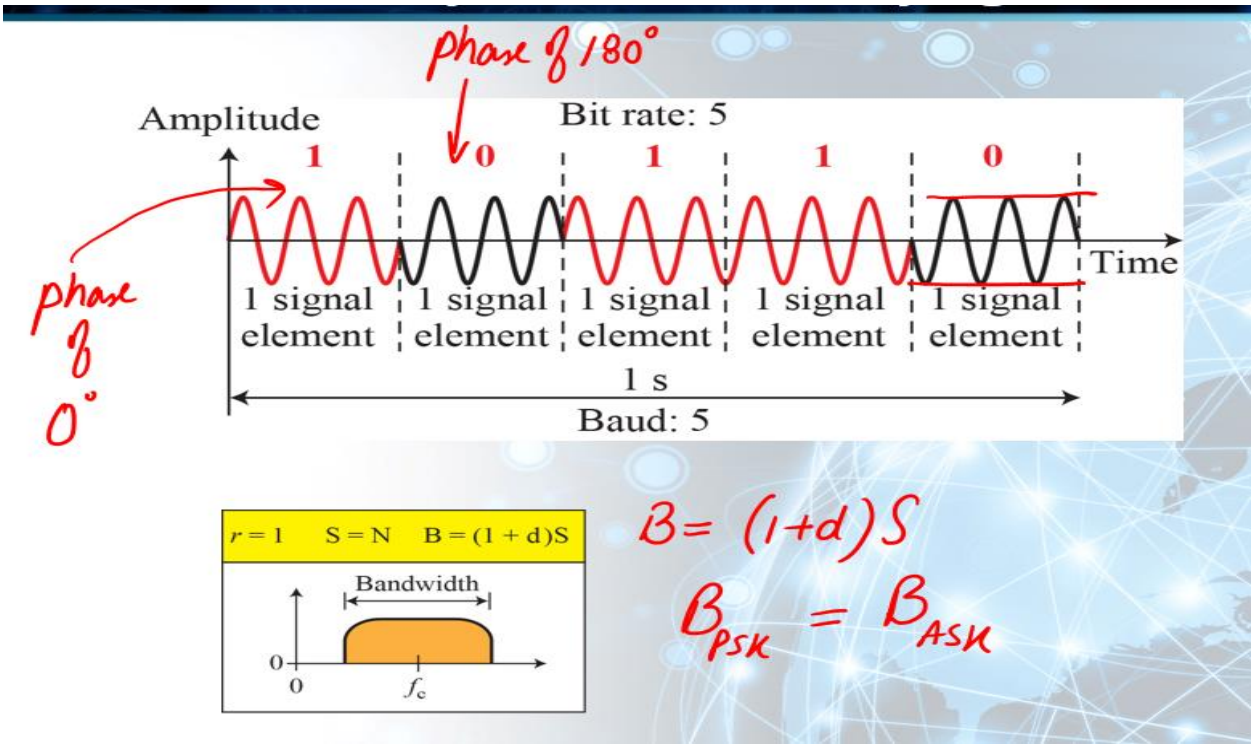


Phase Shift Keying (PSK)

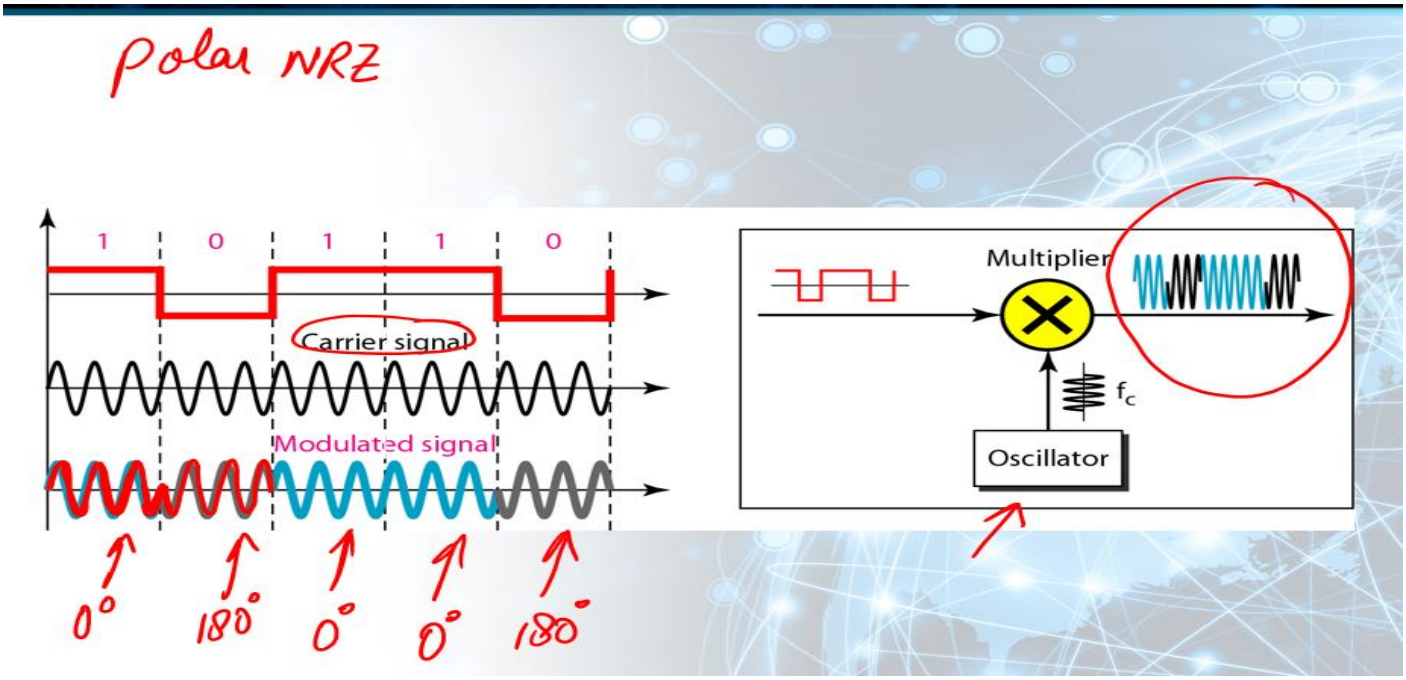
- The phase of the carrier is varied to represent two or more different signal elements
- Both peak amplitude and frequency remain constant
- PSK is relatively common than ASK or FSK

Quadrature Phase Shift Keying (QPSK) is a form of Phase Shift Keying in which two bits are modulated at once, selecting one of four possible carrier phase shifts (0, 90, 180, or 270 degrees).

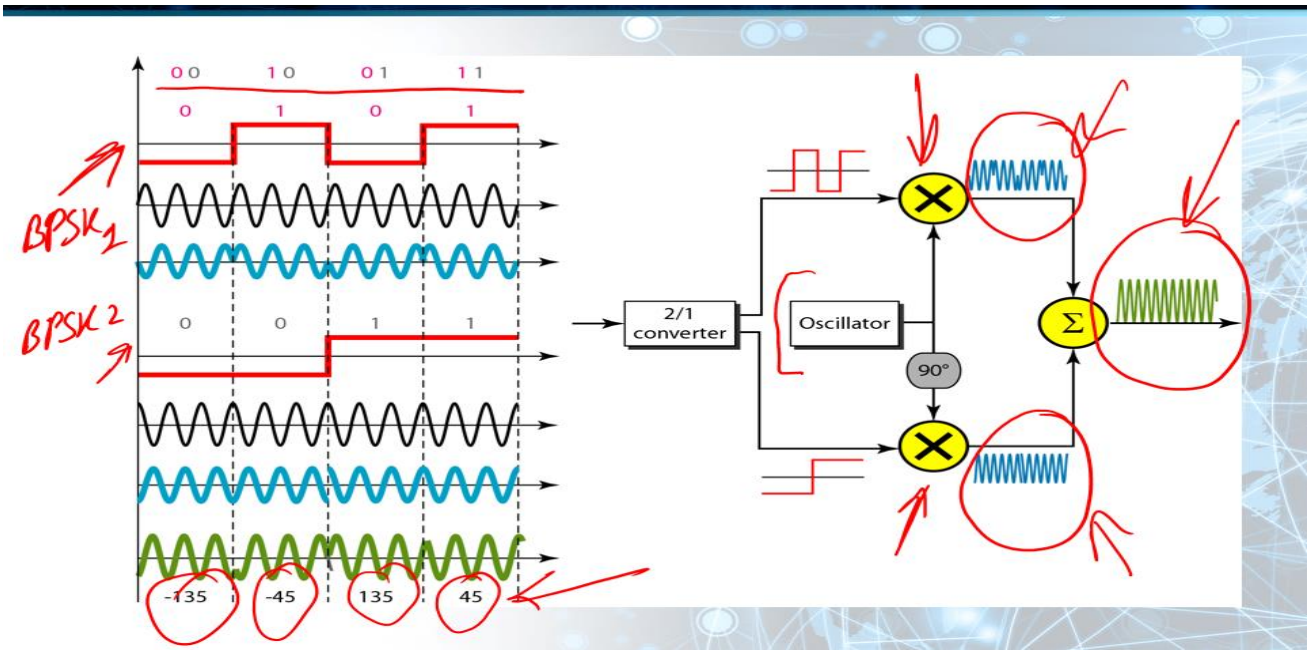
Binary Phase Shift Keying



Implementation of Binary Phase Shift Keying



QPSK and its Implementation



Example :Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of d = 0.

$r = 2$

$$S = \frac{N}{r}$$

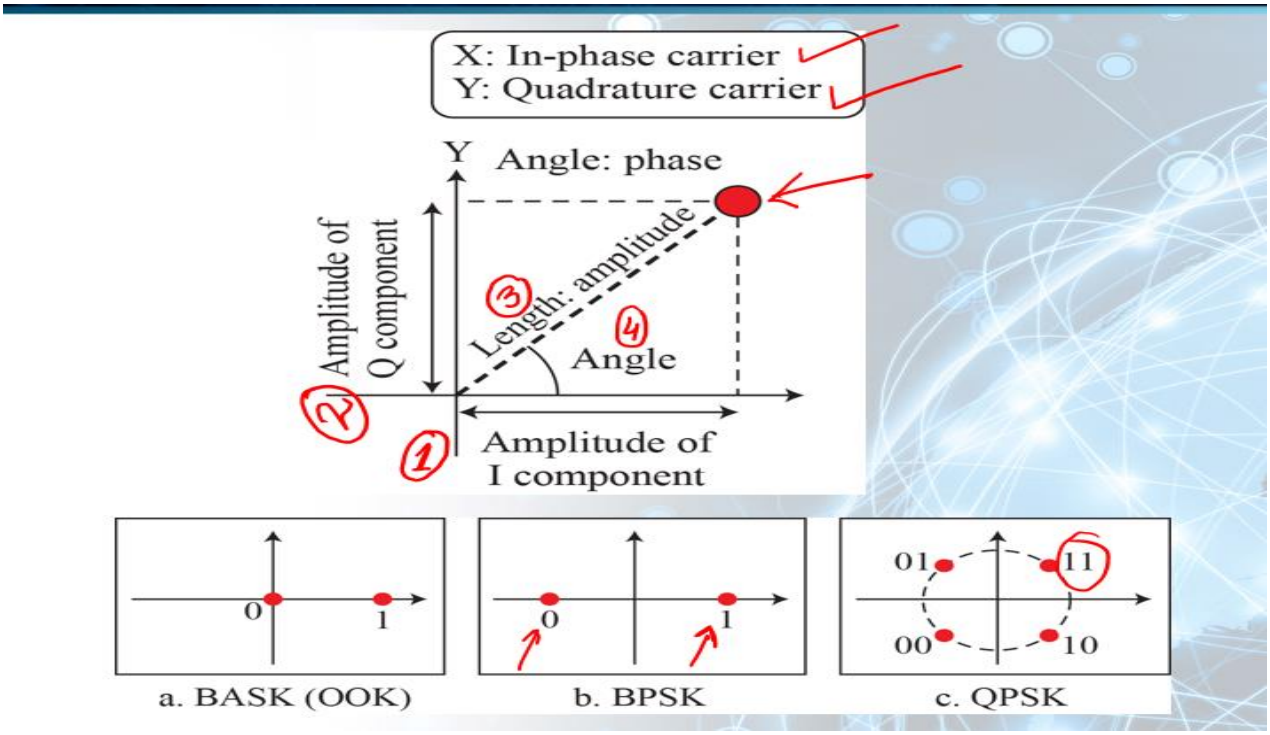
$$= \frac{12\ Mbps}{2}$$

$$= 6\ M\ baud$$

$d = 0 \longrightarrow B = S = 6\ MHz$

Constellation Diagram

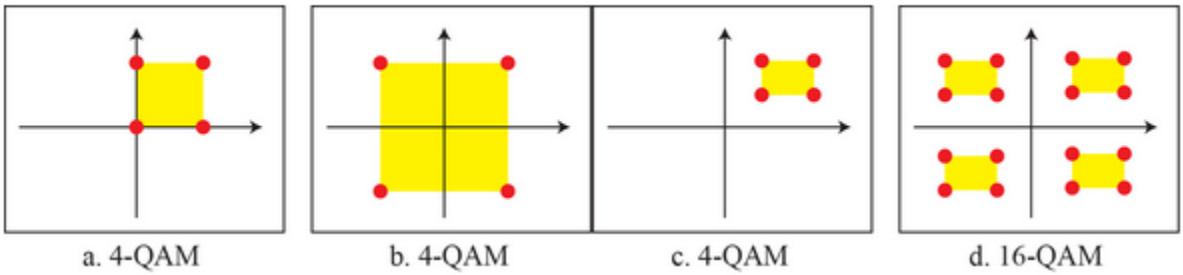
- Helps us define the phase and amplitude of a signal element when we are using two carriers (one in phase and other in quadrature)
- Signal element is represented as a dot



Quadrature Amplitude Modulation (QAM)

- PSK is limited by the ability of the equipment to distinguish small differences in phase which limits its potential bit rate
- We have been altering only one of the three characteristics of a sine wave at a time; but what if we alter two?
- Why not combine ASK and PSK?

Constellation diagrams for some QAMs



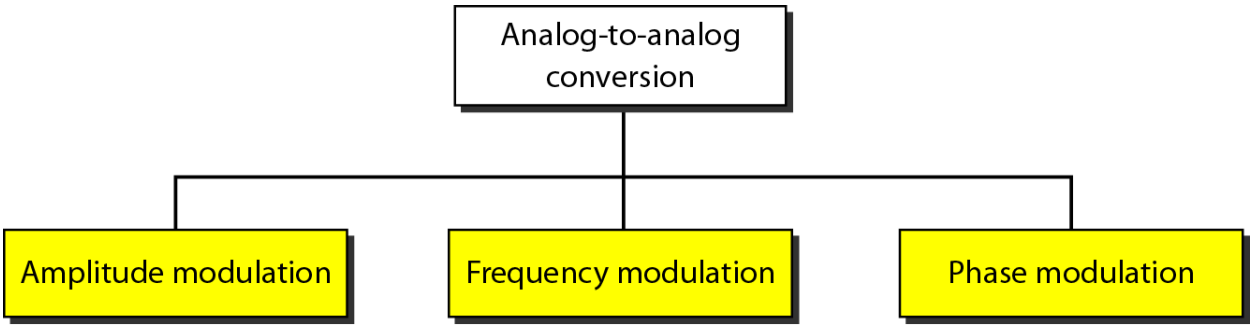
Analog-to-Analog Conversion

▪Representation of Analog information by an Analog signal

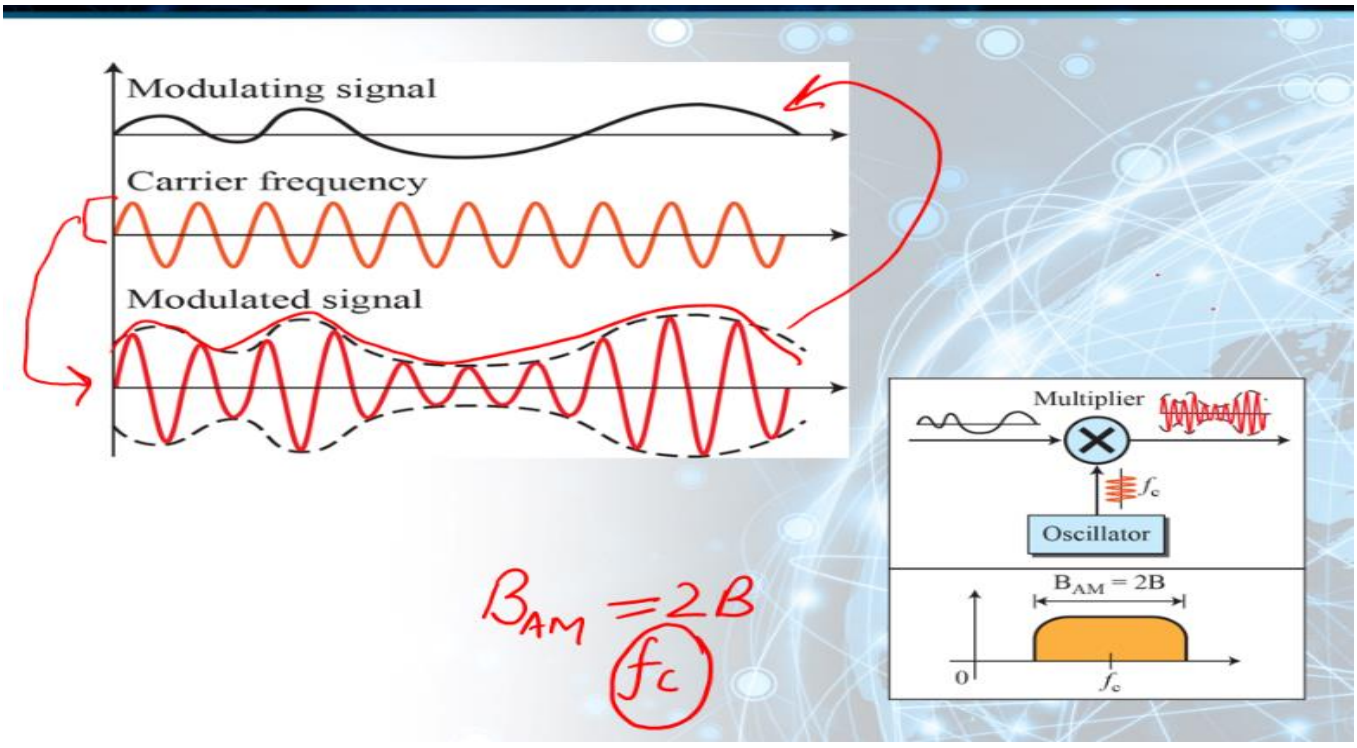
- Amplitude Modulation (AM)
- Frequency Modulation (FM)
- Phase Modulation (PM)

Modulation of analog signal is needed if the medium is bandpass in nature or if only a bandpass channel is available to us .

Types of Analog-to-Analog Modulation



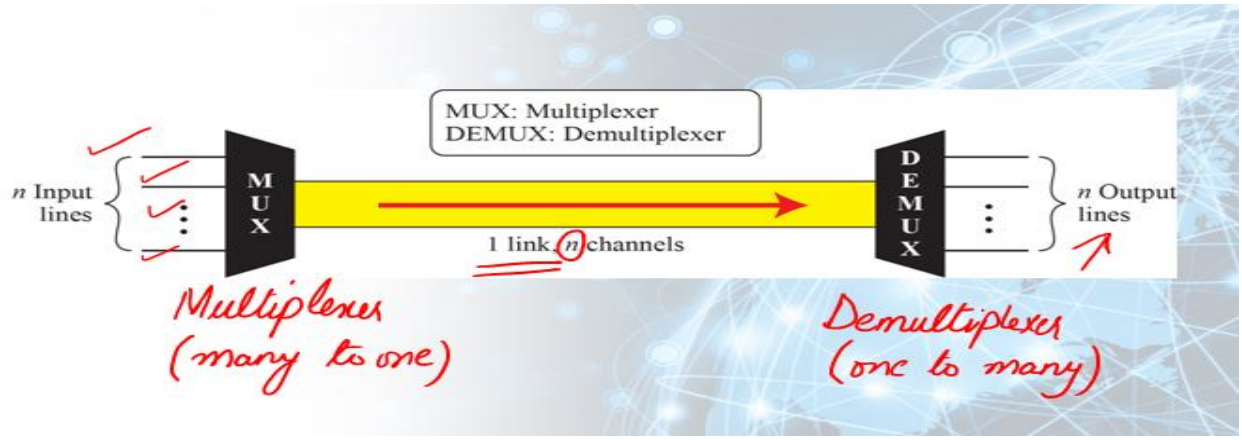
Amplitude modulation



Multiplexing

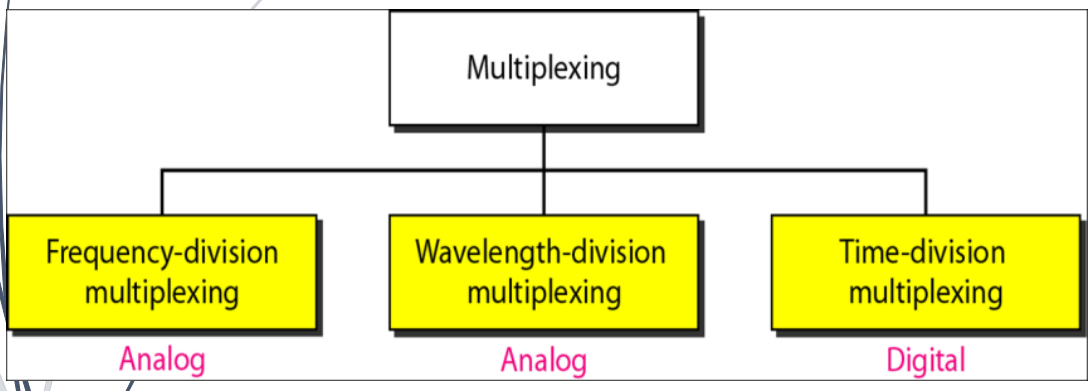
- Simultaneous transmission of multiple signals across a single data link
- As data & telecomm use increases, so does traffic
  - ✓ Add individual links each time a new channel is needed
  - ✓ Install higher-bandwidth links and use each to carry multiple signals

Multiplexing involves combining multiple data streams into a single transmission channel. On the other hand, demultiplexing involves separating a single transmission channel into multiple data streams at the receiving end.



There are three types of multiplexing:

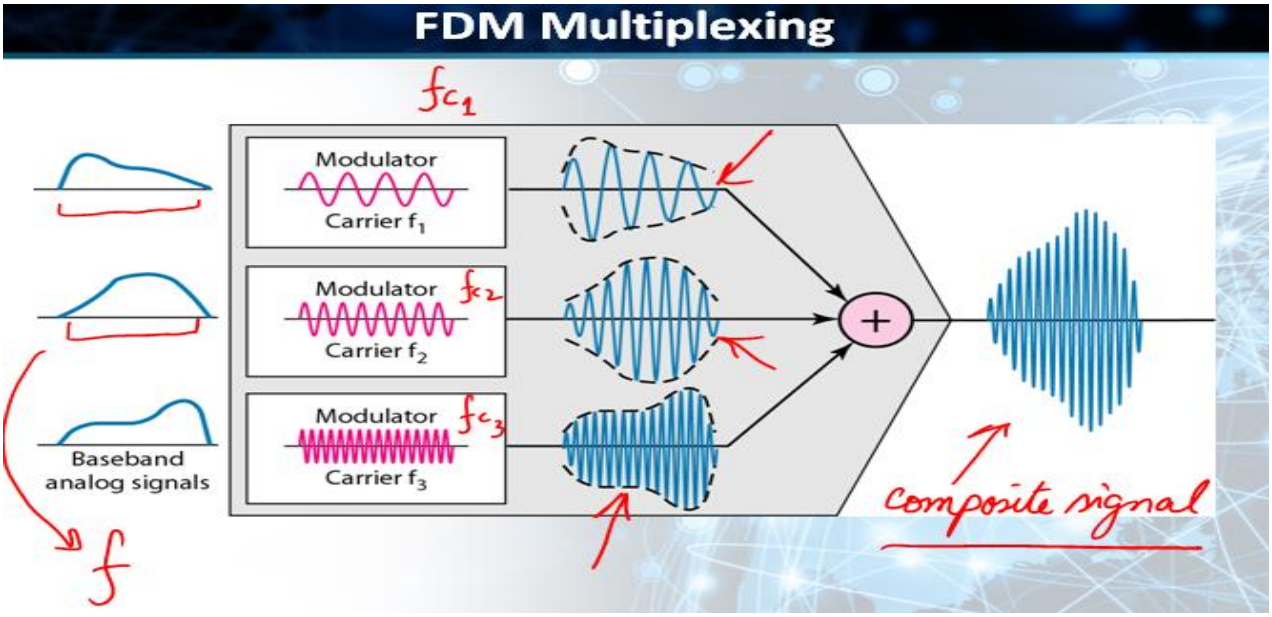
- ❖ Frequency Division Multiplexing (FDM)
- ❖ Time-Division Multiplexing (TDM)
- ❖ Wavelength Division Multiplexing (WDM)

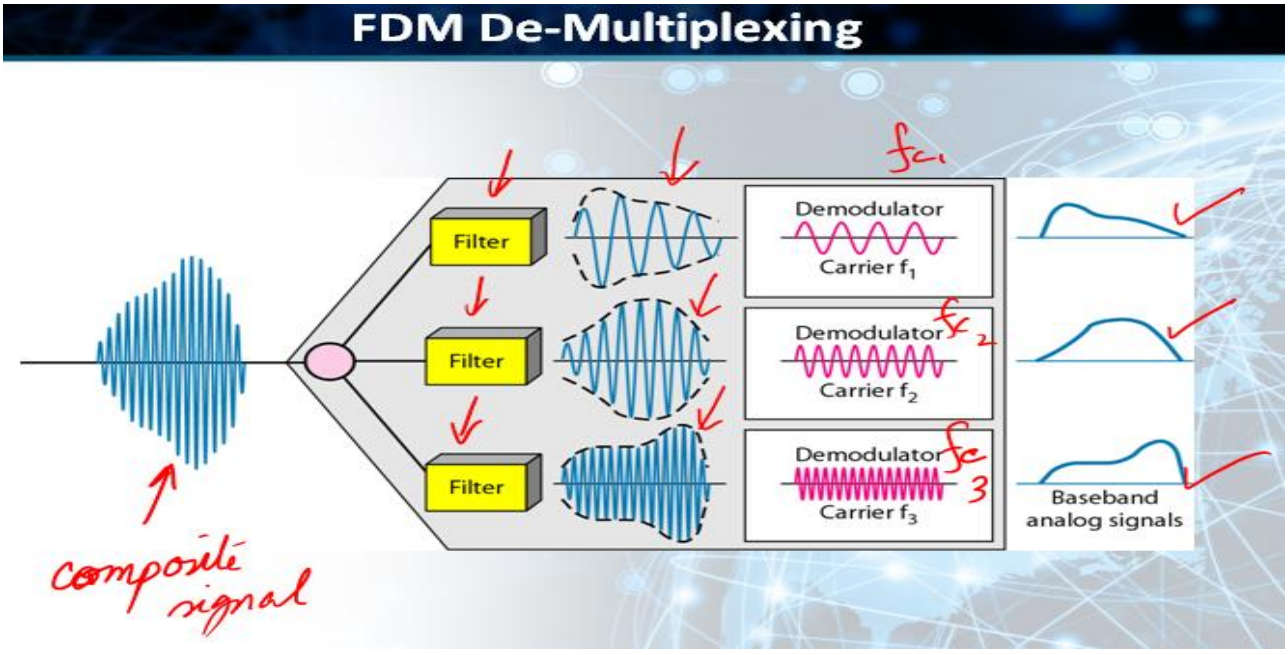


Frequency-Division Multiplexing

- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted
- Signals generated by each sending device modulate different carrier frequencies

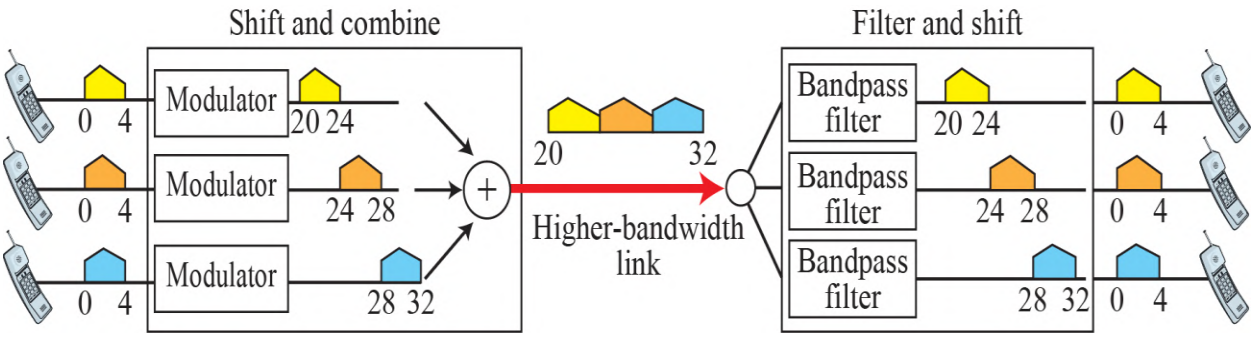
- These modulated signals are then combined into a single composite signal that can be transported by the link
- An analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted





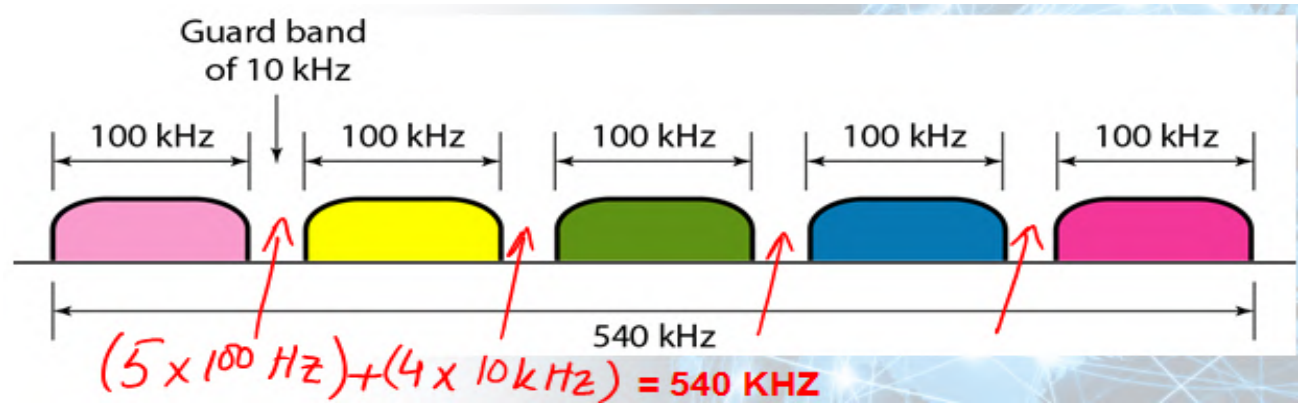
Example

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.



Example

Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?



The Analog Carrier System

- Telephone companies multiplex signals from lower-bandwidth lines on to higher-bandwidth lines
- For Analog, FDM is used

Bandwidth per channel: 100 kHz
Number of guard bands (between channels): 4 (since there are 5 channels)
Guard band width: 10 kHz

Total bandwidth required is given by:

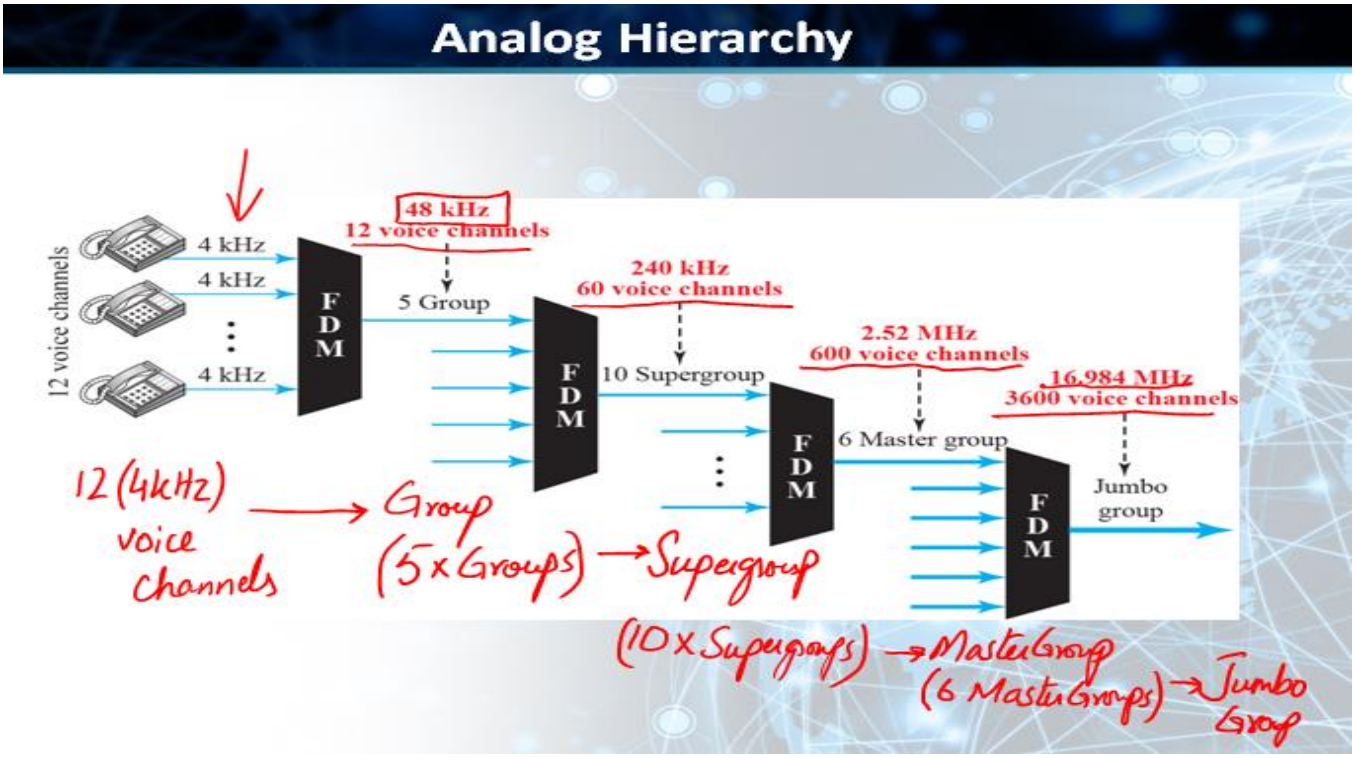
Total Bandwidth=(Number of channels×Channel bandwidth)+(Number of guard bands×Guard band width)

Total Bandwidth=(5×100kHz)+(4×10kHz)

Total Bandwidth=500kHz+40kHz

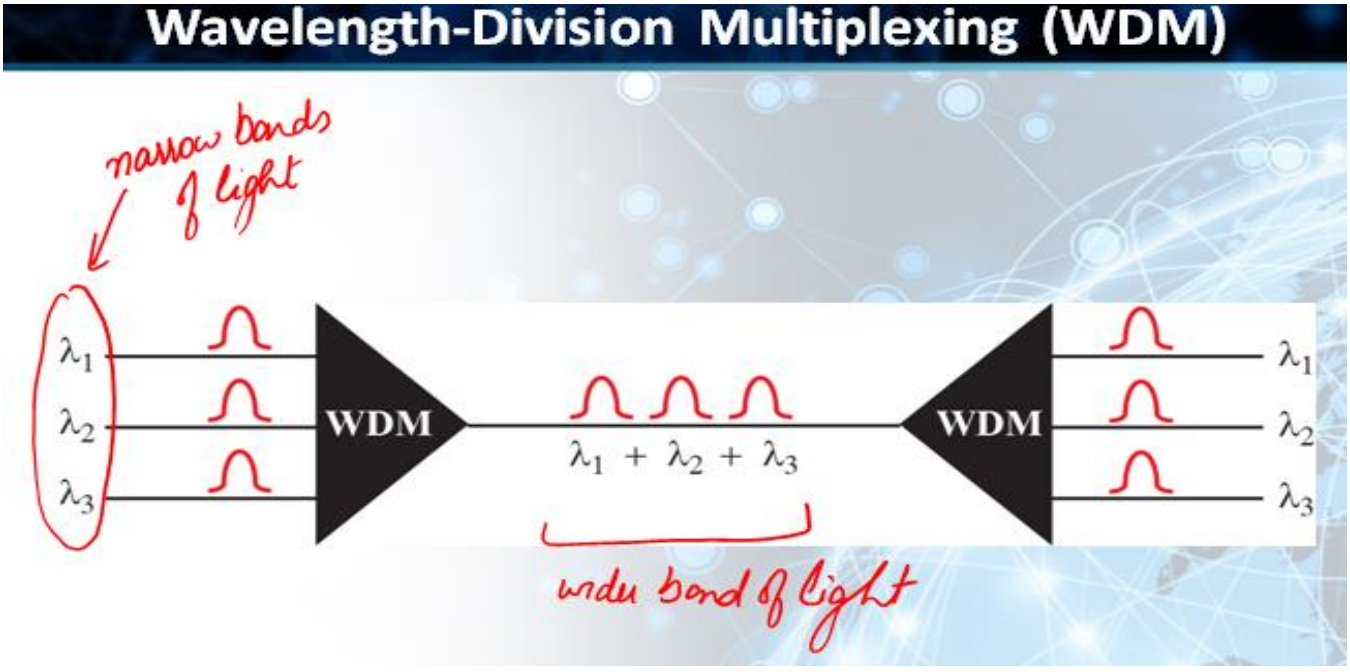
Total Bandwidth=540kHz

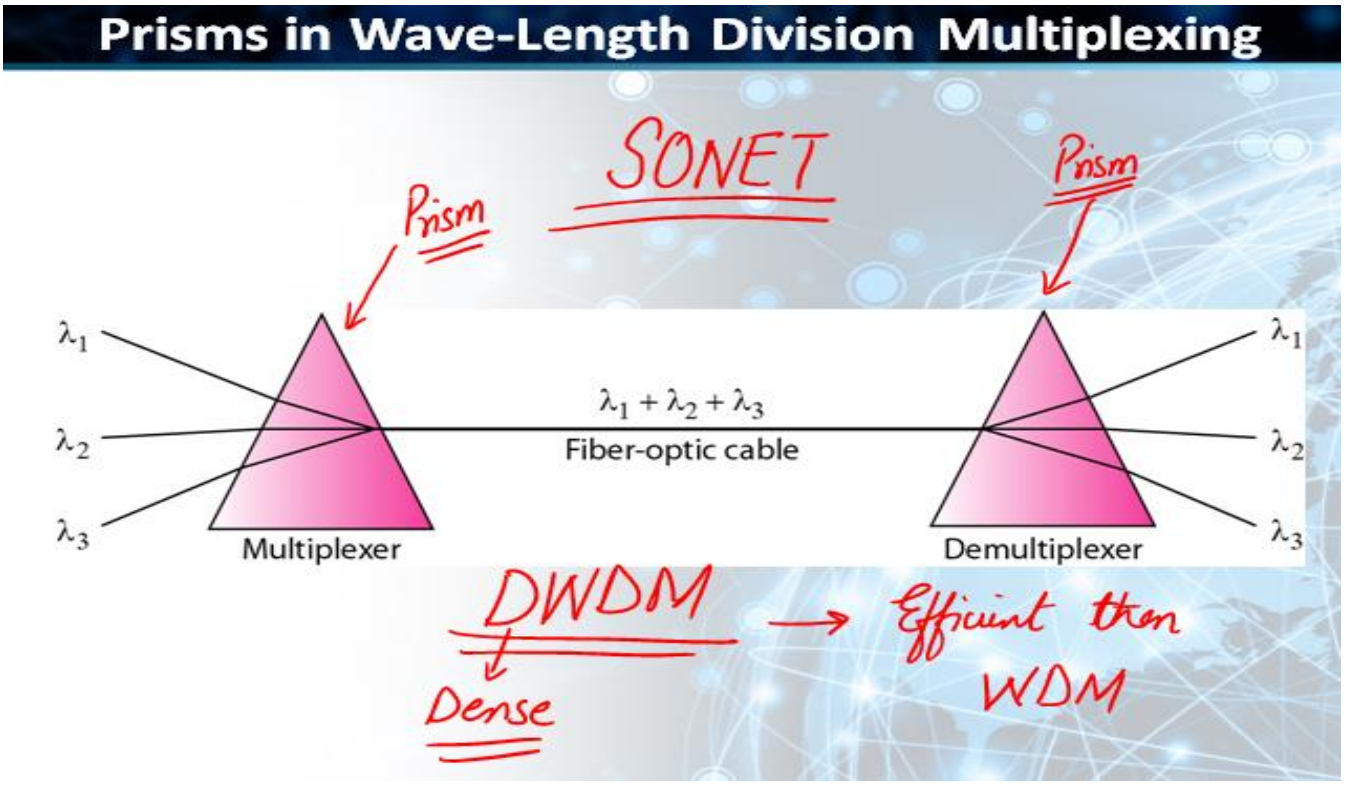
Therefore, the minimum bandwidth of the link, considering the 10 kHz guard bands between channels, is 540 kHz.



Wavelength-Division Multiplexing

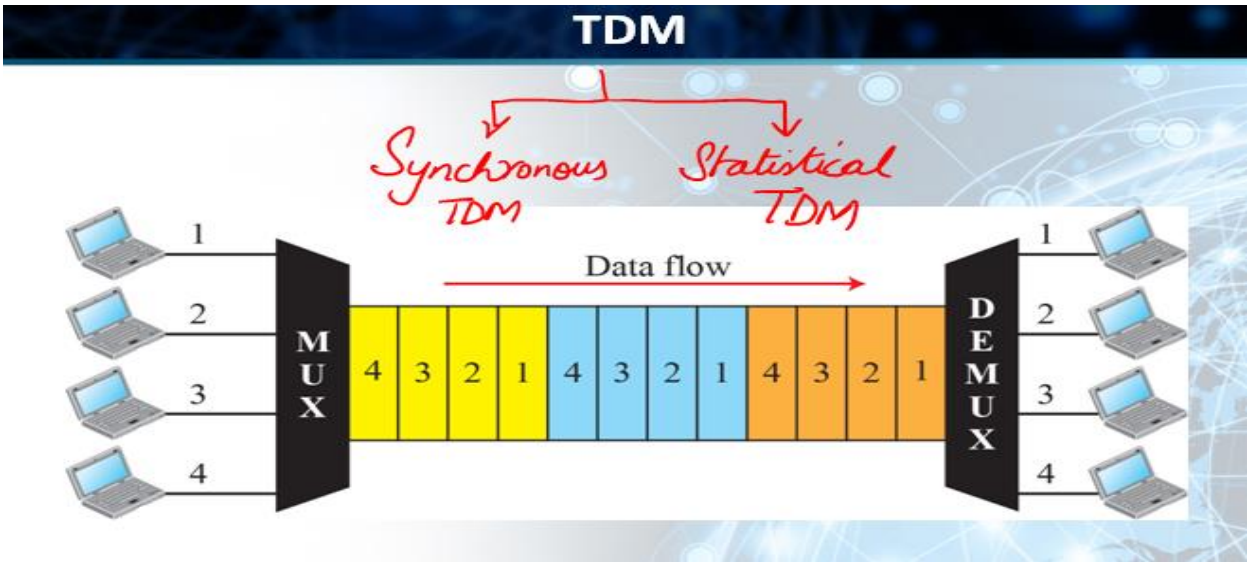
- Designed to use the high-data-rate capability of fiber-optic cable
- Fiber data rate is higher than the data rate of metallic transmission cable
- Using a fiber-optic cable for a single line wastes the available bandwidth
- Multiplexing allows us to combine several lines into one





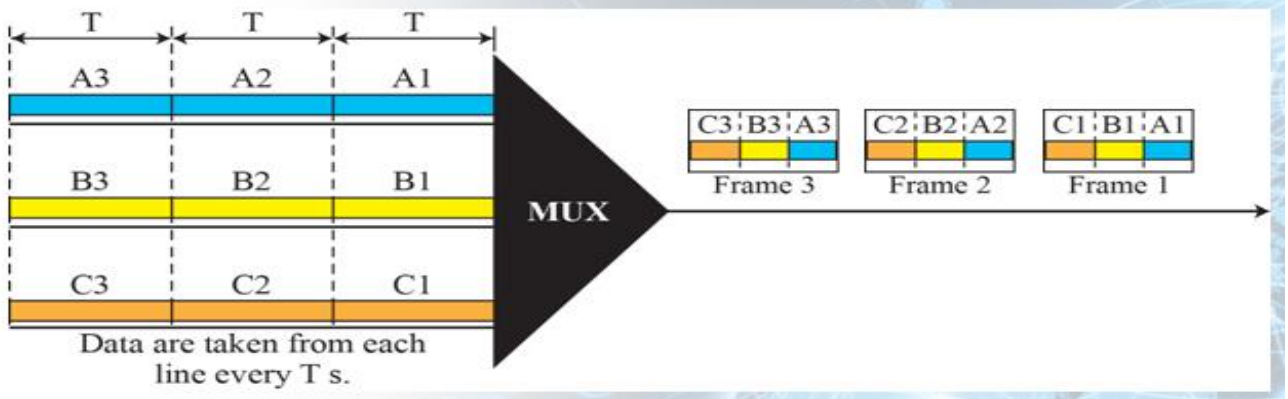
Time-Division Multiplexing

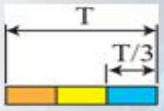
- Digital process that allows several connections to share the high bandwidth of a link
- Time is shared i.e. each connection occupies a portion of time in the link



Synchronous time division multiplexing assigns a fixed time slot to each connected device, whether the device transmits data or not. With asynchronous time division multiplexing, time slots are flexible, and assigned when connected devices have data that is ready to send.

Synchronous Time-Division Multiplexing





Each frame is 3 time slots.
Each time slot duration is T/3 s.

Example

In Figure the data rate for each input connection is 1 kbps. If 1 bit at a time is multiplexed (a unit is 1 bit), what is the duration of

- **each input slot,**
- **each output slot, and**
- **each frame?**

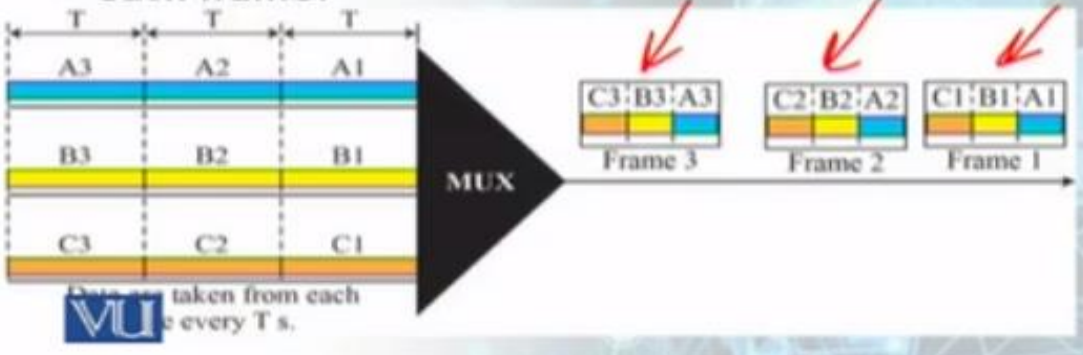
Data Rate = 1 kbps

Bit Duration = $\frac{1}{1000}$ s = 1 msec

Duration of i/p slot = 1 msec ✓

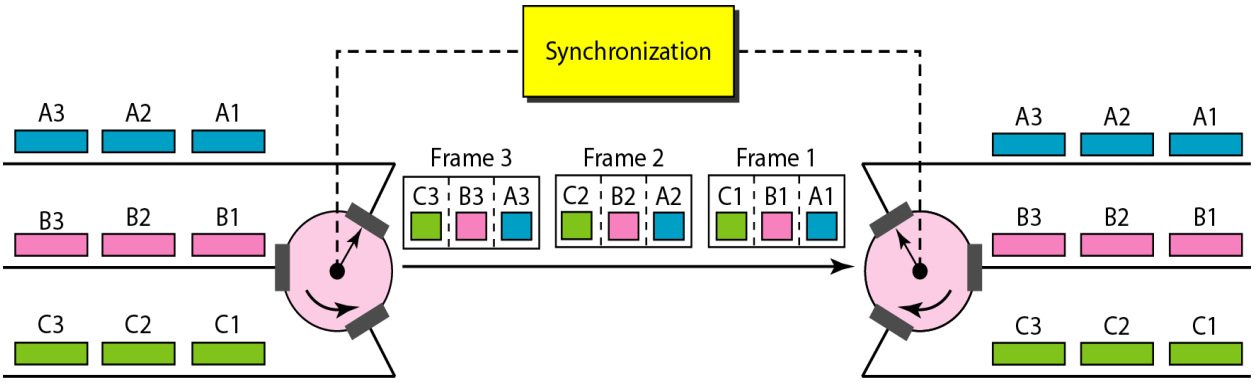
Duration of o/p slot = $\frac{1}{3}$ msec ✓

Duration of Frame = $3 \times \frac{1}{3}$ msec = 1 ms ✓



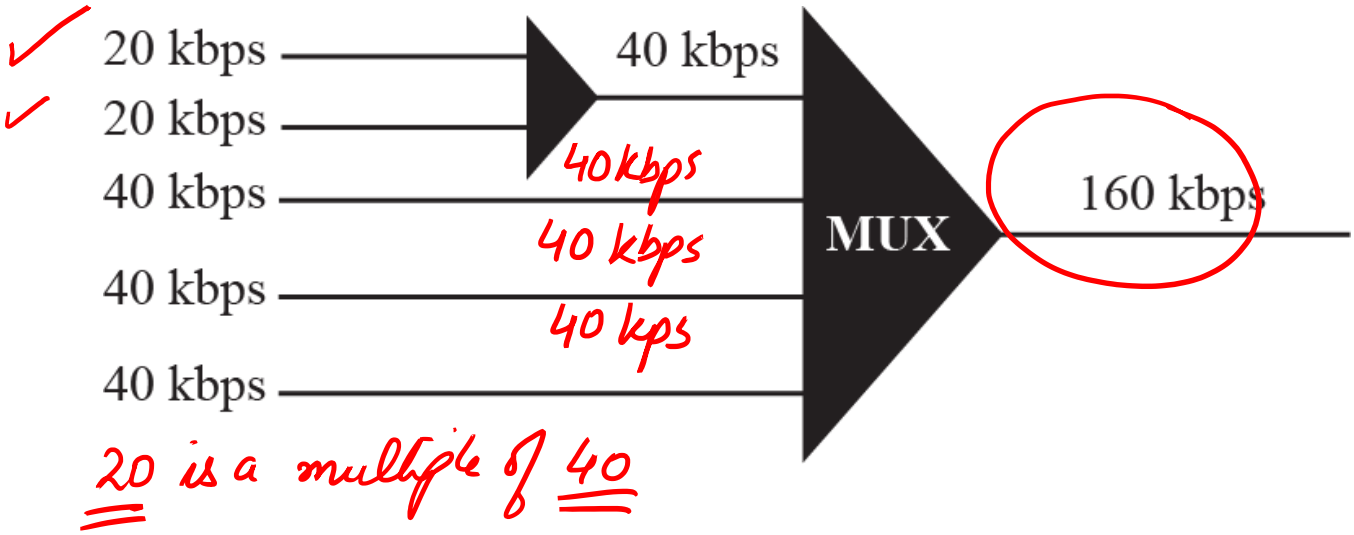
Interleaving

On the multiplexing side, as the switch opens in front of a connection, that connection has the opportunity to send a unit onto the path. This process is called interleaving. On the demultiplexing side, as the switch opens in front of a connection, that connection has the opportunity to receive a unit from the path.



Multilevel Multiplexing

Multilevel multiplexing is a technique used when the data rate of an input line is a multiple of others.

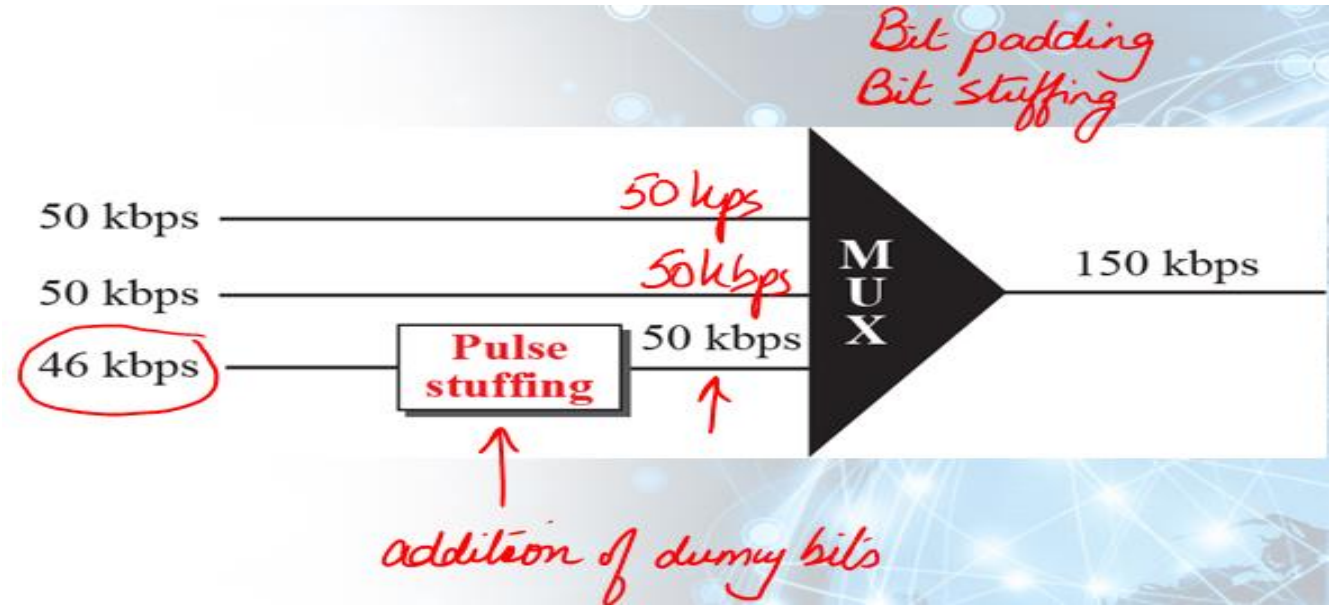


Multiple-SLot Multiplexing

Multiple multiplexing is a technique used to allot more than one slot in a frame to a single input line.

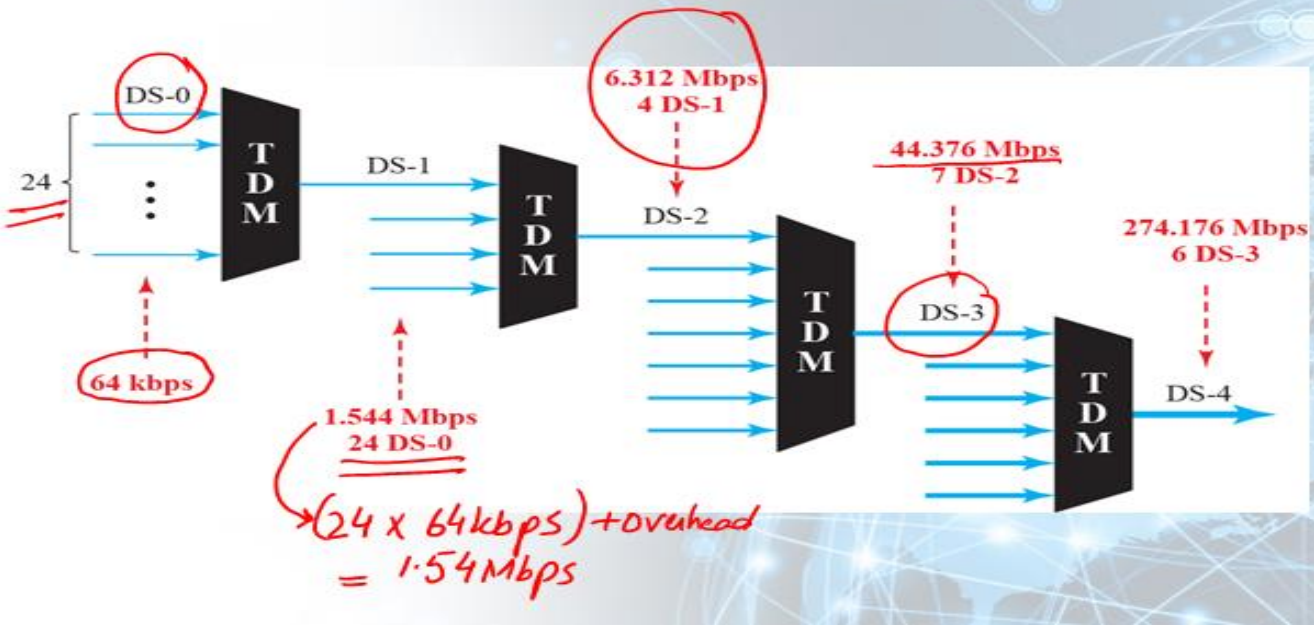
Pulse Stuffing

Sometimes the bit rates of sources are not multiple integers of each other. Then to make the data rate compatible , dummy bits are added to input lines . This technique is called pulse stuffing, bit padding, or bit stuffing.



To maximize the efficiency of their infrastructure, telephone companies have traditionally multiplexed digital signals from lower data rate lines onto higher data rate lines. The digital hierarchy uses DS-0 (64 Kbps), DS-1 (1.544 Mbps), DS-2 (6.312 Mbps), DS-3 (44.376 Mbps), and DS-4 (274.176 Mbps).

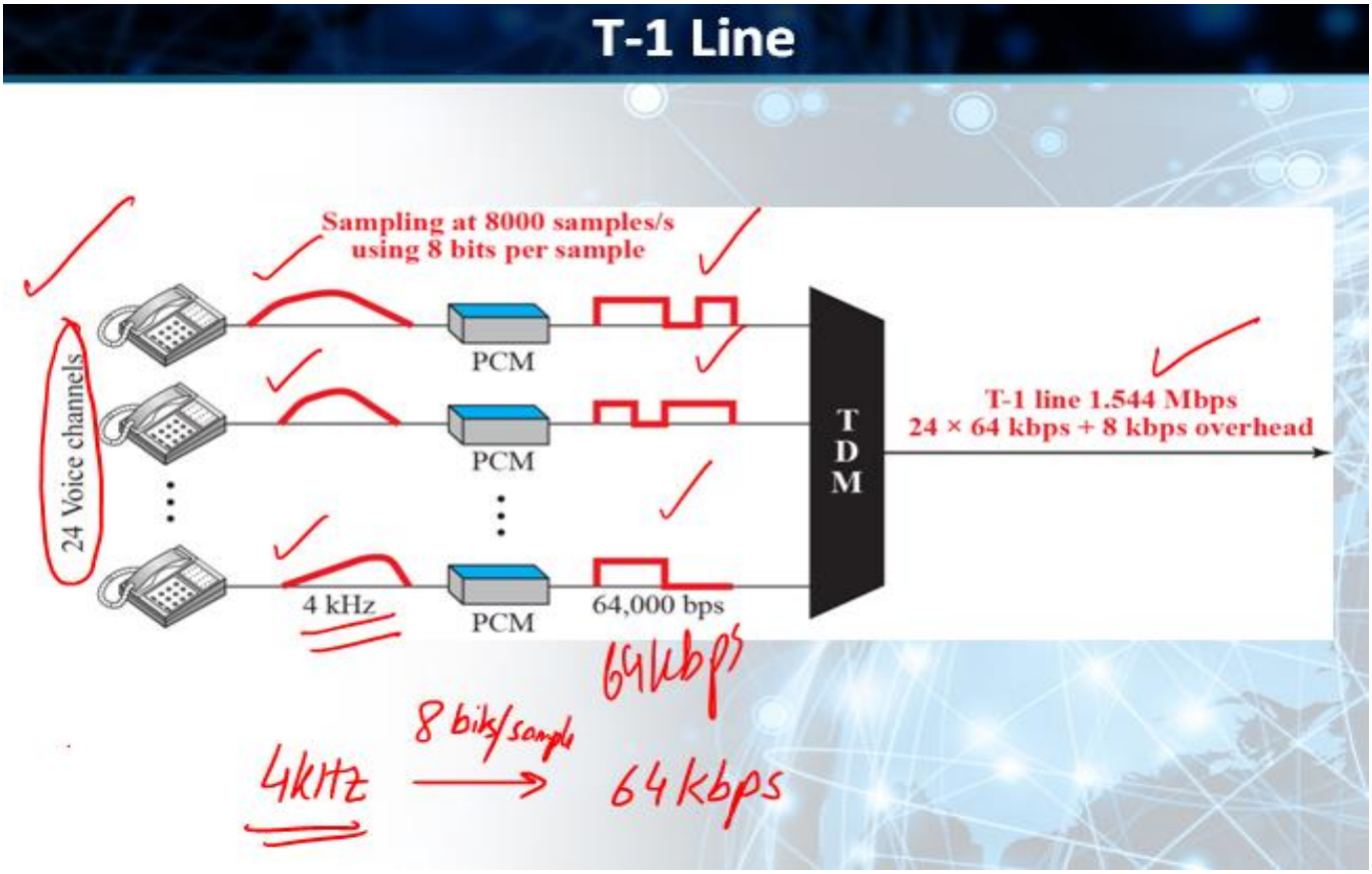
Digital Hierarchy



DS and T Line Rates

DS0

Service	Line	Rate (Mbps)	Voice Channels
DS-1	T-1	1.544	24
DS-2	T-2	6.312	96
DS-3	T-3	44.736	672
DS-4	T-4	274.176	4032



Time-Division Multiplexing

- Synchronous TDM
- Statistical TDM

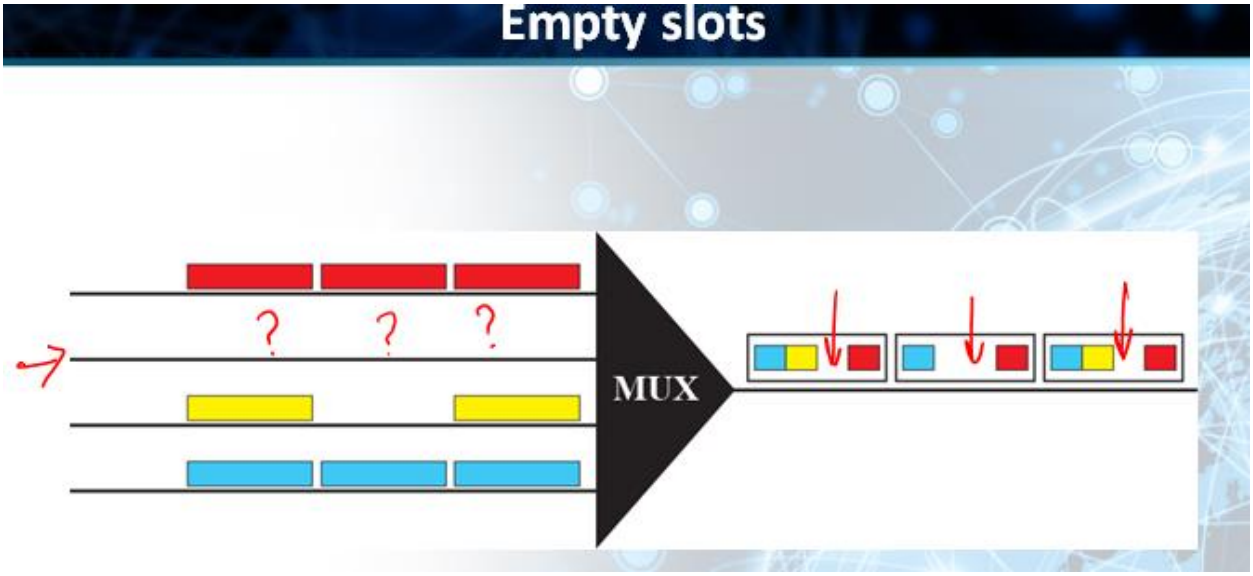
In synchronous TDM, each input has a reserved slot in the output frame. This can be inefficient if some input lines have no data to send.

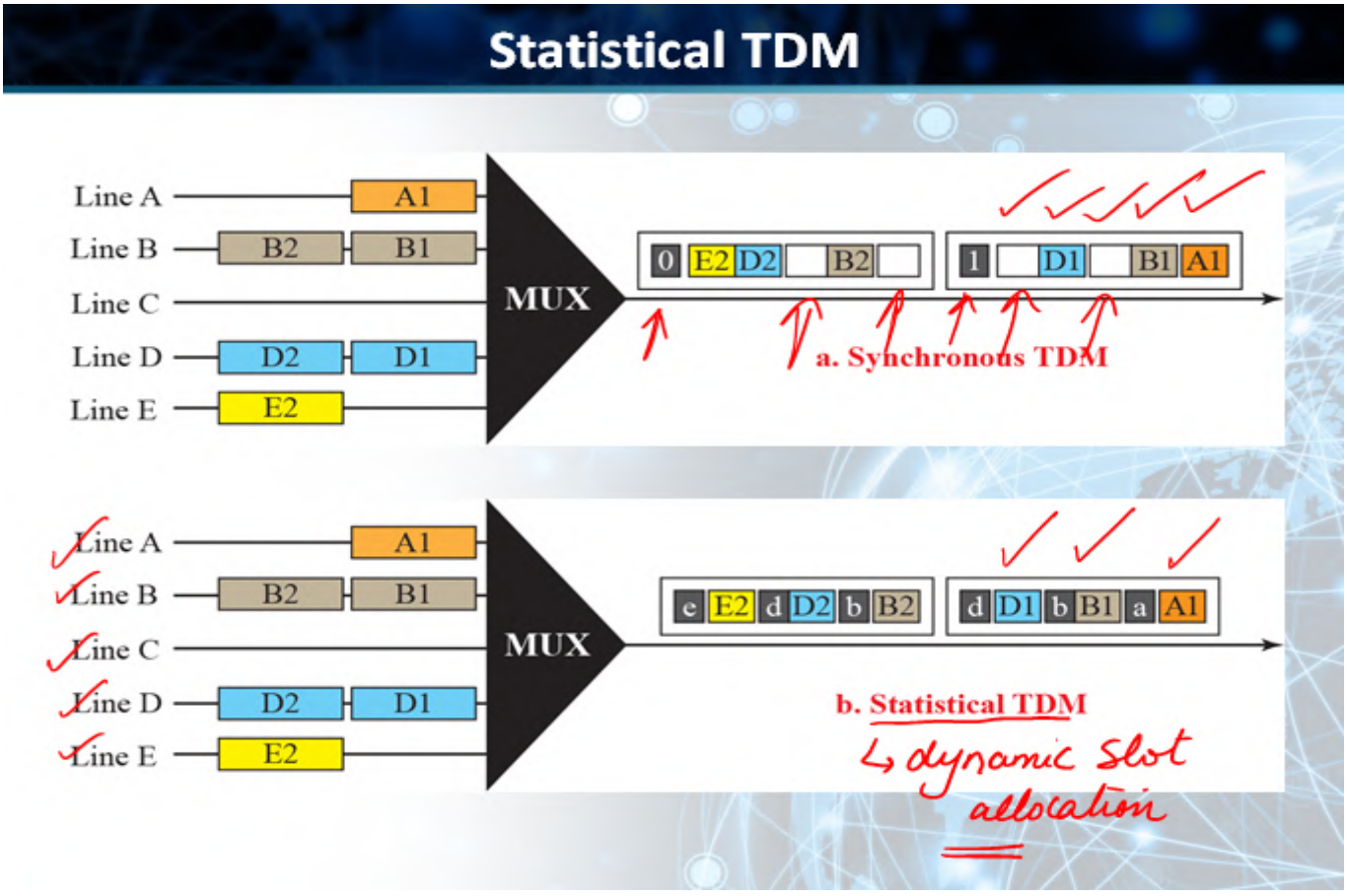
In statistical time-division multiplexing, slots are dynamically allocated to improve bandwidth efficiency.

Only when an input line has a slot's worth of data to send is it given a slot in the output frame.

In statistical multiplexing, the number of slots in each frame is less than the number of input lines.

The multiplexer checks each input line in round-robin fashion; it allocates a slot for an input line if the line has data to send; otherwise, it skips the line and checks the next line.





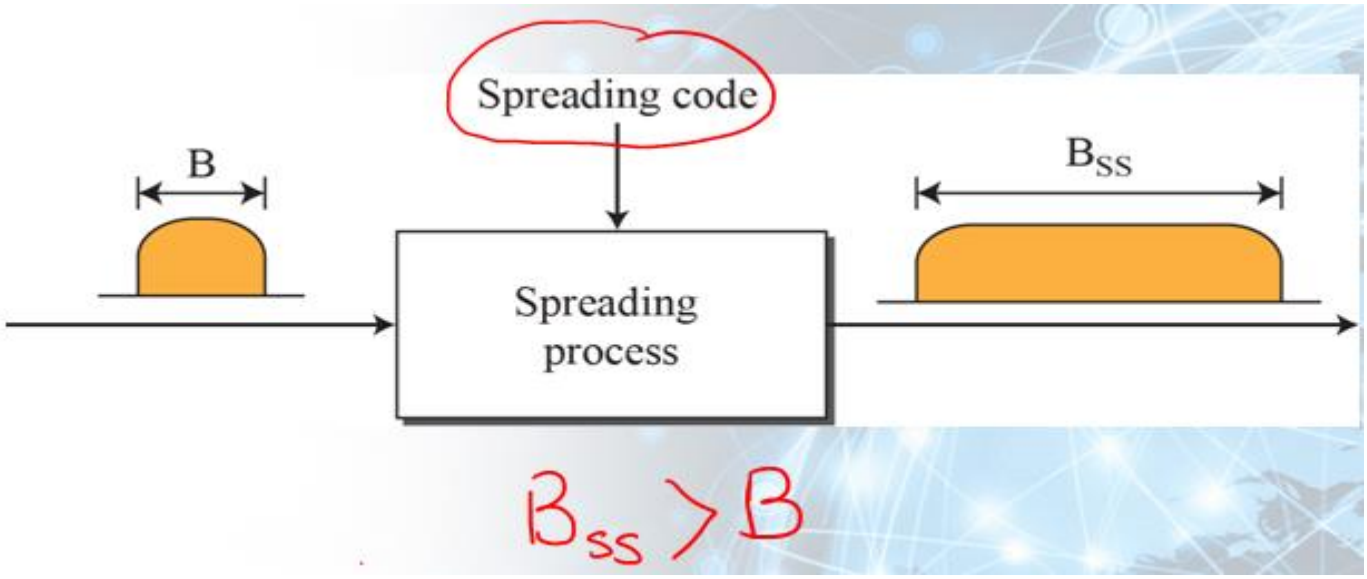
SPREAD SPECTRUM

- In wireless applications, stations must be able to share the medium without interception by an eavesdropper and without being subject to jamming from a malicious intruder

- To achieve these goals, spread spectrum adds redundancy and spread original spectrum needed for each station

SPREAD SPECTRUM – Principles

- Bandwidth allocated to each station needs to be larger than what is needed to allow Redundancy
- Spreading process should be independent of the original signal

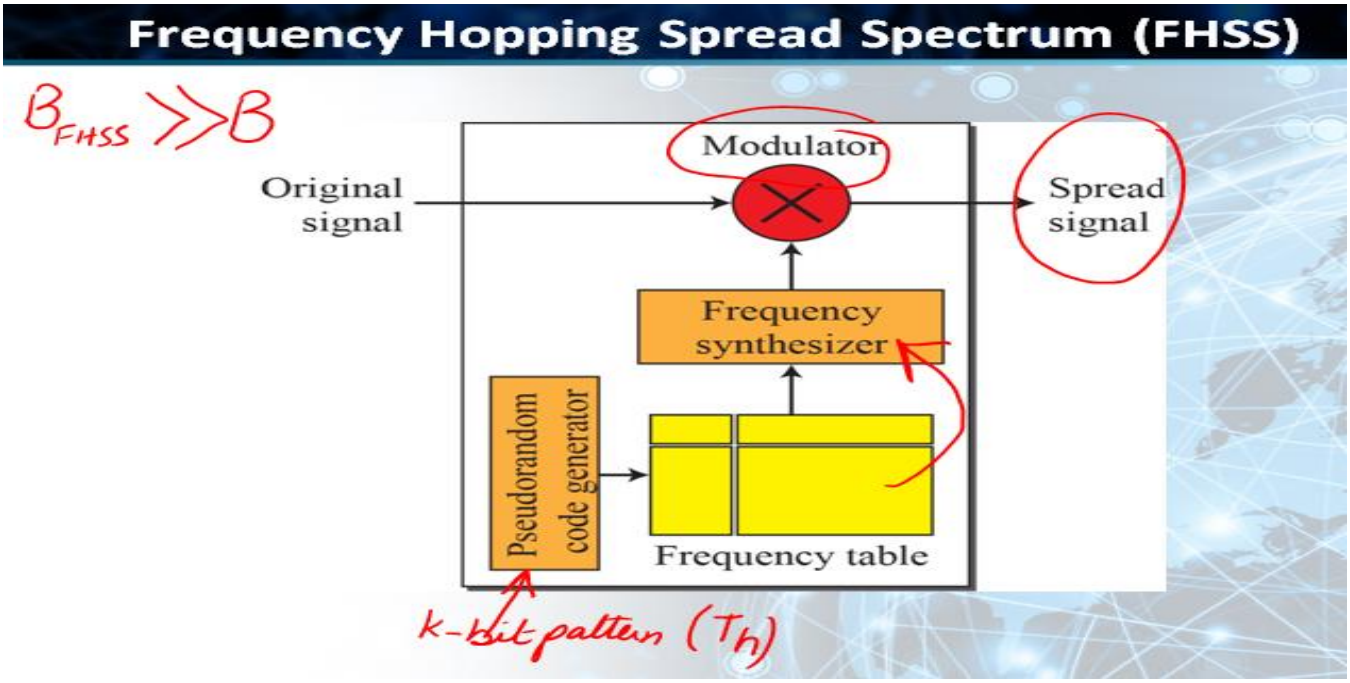


SPREAD SPECTRUM TECHNIQUES

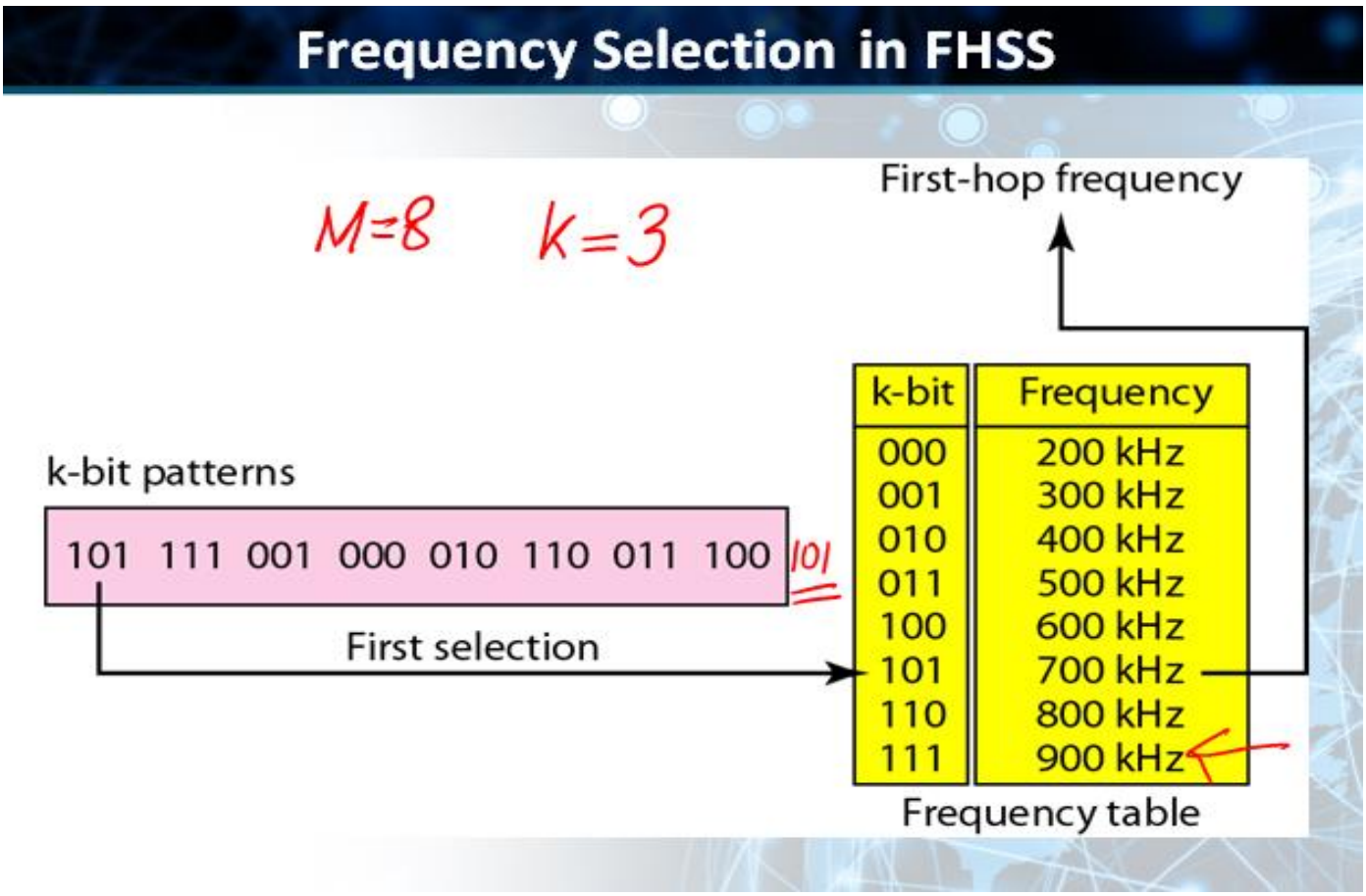
- Frequency Hopping Spread Spectrum (FHSS)
- Direct Sequence Spread Spectrum (DSSS)

Frequency Hopping Spread Spectrum (FHSS)

- 'M' different carrier frequencies that are modulated by the source signal
- At one moment, signal modulates one carrier frequency and at next moment, it modulates another

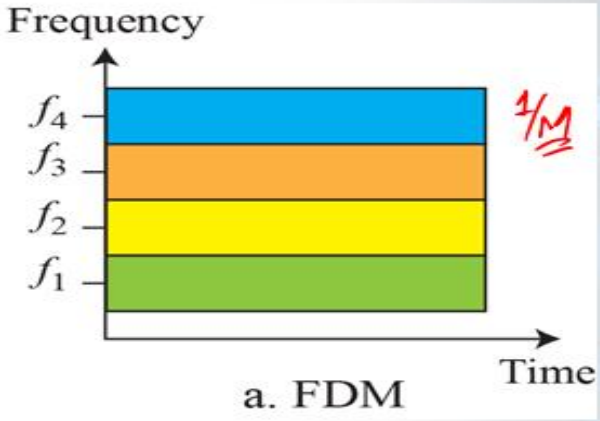


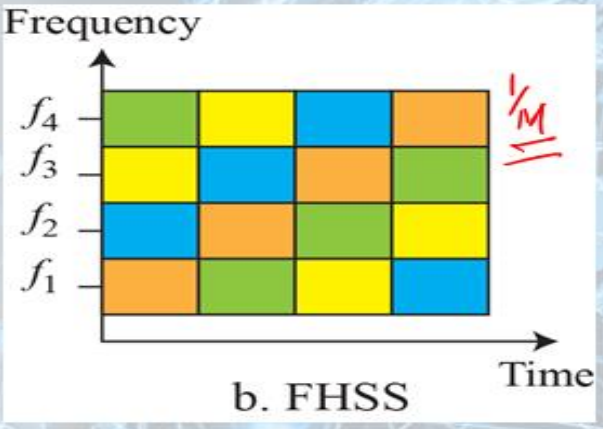
Frequency Selection in FHSS



FHSS Cycles

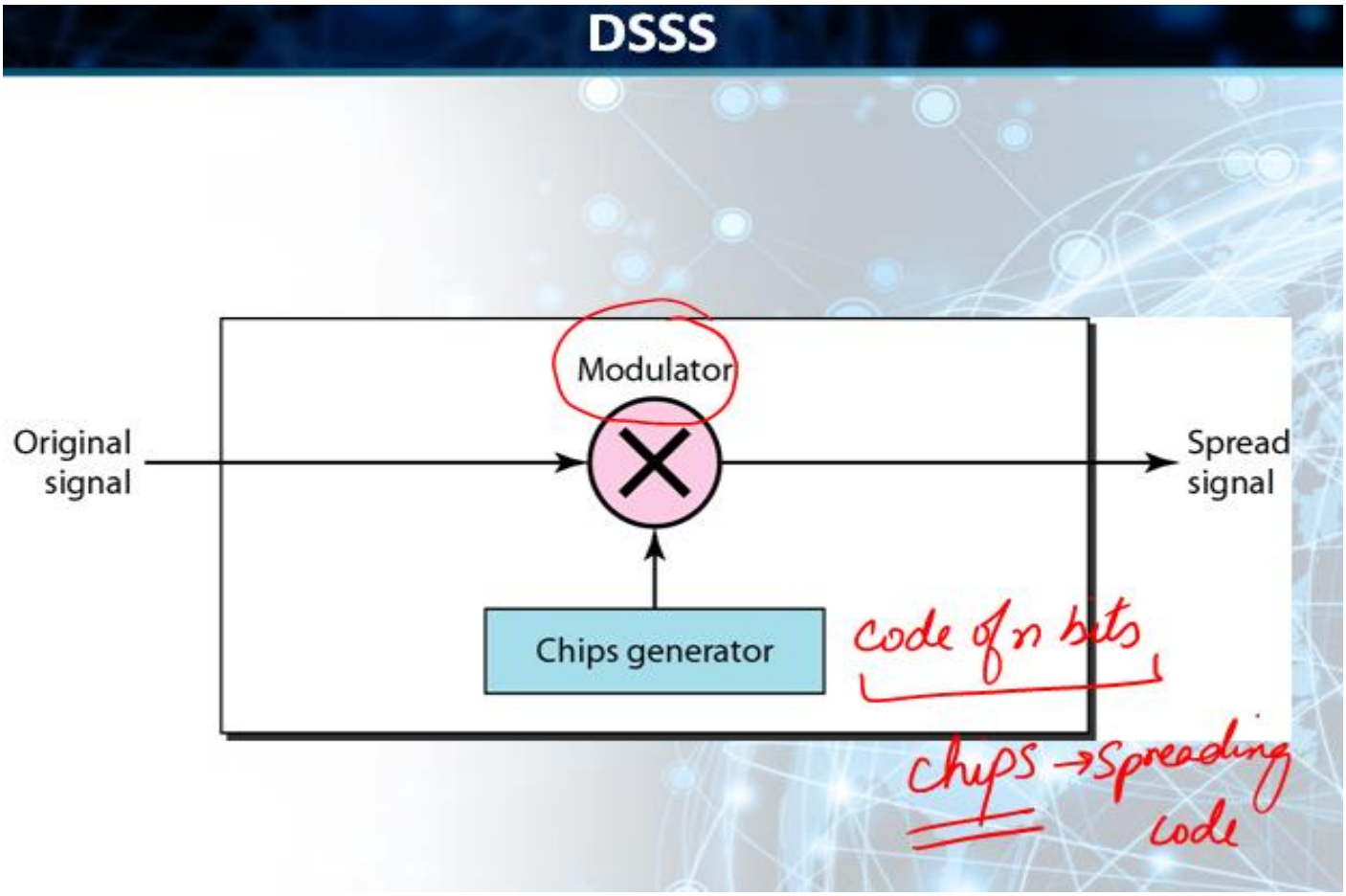
Bandwidth Sharing





DSSS

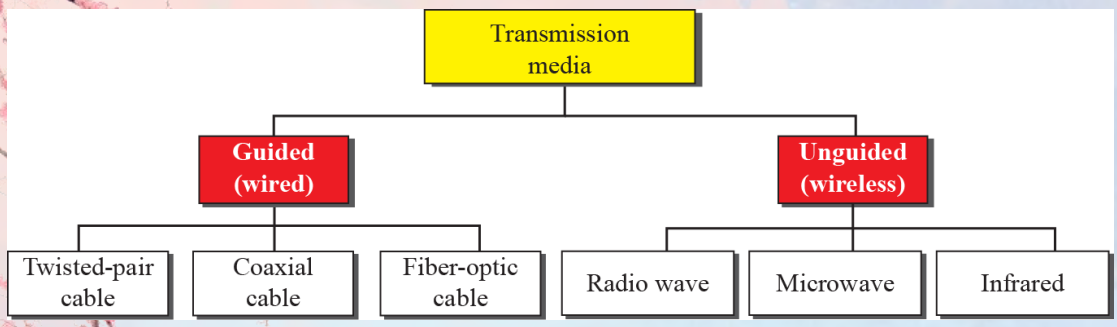
- DSSS also expands the bandwidth of the original signal, but the process is different
- We replace each data bit with ‘n’ bits using a spreading code
- Each bit is assigned a code of ’n’ bits, called chips, where the chip rate is ‘n’ times that of the data bit



WEEK - 7

Transmission media is a communication channel that transfers the information from the sender to the receiver. Data is transmitted through the electromagnetic signals.

It is a physical path between sender/transmitter and receiver in data communication. Data is transmitted in the form of bits.



Transmission Media

- Located below the physical layer and are directly controlled by the physical layer
- Belong to layer zero
- Metallic Media i.e. Twisted pair and Coaxial Cable
- Optical Fiber Cable
- Free Space i.e. Air, Vacuum
- The transmission media is available in the lowest layer of the OSI reference model, i.e., **Physical layer**

Guided Media

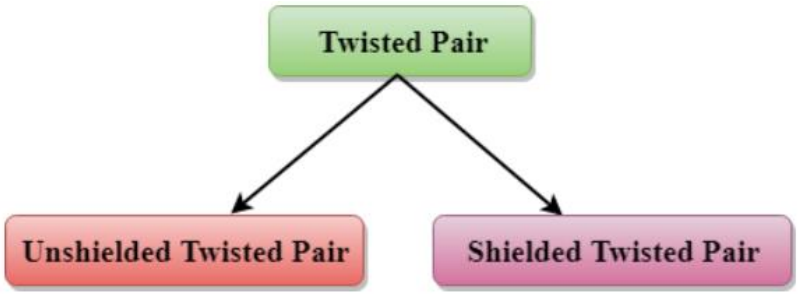
- Media that provides a conduit from one device to another
- Twisted-pair cable, coaxial cable, and fiber-optic cable
- Signal traveling along any of these media is directed and contained by the physical limits of the medium

Twisted-Pair Cable

- Consists of 2 copper conductors, each with its own plastic insulation, twisted together
- One wire carries signals and other is ground reference
- Receiver uses difference between the two
- Interference (Noise) & Crosstalk

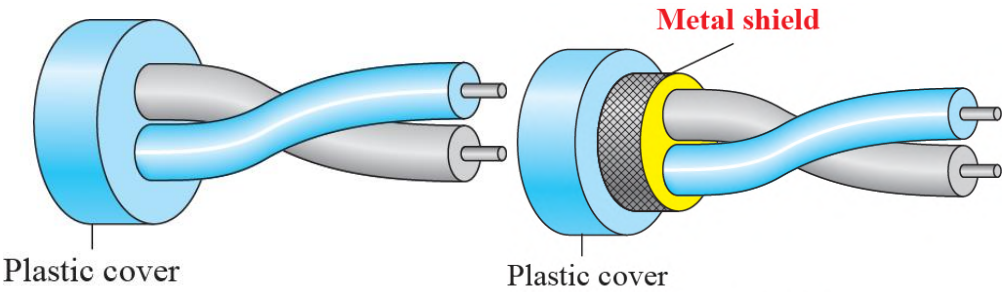
Twisted pair is a guided media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. The frequency range for twisted pair cable is from 0 to 3.5KHz.A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.

Types of Twisted pair:



Types of Twisted Pair cable

- Unshielded twisted pair (UTP). These cables generally comprise wires and insulators.
- Shielded twisted pair (STP). These cables come with a braided, wired mesh that encases each pair of insulated copper wires



a. UTP b. STP

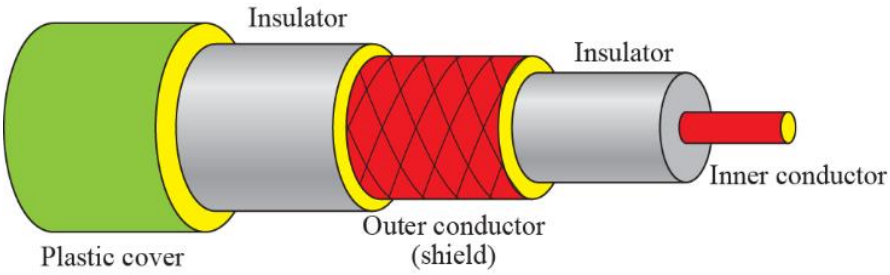
Categories of Unshielded Twisted-Pair Cables

Category	*Specification*	*Data Rate (Mbps)*	*Use*
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs

5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called *SSTP (shielded screen twisted-pair)*. Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Coaxial Cable

The name of the cable is coaxial as it contains two conductors parallel to each other. Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable. It has a higher frequency as compared to Twisted pair cable. The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor. The middle core is responsible for the data transferring whereas the copper mesh prevents from the **EMI**(Electromagnetic interference).



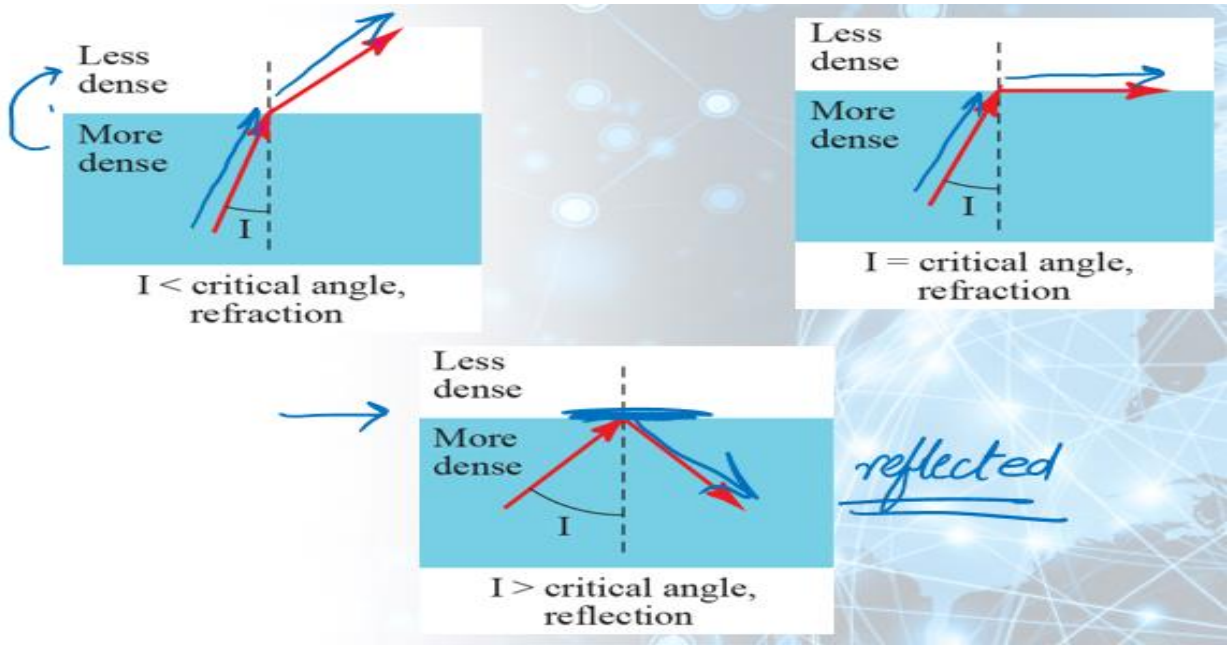
Categories of Coaxial Cables

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

Fiber-Optic Cable

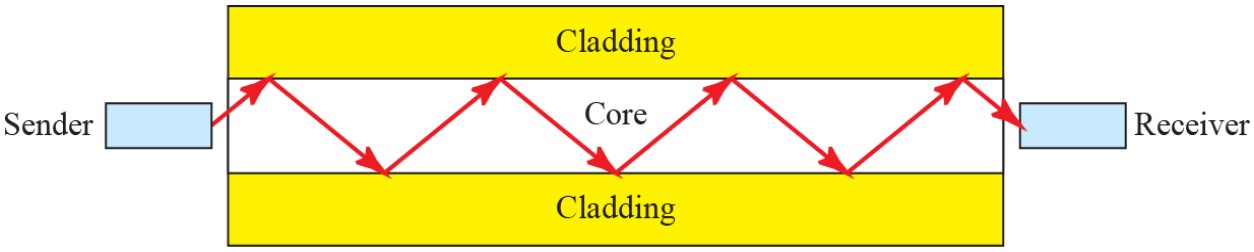
- Made of glass or plastic and transmits signals in the form of light
- Light travels in a straight line as long as it is moving through a single uniform substance
- If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction

Bending of Light Ray



Fiber-Optic Cable

Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light. The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring. Fibre optics provide faster data transmission than copper wires.

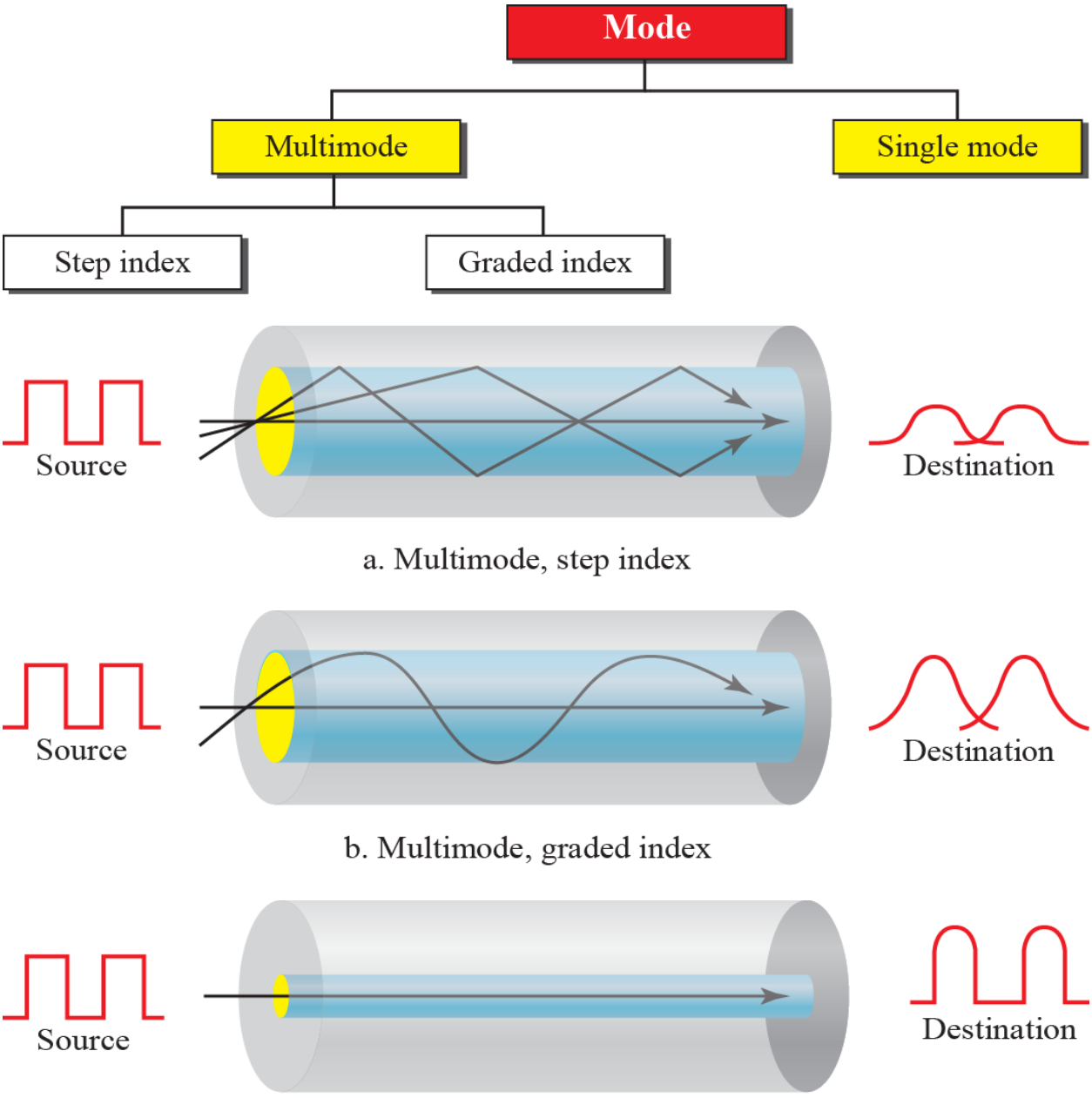


Propagation modes

Fiber-optic cable has two propagation modes: multimode and single mode. They perform differently with respect to both attenuation and time dispersion. The

Multimode step-index fibers have a uniform refractive index profile, while multimode graded-index fibers have a refractive index that decreases gradually from the center to the outer cladding, allowing for reduced modal dispersion.

single-mode fiber-optic cable provides much better performance with lower attenuation.



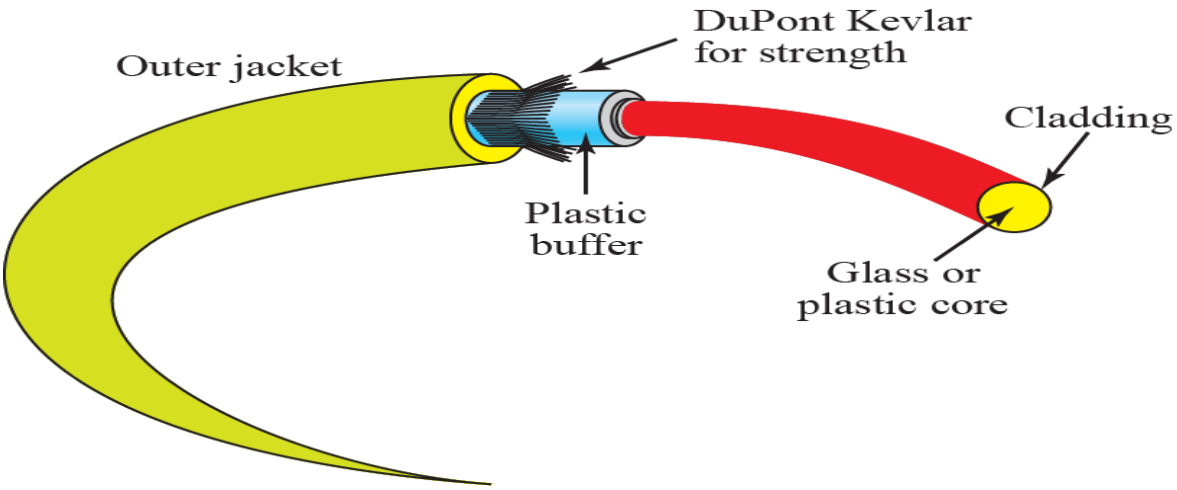
Single-mode fibers have a small core size and allow only one mode of light to propagate

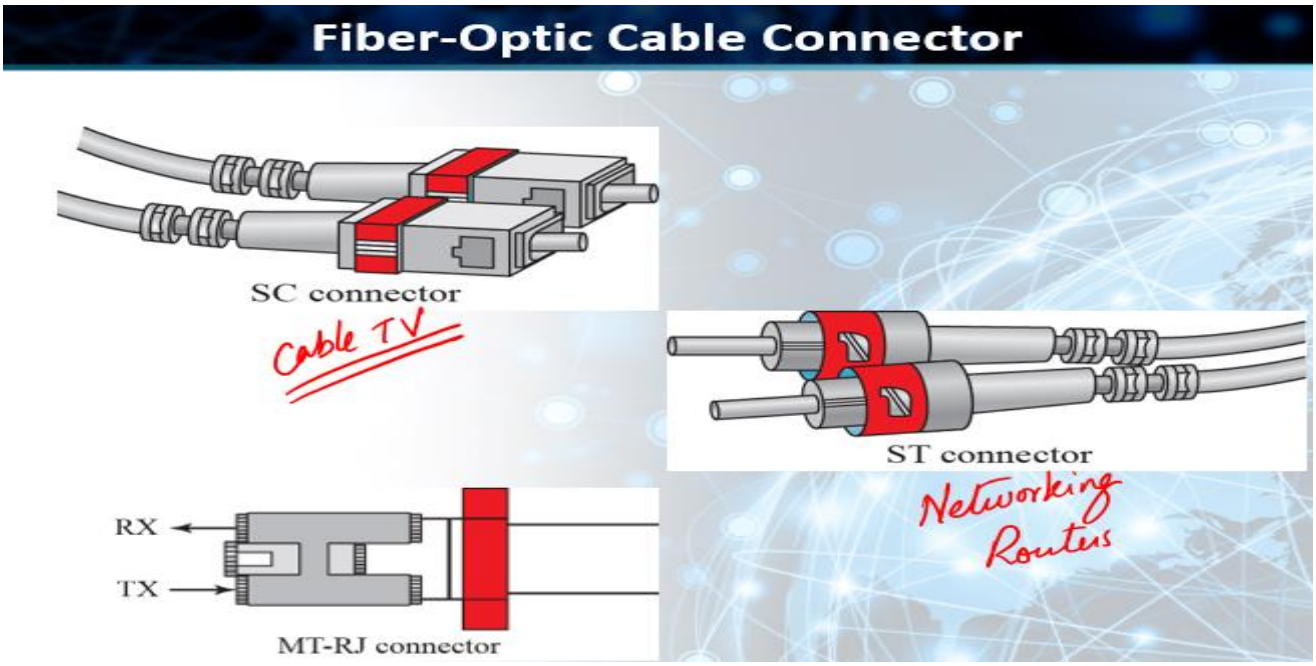
Fiber Types

The common types of fiber optic

Type	Core (μm)	Cladding (μm)	Mode
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Fiber Composition





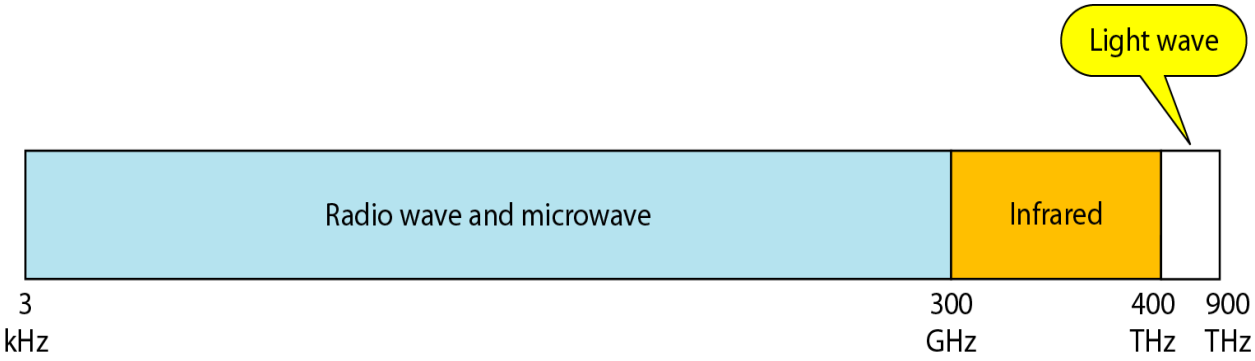
Advantages & Disadvantages

- Higher Bandwidth
- Less Attenuation
- Less EM Interference
- Light Weight
- Less corrosive than copper
- Installation/Maintenance
- Unidirectional
- Cost

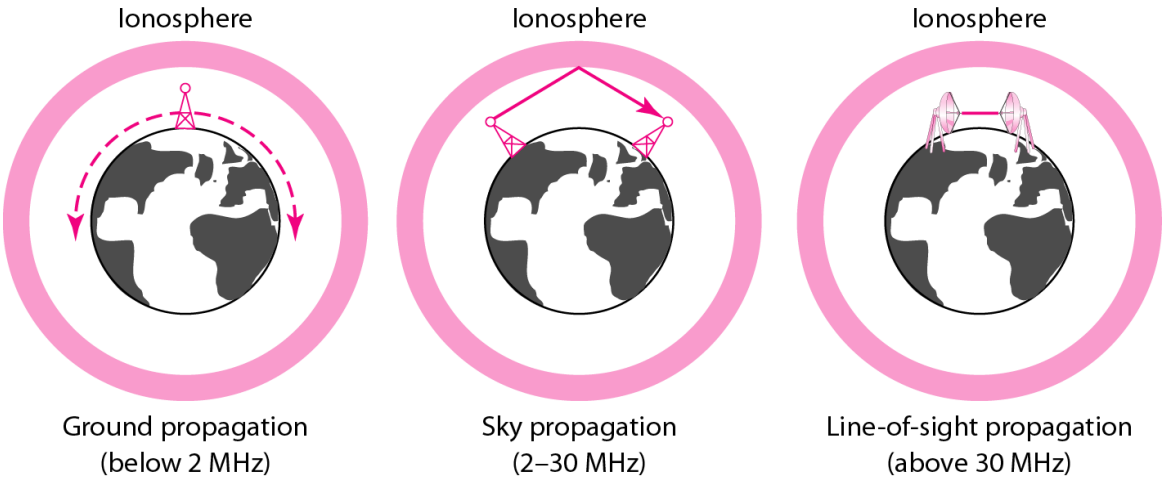
Unguided Media

- Unguided medium transport waves without using a physical conductor
- Often referred to wireless communication
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them

Electromagnetic Spectrum



Propagation Methods



Bands

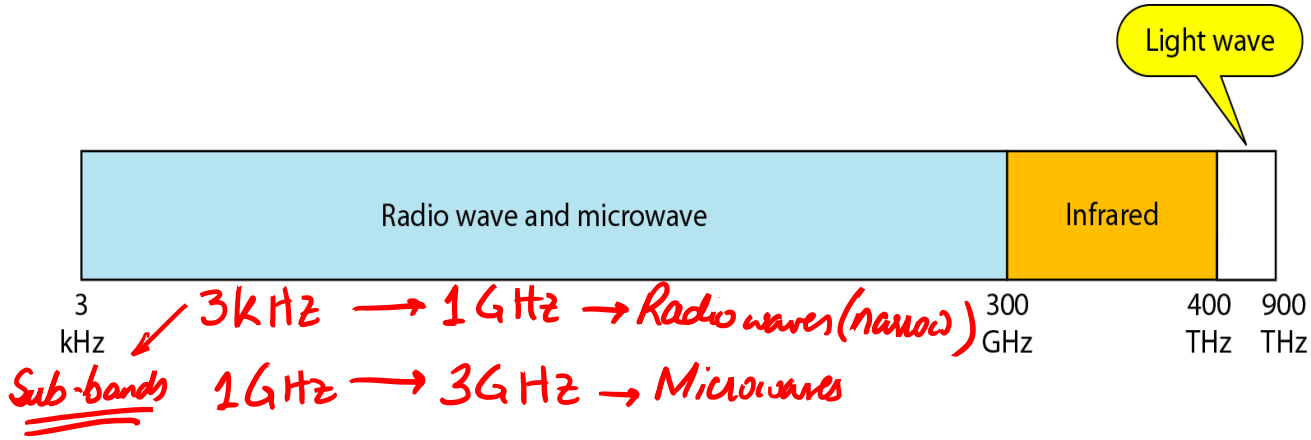
Band	*Range*	*Propagation*	*Application*
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

Radio Waves

- Electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves

here G represents giga prefix whose value is 10^9 = 1,000,000,000.

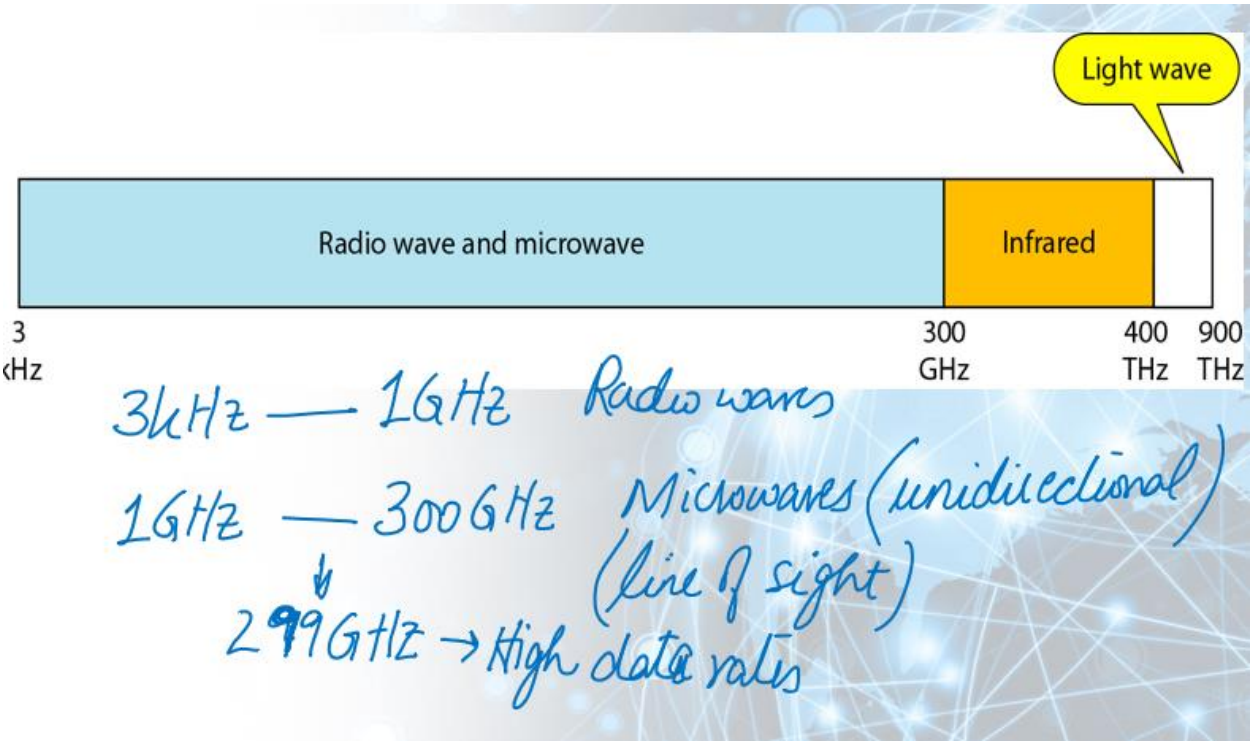
- Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called microwaves



Microwaves

- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves
- Microwaves are unidirectional
- When an antenna transmits microwaves, they can be narrowly focused

Electromagnetic Spectrum



Infrared

- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication
- Infrared waves, having high frequencies, cannot penetrate walls
- Prevents interference between one system and another

WEEK ~ 8

Topic 114

Switching

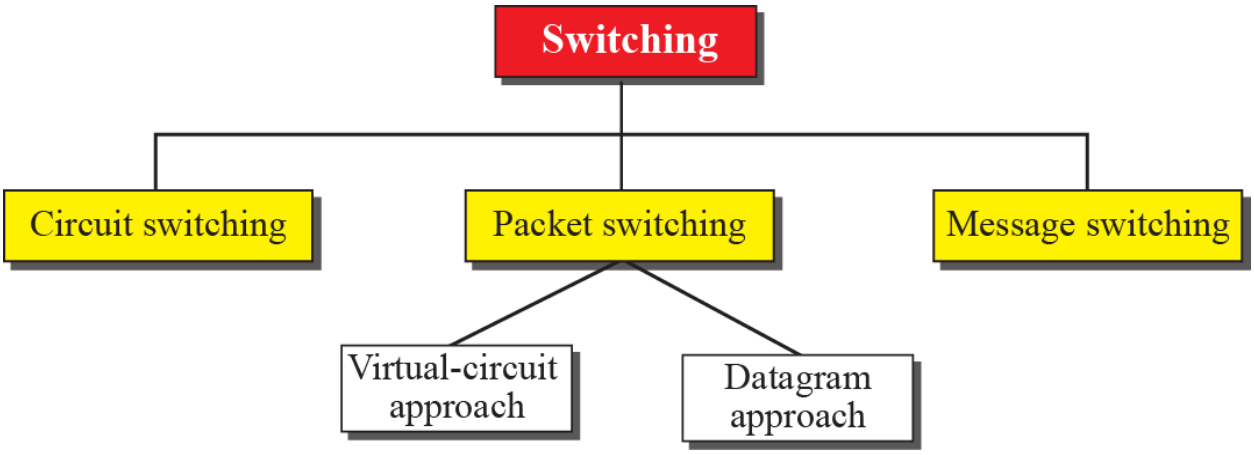
- A network is a set of connected devices
- Problem of how to connect multiple devices to make one-to-one communication possible
- The solution is Switching
- Switched network consists of a series of switches

Three Methods of Switching

- Three Methods:
  - ✓ Circuit Switching
  - ✓ Packet Switching
  - ✓ Message switching
- The first two are commonly used today
- The third has been phased out in general communications

Topic 115

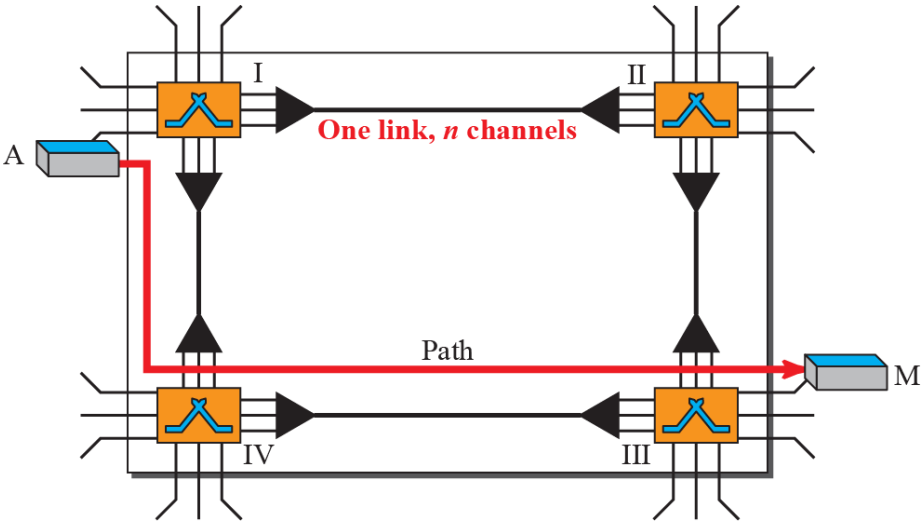
Taxonomy of Switched Networks



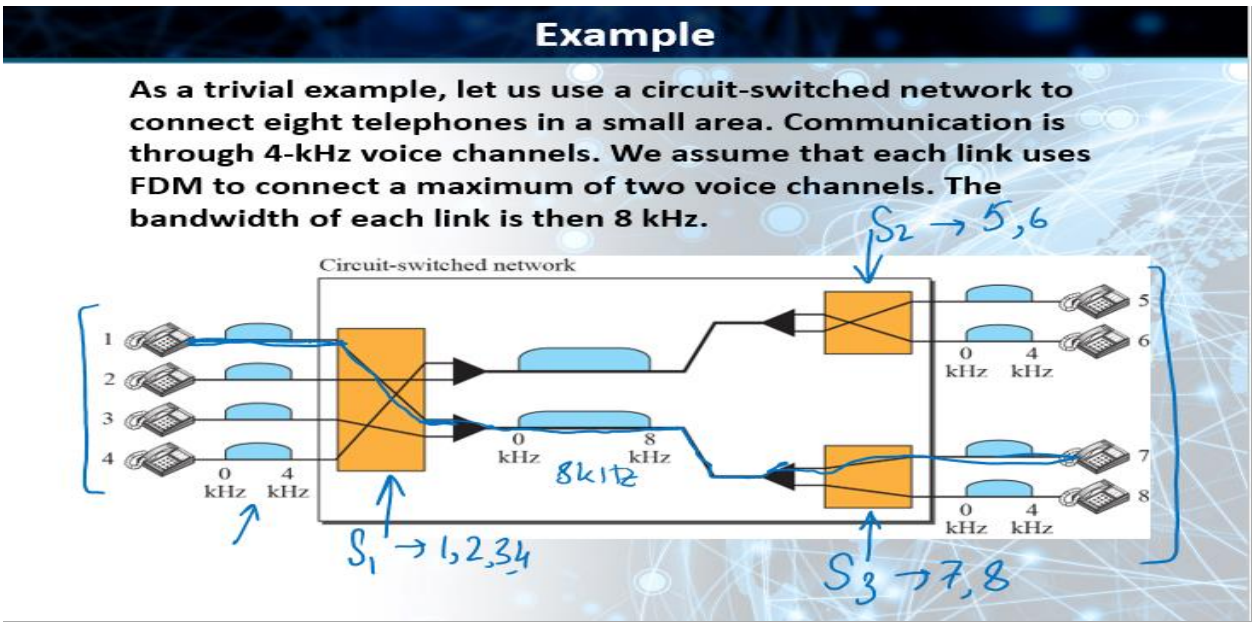
Circuit-switched Networks

- A set of switches connected by physical links
- A connection between two stations is a dedicated path made of one or more links
- Each connection uses only one dedicated channel on each link
- Each link is normally divided into n channels by using FDM or TDM
- A set of switches connected by physical links

A Circuit-Switched Network



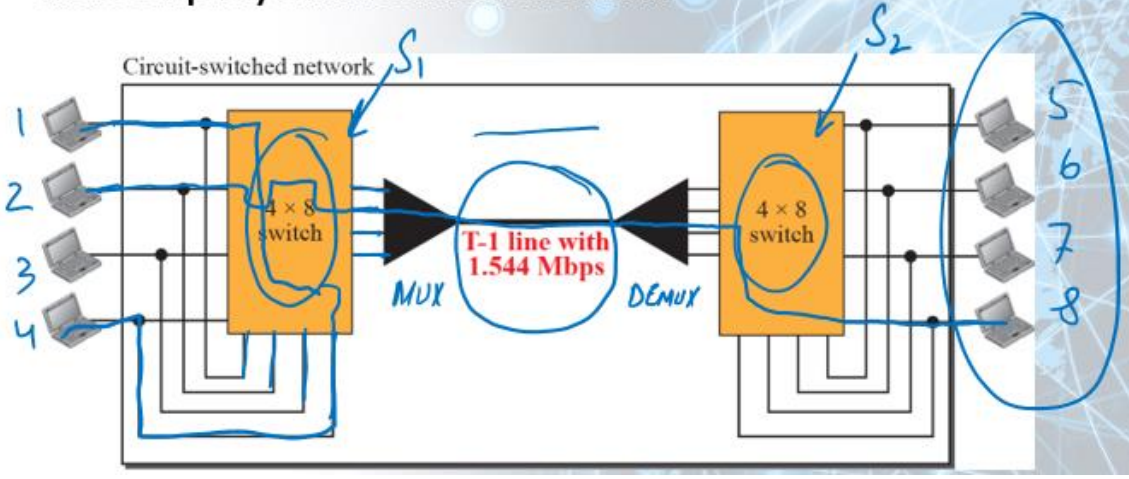
Topic 116



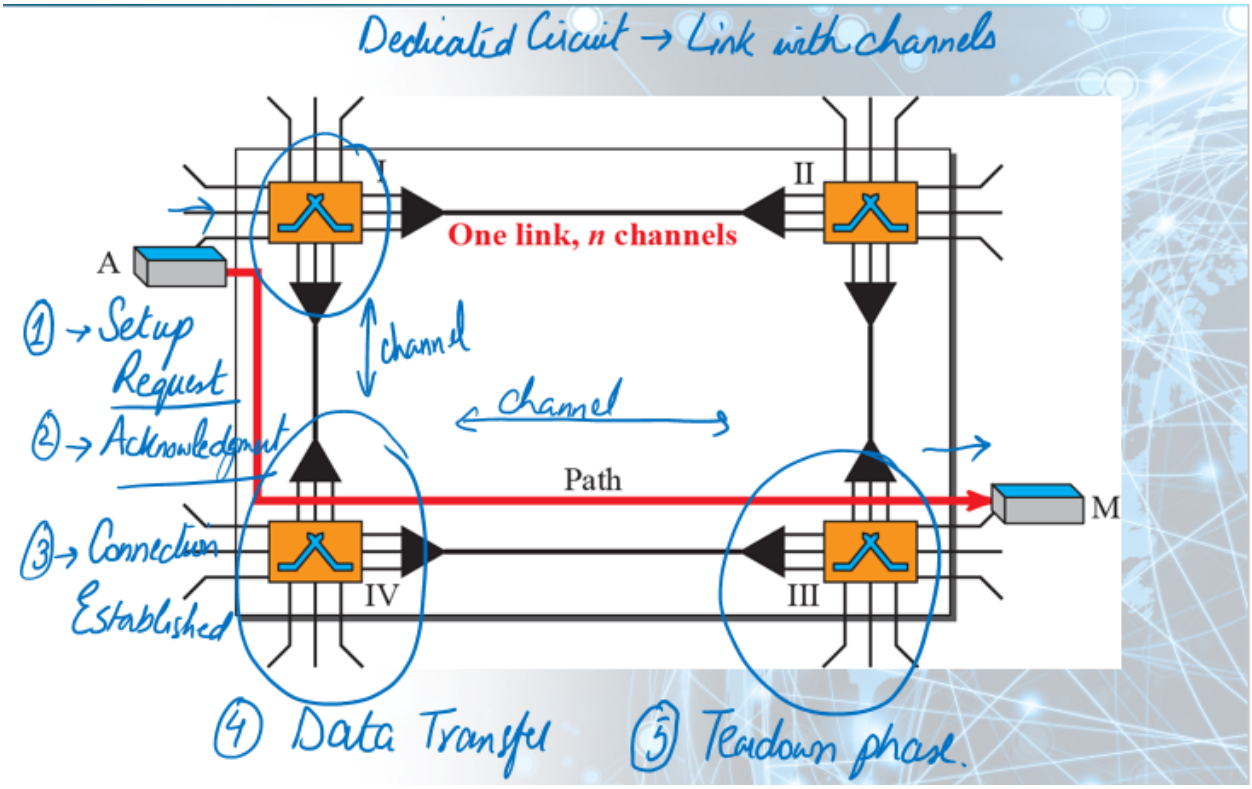
Multiplexer (Mux):Combines multiple input signals into a single output stream for transmission over a shared communication channel.
Demultiplexer (Demux):Separates the multiplexed signal back into its individual constituent signals at the receiving end.

Example

As another example, consider a circuit-switched network that connects computers in two remote offices of a private company. The offices are connected using a T-1 line leased from a communication service provider. There are two 4 × 8 (4 inputs and 8 outputs) switches in this network.



Three Phases in a Circuit Switched Network

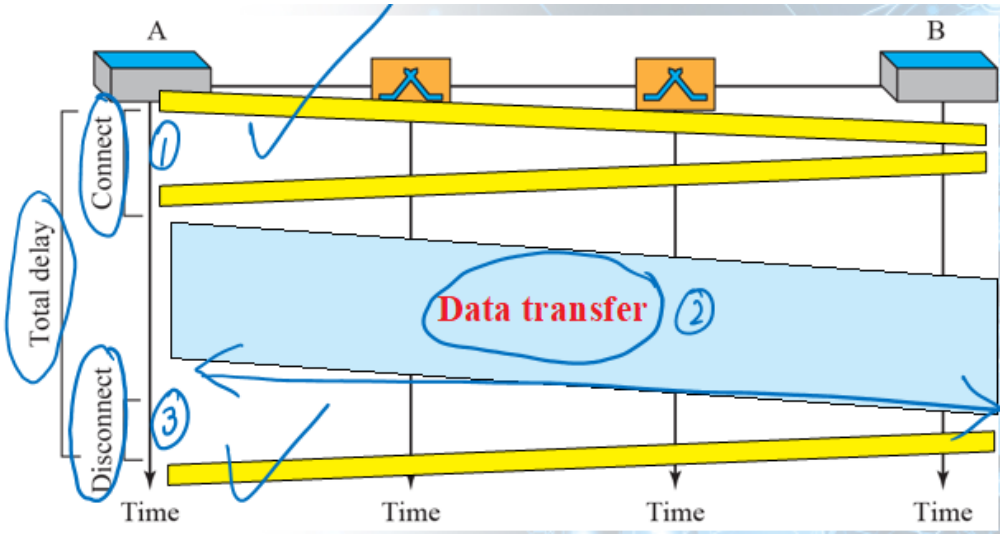


Efficiency of a Circuit-Switched Network

- Not as efficient as packet switching because resources are allocated during the entire duration of the connection and these resources are unavailable to other connections
- In a telephone network, people normally terminate the communication when they have finished their conversation
- Data Network is an issue

Delay in a Circuit-Switched Network

- Circuit switched networks have low efficiency but minimal delay
- Data is not delayed at each switch; the resources are allocated for the duration of the connection



Packet Switching

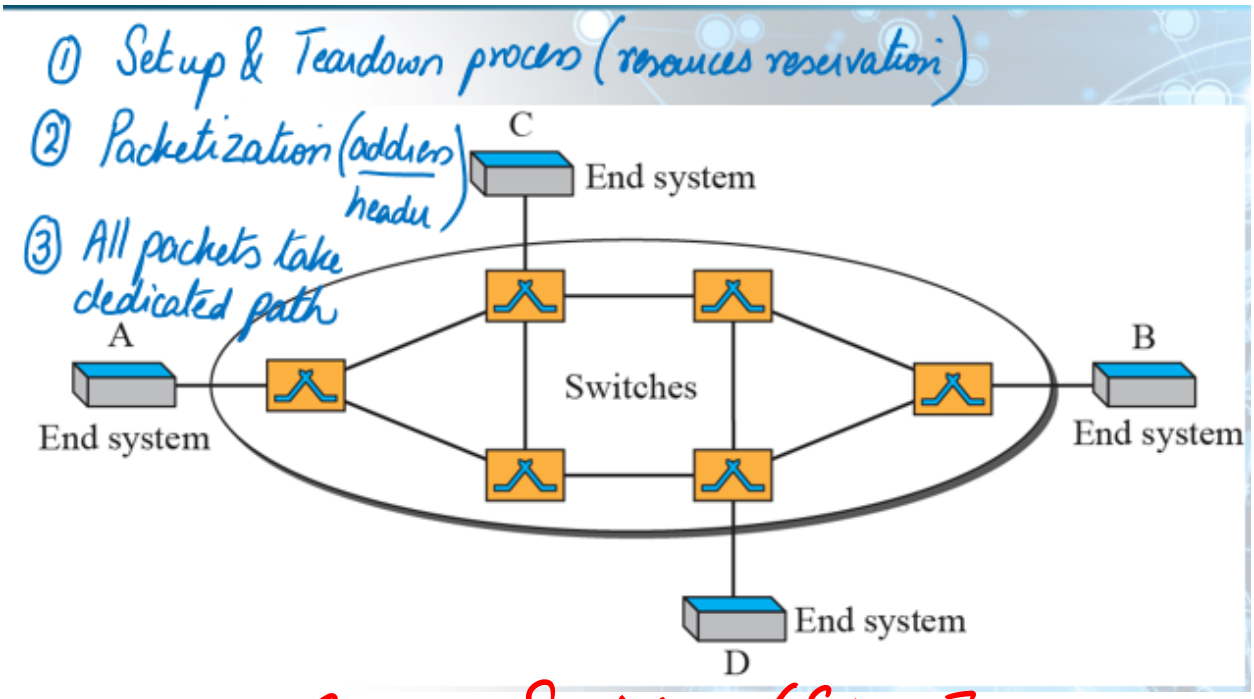
- If the message is going to pass through a packet-switched network, it needs to be divided into packets of fixed or variable size
- The size of the packet is determined by the network and the governing protocol

Datagram Networks

- Each packet is treated independently of all others.
- Even if a packet is part of a multi-packet transmission, the network treats it as though it existed alone
- Packets are referred to as datagram

Virtual-Circuit Networks

- A virtual-circuit network is a cross between a circuit-switched network and a datagram network



Physical → Circuit Switching (Set up Transfer & Teardown)

Data Link Layer → Virtual Circuit Approach (Set up Transfer & Teardown → Packets (Frames))

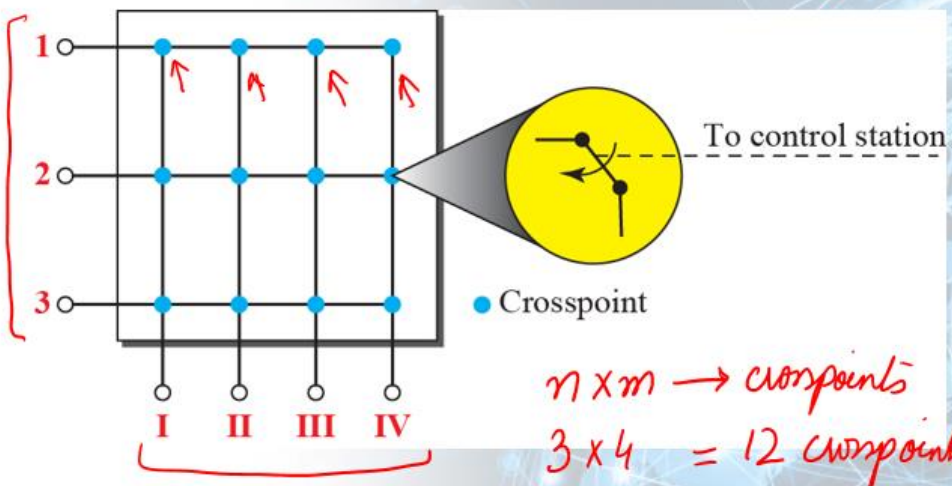
Network Layer → Datagram Switching (Independent Datagrams or Packets)

Structure Of A Circuit Switch

- Circuit switching today can use either of two technologies:
  - ✓ The Space-Division switch
  - ✓ The Time-Division switch

Crossbar switch with 3 inputs & 4 outputs

n inputs m outputs (n=3; m=4)



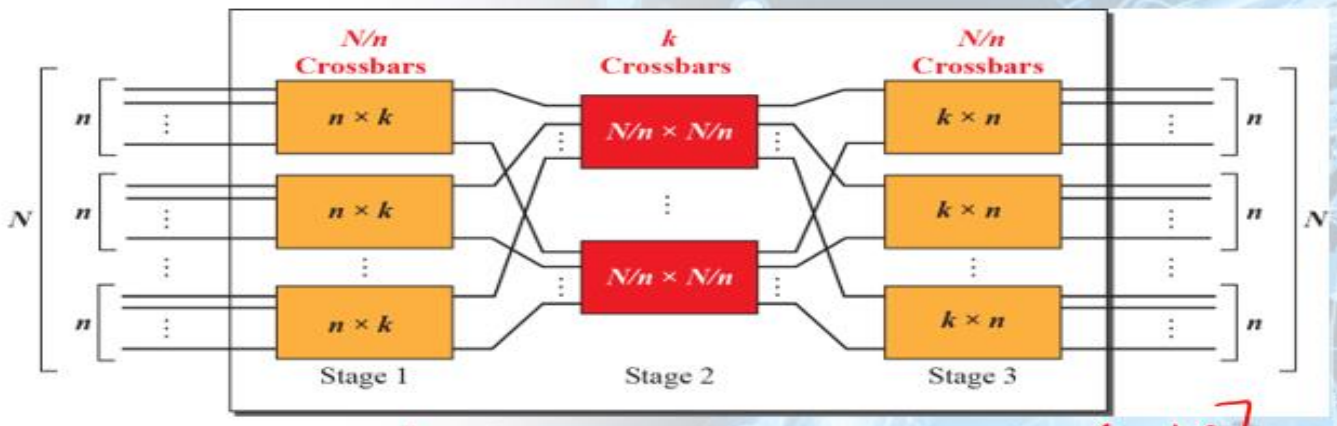
n x m → crosspoints

3 x 4 = 12 crosspoints

25% of crosspoints

Multistage Switch

N x N crosspoints



No of crosspoints ⇒ $2kN + k\left(\frac{N}{n}\right)^2$ → lesser

// // // ⇒ N^2

Time-Division Switch

- Uses TDM inside a switch
- Most popular technology is Time-Slot Interchange (TSI)

Topic 117

Three Phases in a Circuit Switched Network

- The actual communication in a circuit-switched network requires 3 phases:
  - ✓ Connection Setup
  - ✓ Data Transfer
  - ✓ Connection Teardown

Note : Topic 118 , 119 , 120 , 121 , 122 and 123 contains previous Material from Topic 116 .

Topic 118 contains previous topics from Topic 116 which are :

Efficiency of a Circuit-Switched Network

Delay in a Circuit-Switched Network

Topic 119

Topic 119 contains previous topics from Topic 116 which are :

Packet Switching

Datagram Networks

Virtual-Circuit Networks

Topic 120

Topic 120 contains previous topics from Topic 116 which is :

Virtual-Circuit Networks

Topic 121

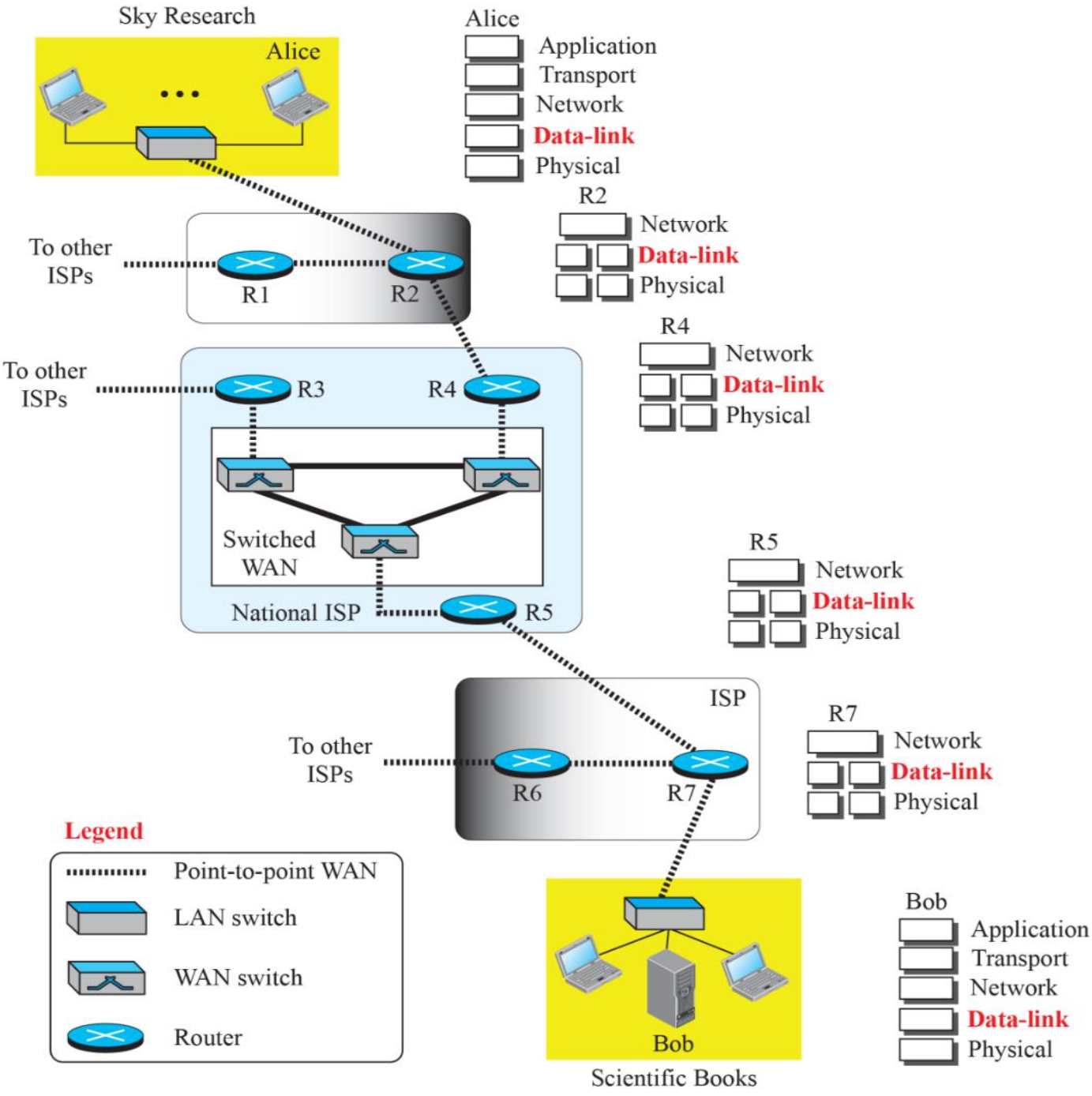
Topic 121 contains previous topics from Topic 116 which is :

Virtual-Circuit Networks

WEEK ~ 9

Topic 124 - 134

Communication at the Data-Link Layer



In the above example:

- The data-link layer on Alice's computer communicates with the data-link layer at router R2.
- The data-link layer at router R2 communicates with the data-link layer at router R4, and so on.
- Finally, the data-link layer at router R7 communicates with the data-link layer at Bob's computer.

Only **one data-link layer** is involved at the **source or the destination**, but **two data-link layers** are involved at each **router**. The reason is that **Alice's and Bob's computers are each connected to a single network**, but **each router takes input from one network and sends output to another network**.

Data-Link Layer

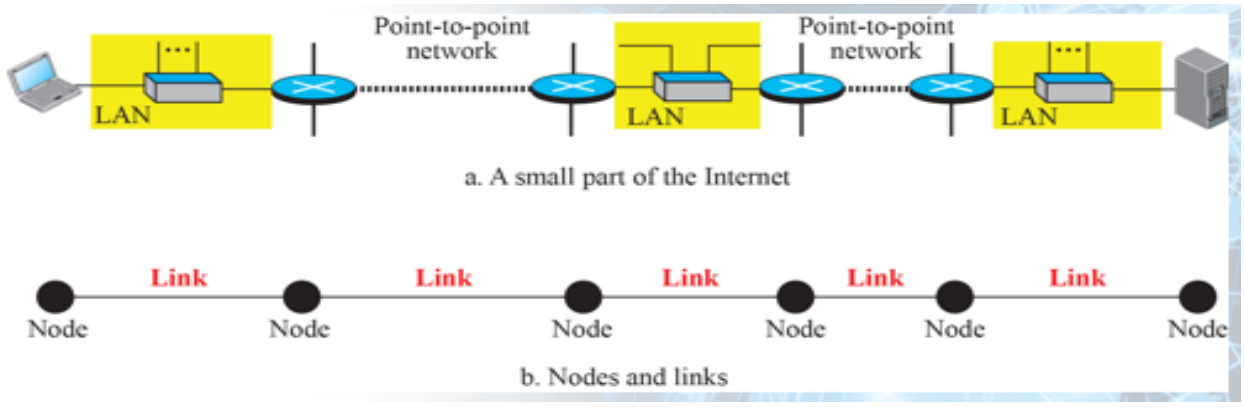
- The Internet is a combination of networks glued together by connecting devices (routers or switches)
- If a packet is to travel from a host to another host, it needs to pass through these networks
- Data Link layer controls node-to-node communication

Nodes and Links

- **Communication at the data-link layer is node-to-node:**

  Communication at the data-link layer occurs between two directly connected nodes in a network.

- **A data unit from one point on the Internet needs to pass through many networks (LANs and WANs) to reach another point.**
- **We refer to the two end hosts and the routers as nodes and the networks in between as links.**
- **Nodes** are devices such as computers or network elements that send, receive, or process data.
- **Links** are the communication pathways that connect these nodes, enabling the transfer of data between them.



The first node is the **source host**; the last node is the **destination host**.

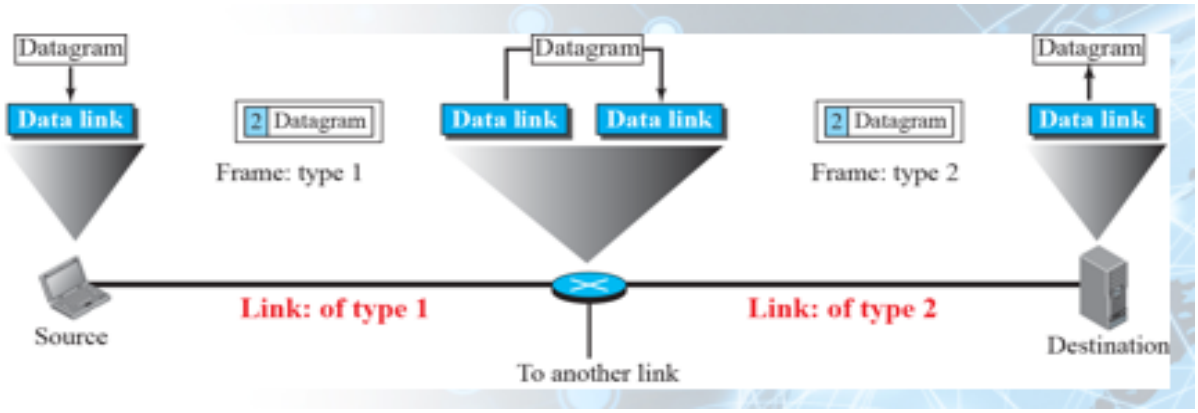
The other four nodes are four **routers**.

The first, the third, and the fifth **links** represent the three LANs; the second and the fourth **links** represent the two WANs.

Services provided by Data-Link Layer

- Data-Link Layer is located between the physical and the network layers.
- Data-Link Layer provides services to Network Layer and receives services from Physical layer.
- **Framing:** is the process of structuring data for transmission by adding start and stop bits, headers, or other delimiters. It helps to set the boundaries of a data frame, enabling the receiver to identify and extract the transmitted information accurately.
- **Encapsulation:** involves wrapping data in a specific format or protocol before transmission. This process adds headers and trailers to the original data, creating a packet for efficient routing and delivery.
- **Decapsulation:** is the reverse process where the recipient removes the added headers and trailers to retrieve the original data from the received packet.
- **Flow Control:** control is the regulation of data transmission between devices to prevent congestion and ensure efficient and reliable information exchange.
- **Error Control:** Error control in data communication involves implementing techniques to detect and correct errors that may occur during the transmission of data. These techniques ensure data integrity and reliability. Common error control methods include **checksums**, **parity bits**, and more advanced techniques like **cyclic redundancy check (CRC)**.
- **Congestion Control:** involves managing and preventing network congestion to maintain optimal performance. It utilizes techniques such as traffic shaping, prioritization, and flow control to regulate data flow and prevent network saturation, ensuring efficient and reliable data transmission.

A Communication with only Three Nodes



In the above diagram router is source of communication between the **source** and **destination**.

- The datagram received by the data-link layer of the source host is encapsulated in a frame.
- The frame is logically transported from the source host to the router.
- The frame is decapsulated at the data-link layer of the router and encapsulated at another frame.
- The new frame is logically transported from the router to the destination host.

Note that, router has separate data link layer for every interface connecting to the physical link.

Services provided by Data-Link Layer

Flow control

- **Flow control** is a mechanism used to manage the rate of data transmission between two devices to prevent congestion or data loss. It ensures that a fast sender does not overwhelm a slower receiver.
- **Flow control** can be implemented through **techniques** such as **buffering**, **acknowledgement signals**, and **windowing**.
    - **Buffers** temporarily store data during transmission, allowing the sender and receiver to operate at different speeds.
    - **Acknowledgement signals** inform the sender about the successful receipt of data, enabling it to adjust its transmission rate accordingly.
    - **Windowing** involves specifying the number of unacknowledged packets that can be in transit at any given time, controlling the flow of data between sender and receiver.
- **Flow control** is crucial for optimizing data transfer reliability and efficiency in communication networks.

Error Control:

Error control in data communication involves implementing techniques to detect and correct errors that may occur during the transmission of data. These techniques ensure data integrity and reliability. Common error control methods include **checksums**, **parity bits**, and ore advanced techniques like **cyclic redundancy check (CRC)**.

Congestion Control:

Involves managing and preventing network congestion to maintain optimal performance. It utilizes techniques such as traffic shaping, prioritization, and flow control to regulate data flow and prevent network saturation, ensuring efficient and reliable data transmission.

Two Categories of Links

We can have the following two types of links:

Point-to-point link:

Point-to-point link is a dedicated communication channel that connects two specific devices, allowing data to be transferred directly between them without the need for any intermediary devices or shared infrastructure.

Broadcast link:

Broadcast link is a network connection that allows data to be sent from one sender to multiple receivers simultaneously. It's like a radio broadcast where one station sends signals that many people can tune into and receive at the same time. In a broadcast link, information is shared widely to all connected devices on the network.

Two Sublayers of Data-Link Layer

We can divide the data-link layer into two sublayers:

Data Link Control (DLC):

Data Link Control (DLC) is a sublayer of the Data Link Layer.

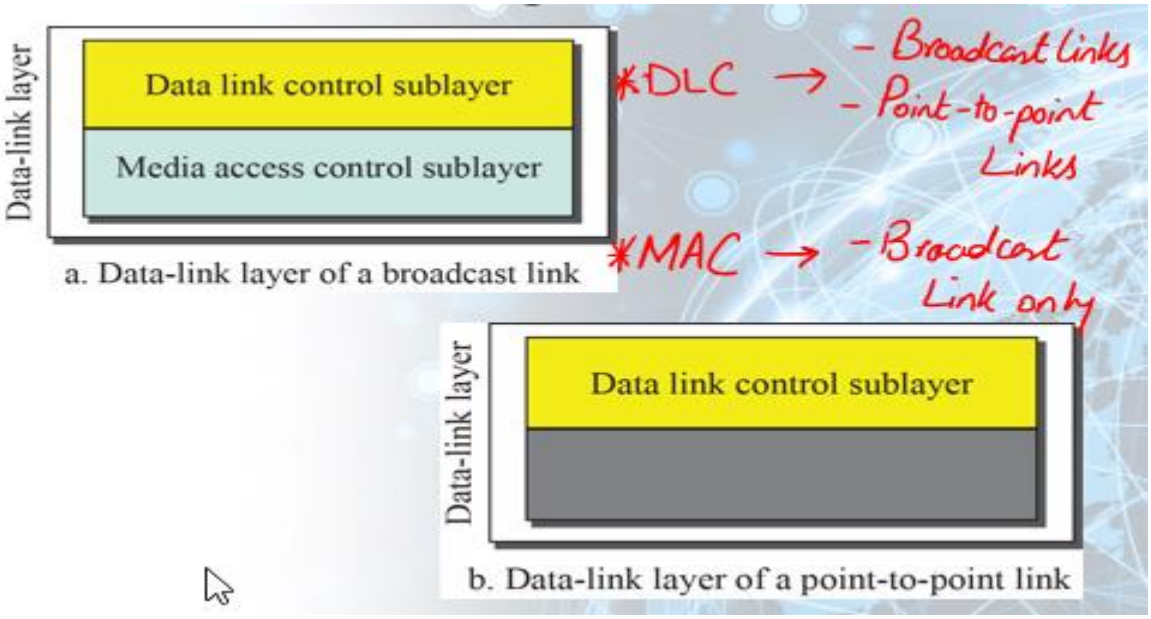
- It manages the reliable and efficient transfer of data frames between devices on a network, providing **error detection and correction**, **flow control**, and **addressing**.
- DLC ensures that data is transmitted accurately and in the correct order over the physical layer.

Media Access Control (MAC):

Media Access Control (MAC) is typically implemented in the data link layer of the OSI model and used to manage access to a shared communication medium.

- It is responsible for assigning unique addresses to network devices and controlling their access to the communication channel, ensuring that data is transmitted efficiently and without collisions.
- MAC addresses are hardware-based and serve as unique identifiers for devices on a network.
- The MAC protocol is typically implemented in the data link layer of the OSI model and plays a crucial role in facilitating reliable and orderly communication within a network.

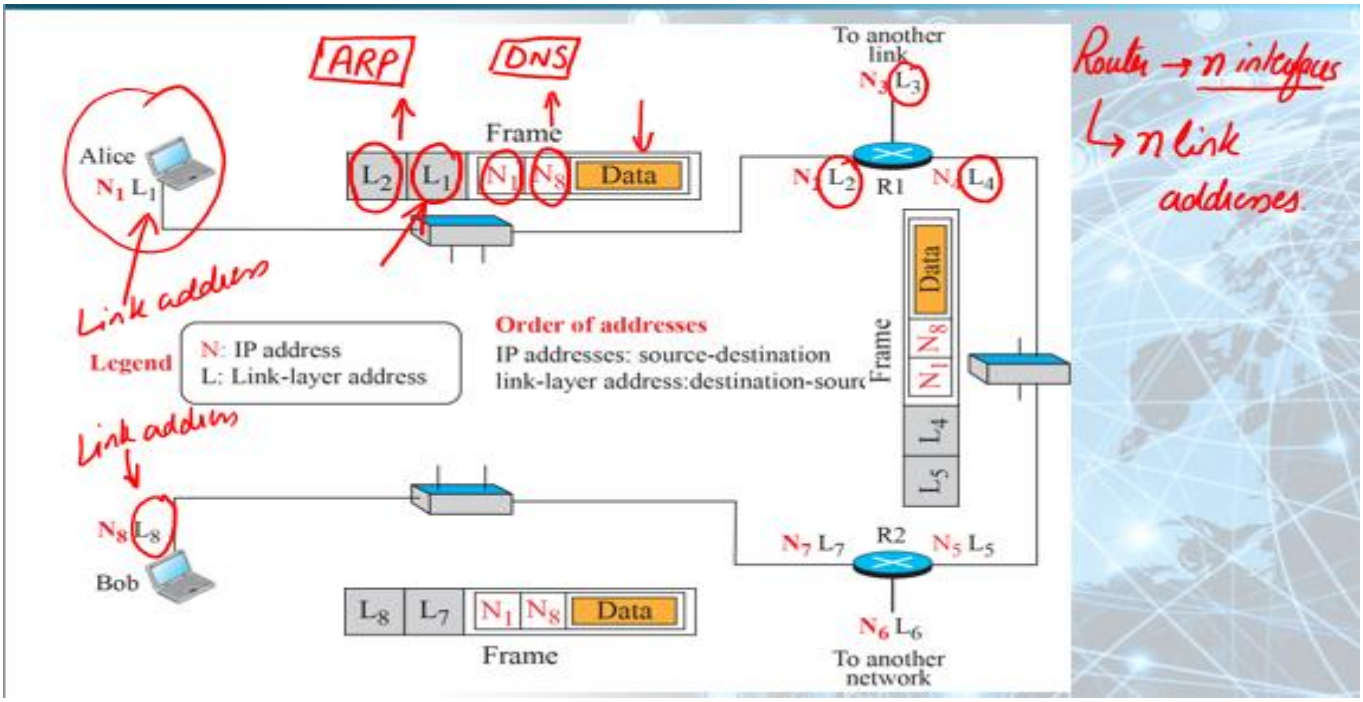
Dividing the data-link layer into two sublayers



Why LINK-LAYER ADDRESSING?

- Each datagram on the Internet, from the same source host to the same destination host, may take a **different path**.
- We need to remember that the IP addresses in a datagram should not be changed. If the destination IP address in a datagram changes, the packet never reaches its destination.
- We cannot make a datagram reach its destination **using only IP addresses** on the Internet. For that purpose, we need another addressing mechanism in a connectionless internetwork: the **link-layer addresses of the two nodes**. A link-layer address is sometimes called a **link address**, sometimes a **physical address**, and sometimes a **MAC address**.
- Since a **link is controlled** at the data-link layer, the addresses need to belong to the data-link layer.
- When a datagram passes **from the network layer to the data-link layer**, the datagram will be **encapsulated in a frame** and **two data-link addresses are added to the frame header**.
- These two **addresses are changed every time** the frame moves from one link to another.

IP addresses & Link-Layer Addresses

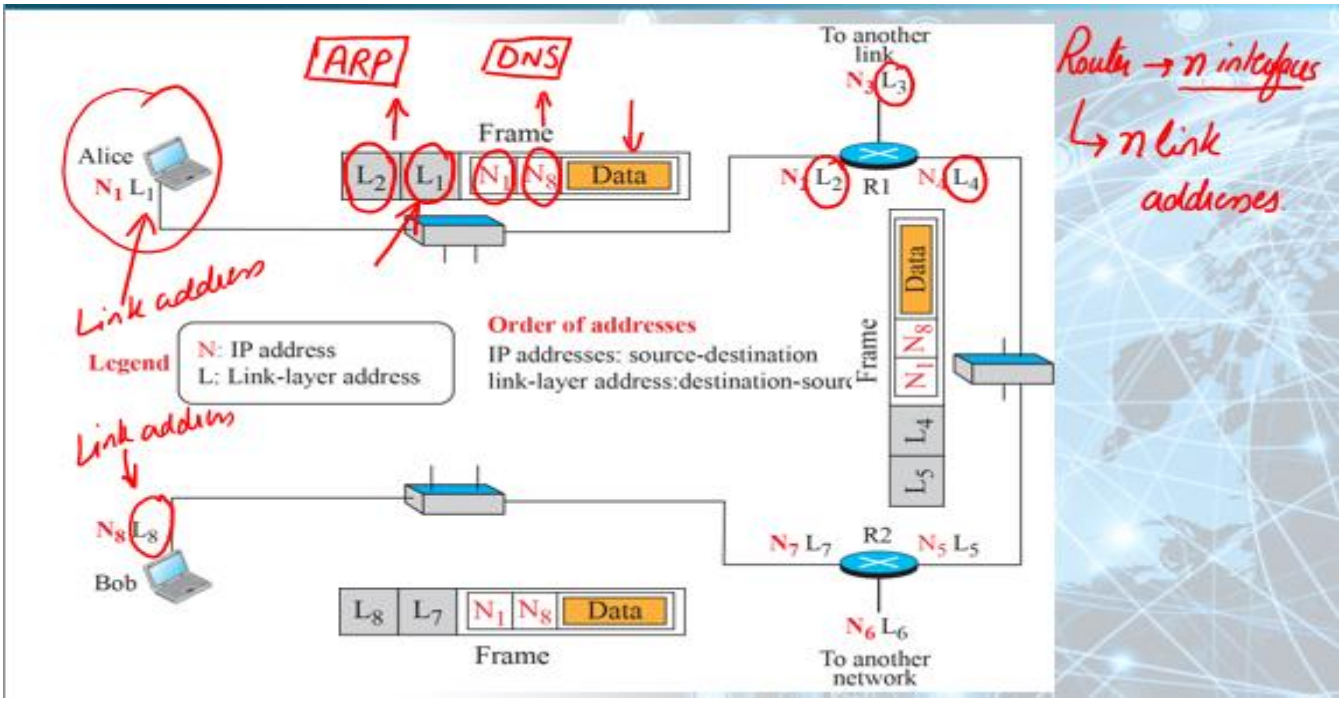


In the above figure:

- We have **three links** and **two routers**.
- We also have shown only **two hosts**: **Alice (source)** and **Bob (destination)**.
- For each host, we have shown **two addresses**, the **IP addresses (N)** and the **link-layer addresses (L)**.
- Note that a **router** has as many **pairs of addresses** as the **number of links** the router is connected to.
- Each **frame** carries the **same datagram** with the **same source and destination** addresses **(N1 and N8)**, but the **link-layer addresses** of the **frame** change from **link to link**.
- In link 1, the link-layer addresses are L1 and L2.
- In link 2, they are L4 and L5.
- In link 3, they are L7 and L8.
- Note that the **IP addresses** and the **link-layer addresses** are not in the same order.
- For **IP addresses**, the **source address comes before the destination address**; for **link-layer addresses**, the **destination address comes before the source**.

Why Link Layer Addressing -2

IP addresses & Link-Layer Addresses



Why do we need more than one IP address in a router, one for each interface?

- The answer is that an interface is a connection of a router to a link.
- We will see that an IP address defines a point on the Internet at which a device is connected.
- A router with n interfaces is connected to the Internet at n points.
- This is the **situation of a house at the corner of a street with two gates**; **each gate has the address related to the corresponding street**.

How are the source and destination IP addresses in a packet determined?

- The answer is that the host should know its own IP address, which becomes the source IP address in the packet.
- The application layer uses the services of DNS to find the destination address of the packet and passes it to the network layer to be inserted in the packet.

How are the source and destination link-layer addresses determined for each link?

Again, each hop (router or host) should know its own link-layer address.

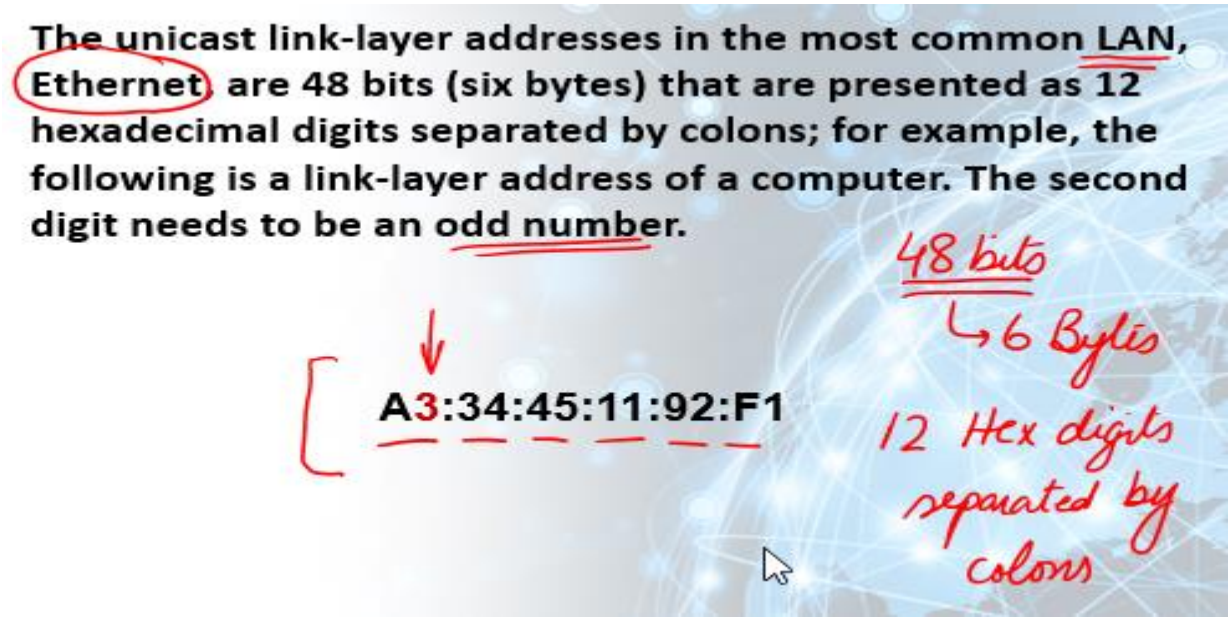
The destination link-layer address is determined by using the Address Resolution Protocol.

Three Types of addresses

Unicast:

- A unicast address refers to a unique identifier assigned to a single network interface or device.
- It allows communication between a specific sender and a specific receiver in a network, ensuring that the data is delivered only to the intended destination.
- Unicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons.

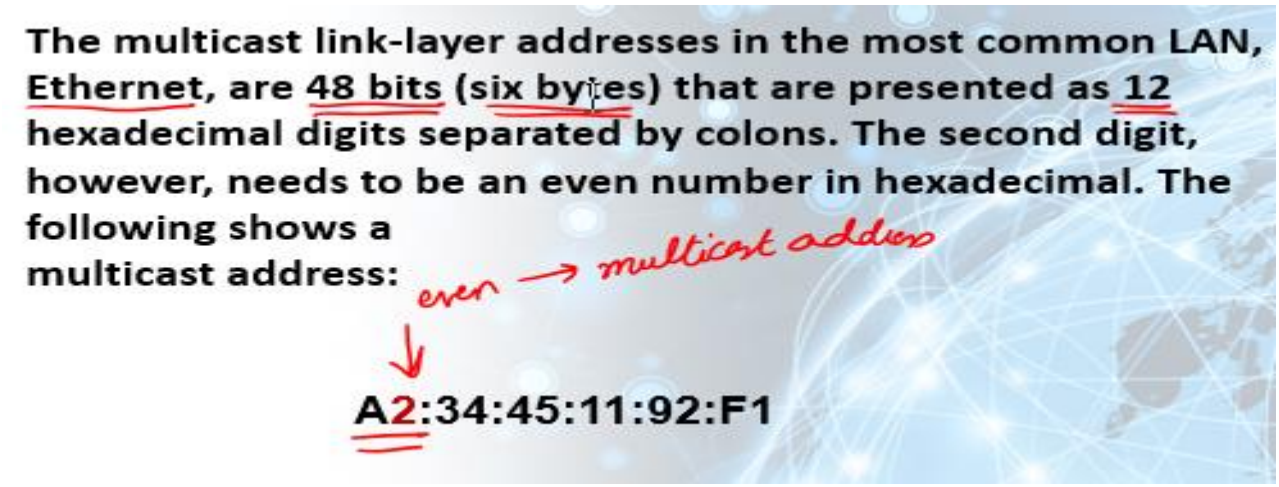
Example



Multicast:

- A multicast address in data communication is a destination address used for communication where a single sender can transmit data to multiple receivers simultaneously.
- It is a unique identifier that represents a group of devices interested in receiving the same data.
- Multicast communication enables efficient distribution of information to a specific group rather than broadcasting to all devices on the network.
- Multicast link-layer addresses in the most common LAN, Ethernet, are 48 bits (six bytes) that are presented as 12 hexadecimal digits separated by colons.
- **The second digit, however, needs to be an even number in hexadecimal.**

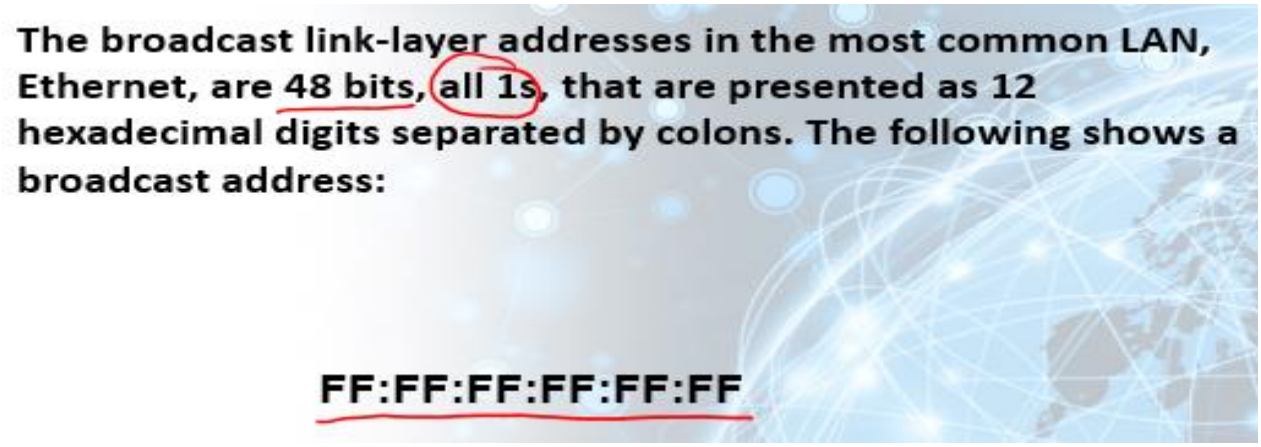
Example



Broadcast:

- Broadcast address is a special address that allows a message to be sent to all devices within a network.
- It is typically the highest address in the network range and is used to transmit data to all devices simultaneously.
- Broadcast link-layer addresses in the most common LAN, Ethernet, are 48 bits, **all 1s**, that are presented as 12 hexadecimal digits separated by colons.

Example



Address Resolution Protocol (ARP) -1

- Anytime a node has an IP packet to send to another node in a link, it has the IP address of the receiving node.
- However, the IP address of the next node is not helpful in moving a frame through a link; we need the **link-layer address** of the next node.
- This is the time when the **Address Resolution Protocol (ARP)** becomes helpful.
- ARP maps an IP address to a logical-link address.

- ARP accepts an IP address from the IP protocol, maps the address to the corresponding link-layer address, and passes it to the data-link layer.
- Anytime a host or a router needs to find the link-layer address of another host or router in its network, it sends an ARP request packet.
- The packet includes the link-layer and IP addresses of the sender and the IP address of the receiver.
- Because the sender does not know the link-layer address of the receiver, the query is **broadcast** over the link using the **link-layer broadcast address**.

Address Resolution Protocol (ARP):

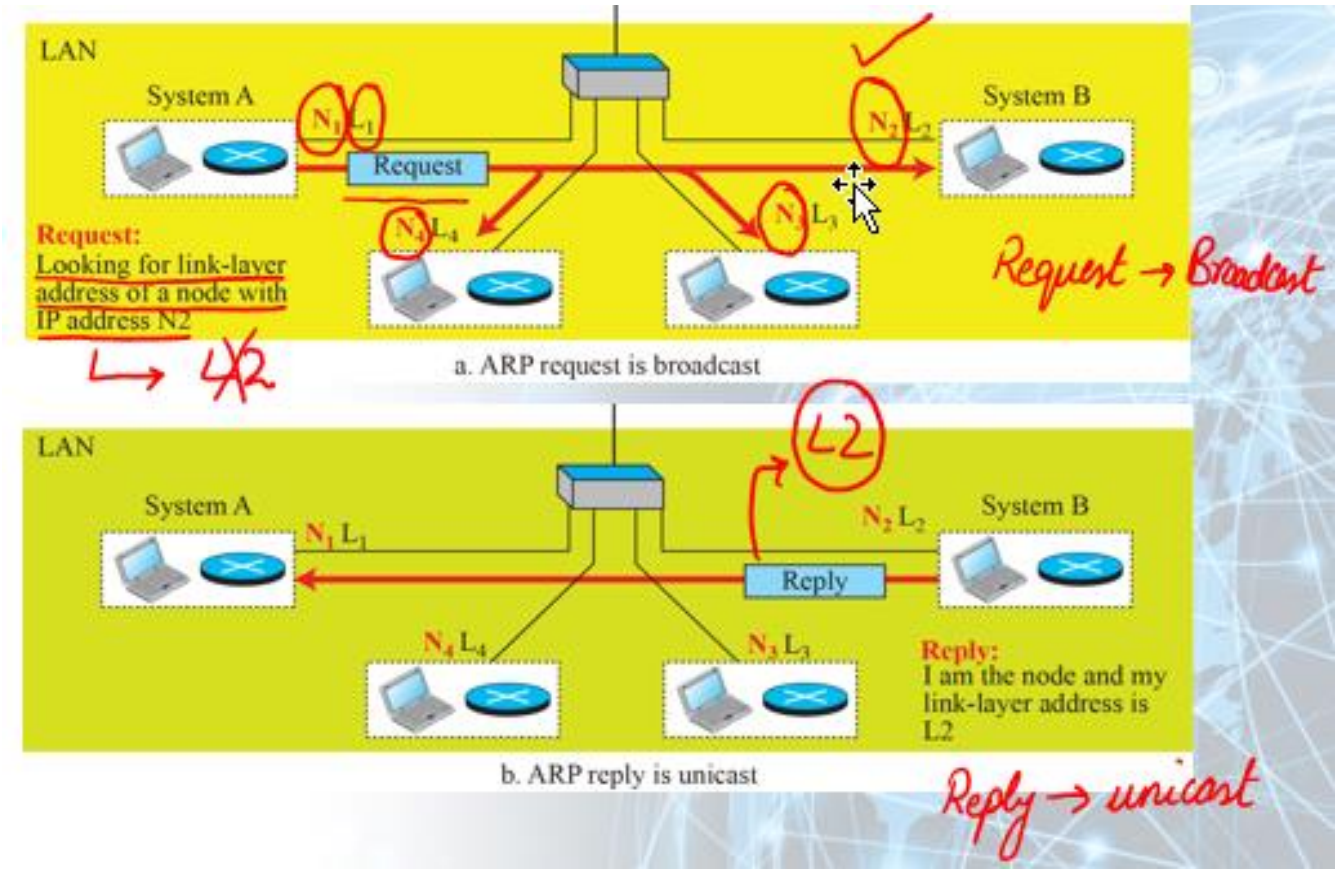
- Address Resolution Protocol (ARP) is a protocol used to map an IP address to a physical (MAC) address in a local network.
- It enables devices to discover each other's hardware address, facilitating proper communication within the network.

Position of ARP in TCP/IP protocol suite

Address Resolution Protocol (ARP) -2

Every host or router on the network receives and processes the ARP request packet, but only the intended recipient recognizes its IP address and sends back an ARP response packet.

The response packet contains the recipient's IP and link-layer addresses. The packet is unicast directly to the node that sent the request packet.



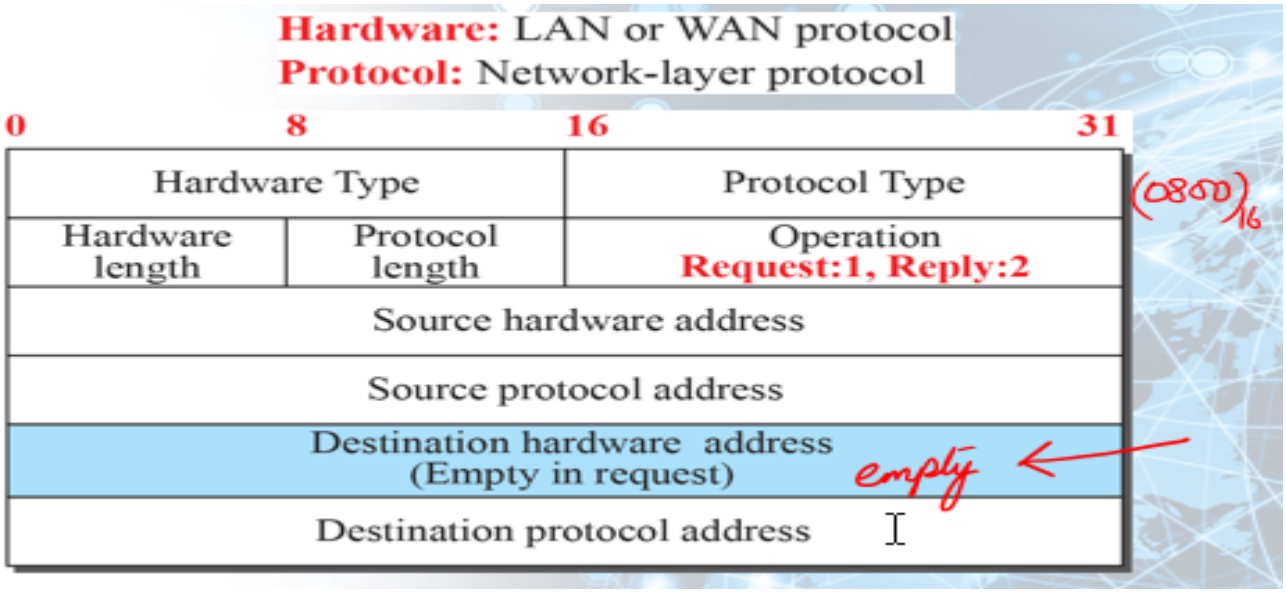
- The system on the left (A) has a packet that needs to be delivered to another system (B) with IP address N2.
- System A needs to pass the packet to its data-link layer for the actual delivery, but it does not know the physical address of the recipient.
- It uses the services of ARP by asking the **ARP protocol to send a broadcast ARP request packet** to ask for the **physical address** of a system with an IP address of N2.
- This packet is received by every system on the physical network, but only **system B** will answer it. **System B** sends an **ARP reply** packet that includes **its physical address**.
- Now system **A can send all the packets it has for this destination using the physical address it received**.

ARP (Address Resolution Protocol) caching:

- Refers to the **temporary storage of mappings** between **IP addresses and corresponding MAC addresses** in a device's memory.
- This caching mechanism helps optimize network performance by reducing the need for frequent **ARP broadcasts**.
- When a device needs to communicate with another **on the same network**, it **checks its ARP cache** to find the MAC address associated with the destination IP address.
- If the **mapping is not found or has expired**, the device **initiates an ARP request** to obtain the correct mapping.

Address Resolution Protocol (ARP) -3

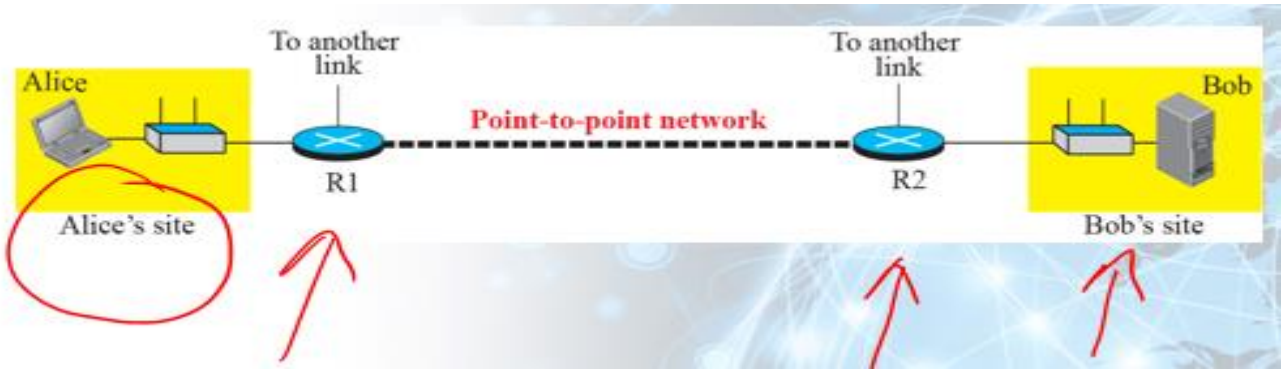
ARP Packet



In the format of an ARP packet, the names of the fields are self-explanatory.

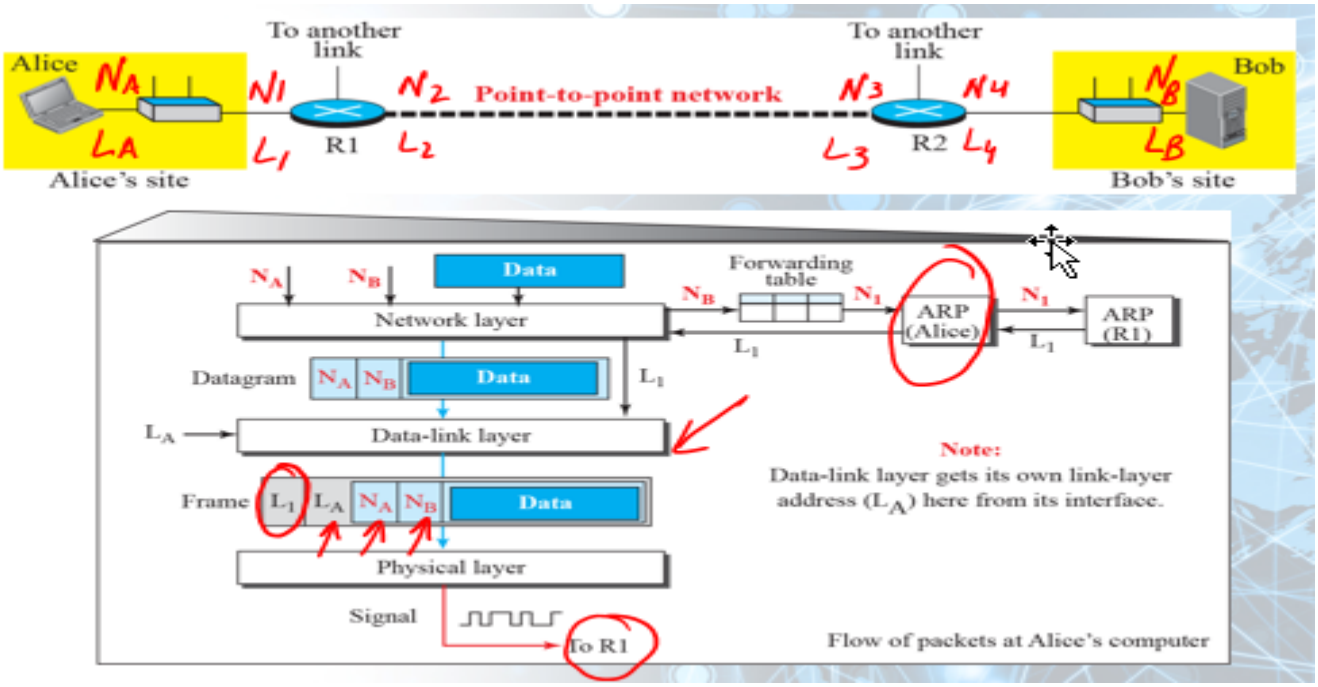
- The **hardware type field** defines the type of the link-layer protocol; Ethernet is given the type 1.
- The **protocol type field** defines the network-layer protocol: IPv4 protocol is (0800)16.
- The **source hardware** and **source protocol addresses** are variable-length fields defining the **link-layer and network-layer addresses** of the **sender**.
- The **destination hardware address** and **destination protocol address** fields define the **receiver link-layer and network-layer addresses**.
- An ARP packet is **encapsulated** directly into a **data-link frame**.
- The frame needs to have a field to show that the **payload** belongs to the ARP and not to the network-layer datagram.

The internet for our example



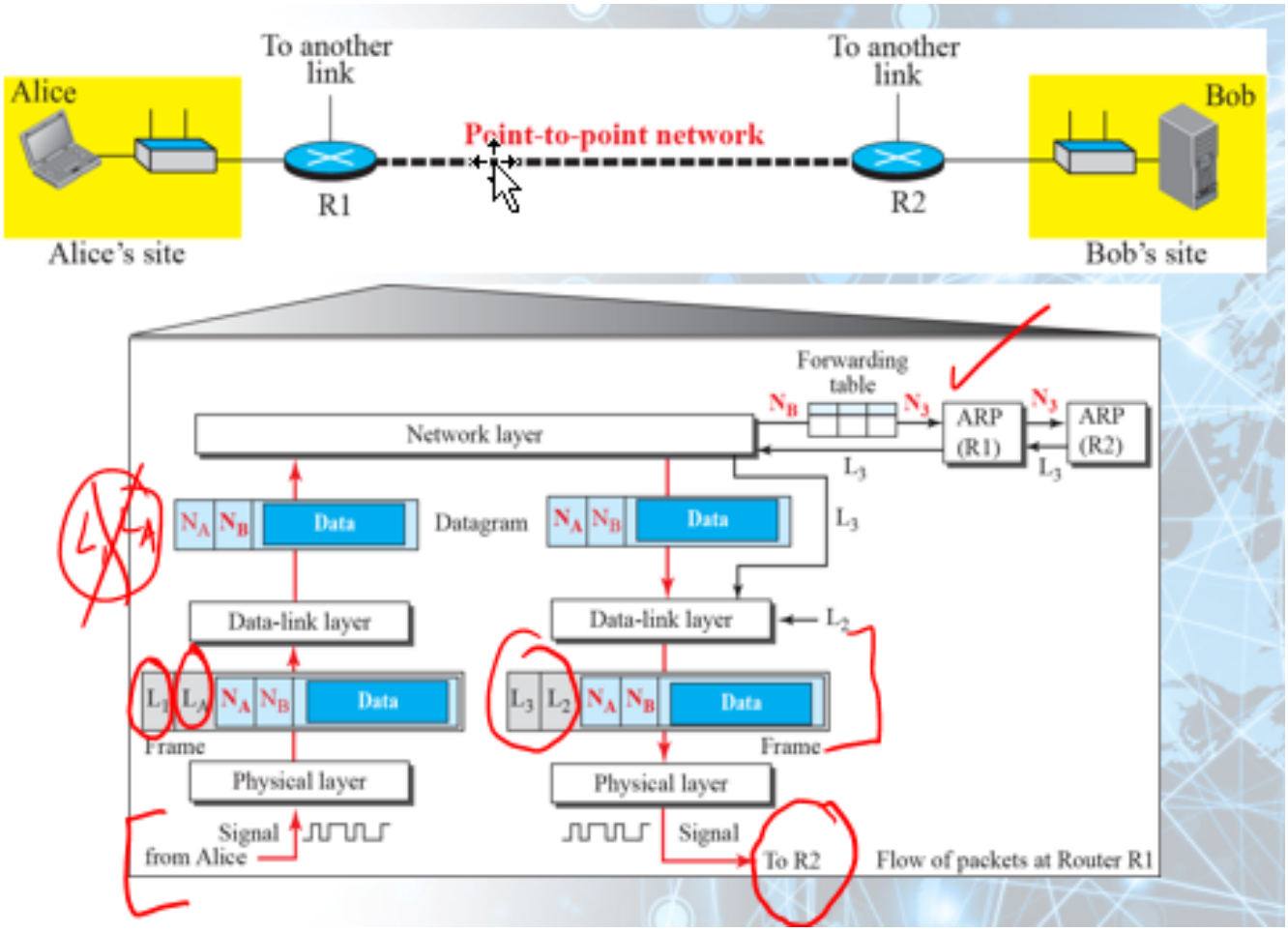
Assume Alice needs to send a datagram to Bob, who is three nodes away on the Internet.

Flow of packets at Alice site



- The network layer knows it's given **NA, NB, and the packet**, but it needs to find the **link-layer address** of the **next node**.
- The network layer **consults its routing table** and tries to find which router is next (the default router in this case) for the destination NB.
- The routing table gives N1, but the **network layer needs to find the link-layer address** of **router R1**.
- It uses its ARP to find the **link-layer address L1**.
- The network layer **can now pass the datagram with the link-layer address to the data-link layer**.
- The data-link layer knows its own link-layer address, LA.
- It creates the frame and passes it to the **physical layer**, where the **address is converted to signals and sent through the media**.

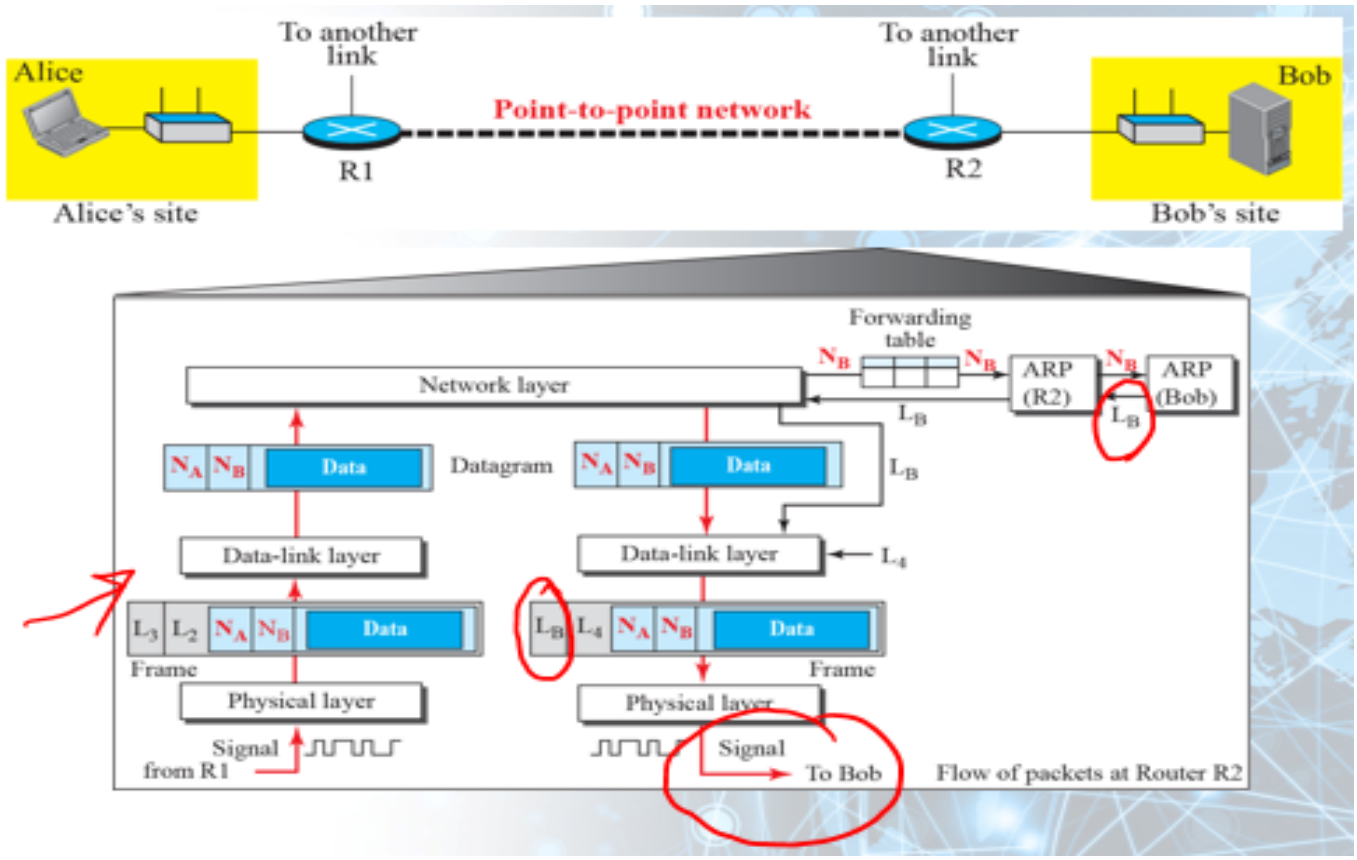
Flow of activities at router R1



Router R1, as we know, has only three lower layers.

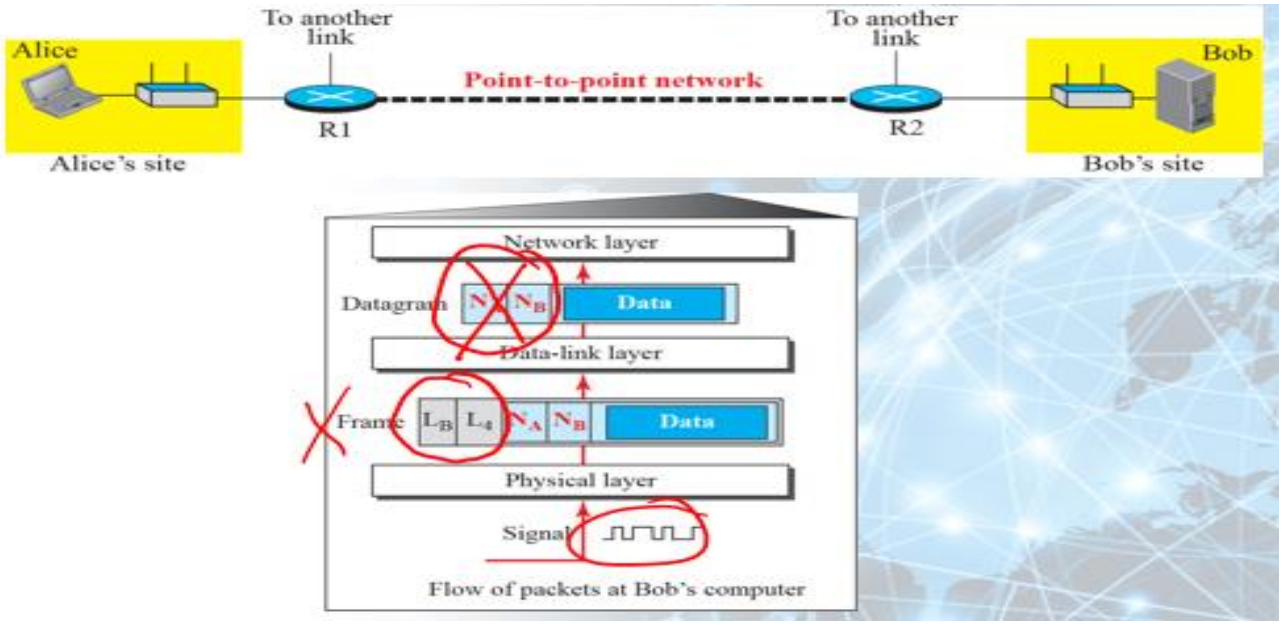
- The packet received needs to go up through these three layers and come down.
- At arrival, the **physical layer of the left link** creates the frame and **passes it to the data-link layer**.
- The data-link layer **decapsulates the datagram** and **passes it to the network layer**.
- The network layer **examines the network-layer address** of the datagram and finds that the datagram **needs to be delivered to the device with IP address NB**.
- The network layer **consults its routing table** to find out which is the next node (router) in the path to NB.
- The forwarding table returns to N3.
- The IP address of router R2 is in the same link as R1.
- The network layer **now uses the ARP to find the link-layer address of this router**, which comes up as L3.
- The network layer passes the datagram and L3 to the data-link layer belonging to the link at the right side.
- The link layer encapsulates the datagram, adds L3 and L2 (its own link-layer address), and passes the frame to the physical layer.
- The physical layer encodes the bits to signals and sends them through the medium to R2.

Flow of activities at router R2



Activities at router R2 are almost the same as in R1

Activities at Bob's site



Now let us see what happens at Bob's site, how the signals at Bob's site are changed to a message.

- At Bob's site there are no more addresses or mapping needed.
- The signal received from the link is changed to a frame.
- The frame is passed to the data-link layer, which decapsulates the datagram and passes it to the network layer.
- The network layer decapsulates the message and passes it to the transport layer.

Topic 135 – 148

Types of Errors -1

Data transmission suffers unpredictable changes because of interference.

Single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.

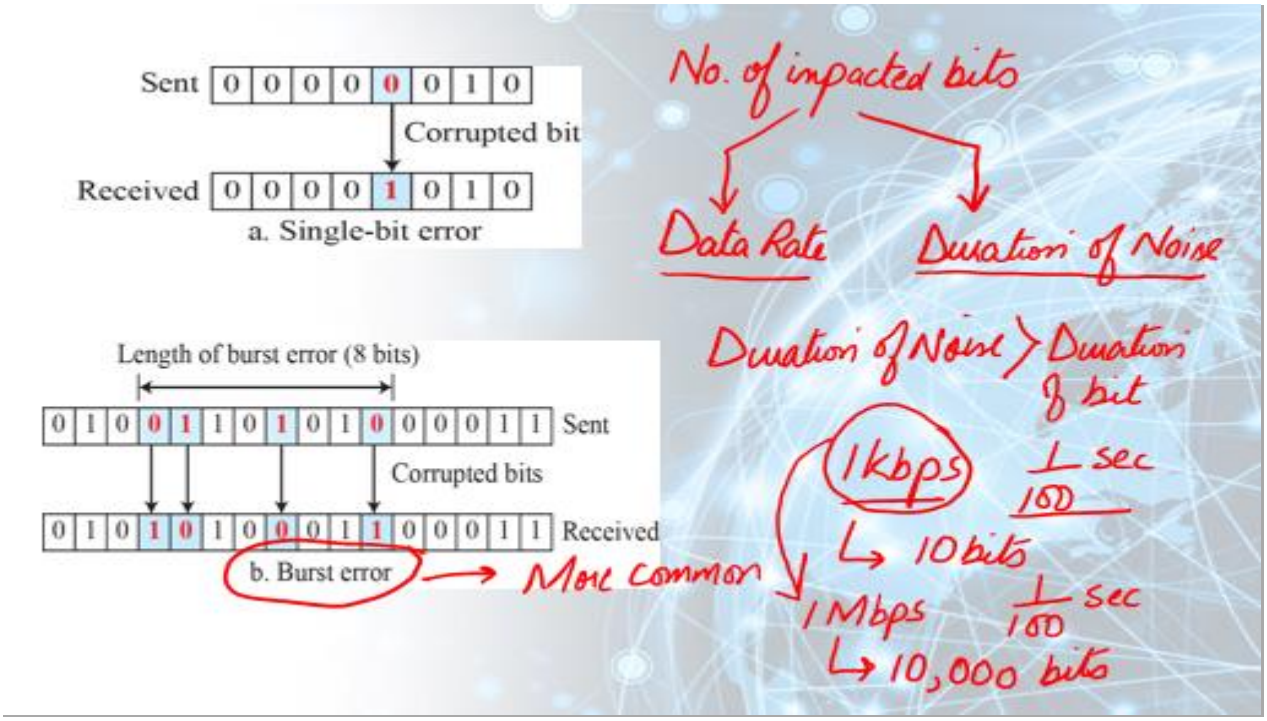
Burst Error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1.

Whenever bits flow from one point to another, they are subject to **unpredictable** changes because of **interference**. This interference can change the **shape of the signal**.

The term single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.

The term burst error means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1 **Types of Errors -2**

Types of Errors -2



- The number of bits affected depends on the **data rate** and **duration of noise**.
- If we are sending data at 1 kbps, a noise of 1/100 second can affect 10 bits.
- If we are sending data at 1 Mbps, the same noise can affect 10,000 bits.

Redundancy

- The central concept in detecting or correcting errors is redundancy.
- To be able to detect or correct errors, we need to send some extra bits with our data.
- These redundant bits are added by the sender and removed by the receiver.
- Their presence allows the receiver to detect or correct corrupted bits.

Detection versus Correction

- **Correction** is more difficult than **detection**.
- **In error detection**, we are only looking to see if any error has occurred (Yes or No)
- We are not interested in the number of corrupted bits in **Detection**.
- **Single-bit error** is the same as a **Burst error**.
- In **Error Correction**, we need to know:
    - The **exact number** of bits that are corrupted and,
    - Their **location** in the message

The **number of errors** and the **size of the message** are important factors.

If we need to correct a **single error** in an **8-bit data unit**, we need to **consider eight possible error locations**.

If we need to correct **two errors** in a **data unit of the same size**, we need to **consider 28 (permutation of 8 by 2) possibilities**.

You can imagine the receiver's difficulty in finding 10 errors in a data unit of 1000 bits.

Coding

- **Redundancy** is achieved through various coding schemes. Sender adds redundant bits through a process that creates a relationship between redundant bits and the actual data bits.
- The receiver checks the relationships between the two sets of bits to detect errors.
- The ratio of redundant bits to data bits and the robustness of the process are important factors in any coding scheme.

Types of Coding Schemes

We concentrate on block coding; convolution coding is more complex and beyond the scope of this book.

Block Coding

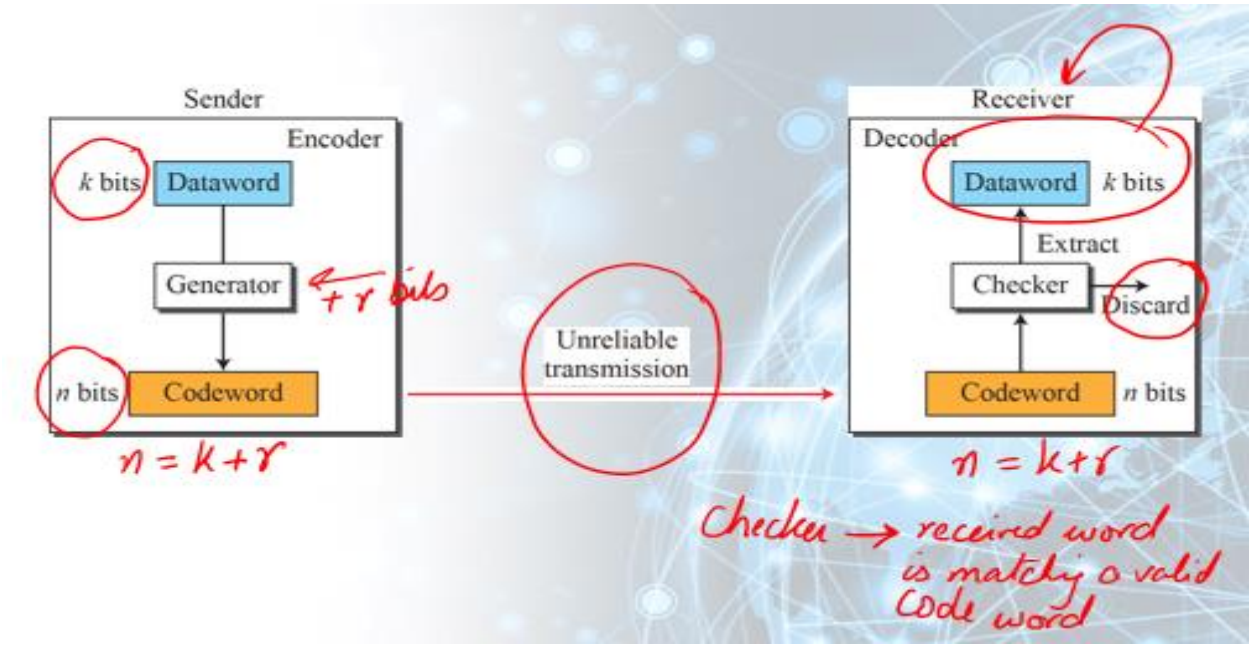
- We divide our message into blocks, each of 'k' bits, called data words.
- We add 'r' redundant bits to each block to make the length 'n = k + r'
- The resulting 'n-bit' blocks are called codewords.

BLOCK CODING in Error Detection

If the following two conditions are met, the receiver can detect a change in the original code word:

-
  - The receiver has (or can find) a list of valid codewords.
  - The original code word has changed to an invalid one.

Process of Error Detection in Block Coding



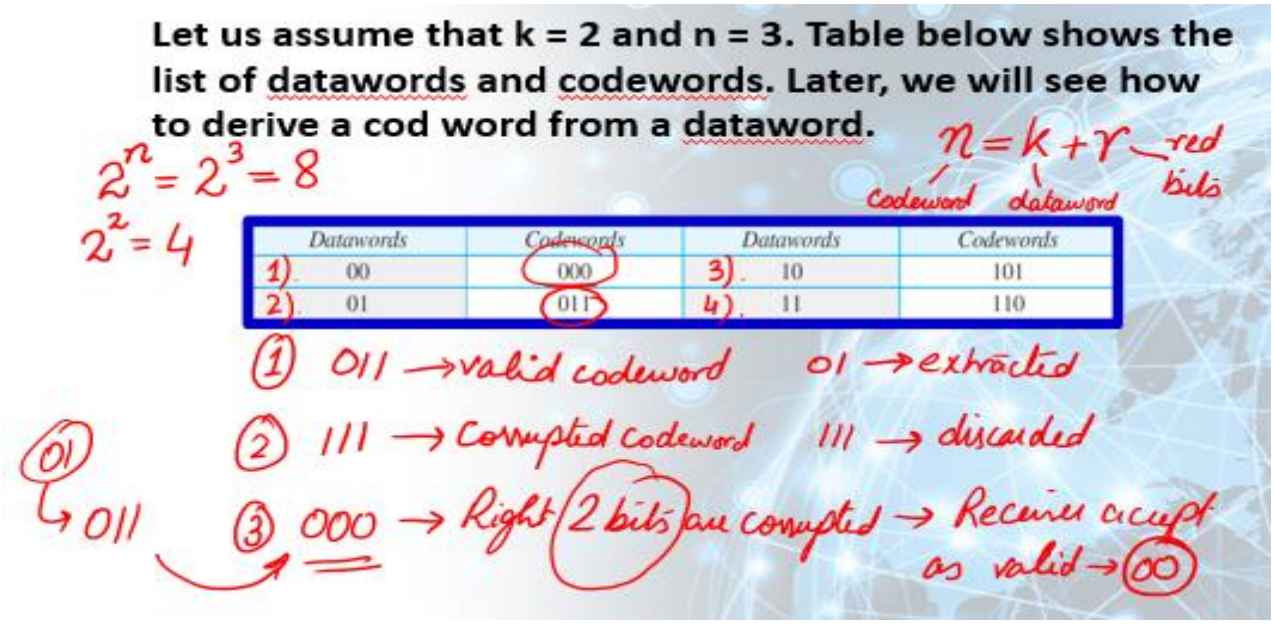
The above figure shows the role of block coding in error detection.

- The sender creates code-words out of data words by using a generator that applies the rules and procedures of encoding (discussed later).
- Each code word sent to the receiver may change during transmission.
- If the received code word is the same as one of the valid codewords, the word is accepted; the corresponding data word is extracted for use.
- If the received code word is not valid, it is discarded.
- However, if the code word is corrupted during transmission but the received word still matches a valid code word, the error remains undetected.

Block Coding:

- We divide our message into blocks, each of ‘k’ bits, called data words.
- We add ‘r’ redundant bits to each block to make the length ‘n = k + r’
- The resulting ‘n-bit’ blocks are called codewords.

Example



In the above example, assume the sender encodes the data word 01 as 011 and sends it to the receiver.

Consider the following cases:

Case:1

1. The **receiver receives 011**. It is a **valid code word**. The receiver **extracts the data word 01 from it**.

Case:2

2. The **code word is corrupted during transmission**, and **111 is received** (the **leftmost bit is corrupted**). This is **not a valid code word** and is **discarded**.

Case:3

3. The code word is **corrupted during transmission**, and **000 is received** (**the right two bits are**

corrupted). This is a **valid code word**. The receiver **incorrectly extracts** the data word 00.

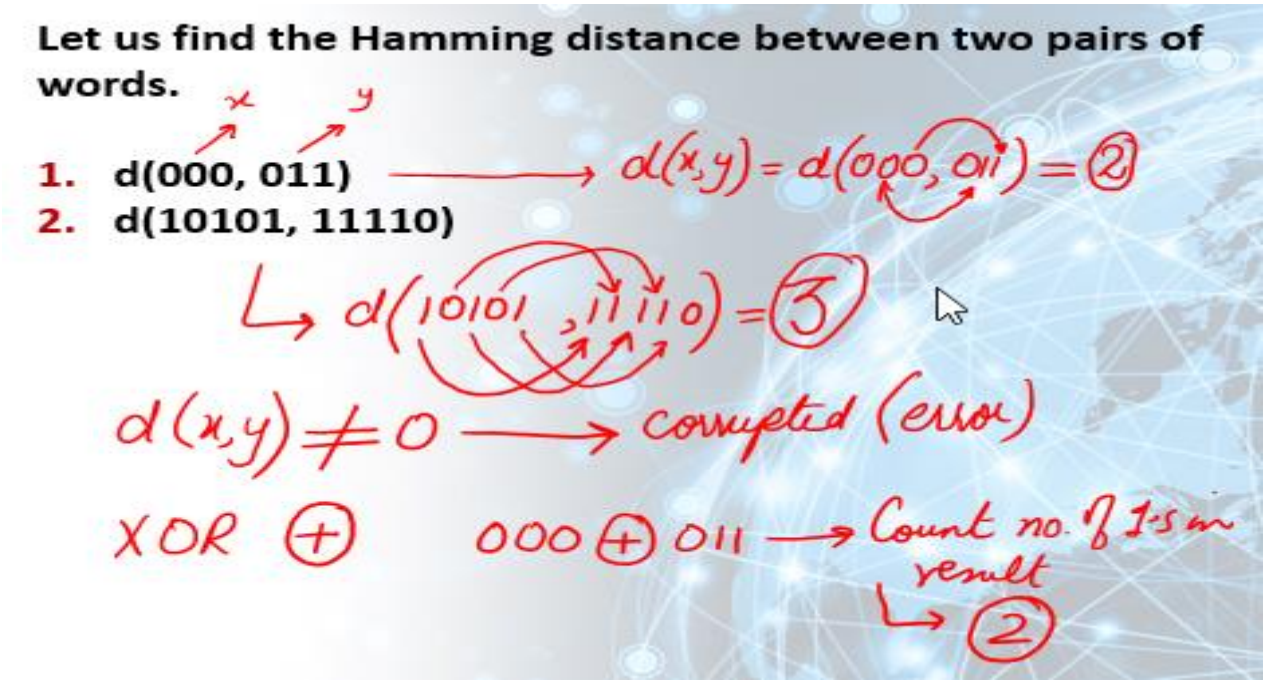
Two corrupted bits have made the error undetectable.

Hamming Distance

- The Hamming Distance between two words of the same size is the number of differences between the corresponding bits.
- We show Hamming Distance between two words x and y is *d*(x,y)

- The humming distance between received code word and sent code word is a number of bits corrupted during transmission.

Example



Let us find the Hamming distance between two pairs of words.

1. The Hamming distance d(000, 011) is 2 because (000 ⊕ 011) is 011 (two 1s).

2. The Hamming distance d(10101, 11110) is 3 because (10101 ⊕ 11110) is 01011 (three 1s)

If the Hamming distance between the sent and the received code word is not zero, the code word has been corrupted during transmission.

The Hamming distance can easily be found if we apply the XOR operation (⊕) on the two words and count the number of 1s in the result.

Note that the Hamming distance is a value greater than or equal to zero.

Minimum Hamming distance is a concept used to measure the similarity between two sequences, often represented as binary strings. Imagine comparing two binary sequences of equal length, bit by bit.

- Minimum Hamming Distance is smallest hamming distance between all possible pairs of codewords
- dmin= s+1

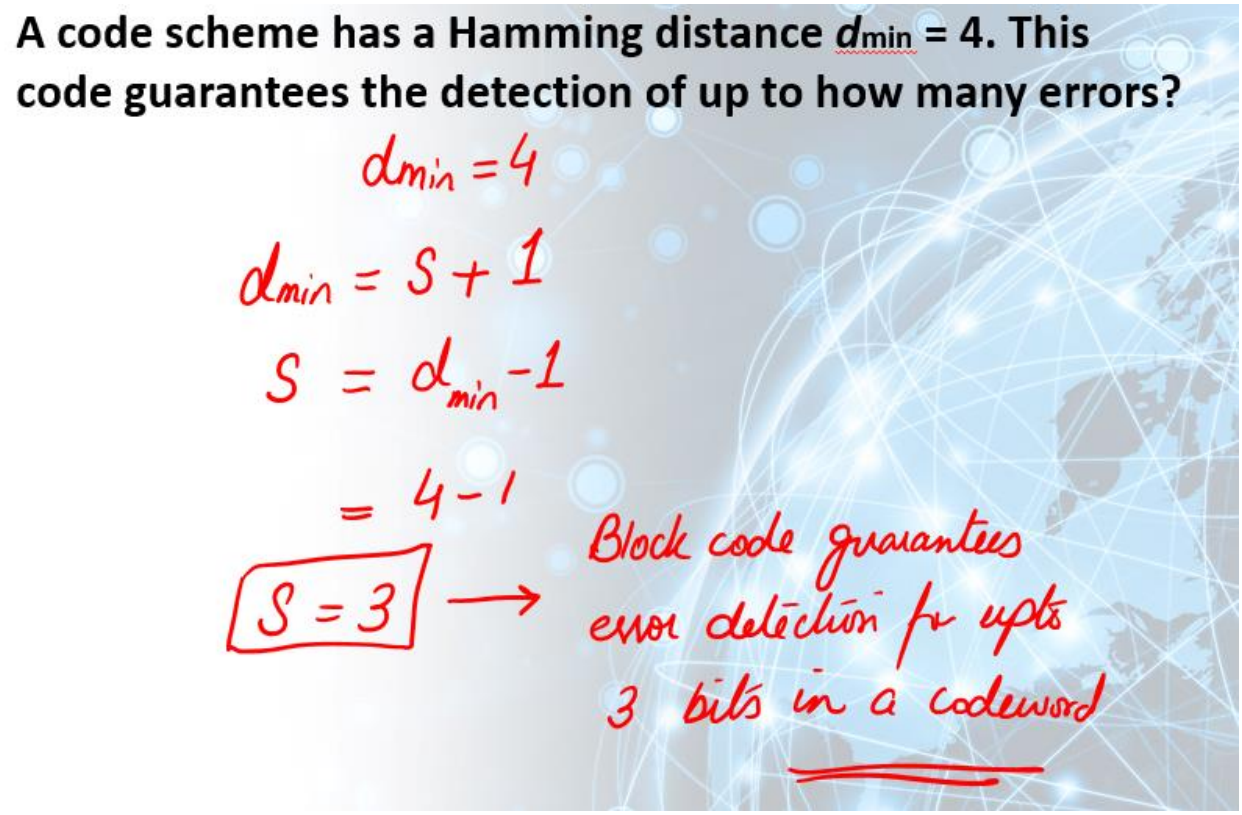
where,

s →no. of detectable errors

dmin→ minimum hamming distance

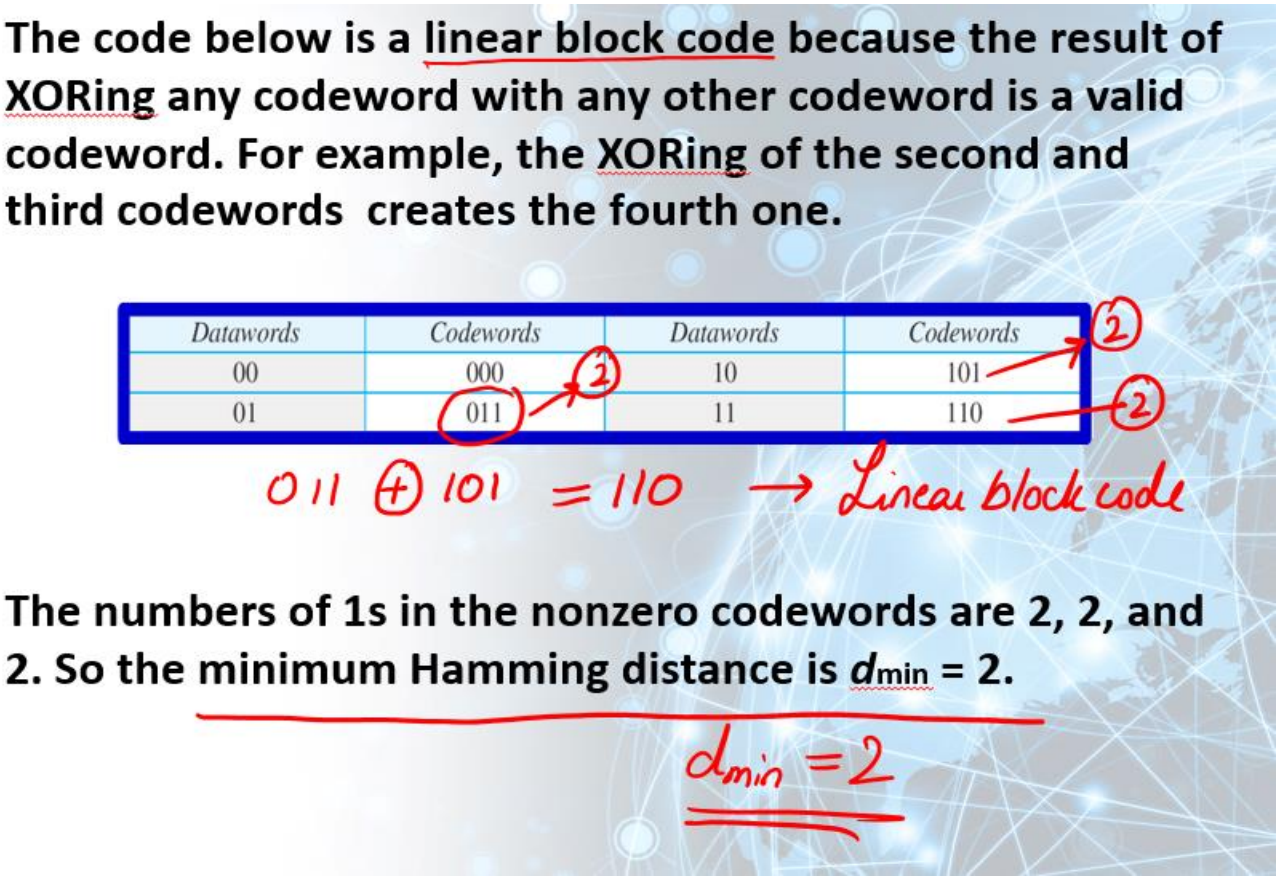
The Hamming distance is the count of differing bits between these sequences. The minimum Hamming distance specifically refers to the smallest number of bit changes needed to make the two sequences identical.

For example, if we have binary strings '10101' and '10011,' the Hamming distance is 2, but the minimum Hamming distance is 2, as changing two bits is the smallest modification required for them to match. This concept finds applications in various fields, including error detection in computer systems and DNA sequence analysis.



Linear Block Codes

Linear block codes are a specific subset of block codes that possess the additive property you described. In linear block codes, the exclusive OR (XOR) of two valid codewords results in another valid codeword. This property is a fundamental characteristic of linear block codes and is closely related to their algebraic structure.

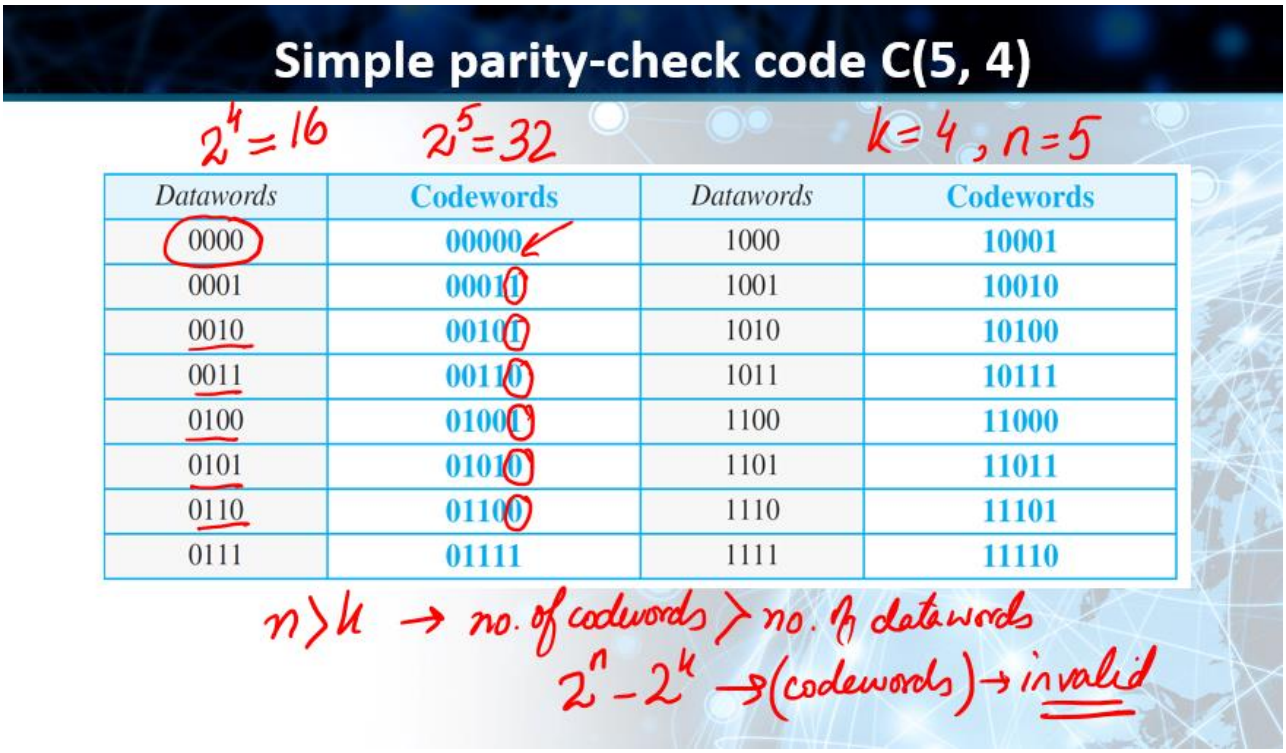


Parity-Check Code

- Most common error-detecting code
- Linear block code (n=k+1)
- The extra parity bit is selected to make total number of 1s in codeword even

A parity check code is a simple error-detecting code used in digital communication.

- It involves adding a single parity bit to a data message, ensuring the total number of bits with a value of 1 is either even (even parity) or odd (odd parity).
- This allows the detection of single-bit errors during transmission.
- Even parity makes the total 1s even, while odd parity makes it odd. The receiver checks the parity of the received message, and a mismatch indicates a single-bit error.
- Parity checks cannot correct errors but serve as a basic error-detection method.



Datawords	Codewords	Datawords	Codewords
0000	00000	1000	10001
0001	00011	1001	10010
0010	00101	1010	10100
0011	00110	1011	10111
0100	01001	1100	11000
0101	01010	1101	11011
0110	01100	1110	11101
0111	01111	1111	11110

CYCLIC CODES

Cyclic codes are a class of linear block codes with a specific algebraic property related to cyclic shifts. These codes exhibit a remarkable structure where, if a codeword is valid, any cyclic shift (rotation) of that codeword is also a valid codeword.

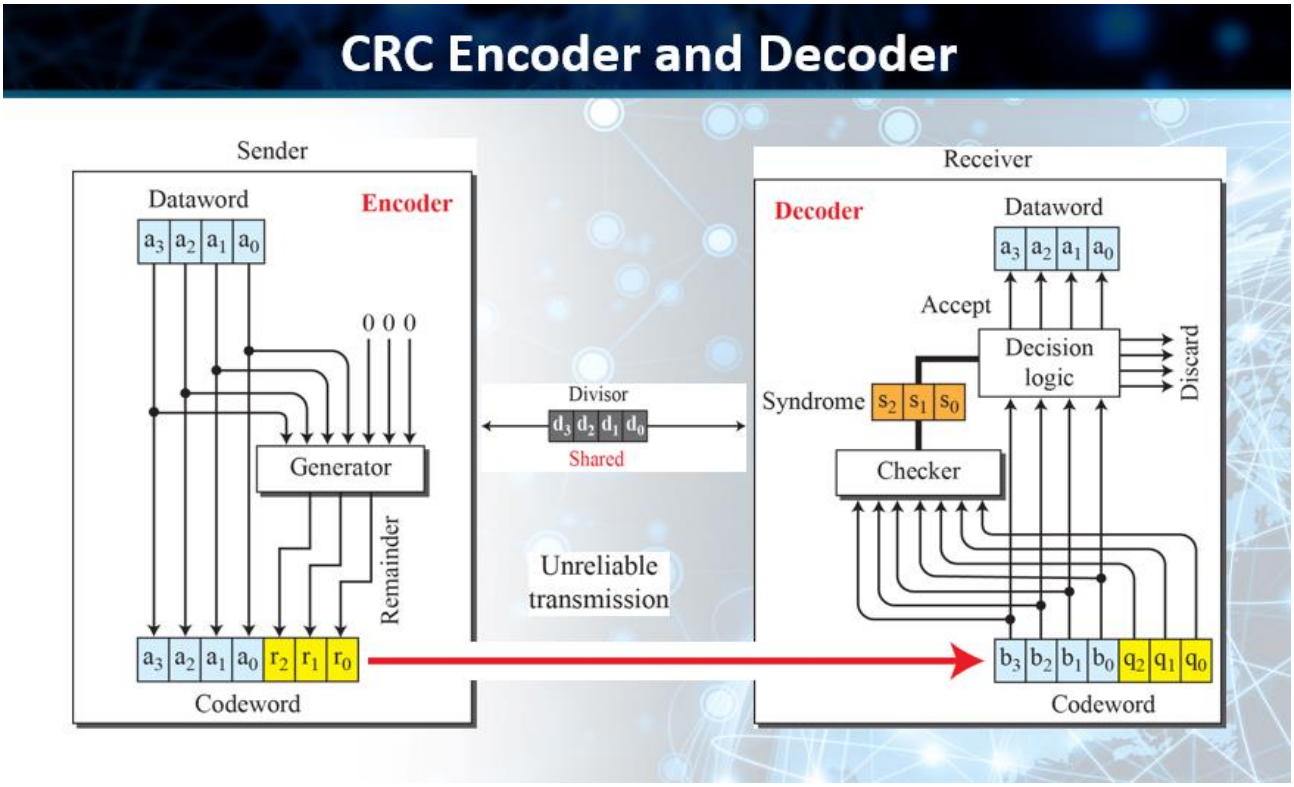
- This property simplifies the encoding and decoding processes and makes cyclic codes particularly useful in various communication systems.
- If 1011000 is a codeword and we cyclically left-shift, then 0110001 is also a codeword

Cyclic Redundancy Check (CRC)

- Subset of Cyclic Codes
- Cyclic redundancy check (CRC) is used in networks such as LANs and WANs

Cyclic Redundancy Check (CRC):

- CRC codes, commonly used in error detection in network communication and storage systems, are a specific type of cyclic code.
- CRC codes use polynomial division to generate a check sequence, and their cyclic properties allow them to detect burst errors effectively.



Advantages of Cyclic Codes:

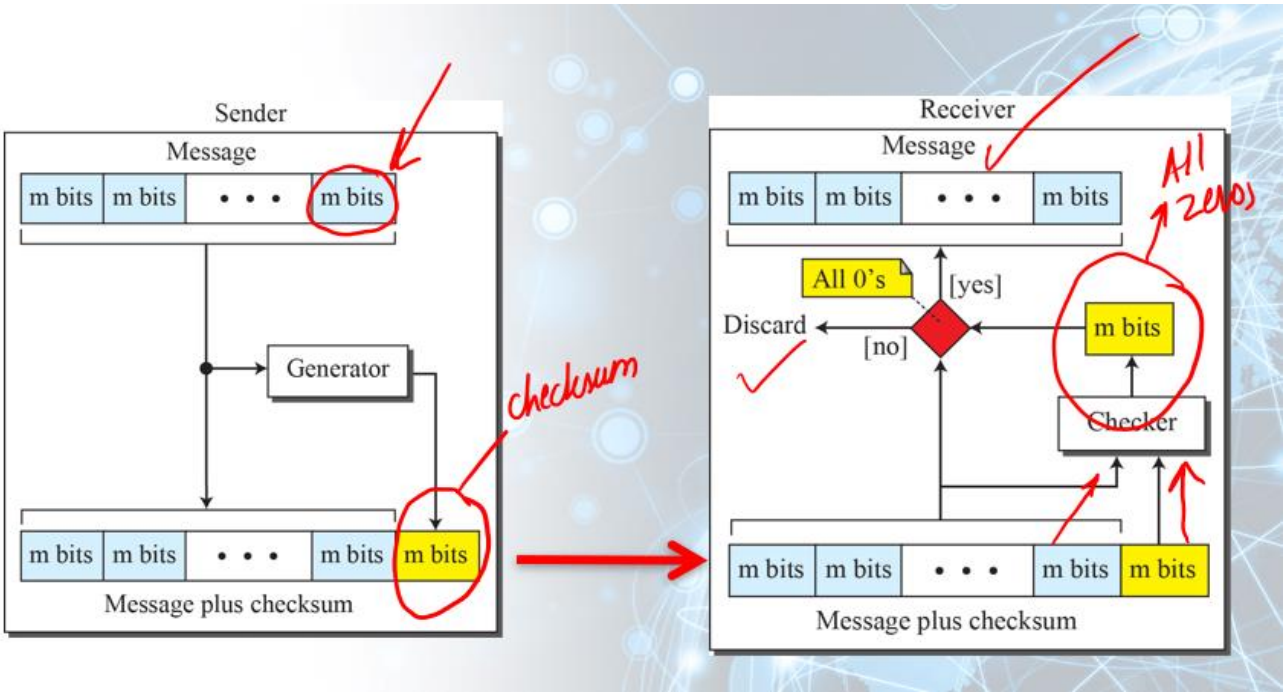
- Good performance in detection:
  - Single-bit errors
  - Double errors
  - Odd number of errors
  - Burst errors
- Easy Implementation
- Fast Implementation

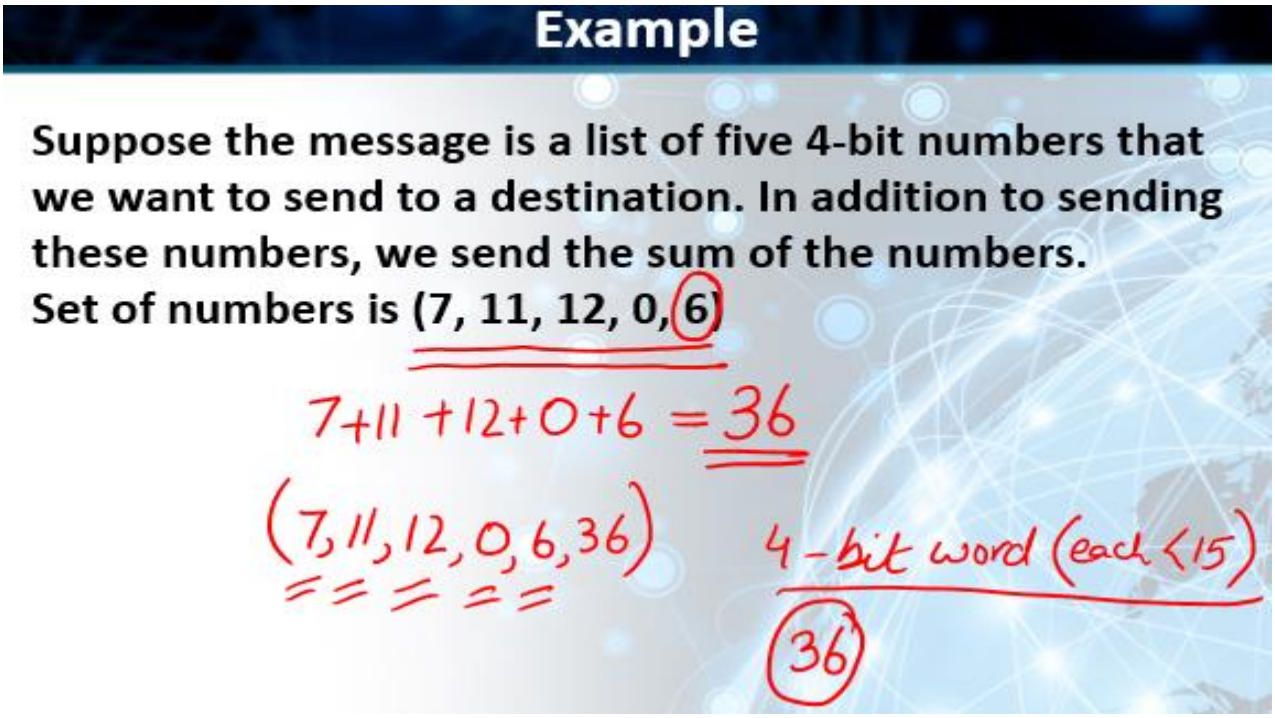
CHECKSUM

A checksum is a value derived from the digital data of a message or file, used to verify the integrity of the data during transmission or storage.

- The checksum is calculated using a specific algorithm and is typically a fixed-size string of characters or a numerical value.

- Error-detection technique that can be applied to a message of any length
- Checksum mostly used at the network and transport layer rather than the data-link layer





Forward Error Correction

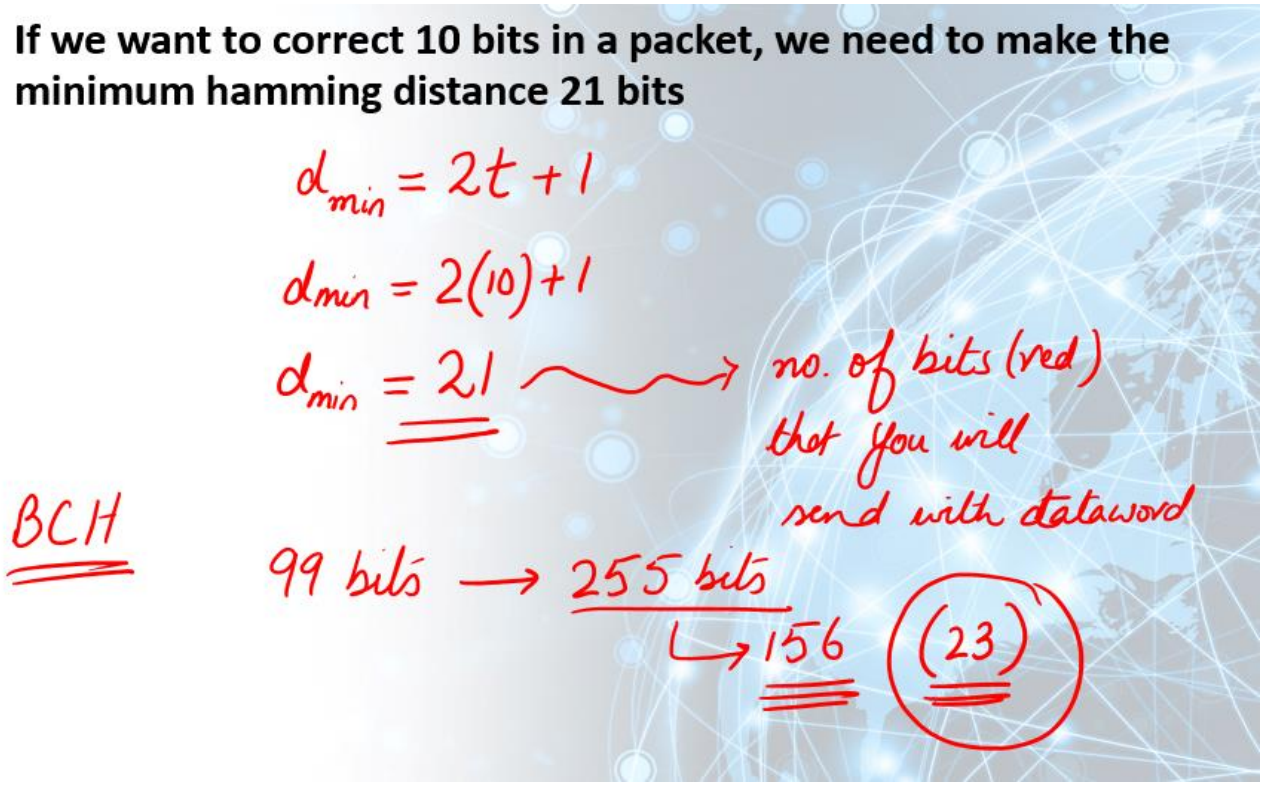
If packets of data get corrupted or lost during real-time multimedia transmission, just asking for them again (retransmission) is not helpful because it can cause delays. Instead, we want to fix errors or recreate the missing data right away. Various methods for doing this are called Forward Error Correction techniques.

Using Hamming Distance

- For error detection, we definitely need more distance
- It can be shown that to correct 't' errors, we need to have:

$$d_{min} = 2t + 1$$

- If we want to correct 10 bits in a packet, we need to make the minimum hamming distance 21 bits
- A lot of redundant bits need to be sent with the data



Another way to achieve FEC in multimedia is to allow some small chunks to be missing at the receiver

We cannot afford to let all the chunks belonging to the same packet be missing; however, we can afford to let one chunk be missing in each packet

Combining Hamming Distance & Interleaving

- Hamming distance and interleaving can be combined
- We can first create n-bit packets that can correct t-bit errors
- Then we interleave m rows and send the bits column by column
- Possible to correct burst errors up to m × t bits of errors

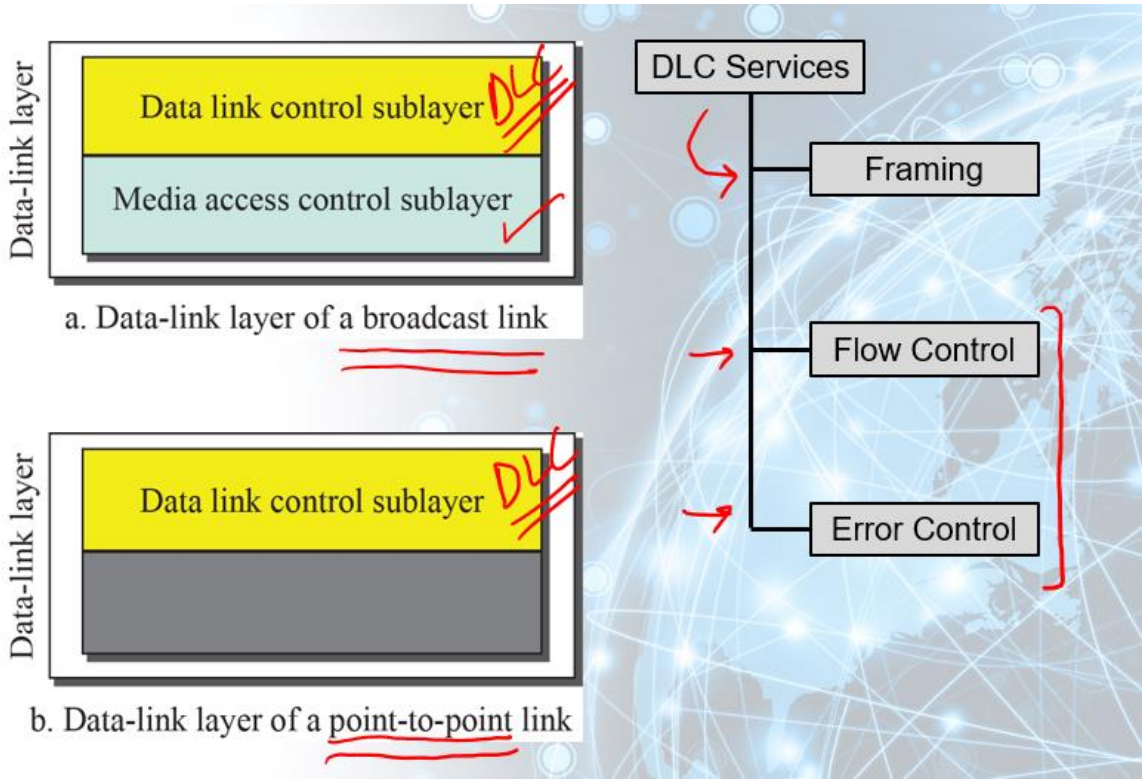
Compounding High & Low Resolution Packets

- **Creation of a duplicate of each packet with a low-resolution redundancy and combine the redundant version with the next packet**
- **For example, we can create four low-resolution packets out of five high-resolution packets and send them**

Topic 149 – 165

Data Link Control (DLC) Services

- The data link control (DLC) deals with procedures for communication between two adjacent nodes no matter whether the link is dedicated or broadcast
- Data link control functions include framing, flow control and error control

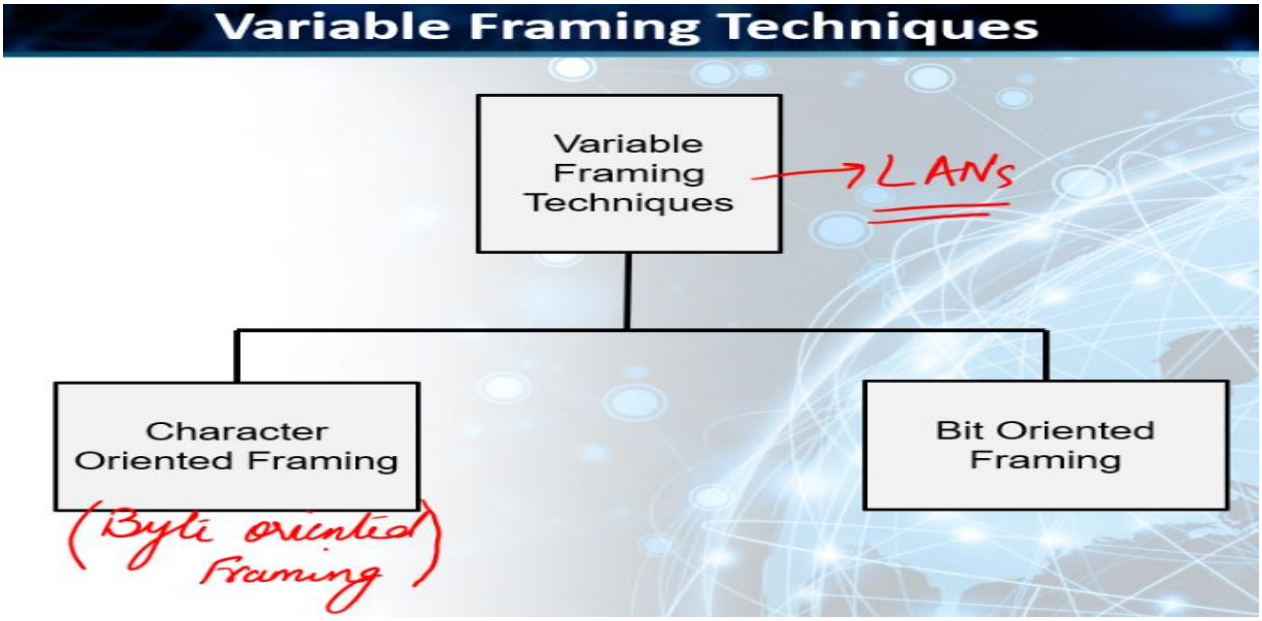


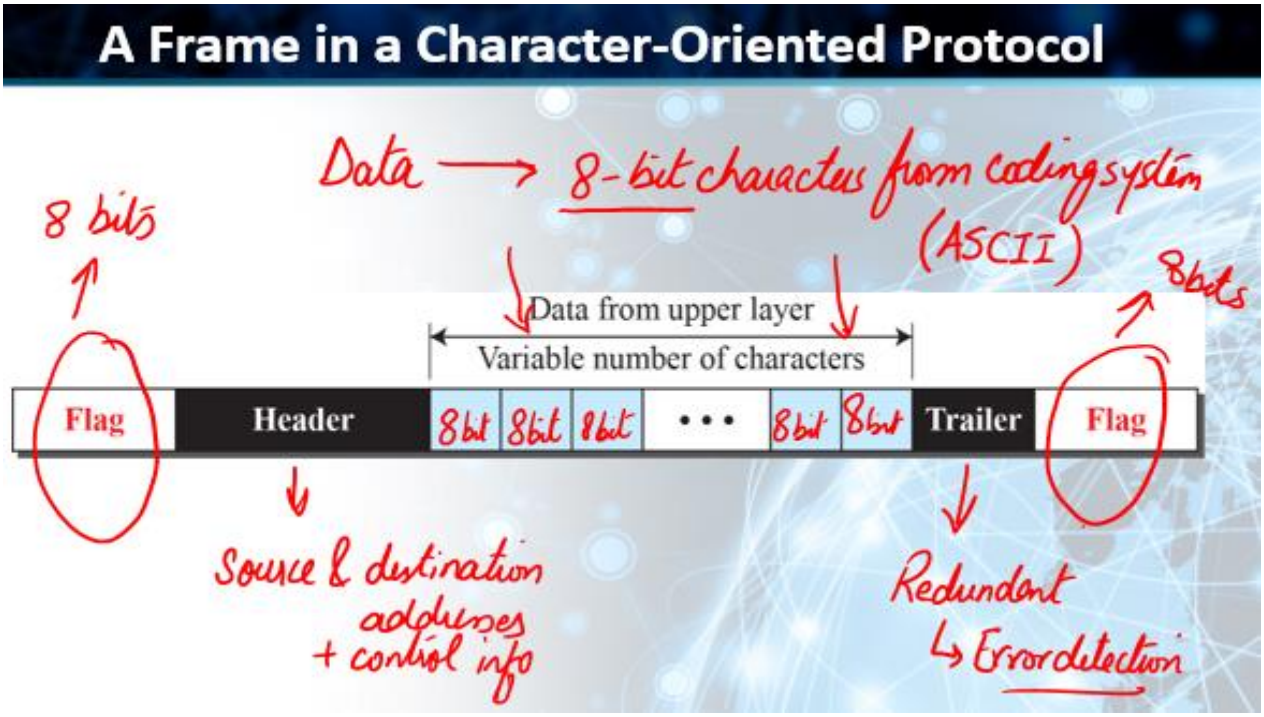
Framing

- Data-Link layer needs to pack bits into frames, so that each frame is distinguishable from another
- Our postal system practices a type of framing
- Framing separates a message by adding a sender address and a destination address
- The destination address defines where the packet is to go; the sender address helps the recipient acknowledge the receipt

Frame Size

- **Why not one BIG Frame?**
- **Frames can be of:**
  - ✓ **Fixed Size**
    - ➢ **Size acts as a boundary/delimiter**
  - ✓ **Variable Size**
    - ➢ **How to define Beginning and End of a Frame?**





Connection Oriented Framing:

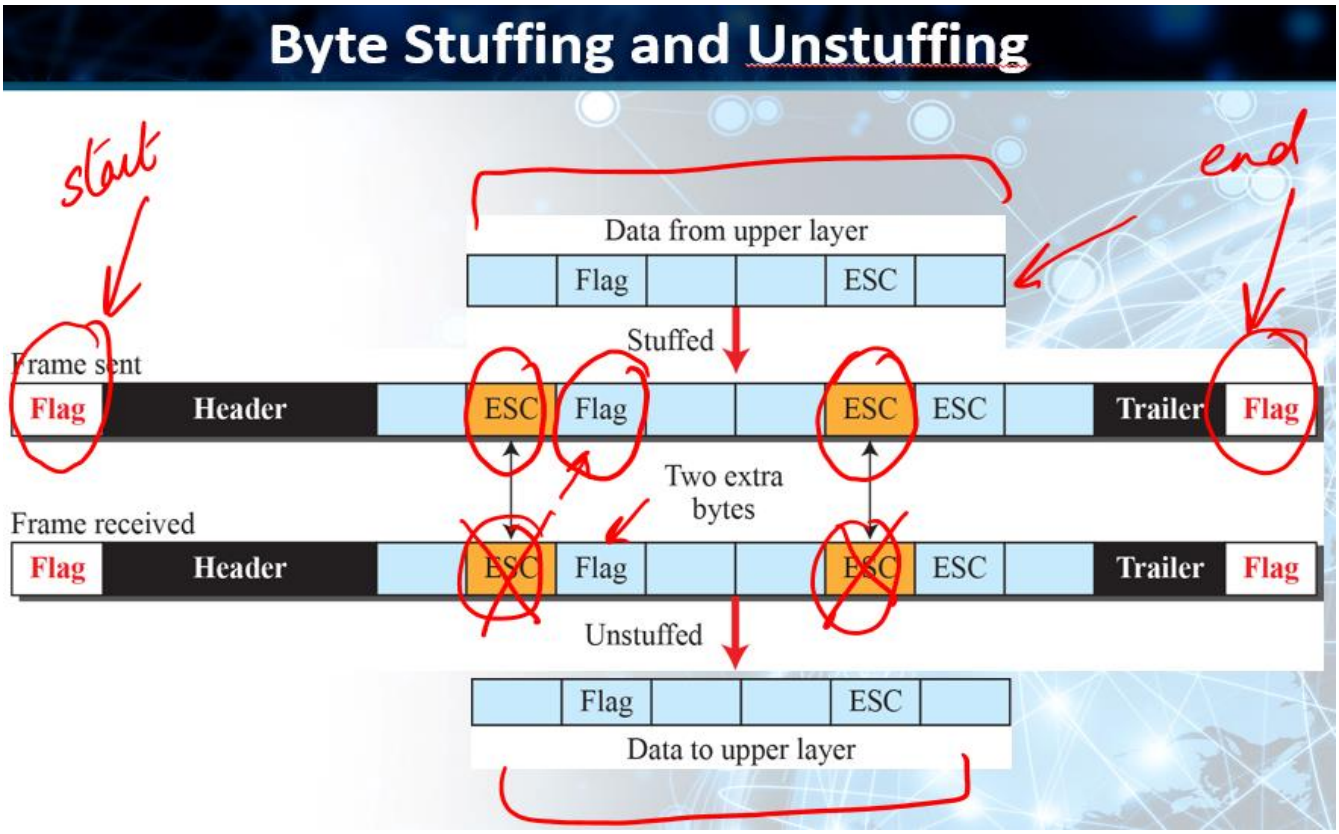
- **Data to be carried are 8-bit characters**

Byte Stuffing in Connection-Oriented Framing:

- Connection-oriented Framing used text characters as flags
- Nowadays any character used for flag can also be a part of the data
- In order to avoid confusing the receiver, we use Byte Stuffing

Several Issues:

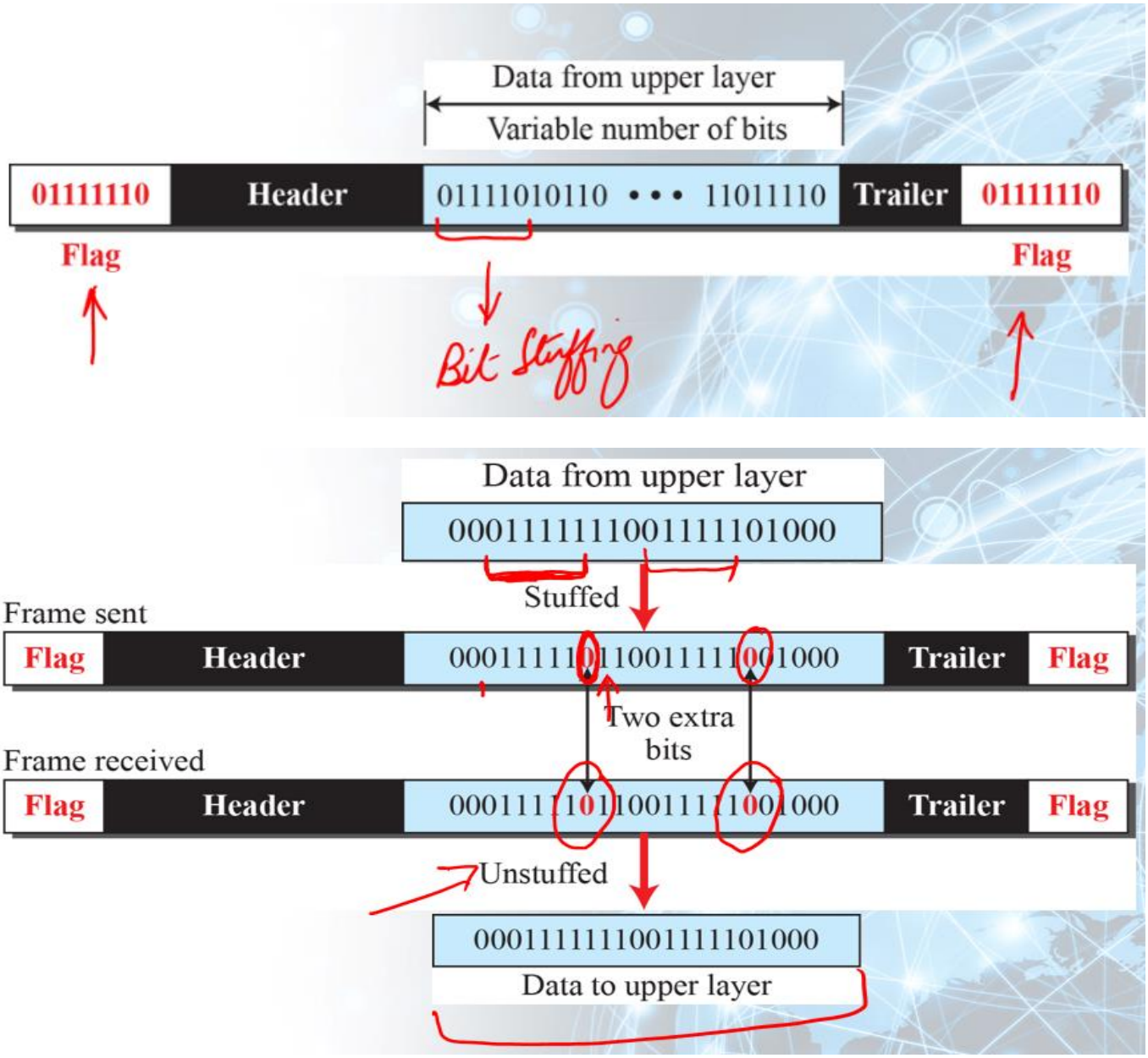
  - One or more escape characters followed by a byte with same pattern as a flag?
  - Unicode (16/32 bit) vs. 8-bit characters
- Data is stuffed with a pre-defined Escape Character (byte) when there is a character with same pattern as a flag



Bit-Oriented Framing

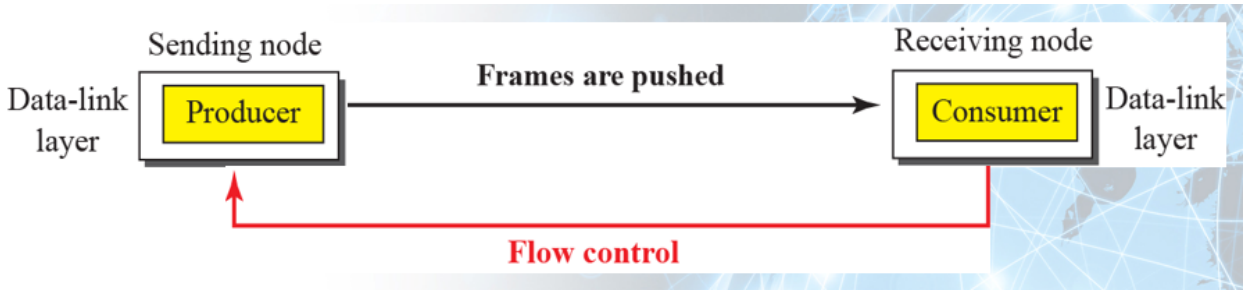
- Data section of frame is a sequence of bits
- We need a delimiter to separate one frame from the other
- A special 8-bit pattern (01111110) to define beginning and end of a frame
- Same issue as Connection-oriented Framing

A Frame in a Bit-Oriented Protocol:



Flow and Error Control

- One of the responsibilities of the data-link control sublayer is flow and error control at the data-link layer
- Balance between production and consumption rates
- If frames are produced faster than they are consumed at the receiving data link layer, the frames will be discarded
- Use of buffers; one at sending end and other at receiving end



Example:

- Consumers need to communicate with the producers on two occasions:
  - When the buffer is full; &
  - When there are vacancies
  - If the two parties use a buffer with only one slot, the communication can be easier

Error Control

- Error Control at Data Link layer uses CRC in one of the two ways:
  - If a frame is corrupted, it is silently discarded and if it is good, it is delivered to network layer
  - If frame is corrupted, it is silently discarded and if it is good, an acknowledgement is sent to sender
- A DLC protocol can be either connectionless or connection-oriented
- Connectionless: No relationship between the frames
- Connection-Oriented: Frames are numbered and sent in order

Error Control

- Error Control at Data Link layer uses CRC in one of the two ways:
  - ✓ If a frame is corrupted, it is silently discarded and if it is good, it is delivered to network layer
  - ✓ If frame is corrupted, it is silently discarded and if it is good, an acknowledgement is sent to sender

Connectionless and Connection-Oriented

- A DLC protocol can be either connectionless or connection-oriented
- Connectionless: No relationship between the frames
- Connection-Oriented: Frames are numbered and sent in order

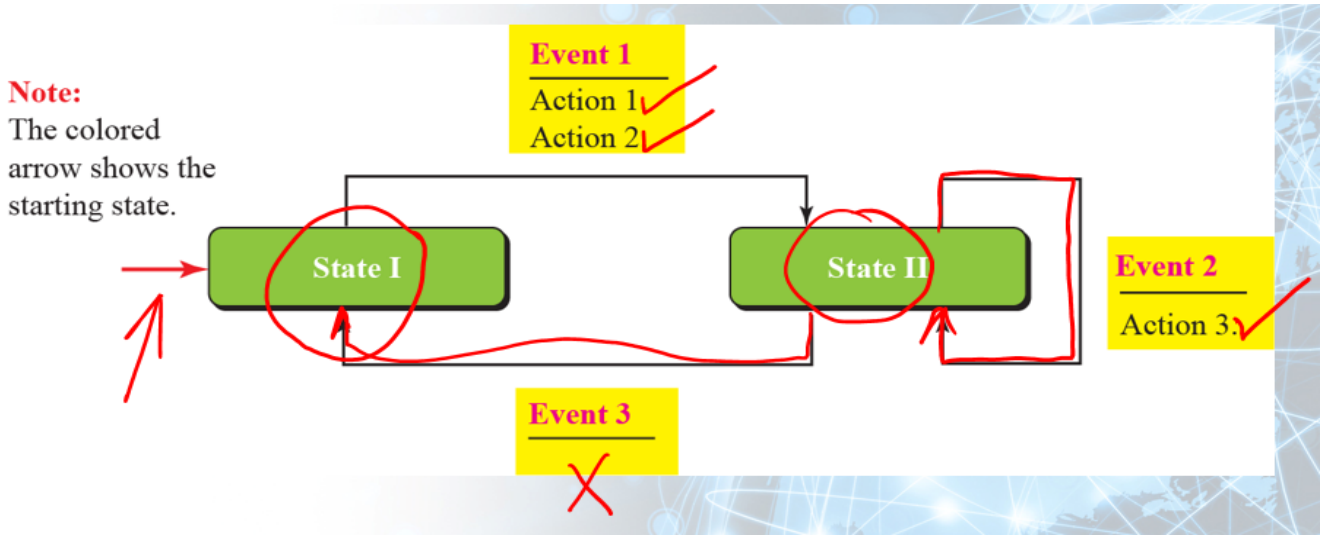
DATA-LINK LAYER PROTOCOLS:

- Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:
  - Simple Protocol

- Stop-and-Wait Protocol
- Go-Back-N Protocol
- Selective-Repeat Protocol
- Last two protocols have almost disappeared completely

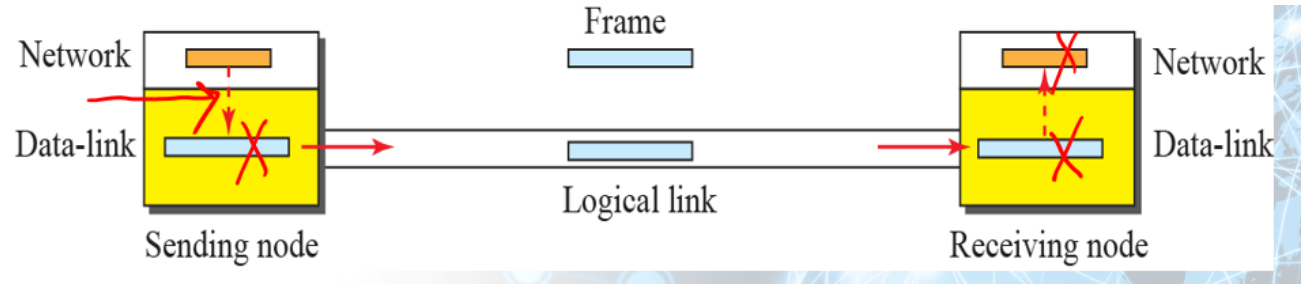
Finite State Machine (FSM):

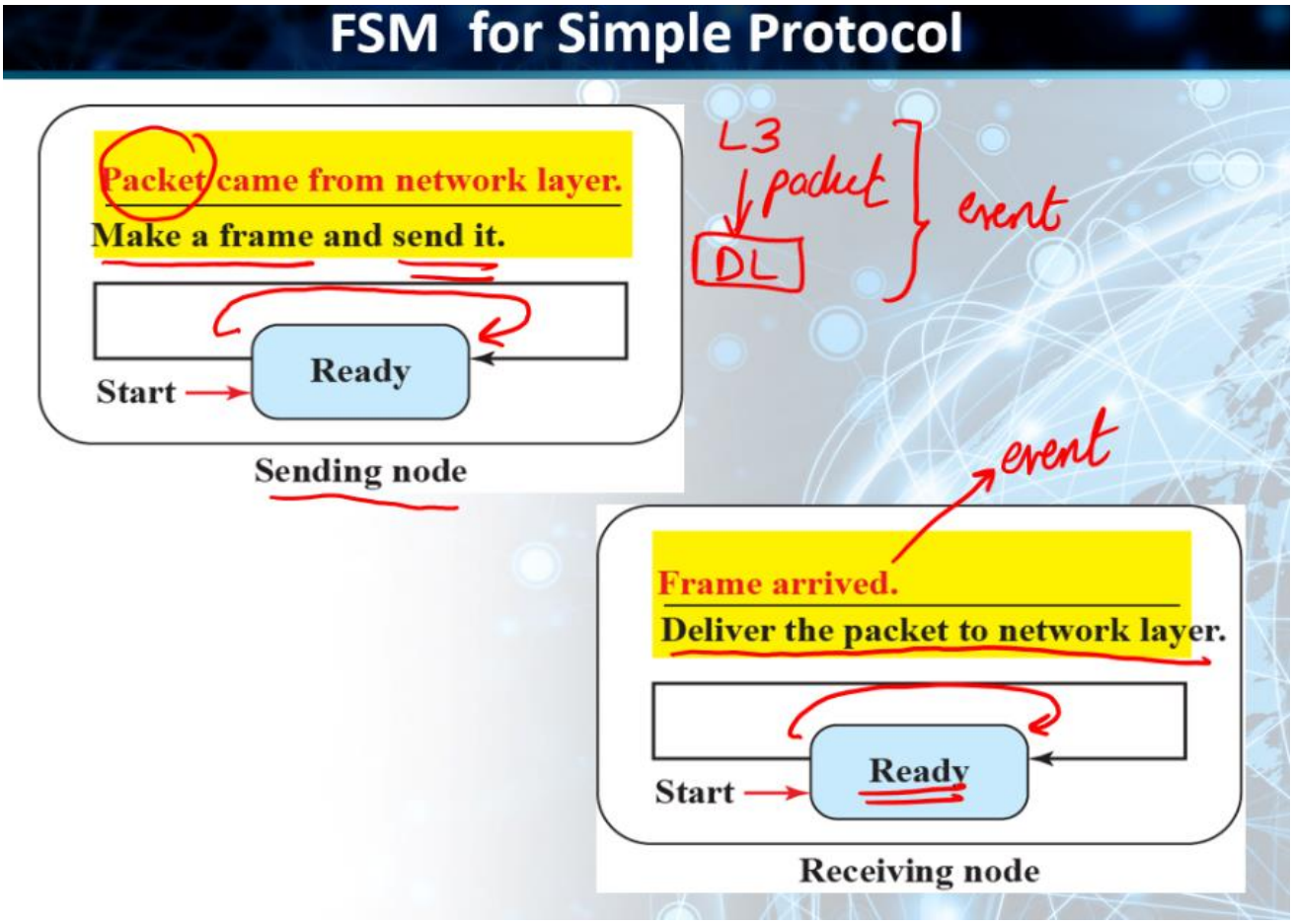
- A machine with a finite number of states
- Machines stays in one of the states until an event occurs
- Each event is associated with 2 reactions:
  - List of actions to be performed
  - Determining the next state



DATA-LINK LAYER PROTOCOLS

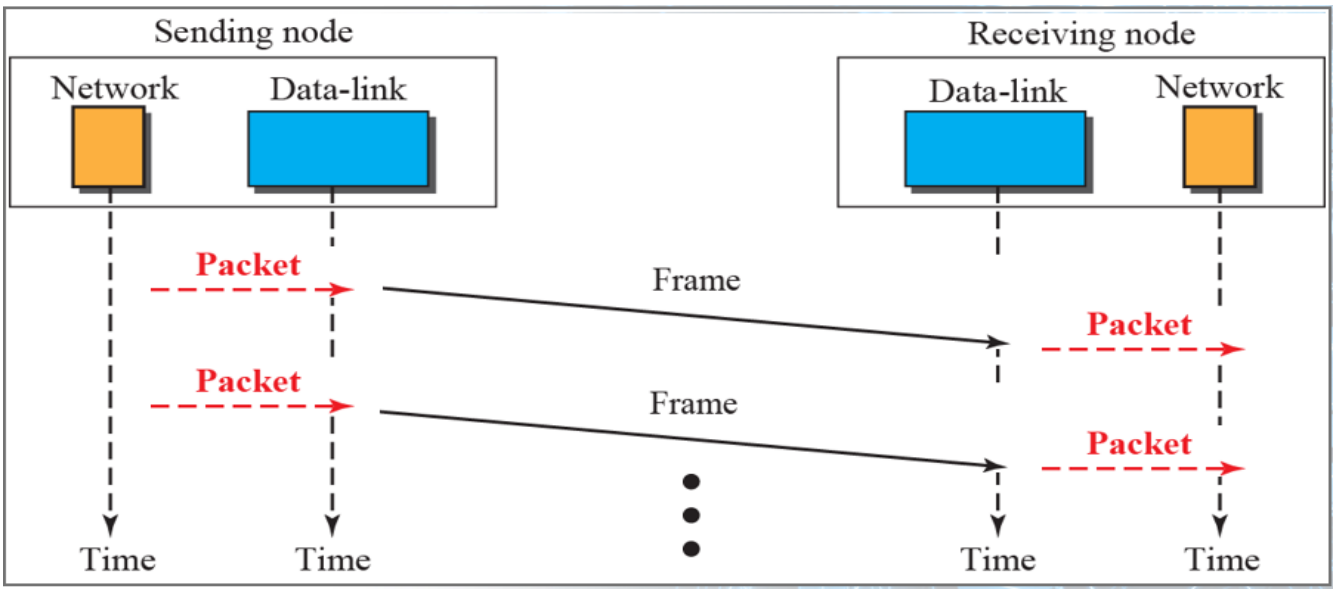
- Traditionally four protocols have been defined for the data-link layer to deal with flow and error control:
  - Simple Protocol
  - Stop-and-Wait Protocol
  - Go-Back-N Protocol
  - Selective-Repeat Protocol
- Simple protocol has neither flow nor error control
- **Assumption:** The receiver can immediately handle any frame it receives
- The receiver can never be overwhelmed with incoming frames





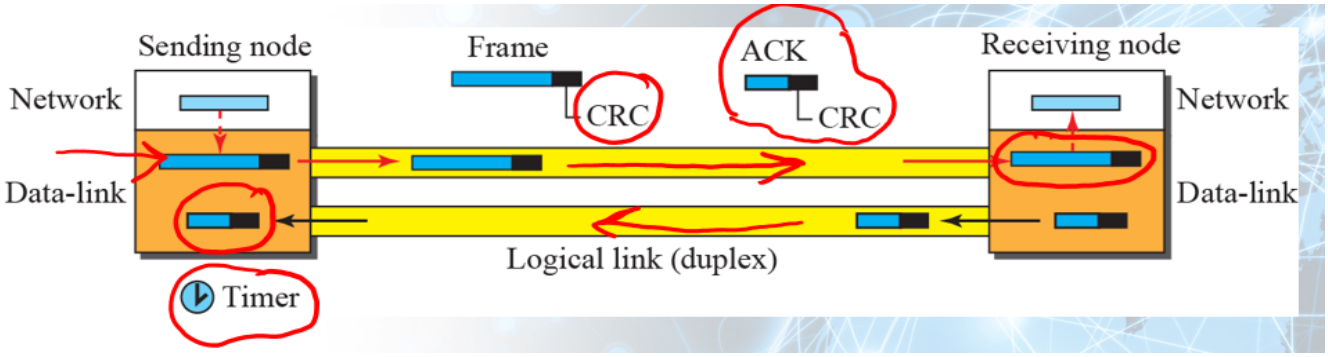
Example:

Here is an example of communication using this protocol. It is very simple. The sender sends frames one after another without even thinking about the receiver.



Stop-and-Wait Protocol

- Stop-and-Wait protocol uses both flow and error control
- The sender sends one frame at a time and waits for an acknowledgment before sending the next one
- To detect corrupted frames, we add a CRC code



Example

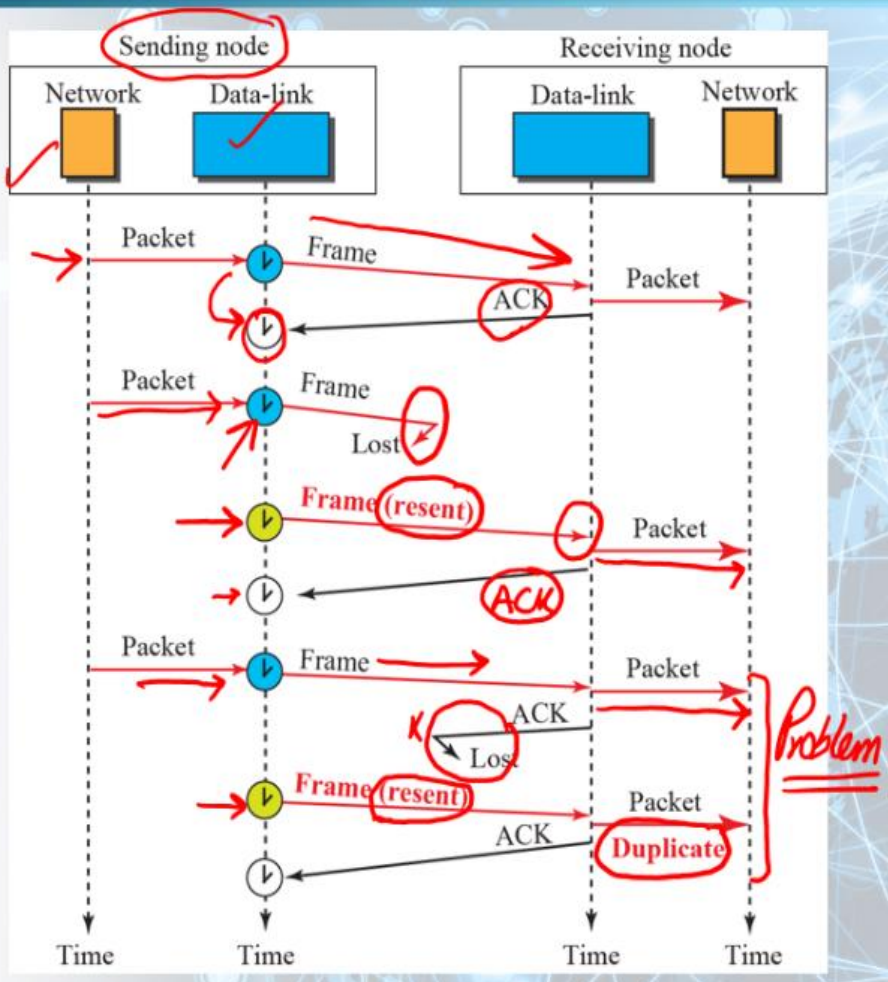
Legend

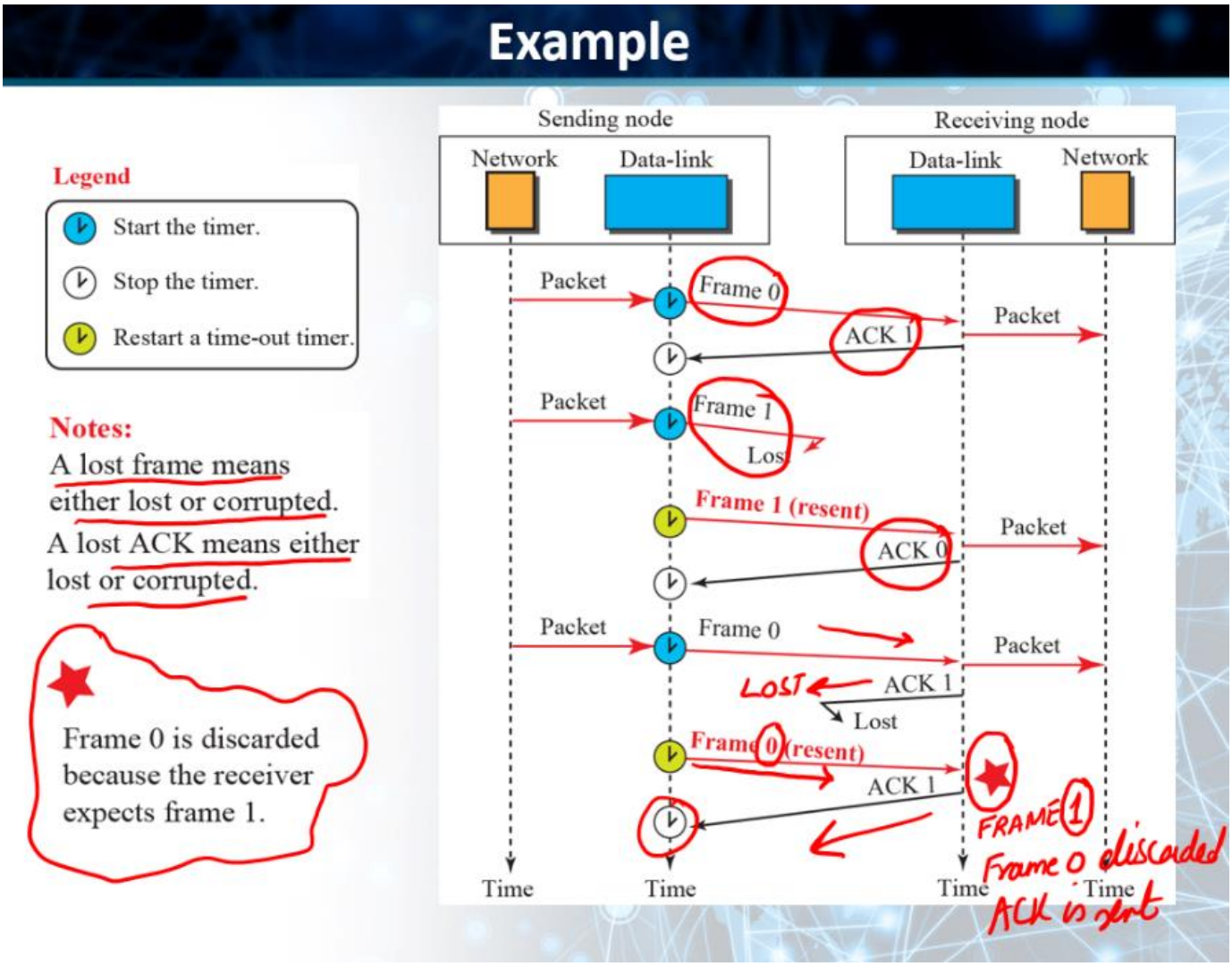
- Start the timer.
- Stop the timer.
- Restart a time-out timer.

Notes:

A lost frame means either lost or corrupted.

A lost ACK means either lost or corrupted.





Piggybacking

- Both Simple and Stop-and-wait protocols are designed for unidirectional communication
- Data flows in one direction and ACK travels in the other
- To make the system efficient, the data in one direction is piggybacked with the acknowledgment in the other direction

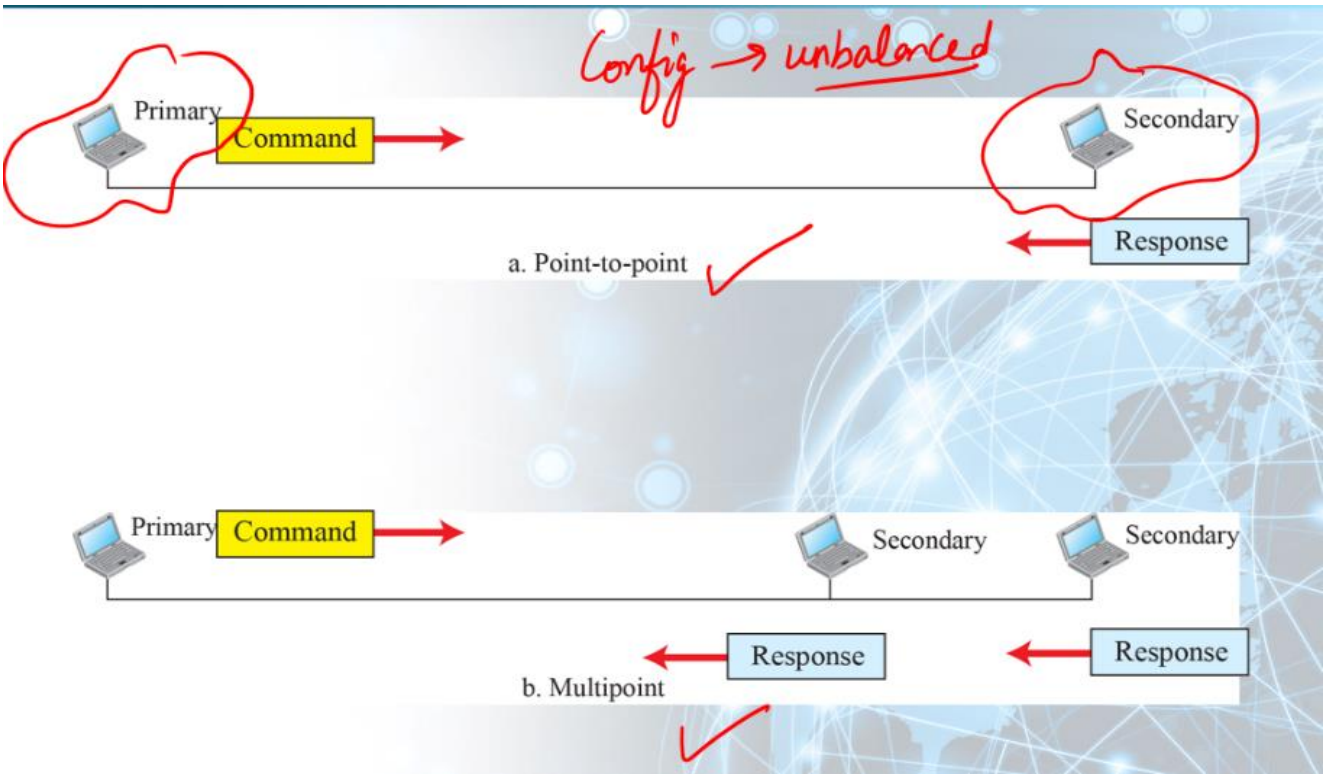
High-level Data Link Control (HDLC)

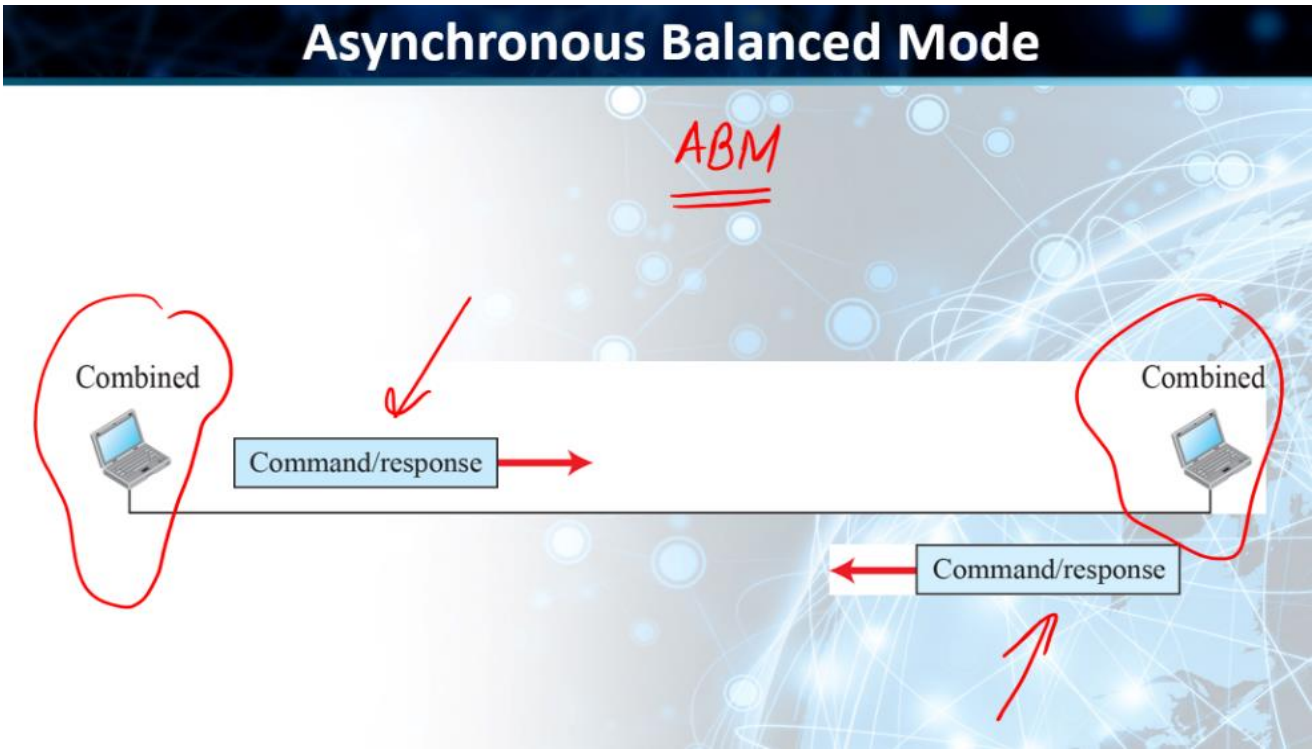
- Bit -oriented protocol for communication over point-to-point and multipoint links
- It implements Stop-and-Wait protocol
- Most of the concepts defined in this protocol is the basis for other protocols such as PPP, Ethernet, or wireless LANs

Configurations & Transfer Modes in HDLC

- **HDLC provides two common transfer modes that can be used in different configurations:**
    - Normal Response Mode (NRM) &
    - Asynchronous Balanced Mode (ABM)

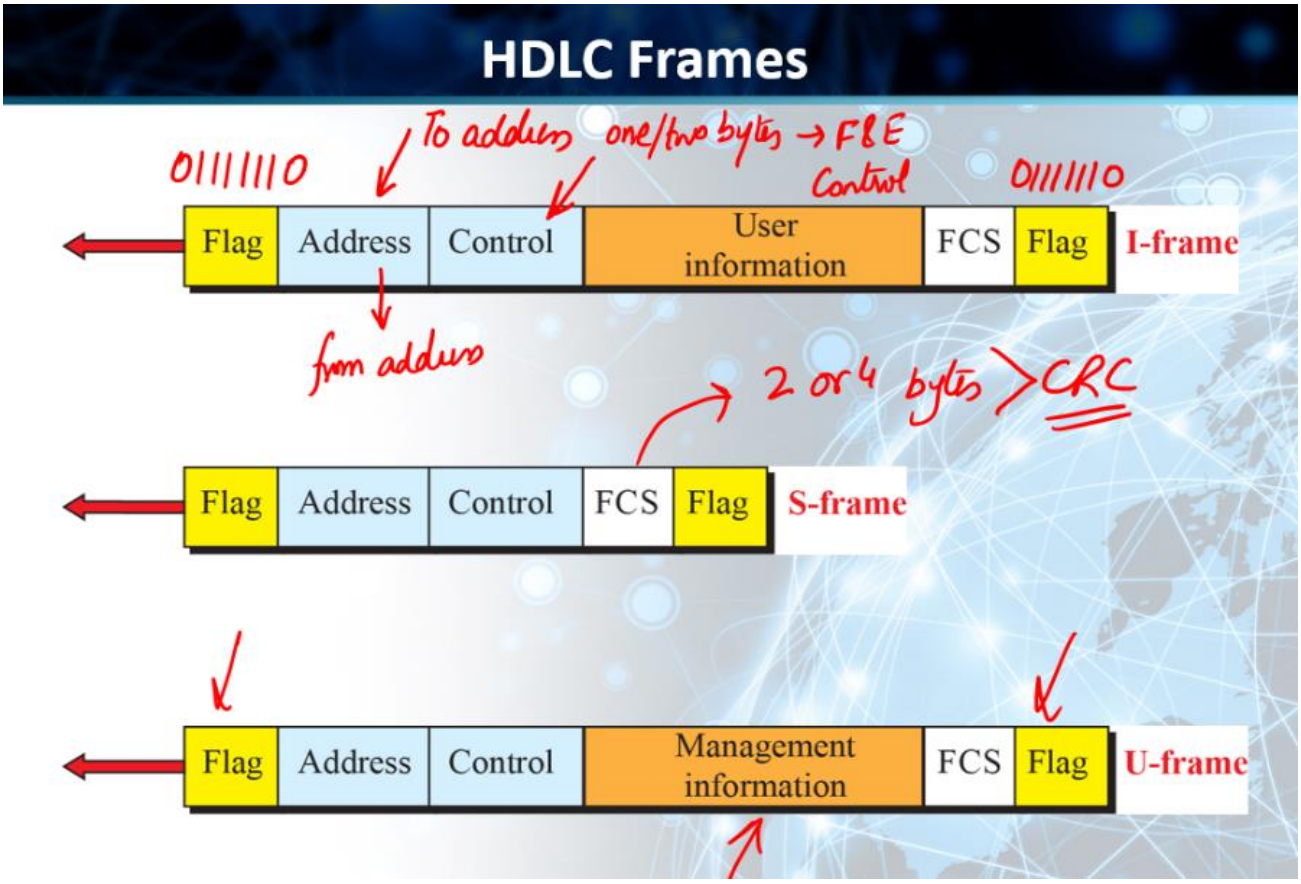
- **HDLC defines three types of frames:**
    - information frames (I-frames)
    - Supervisory frames (S-frames)
    - Unnumbered frames (U-frames)





Framing

- HDLC defines three types of frames:
  - ✓ information frames (I-frames)
  - ✓ Supervisory frames (S-frames)
  - ✓ Unnumbered frames (U-frames)



Single Stream

- A single stream session comes with
- A given bandwidth requirement
- A specified end-to-end delay requirement
- Arrives at the network
- QoS routing is to find a “good” route for the session

Network Operator

- Wider and holistic objectives

  – Minimization of total bandwidth consumed

  – Maximization of the smallest spare capacity on the links of the network

Tradeoff

- End to end
- Hop by hop
- Two QoS models for IP packet networks
- IntServ

  – Simulate the “virtual circuit” of ATM or frame relay on layer-3
- Sets up an end-to-end route with fixed QoS parameters
- DiffServ
- Defining several common classes of service with associated queue priorities and drop precedence on a per-hop basis hops

Point-to-Point Protocol (PPP)

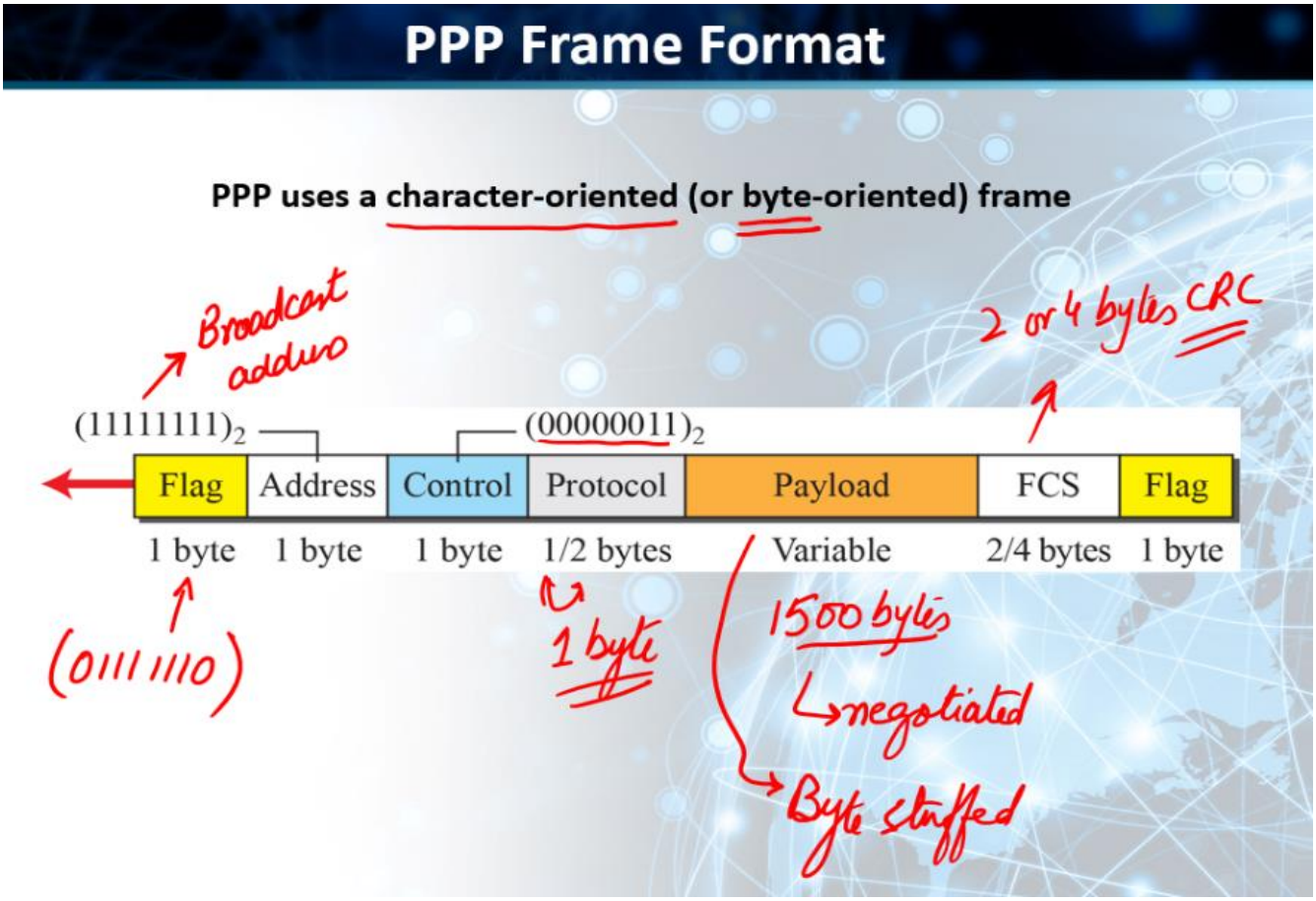
- Most common protocol for point-to-point access
- Millions of Internet users who need to connect their home computers to the server of an Internet service provider use PPP
- To control and manage the transfer of data, there is a need for a PPP at the data-link layer

Services provided by PPP

The designers of PPP have included several services to make it suitable for a point-to-point protocol, but have ignored some traditional services to make it simple

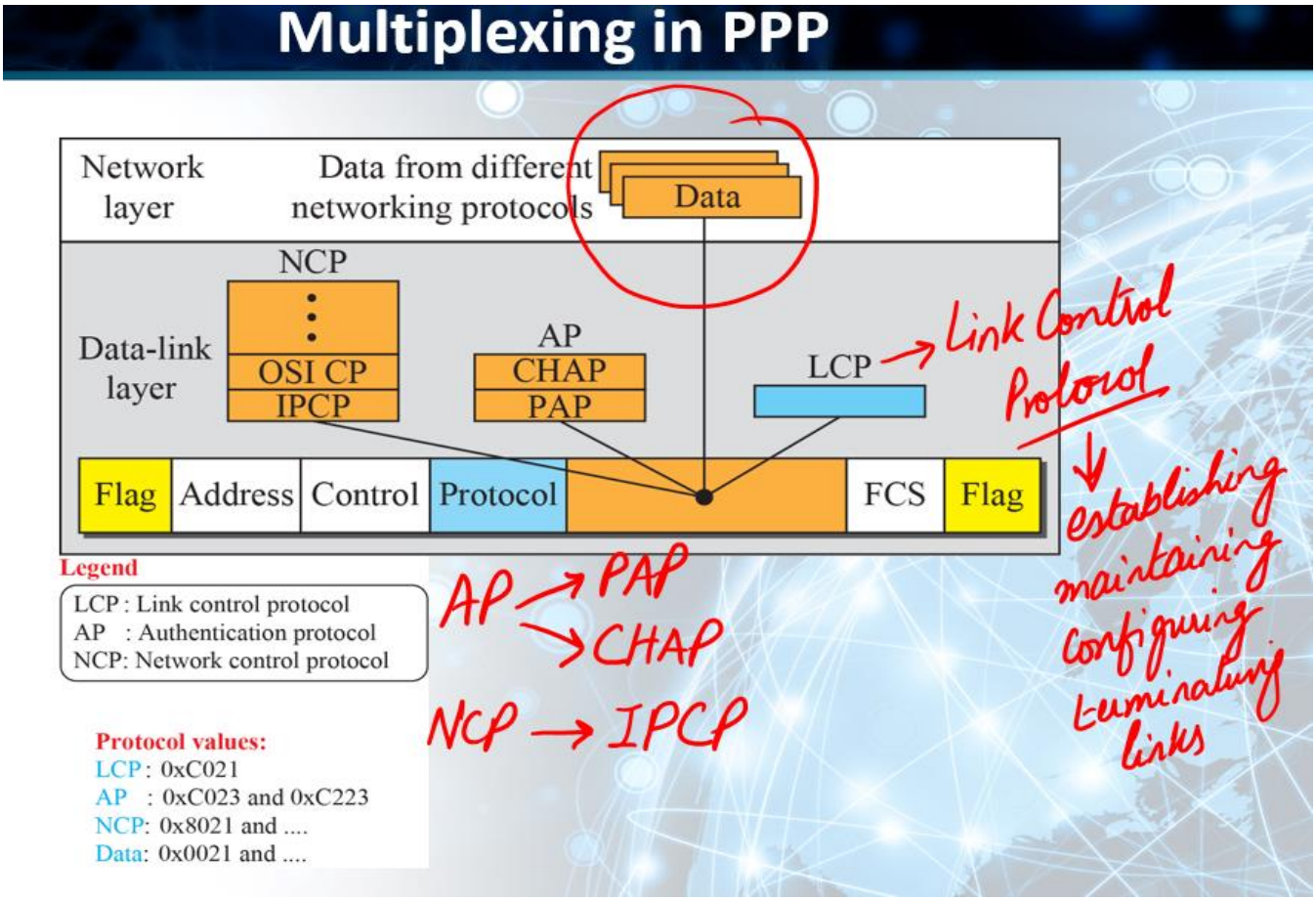
Services Included	Services Not Included
Framing	Flow Control
Link Establishment and Data Exchange	Error Correction (PPP has CRC detection only)
Authentication	No Sequence Numbering
Multilink PPP Address configuration	Absence of sophisticated Addressing Mechanism
Network Address configuration	

PPP Frame Format

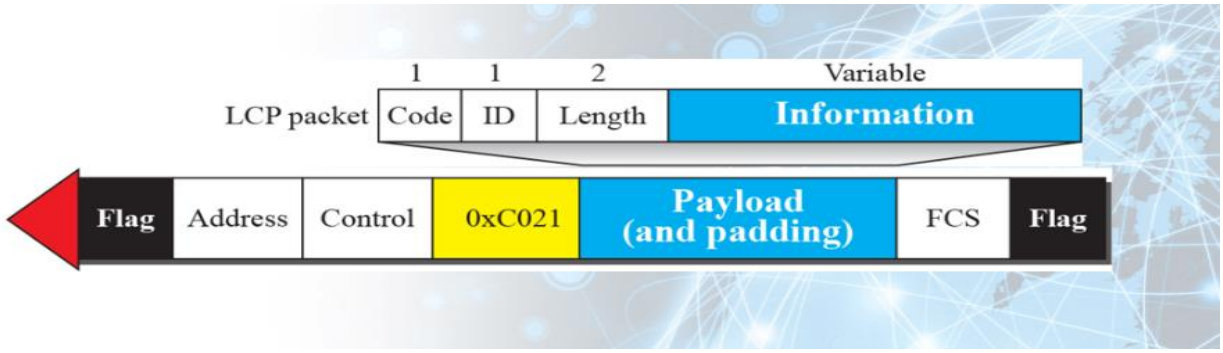


Multiplexing in PPP

- Although PPP is a link-layer protocol, it uses another set of protocols to establish the link, authenticate and carry the network-layer data
- Three sets of protocols are:
    - Link Control Protocol (LCP)
    - Two Authentication Protocols (APs)
    - Several Network Control Protocols (NCPs)

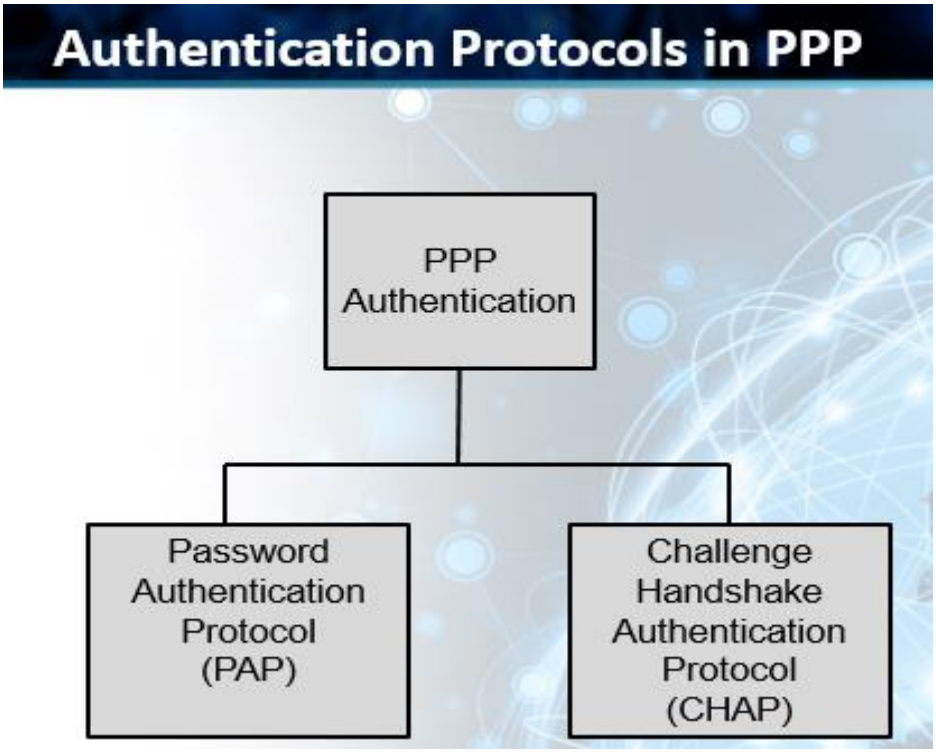


LCP Packets

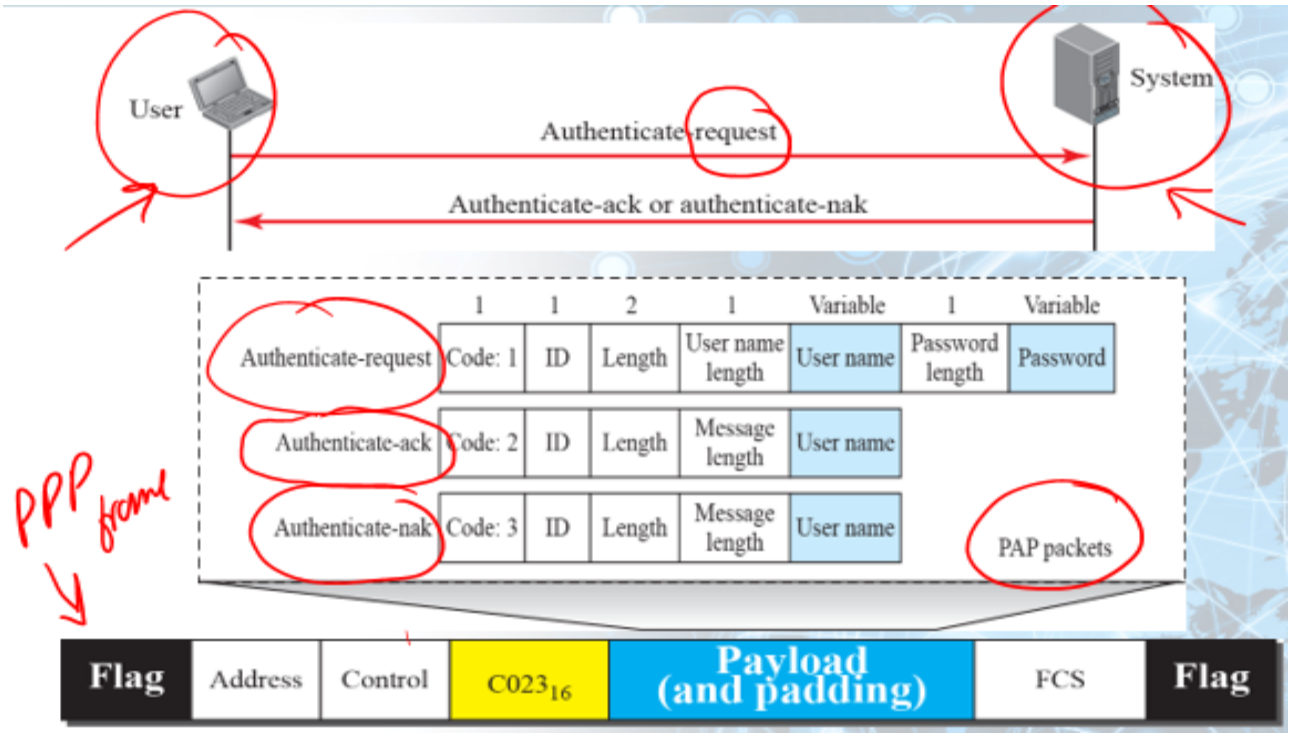


Code	Packet Type	Description
0x01	Configure-request	Contains the list of proposed options and their values
0x02	Configure-ack	Accepts all options proposed
0x03	Configure-nak	Announces that some options are not acceptable
0x04	Configure-reject	Announces that some options are not recognized
0x05	Terminate-request	Request to shut down the line
0x06	Terminate-ack	Accept the shutdown request
0x07	Code-reject	Announces an unknown code
0x08	Protocol-reject	Announces an unknown protocol
0x09	Echo-request	A type of hello message to check if the other end is alive
0x0A	Echo-reply	The response to the echo-request message
0x0B	Discard-request	A request to discard the packet

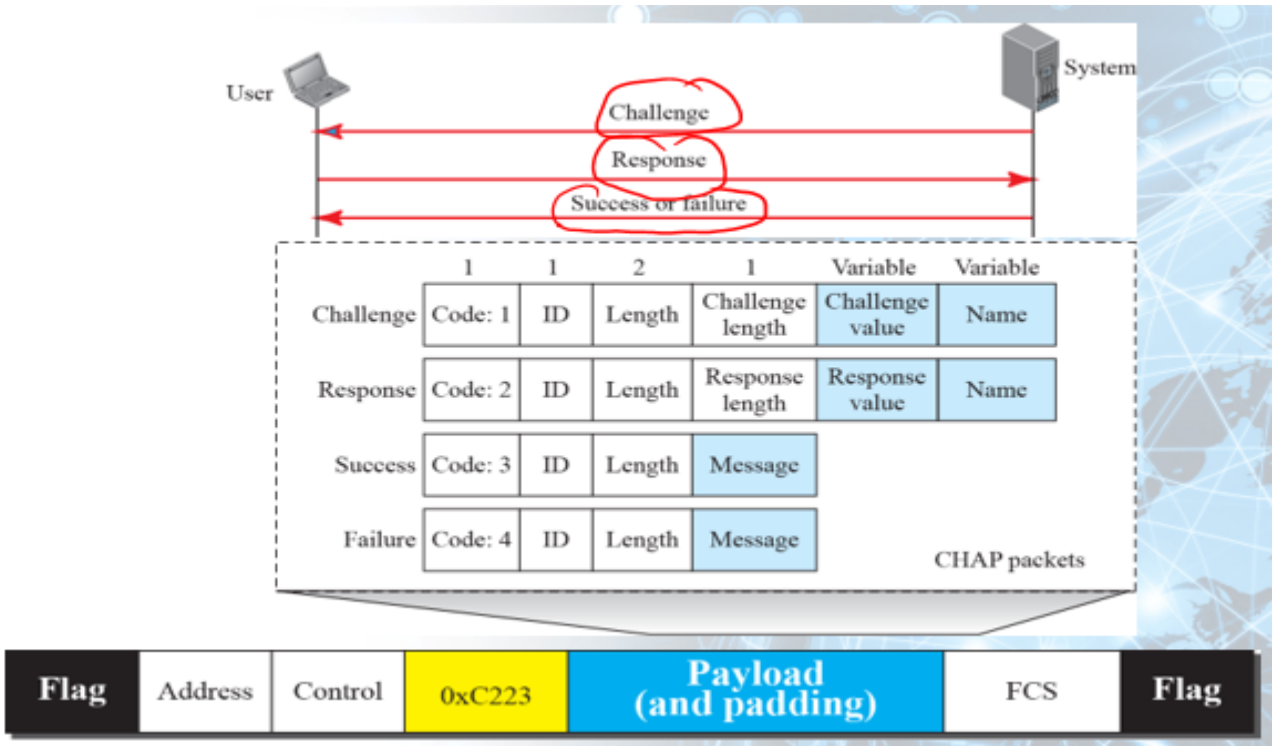
Authentication Protocols in PPP



PAP packets encapsulated in a PPP frame



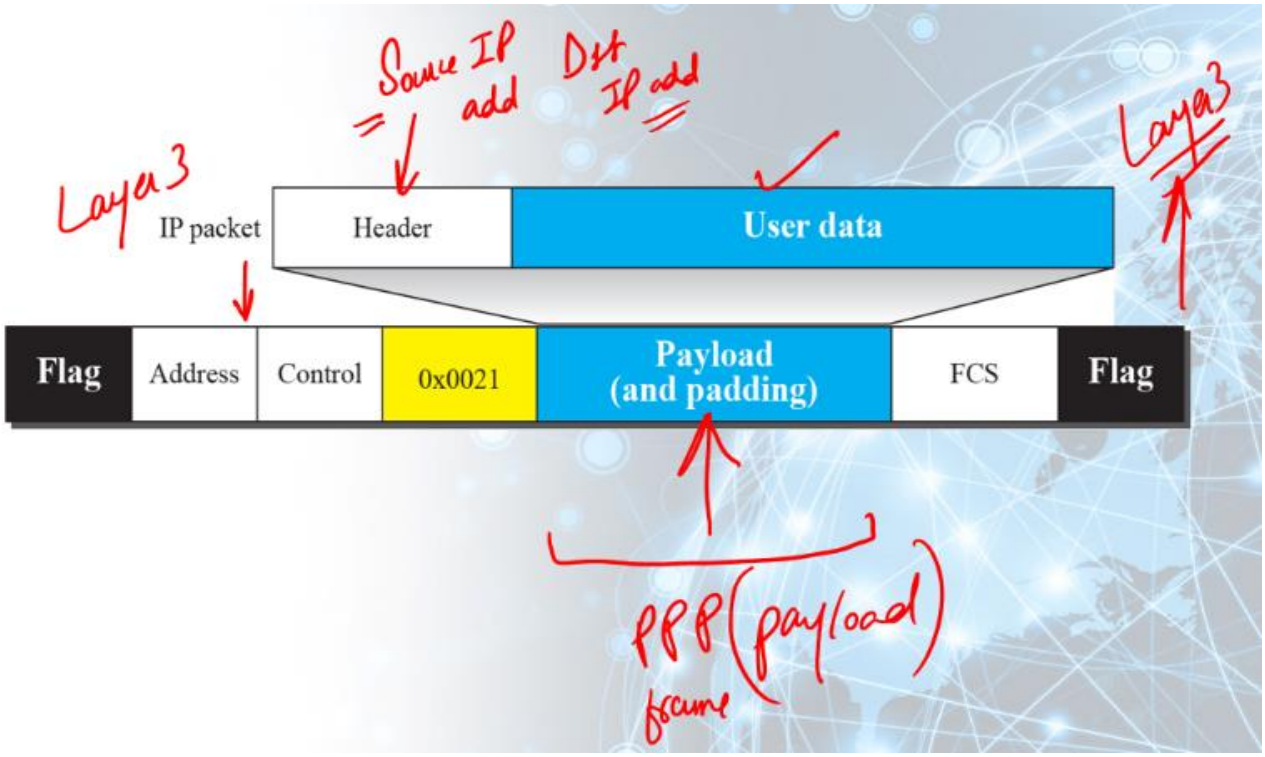
CHAP Packets encapsulated in a PPP frame



Code values for IPCP Packets

Code	*IPCP Packet*
0x01	Configure-request
0x02	Configure-ack
0x03	Configure-nak
0x04	Configure-reject
0x05	Terminate-request
0x06	Terminate-ack
0x07	Code-reject

IP datagram encapsulated in a PPP frame



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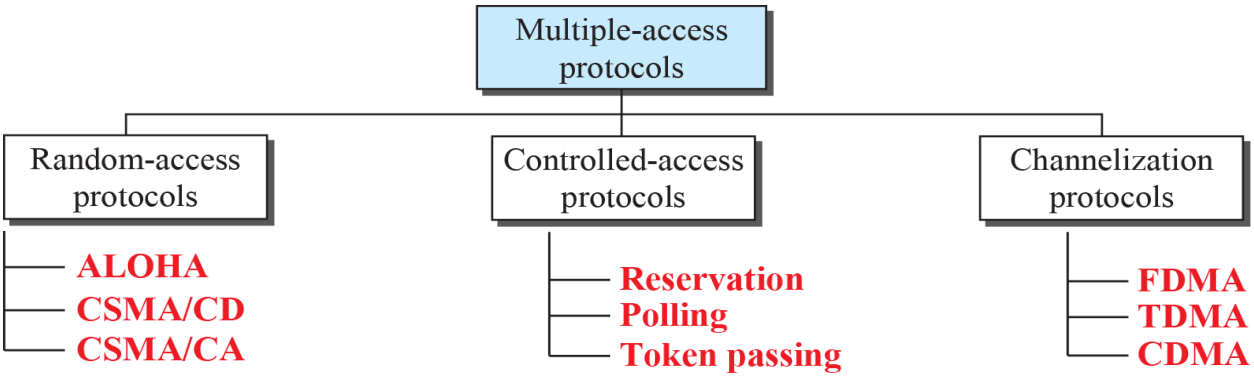
Media Access Control (MAC) Sub-Layer

- The Media Access Control (MAC) sub-layer operates within the data link layer of the OSI model and is responsible for managing access to the physical network medium.
- Key features include:
  - addressing with unique MAC addresses,
  - framing data for transmission,
  - frame filtering based on addresses,
  - various media access methods,
  - flow control, error detection,
  - and support for broadcast and multicast communication.
- The MAC sub-layer plays a crucial role in controlling how devices on a network access and utilize the communication medium.

Taxonomy of Multiple-Access Protocols



Random Access

In a random-access or contention-based network:

- There is no hierarchy among the stations.
- None has inherent superiority or control over others.
- Instead, each station contends equally for the right to access the communication medium.
- The decision of whether a station can send data is based on a procedure defined by the network protocol, and it involves considering the current state of the communication medium.

Characteristics of Random Access:

1. **No Station Superiority:** In a random-access scenario, all stations are considered equal. Unlike in networks with a central coordinator or predetermined time slots, there is no station that holds a superior position or has control over others.

2. **Decision to Send Data:** When a station in a random-access network has data to send, it follows a protocol-defined procedure to determine whether it is allowed to transmit. This procedure typically involves assessing the current state of the communication medium.

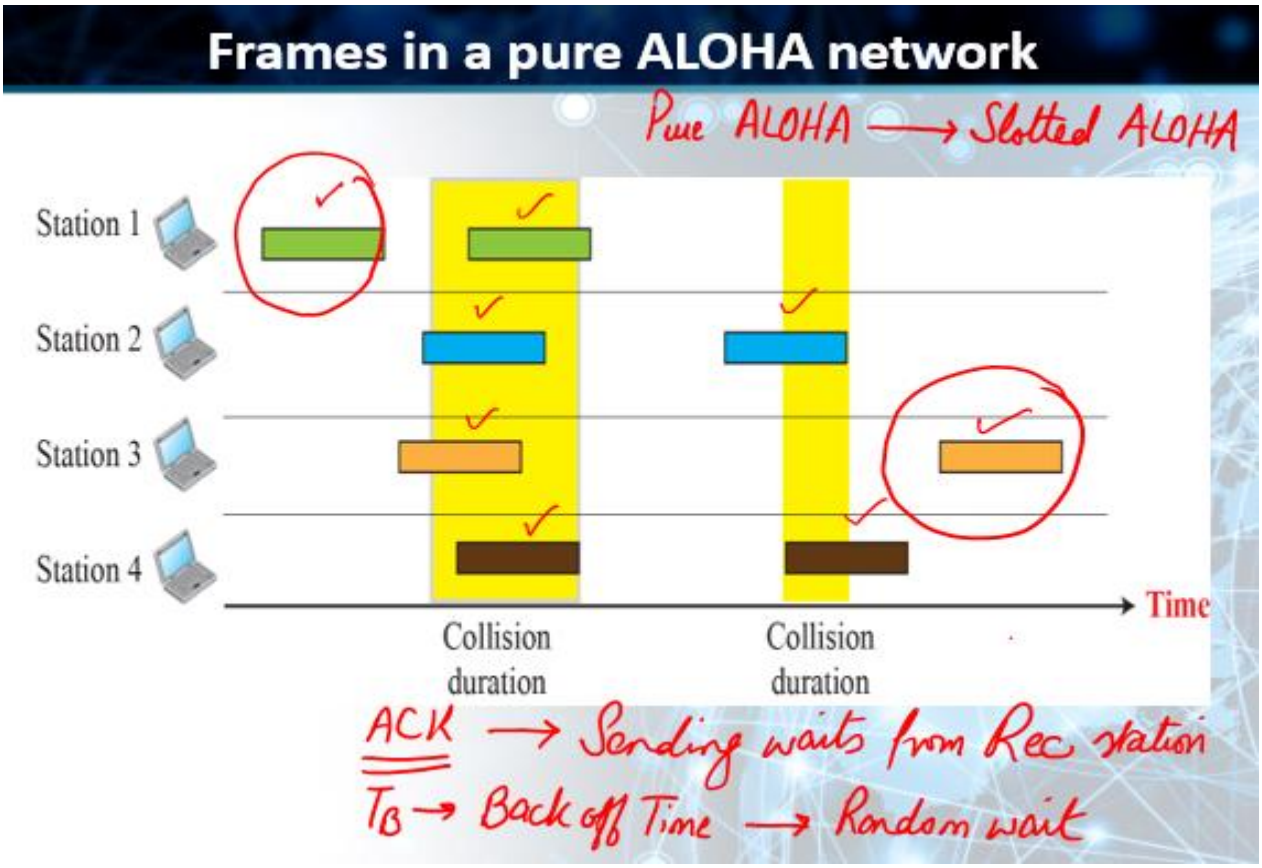
3. **Medium State (Idle or Busy):** The decision to send data depends on the state of the medium—whether it is idle or busy. If the medium is idle, meaning there is no ongoing transmission, the station may decide to initiate its transmission. However, if the medium is currently in use (busy), the station may need to wait until it becomes idle before attempting to send data.

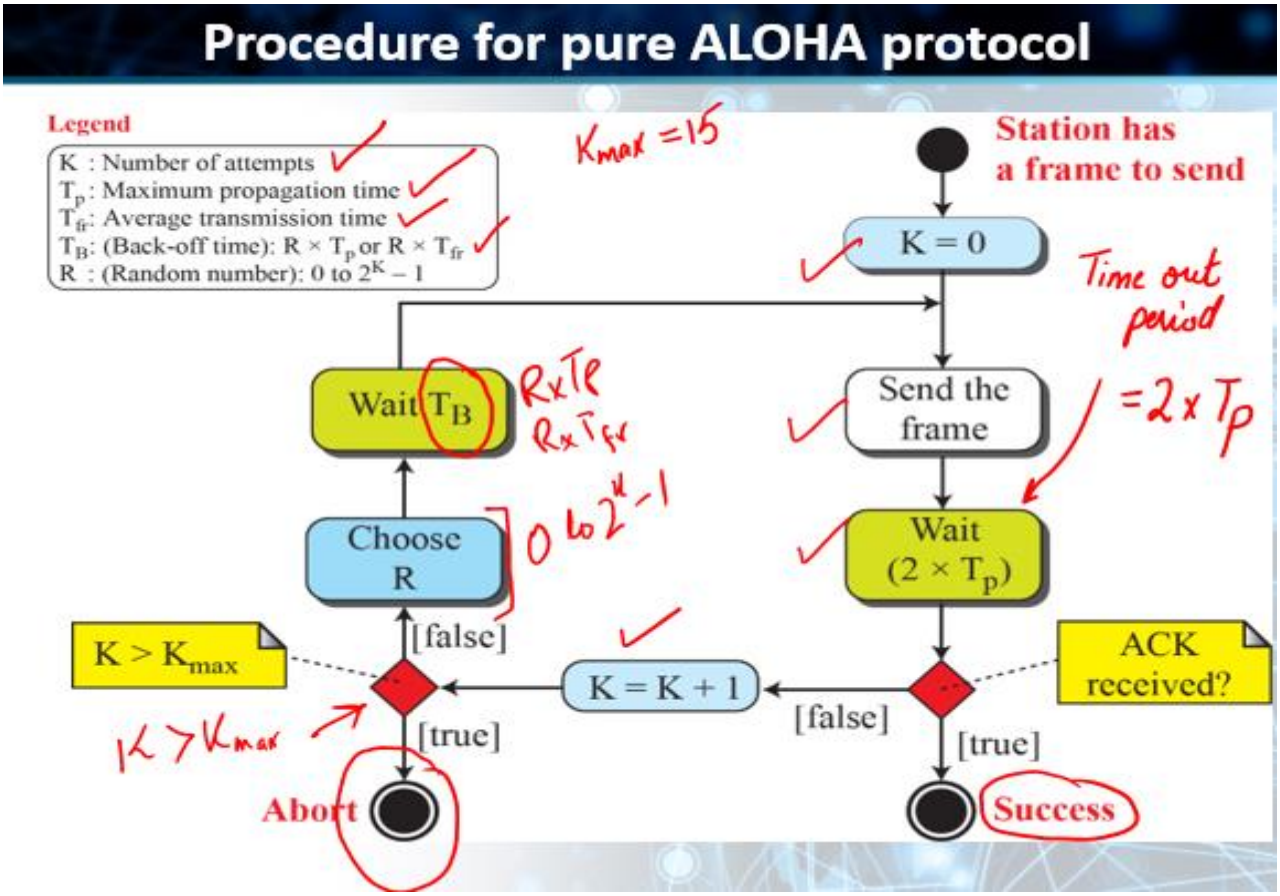
ALOHA

ALOHA protocol is a simple network communication method where devices can transmit data whenever they need, but collisions may occur, requiring retransmission.

- **ALOHA Development (1970s):**
  - Developed in the early 1970s.
  - One of the earliest random access methods.
- **Original Purpose:**
  - Initially designed for radio (wireless) LANs.
  - Applicable to any shared communication medium.
- **Collisions in Shared Mediums:**
  - Stations contend for access to the communication channel.
  - Potential collisions occur when multiple stations attempt to transmit simultaneously.
- **Collision Consequences:**
  - Collisions result in data overlap and garbling.
  - ALOHA employs collision resolution mechanisms to address these issues.
- **Decentralized Nature:**
  - Decentralized approach allows stations to independently decide when to transmit.
  - Provides simplicity and flexibility in shared network environments.
- **Challenges and Efficiency:**

- As network congestion increases, the probability of collisions rises, impacting overall efficiency.
- ALOHA's principles influenced the development of subsequent random access protocols.

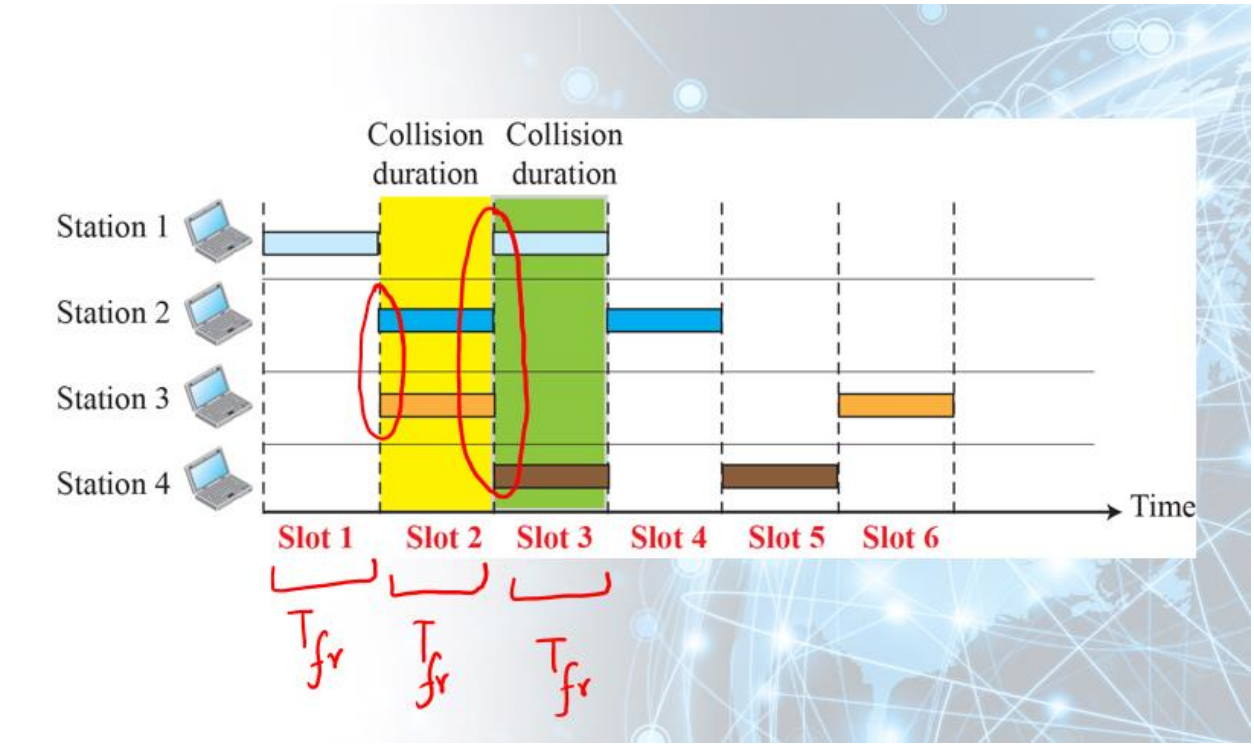


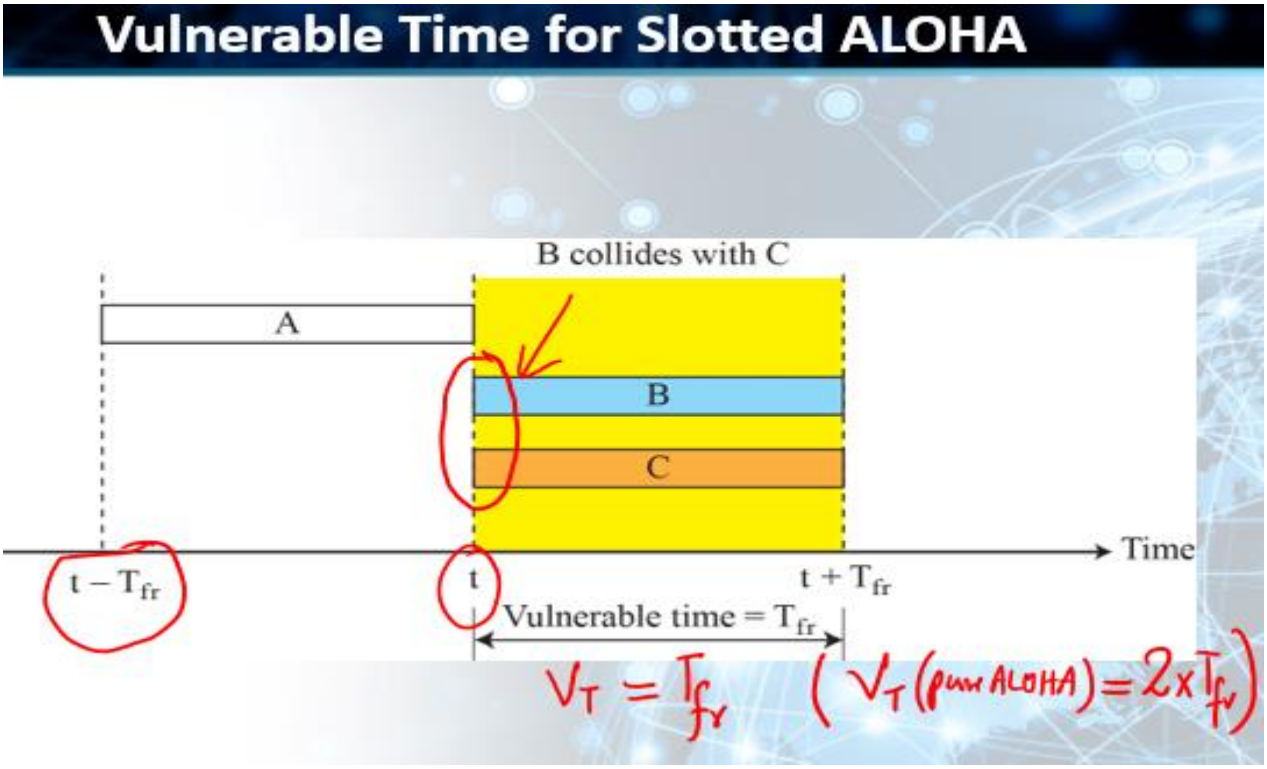


Slotted ALOHA

- We divide time into slots of Tfr sec and force the station to send only at the beginning of the slot
- Invented to improve the efficiency of pure ALOHA
- If a station misses the time slot, it must wait until beginning of next time slot reducing vulnerable time to Tfr (vs. 2 x Tfr for pure ALOHA)

Frames in a Slotted ALOHA Network





Carrier Sense Multiple Access (CSMA)

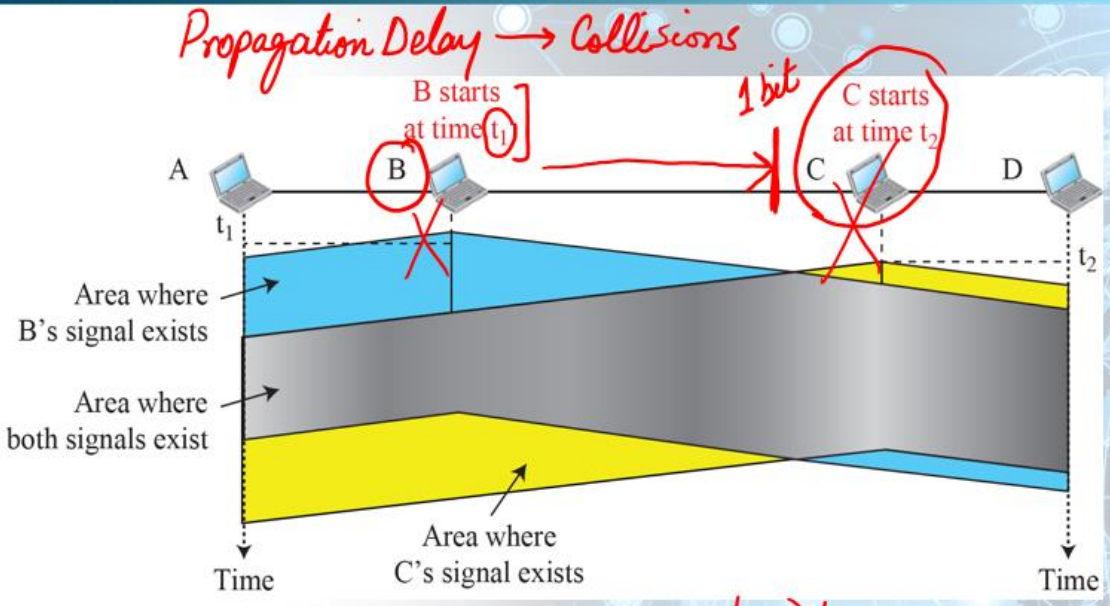
- To minimize the chance of collision and, therefore, increase the performance, CSMA was developed
- The chance of collision is reduced as the station is required to sense/listen to the medium before sending data
- ‘sense before transmit’ or ‘listen before talk’

    - Protocol for shared communication mediums.
    - Devices listen to the medium before transmitting.

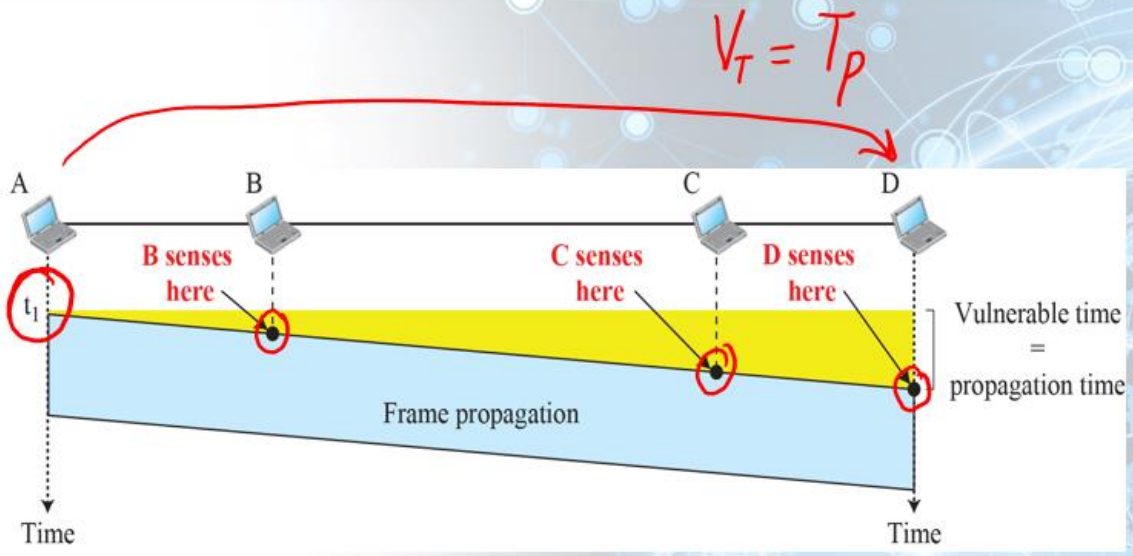
Following are the characteristics:

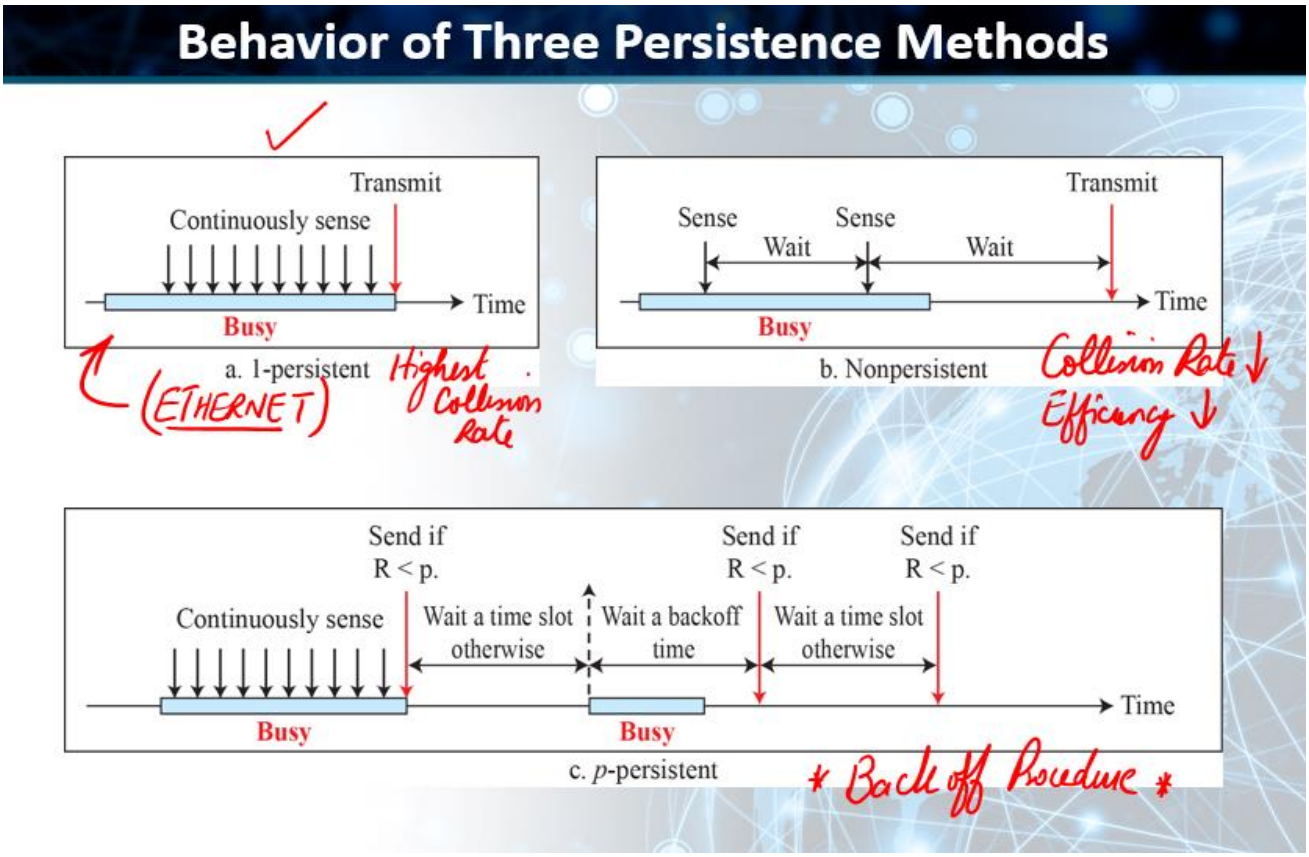
- **Carrier Sensing:**
    - Devices detect the presence of a carrier signal.
    - Ensures the medium is idle before initiating transmission.
- **Collision Avoidance:**
    - Aims to prevent collisions by checking medium availability.
    - Collisions may still occur if multiple devices initiate transmission simultaneously.
- **CSMA Variants:**
    - CSMA/CD (Collision Detection) used in Ethernet networks.
    - CSMA/CA (Collision Avoidance) common in wireless networks.
- **Efficiency and Utilization:**
    - Efficient in moderate traffic conditions.
    - Devices quickly detect ongoing transmissions.
- **Collision Management:**
    - CSMA/CD stops transmission upon collision, initiates backoff, and retries.
    - CSMA/CA employs strategies like waiting before transmission to minimize collisions.

Space/Time Model of a Collision in CSMA



Vulnerable Time in CSMA

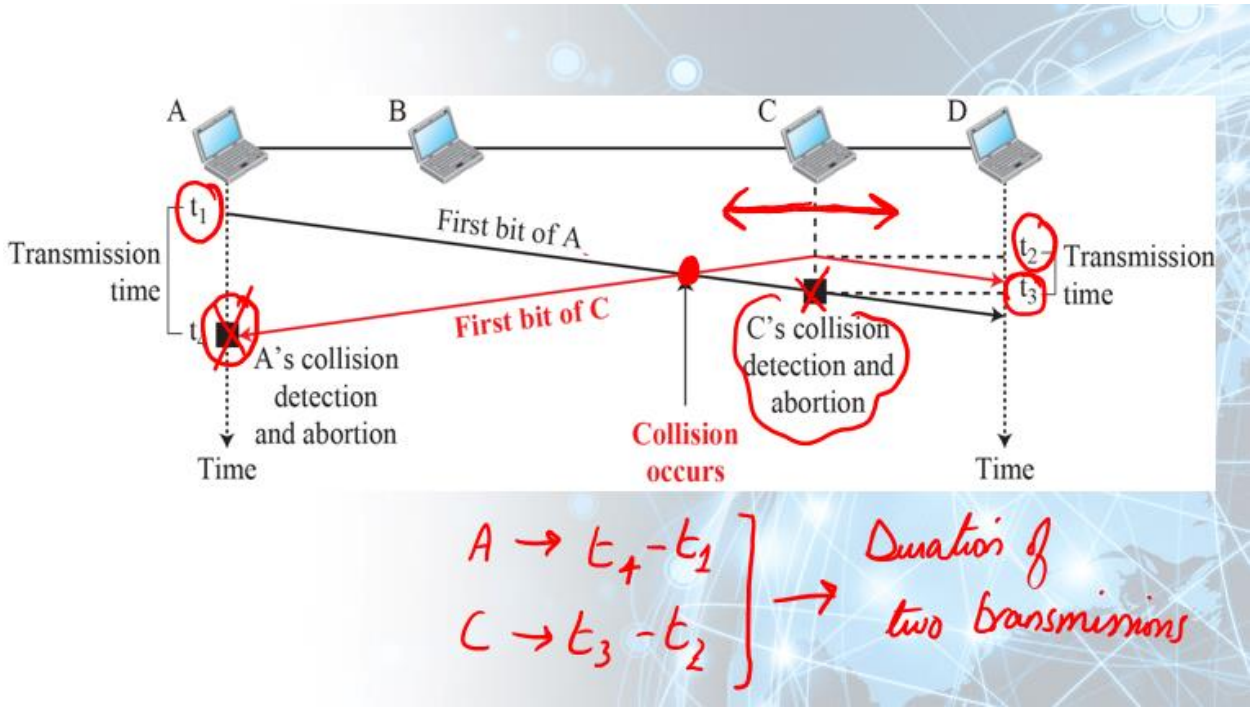


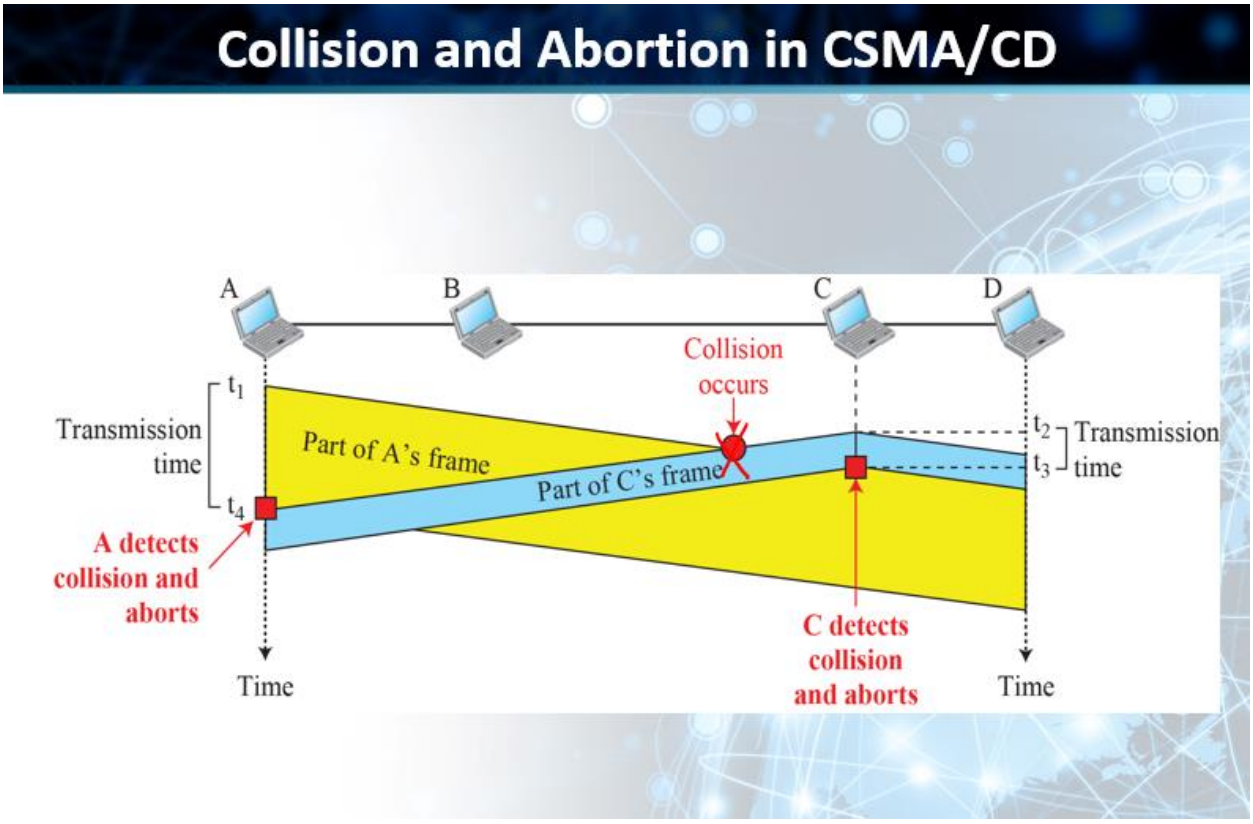


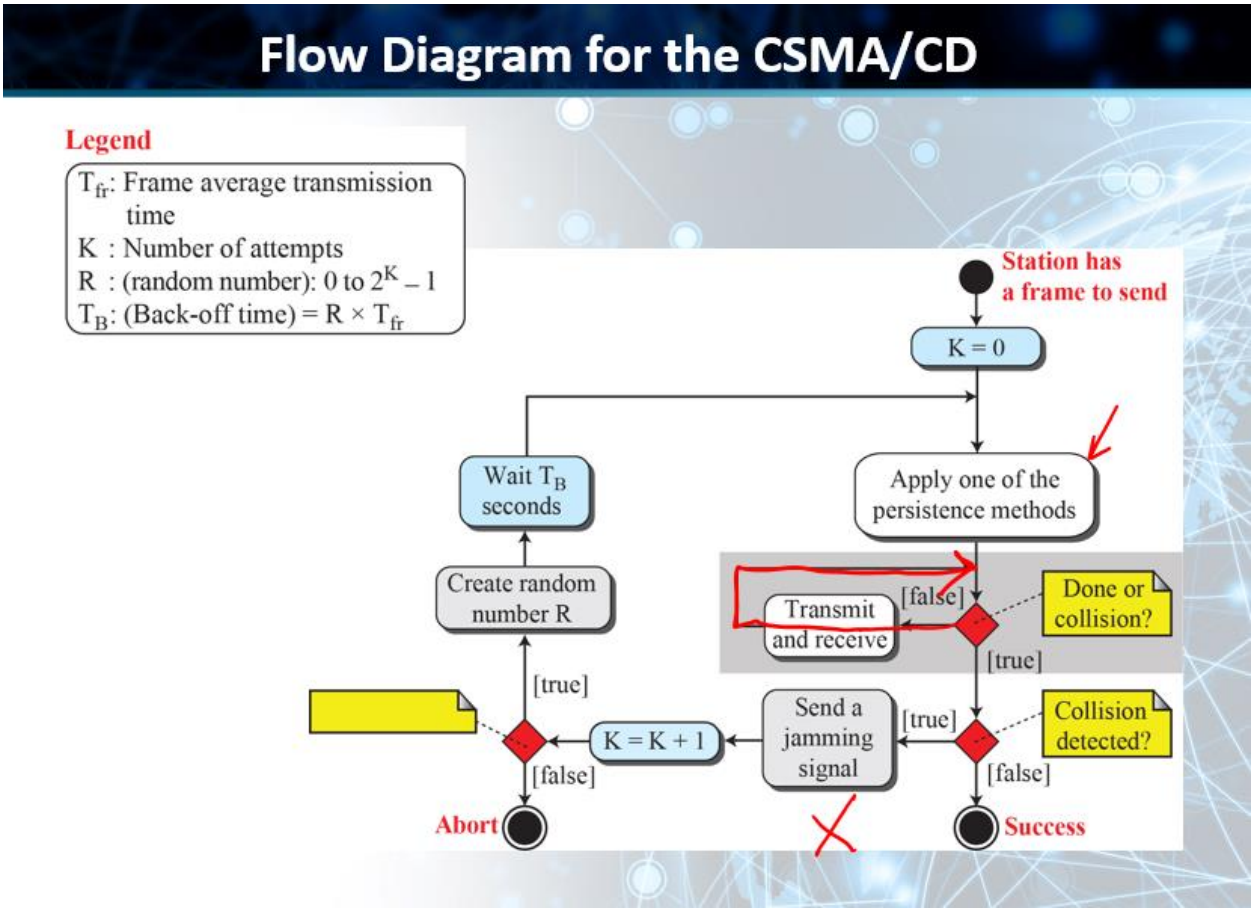
Carrier Sense Multiple Access / Collision Detection

- CSMA method does not specify the procedure following a collision
- CSMA/CD augments the algorithm to handle the collision
- The station monitors the medium after it sends a frame to see if the transmission was successful. If there is a collision, the frame is sent again

Collision of the First Bits in CSMA/CD



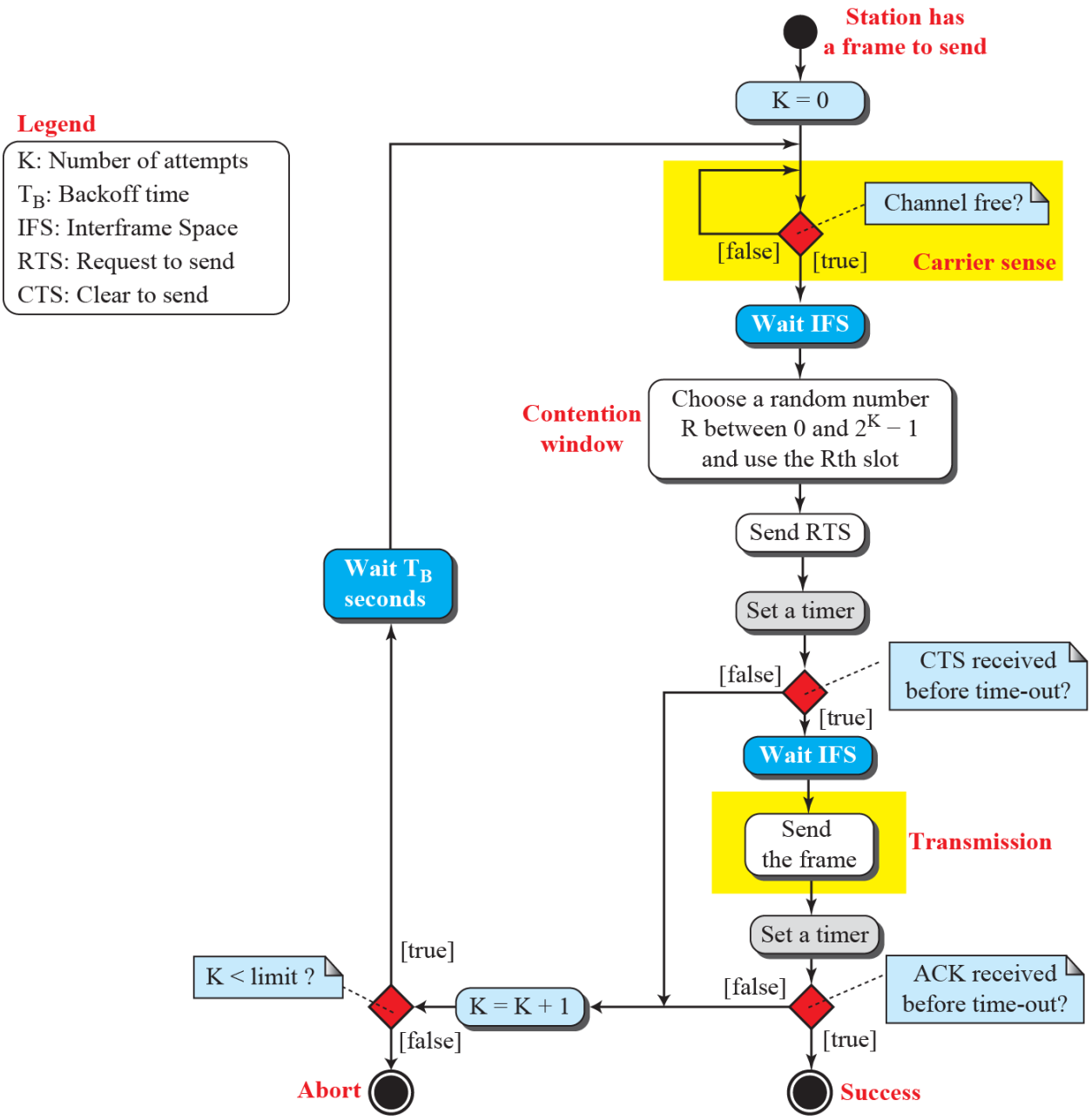




Carrier Sense Multiple Access/Collision Avoidance

- CSMA/CA was invented for Wireless Networks
- Collisions are avoided through the use of three strategies:
  - ✓ The Interfame Space
  - ✓ The Contention Window
  - ✓ Acknowledgements

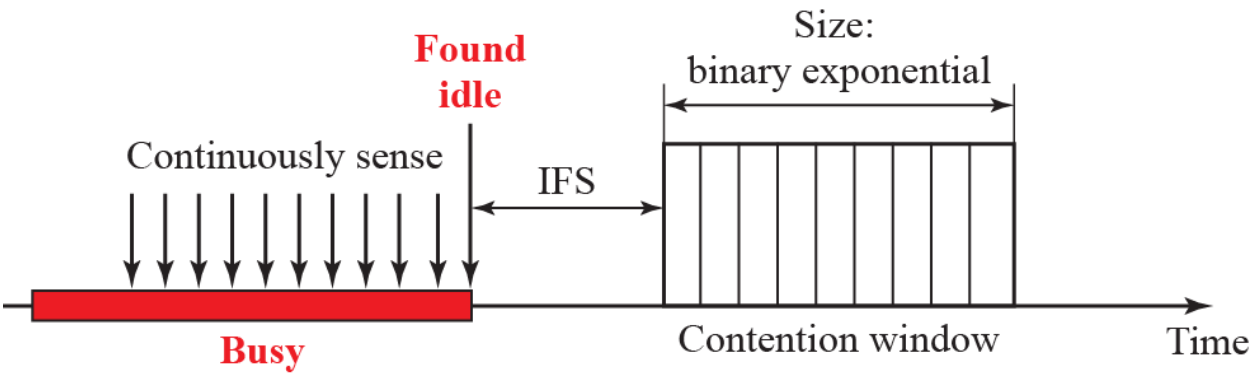
Flow Diagram for CSMA/CA



Interframe Space (IFS). First, collisions are avoided by deferring transmission even if the channel is found idle. When an idle channel is found, the station does not send immediately. It

waits for a period of time called the interframe space or IFS. The IFS variable can also be used to prioritize stations or frame types. For example, a station that is assigned a shorter IFS has a higher priority.

Contention Window. The contention window is an amount of time divided into slots. A station that is ready to send chooses a random number of slots as its wait time. The number of slots in the window changes according to the binary exponential back off strategy. This means that it is set to one slot the first time and then doubles each time the station cannot detect an idle channel after the IFS time. This is very similar to the p-persistent method except that a random outcome defines the number of slots taken by the waiting station. One interesting point about the contention window is that the station needs to sense the channel after each time slot. However, if the station finds the channel busy, it does not restart the process; it just stops the timer and restarts it when the channel is sensed as idle. This gives priority to the station with the longest waiting time. See Figure 12.16

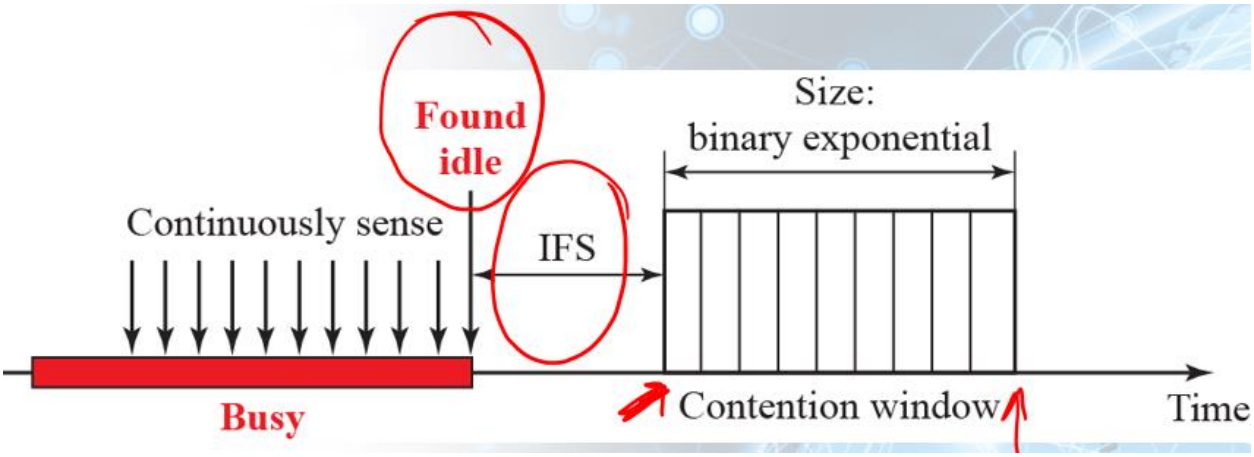


Acknowledgment. With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

CSMA/CA:

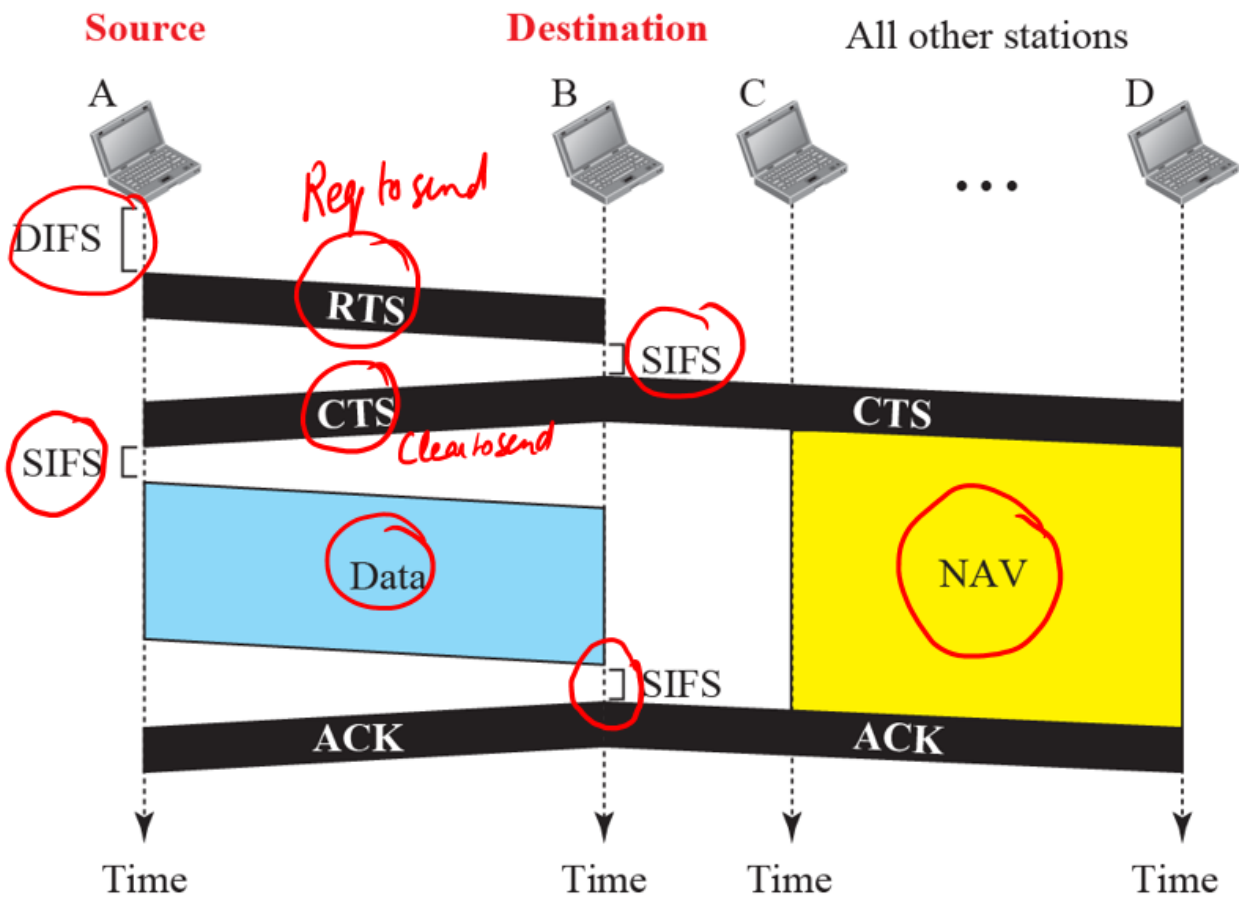
- Interface Space (IFS): Collisions are avoided by deferring transmission even if the channel is idle
- Contention Window: Amount of time divided into slots. Station chooses a random number of slots as its wait time (one slot first time and double each time system cannot detect an idle channel)

Contention Window:



Acknowledgment. With all these precautions, there still may be a collision resulting in destroyed data. In addition, the data may be corrupted during the transmission. The positive acknowledgment and the time-out timer can help guarantee that the receiver has received the frame.

CSMA/CA and Network Allocation Vector (NAV):



CONTROLLED ACCESS:

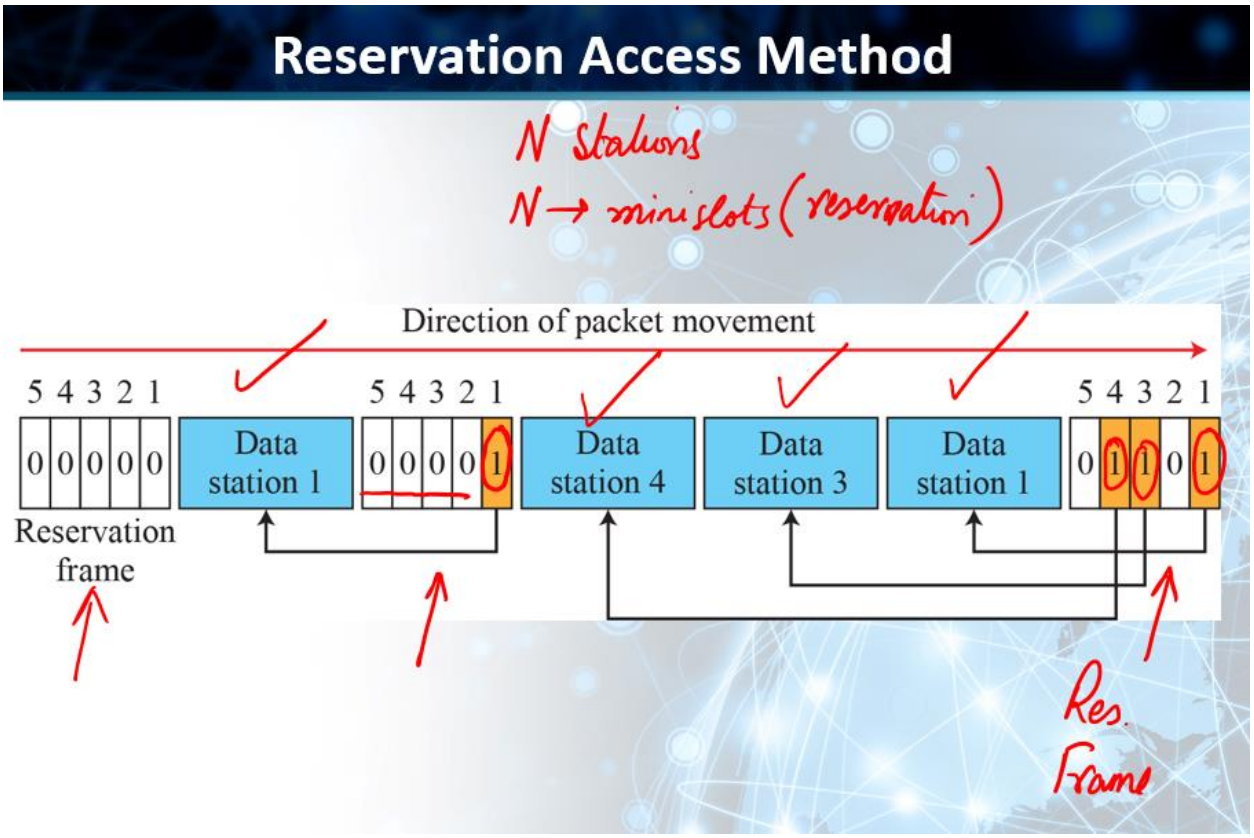
- In the reservation method, a station needs to make a reservation before sending data
- Time is divided into intervals
- In each interval, a reservation frame precedes the data frames sent in that interval
- The stations consult one another to find which station has the right to send
- A station cannot send unless authorized by other stations
- We discuss three controlled-access methods:

1. Reservation
2. Polling
3. Token Passing

Reservation

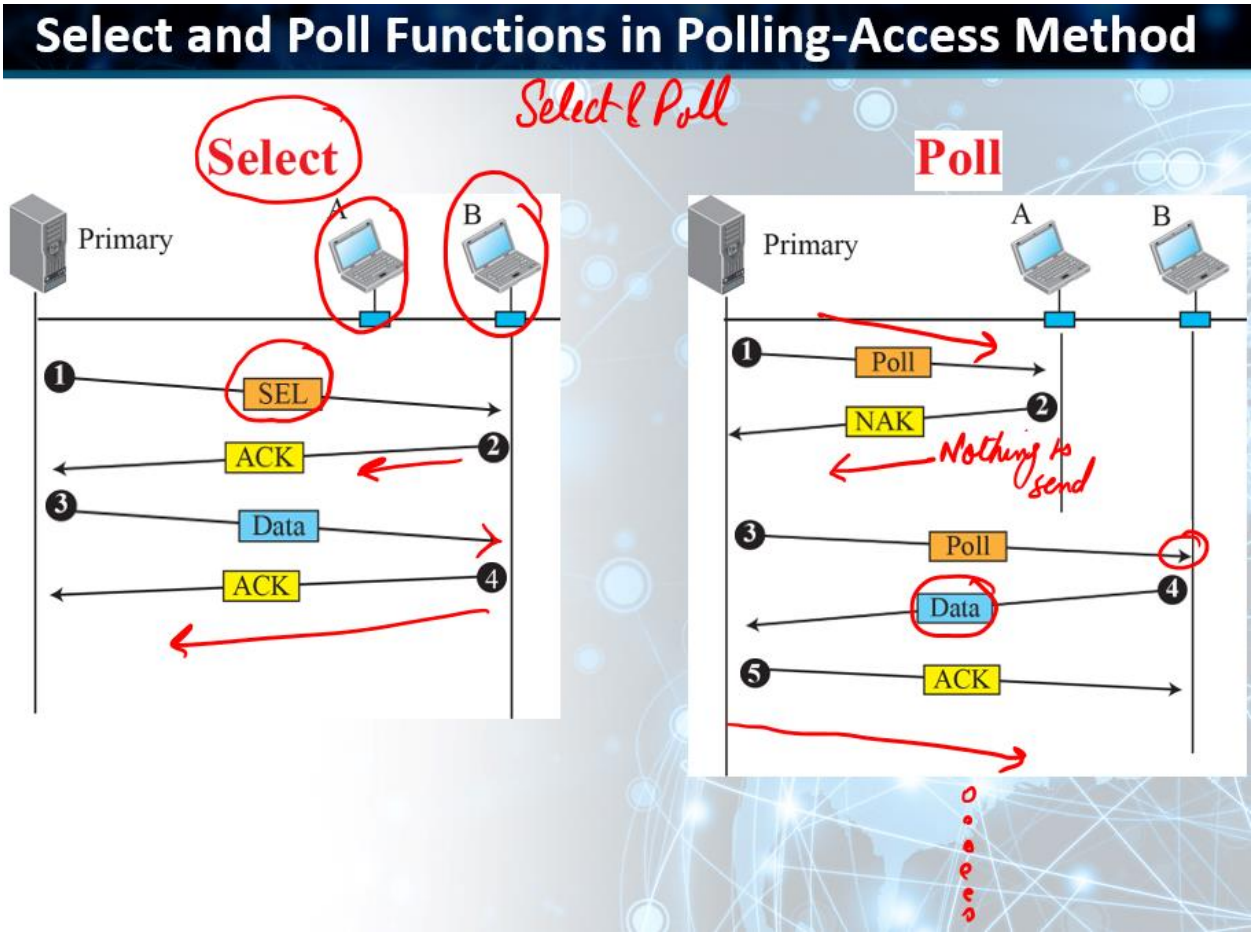
- In the reservation method, a station needs to make a reservation before sending data
- Time is divided into intervals
- In each interval, a reservation frame precedes the data frames sent in that interval

Reservation:



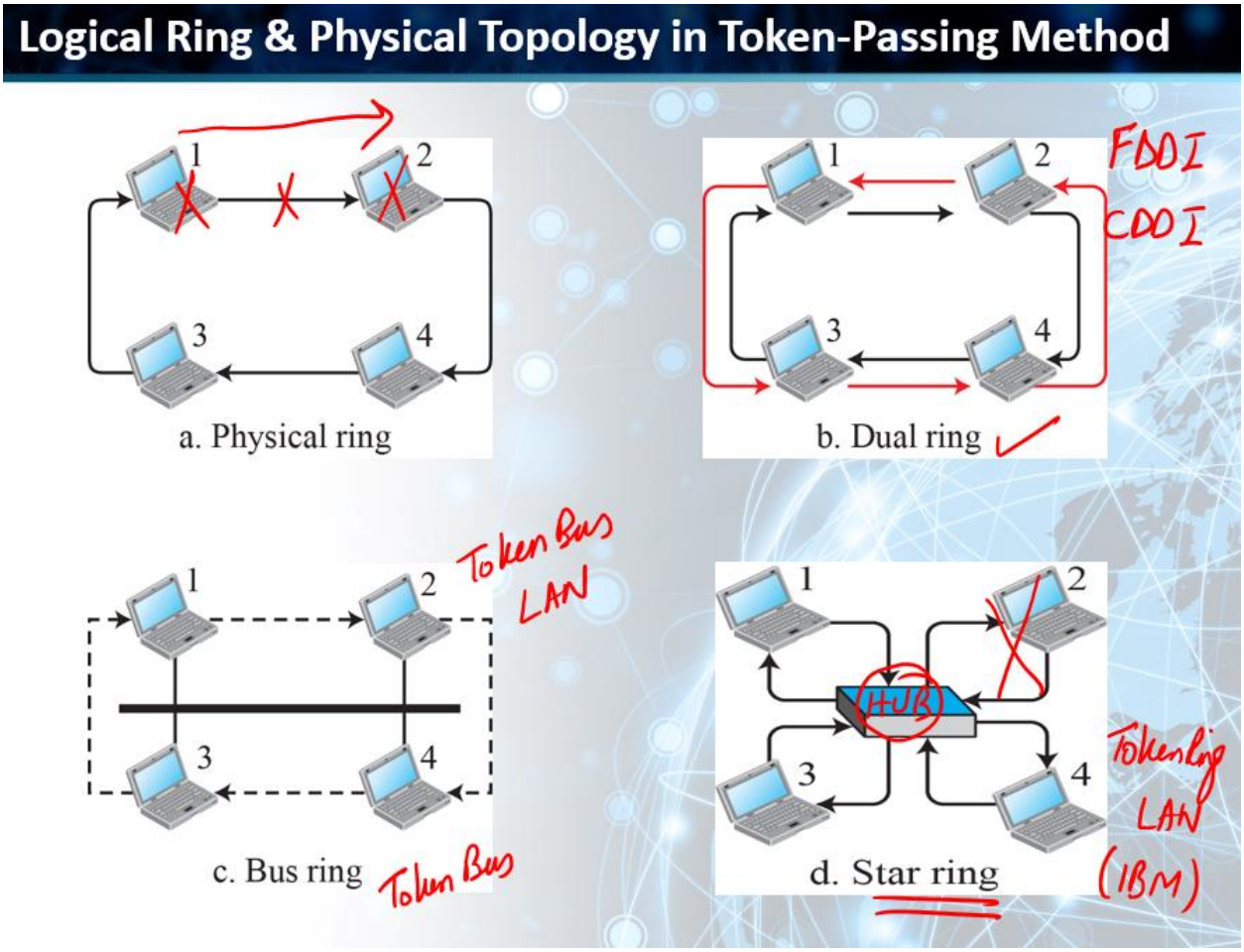
Polling:

- Polling works with topologies in which one device is designated as a primary station and the other devices are secondary stations
- All data exchanges must be made through primary device even when the ultimate destination is a secondary device
- The primary device controls the link; the secondary devices follow its instructions



Token Passing:

- In the token-passing method, the stations in a network are organized in a logical ring
- For each station, there is a predecessor and a successor
- The predecessor is the station which is logically before the station in the ring; the successor is the station which is after the station in the ring
- Special packet called TOKEN circulates through the ring
- Possession of TOKEN gives the station the right to send the data
- TOKEN Management is required to manage possession time, Token monitoring, priority assignment etc.



CHANNELIZATION (Channel Partition)

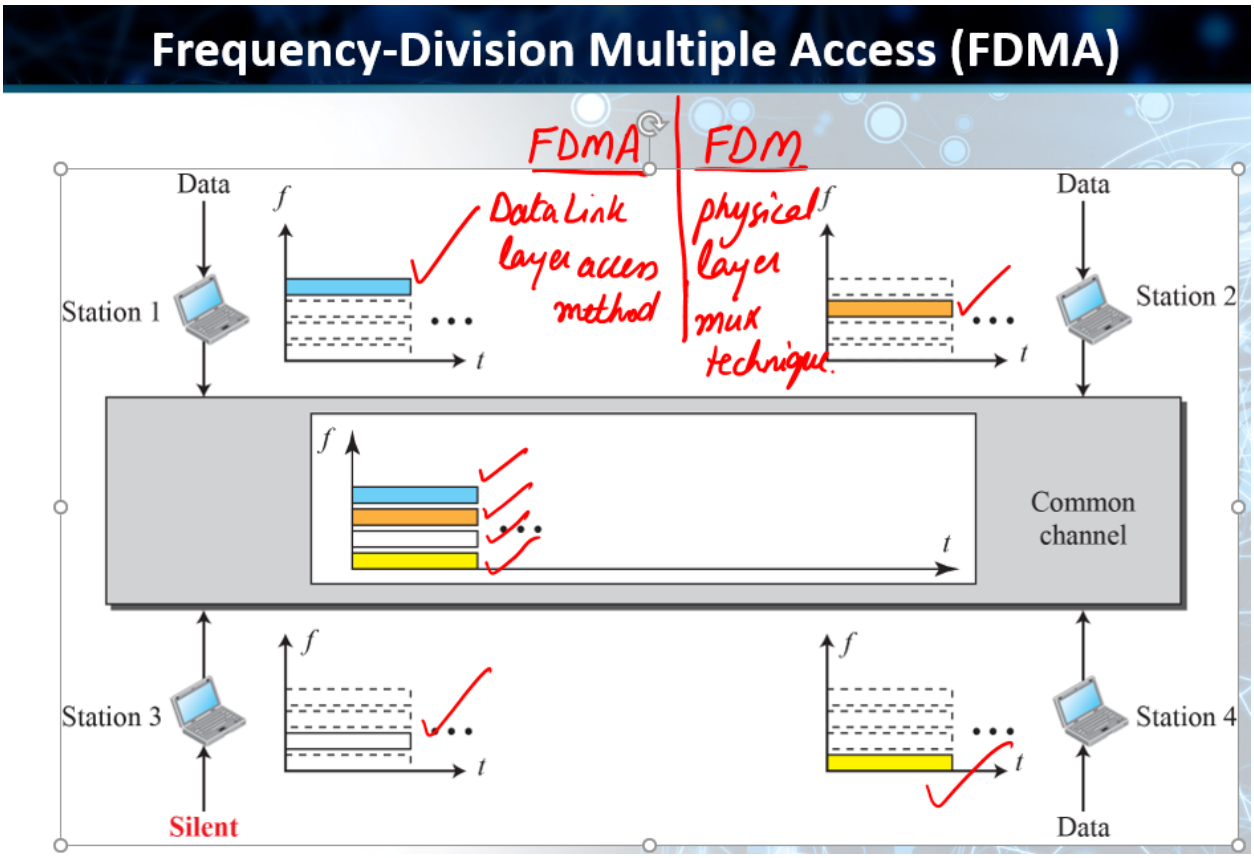
- The available bandwidth of a link is shared in time, frequency, or through code, among different stations
- We discuss three protocols:
  - ✓ Frequency Division Multiple Access (FDMA)
  - ✓ Time Division multiple Access (TDMA)
  - ✓ Code Division Multiple Access (CDMA)

CHANNELIZATION (Channel Partition)

- Three protocols:
  - ✓ Frequency Division Multiple Access (FDMA)
  - ✓ Time Division Multiple Access (TDMA)
  - ✓ Code Division Multiple Access (CDMA)

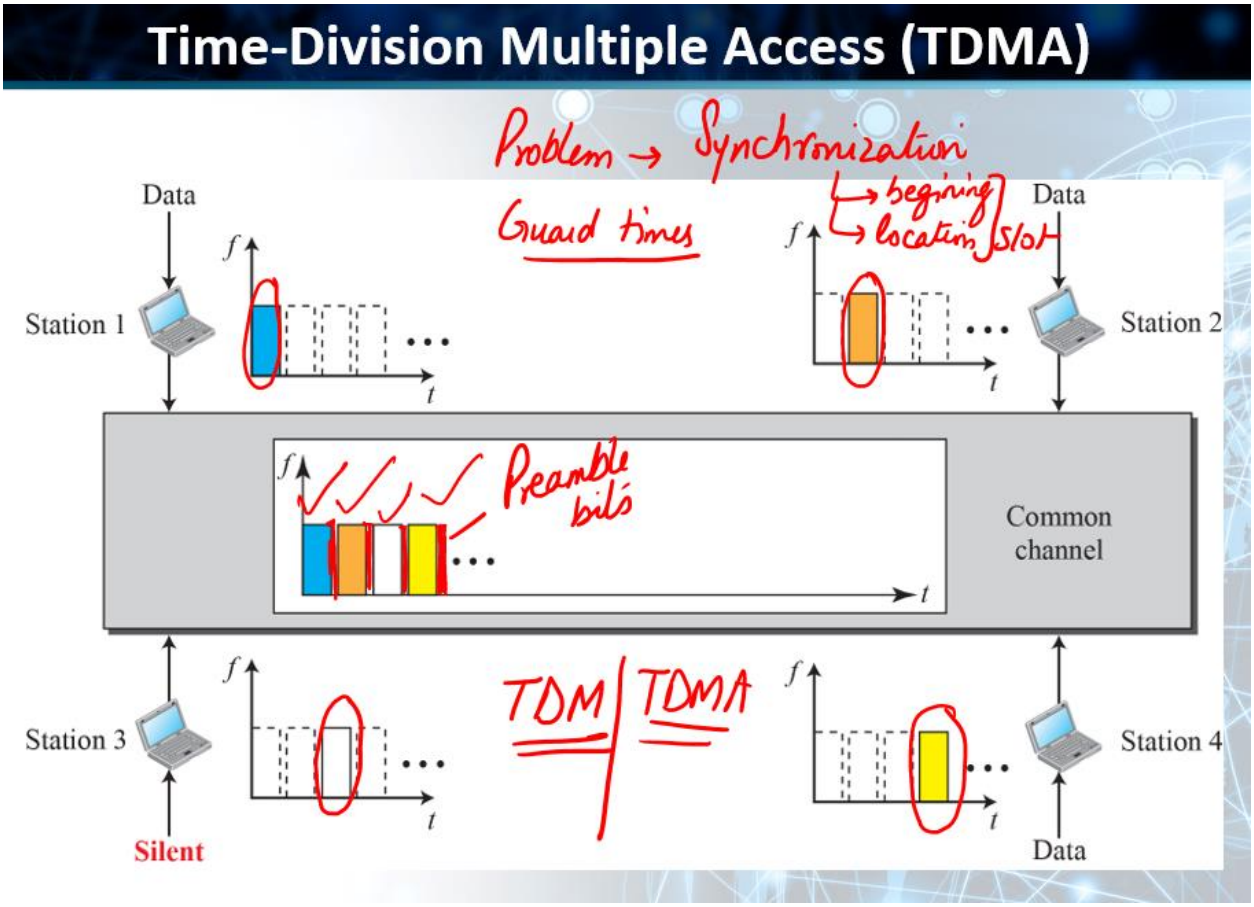
Frequency-Division Multiple Access (FDMA):

- In FDMA, the available bandwidth is divided into frequency bands
- Each station is allocated a band to send its data i.e. each band is reserved for a specific station, and it belongs to the station all the time
- Each station also uses a bandpass filter to confine the transmitter frequencies



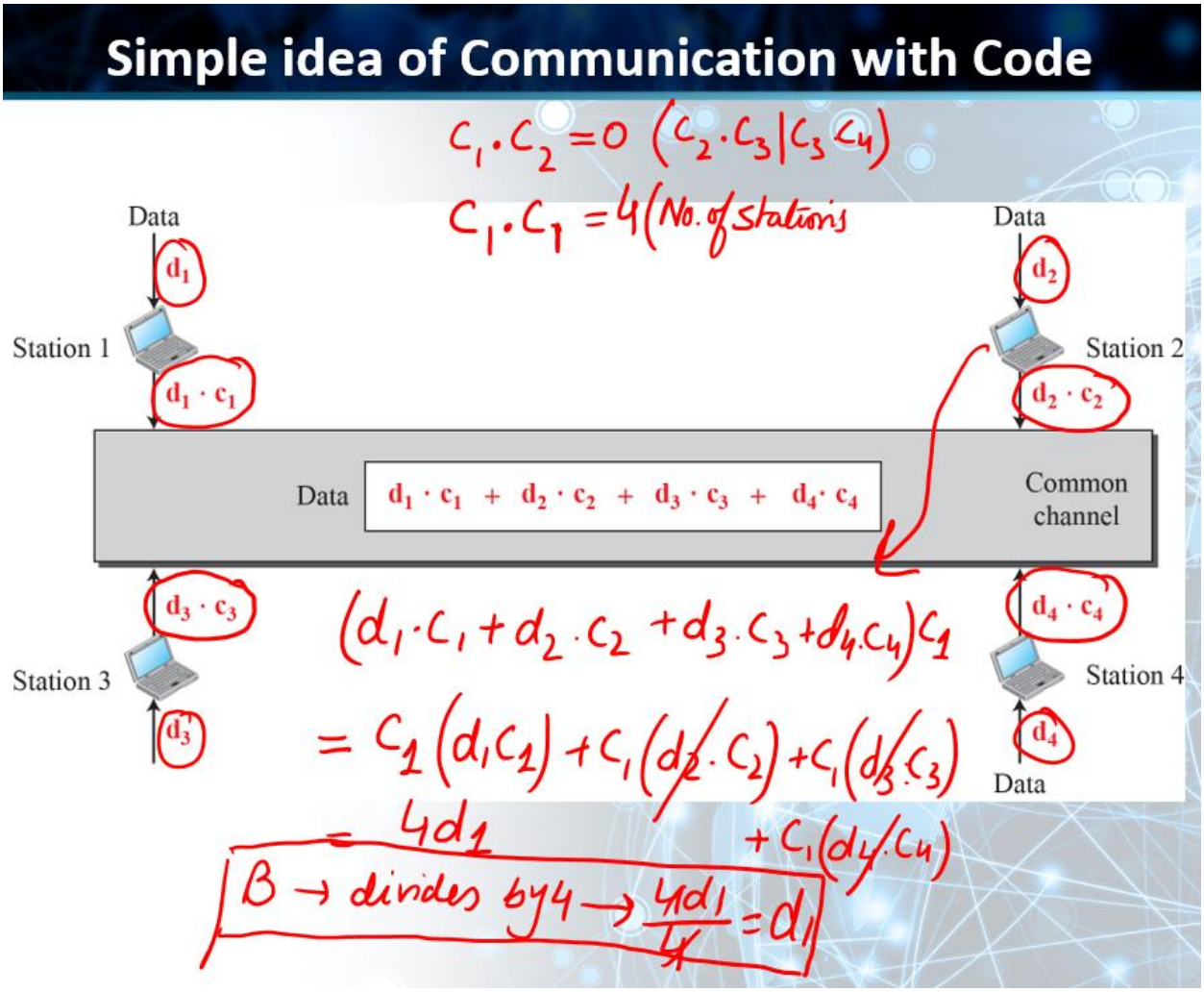
TDMA:

- Stations share the bandwidth of the channel in time
- Each station is allocated a time slot during which it can send data
- Each station transmits its data in its assigned time slot



Code Division Multiple Access (CDMA)

- CDMA differs from FDMA in that only one channel occupies the entire bandwidth of the link
- CDMA differs from TDMA in that all stations can send data simultaneously; there is no timesharing



Data Representation in CDMA:

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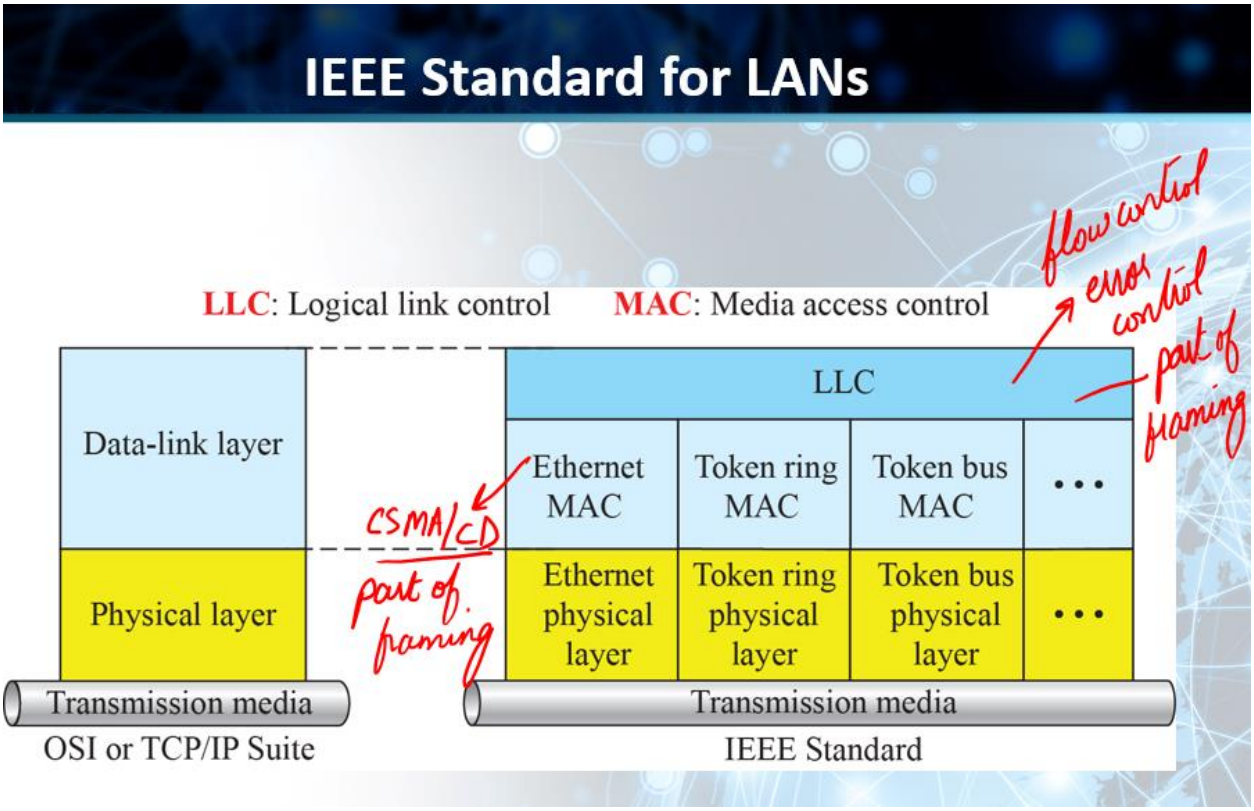
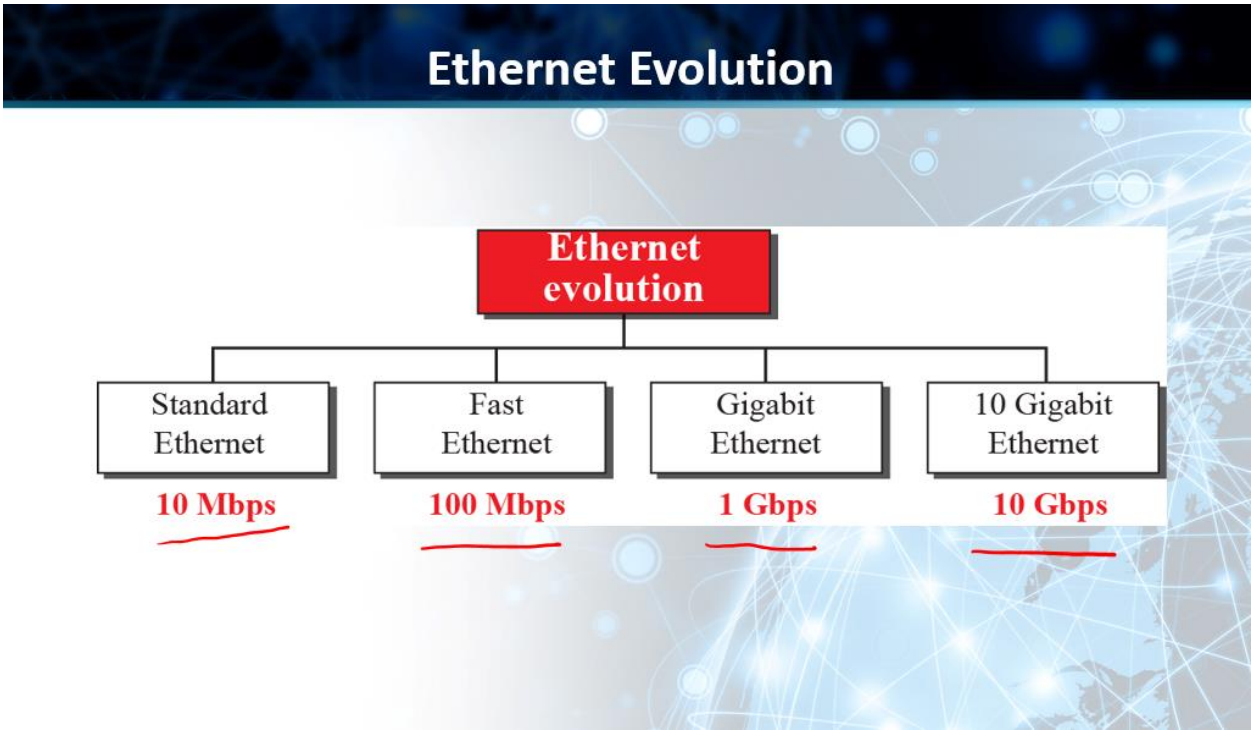
IEEE Project 802

- In 1985, the Computer Society of the IEEE started a project, called Project 802, to set standards to enable inter-communication among equipment from a variety of manufacturers
- Project 802 did not seek to replace any part of the OSI model or TCP/IP protocol suite
- A way of specifying functions of the physical layer and the data-link layer of major LAN protocols



Ethernet Evolution

- **The Ethernet LAN was developed in the 1970s**
- **Since then, it has gone through four generations:**
  - ✓ **Standard Ethernet (10 Mbps)**
  - ✓ **Fast Ethernet (100 Mbps)**
  - ✓ **Gigabit Ethernet (1 Gbps)**
  - ✓ **10 Gigabit Ethernet (10 Gbps)**



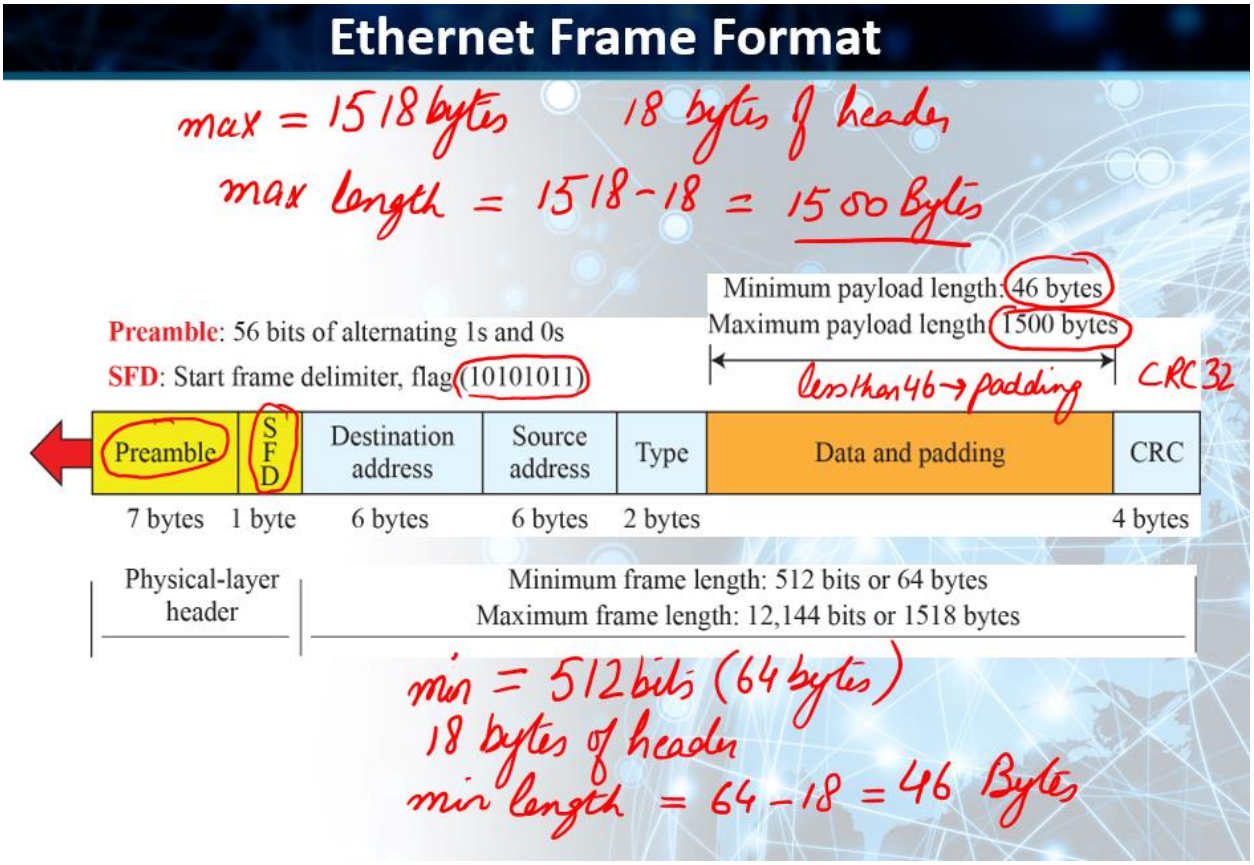
Standard Ethernet:

- The original Ethernet technology with the data rate of 10 Mbps is called Standard Ethernet
- Most implementations have moved to later evolutions
- Still some features of the Standard Ethernet that have not changed during the evolution

Connectionless & Unreliable Service

- Each frame is independent of other
- No connection establishment or tear down process
- The sender may overwhelm receiver with frames and frames are dropped
- If frame drops, sender will not know about it unless we are using TCP (Transport)
- Ethernet is unreliable like IP and UDP
- If a frame is corrupted, receiver silently drops it
- Left to high level protocols to find out abut it

Ethernet Frame Format:



Addressing in Standard Ethernet

- Each station on Ethernet has its own network interface card (NIC)
- The NIC fits inside the station and provides the station with a link-layer/physical address
- The Ethernet address is 6 bytes (48 bits), normally written in hexadecimal notation, with a colon between the bytes

Addressing in Standard Ethernet:

- For example, the following shows an Ethernet MAC address:

4A:30:10:21:10:1A

Transmission of Address Bits

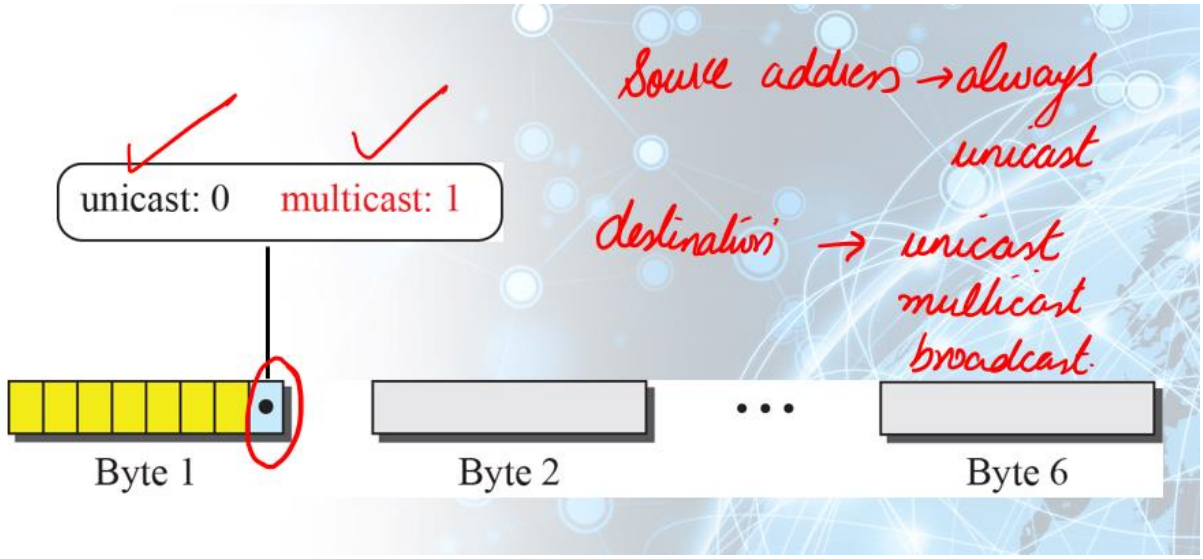
How the address 47:20:1B:2E:08:EE is sent out online. The address is sent left to right, byte by byte; for each byte, it is sent right to left, bit by bit, as shown below:

Hexadecimal	47	20	1B	2E	08	EE
Binarys	01000111	00100000	00011011	00101110	00001000	11101110
Transmitted ←	11100010	00000100	11011000	01110100	00010000	01110111

LS MS

→ unicast
→ multicast
→ broadcast

Unicast and Multicast Addresses:



Example of addresses:

Example 13.2

Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A
b. 47:20:1B:2E:08:EE
c. FF:FF:FF:FF:FF:FF

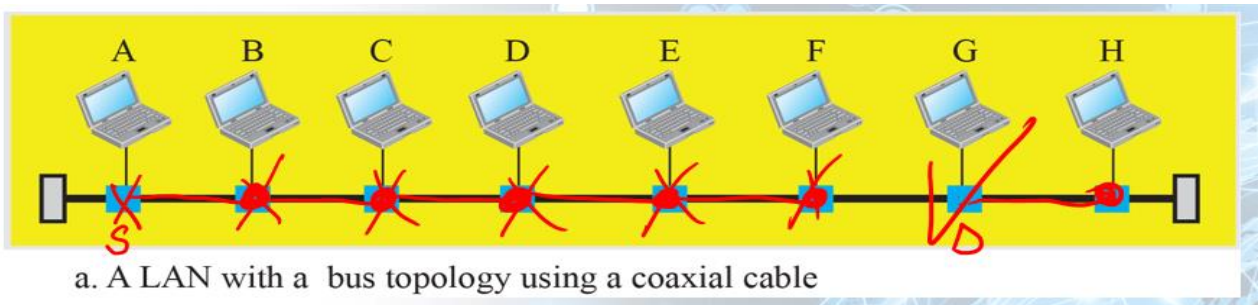
4A A → 1010 (0→LS)

47 7 → 0111 (1→LS)

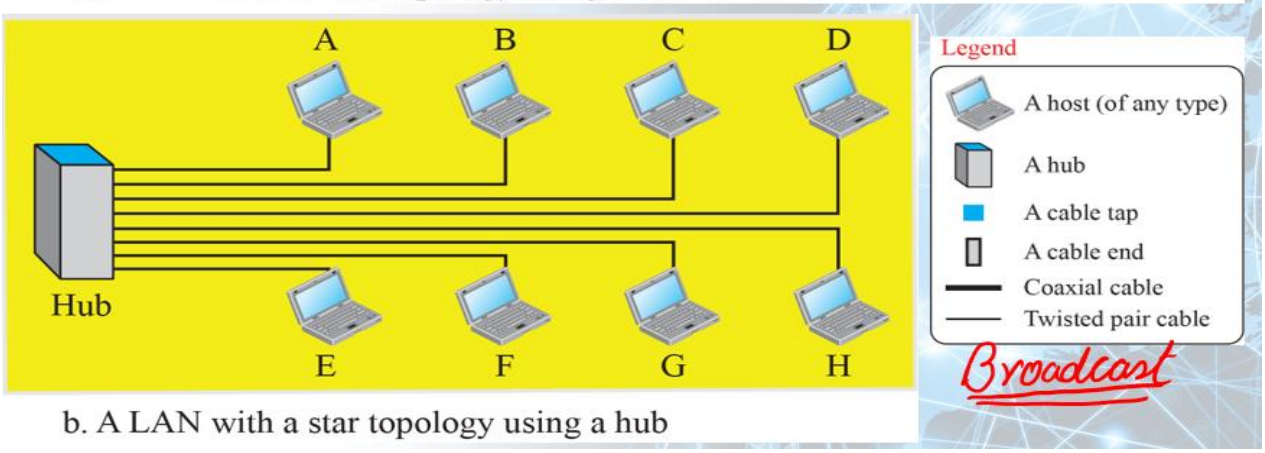
To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are Fs, the address is broadcast. Therefore, we have the following:

a. This is a unicast address because A in binary is 1010 (even).
b. This is a multicast address because 7 in binary is 0111 (odd).
c. This is a broadcast address because all digits are Fs in hexadecimal.

Implementation of Standard Ethernet

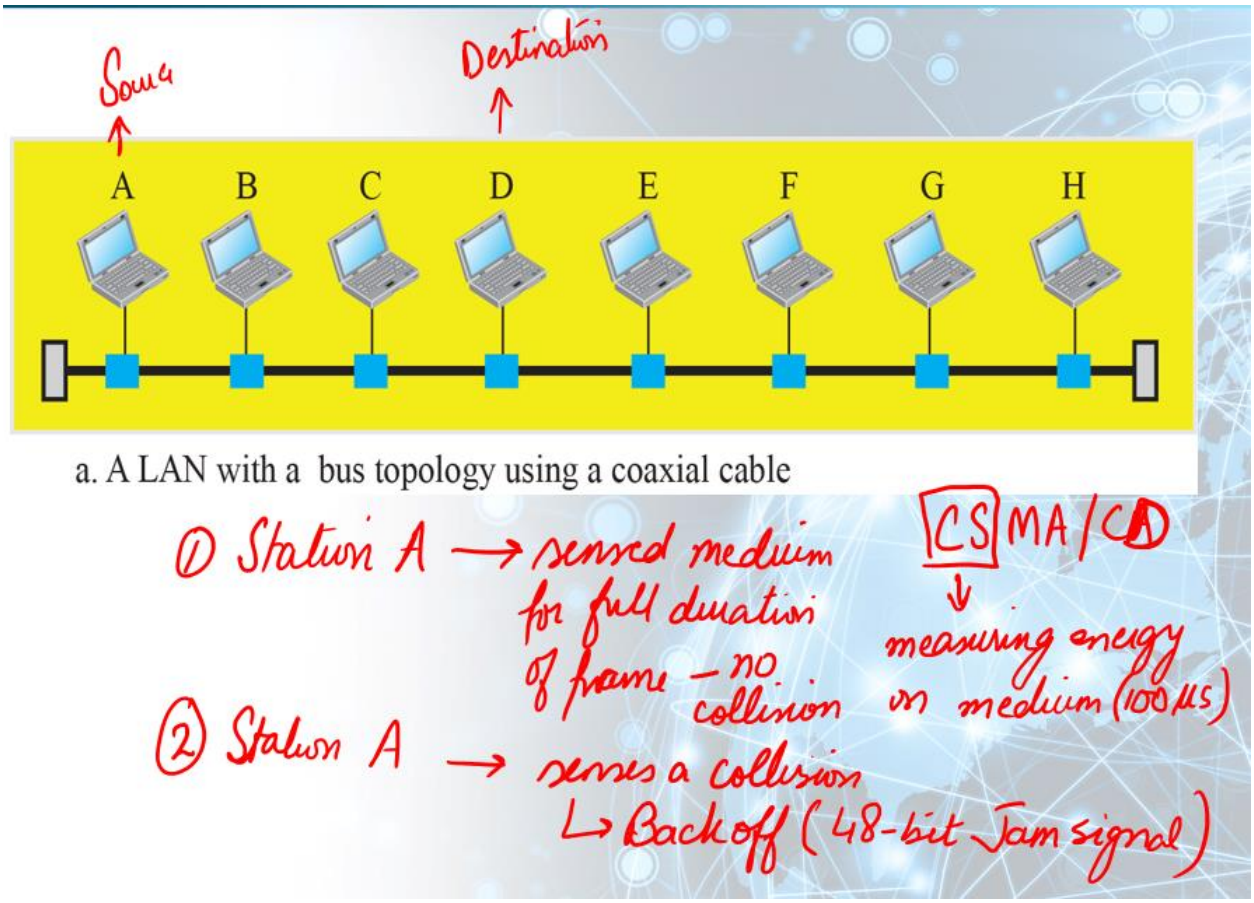


a. A LAN with a bus topology using a coaxial cable



b. A LAN with a star topology using a hub

Access Method in Standard Ethernet:



Access Method in Standard Ethernet

- Since the network that uses the standard Ethernet protocol is a broadcast network, we need to use an access method to control access to the sharing medium
- The standard Ethernet chose CSMA/CD with 1-Persistent Method

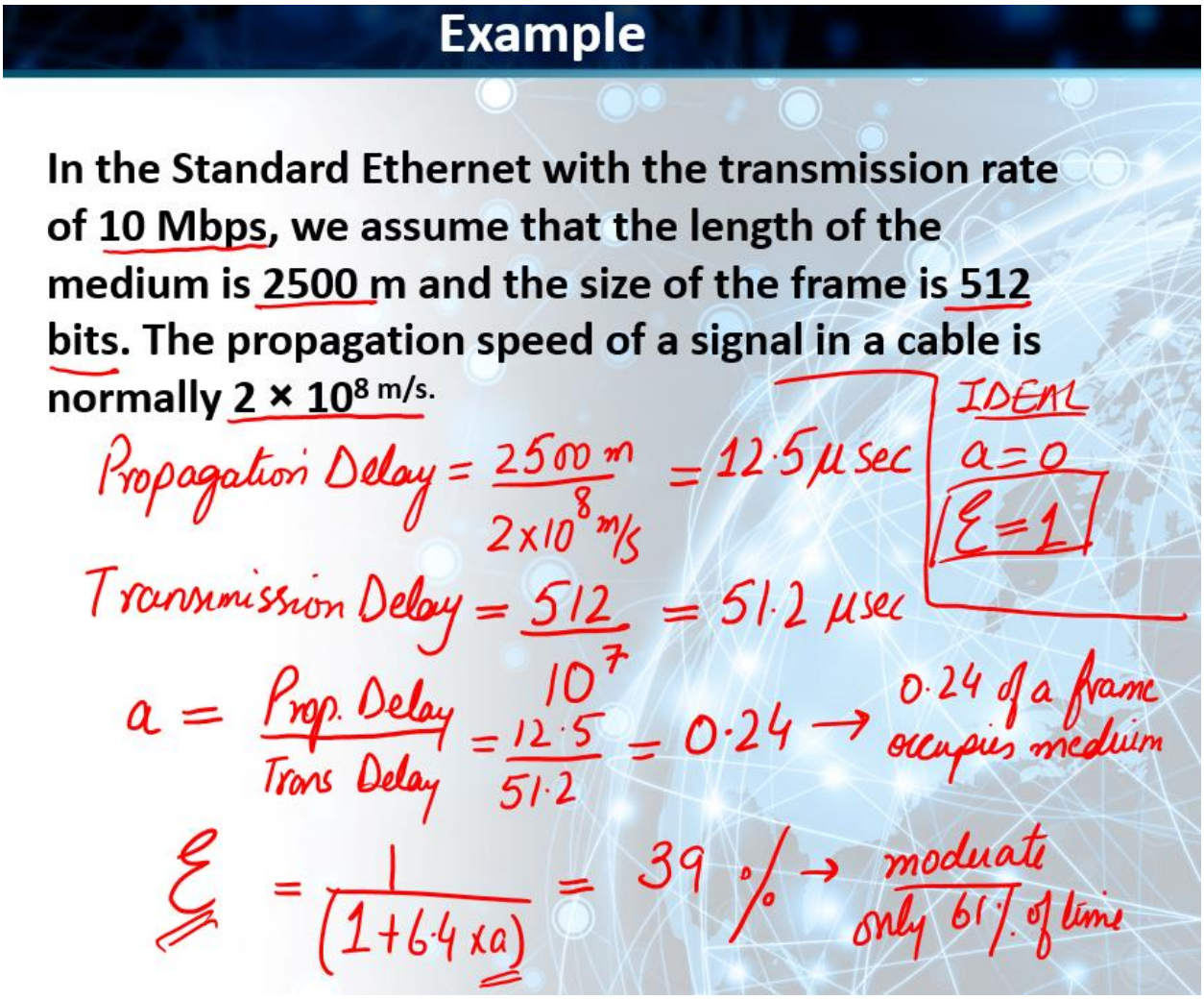
Efficiency of Standard Ethernet

- The ratio of the time used by a station to send data to the time the medium is occupied by this station
- The practical efficiency of standard Ethernet has been measured to be:

Efficiency = 1/(1+ 6.4 x a)

where a = number of frames that can fit on a medium

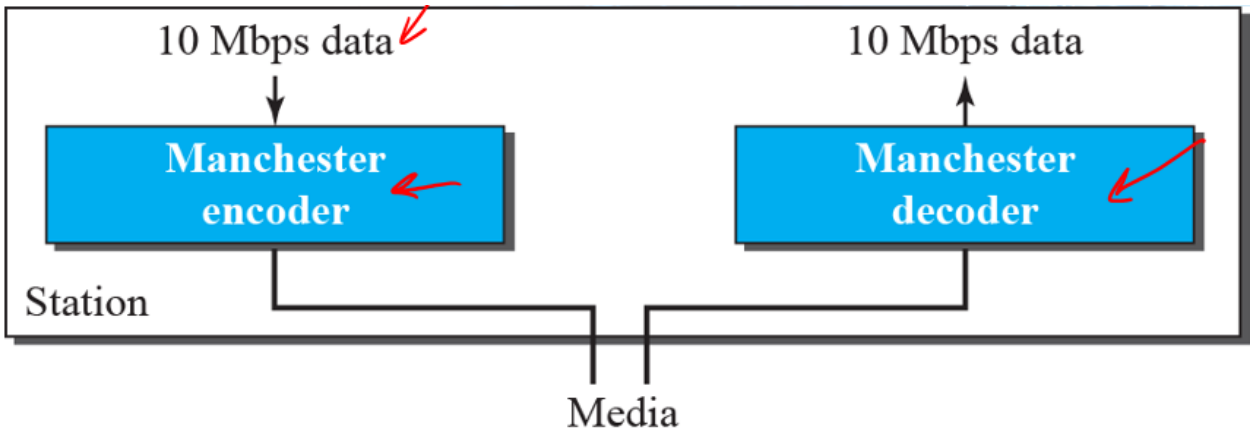
Efficiency of Standard Ethernet:



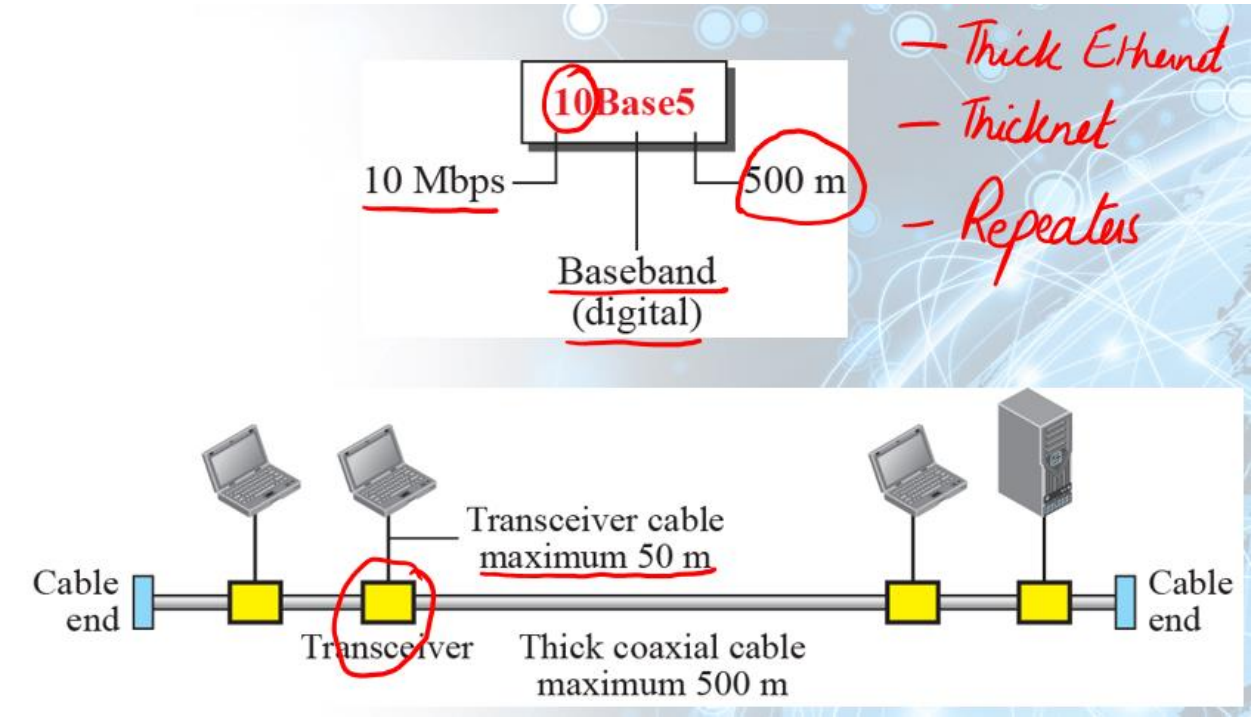
Summary of Standard Ethernet implementations:

Implementation	*Medium*	*Medium Length*	*Encoding*
10Base5	Thick coax	500 m	Manchester
10Base2	Thin coax	185 m	Manchester
10Base-T	2 UTP	100 m	Manchester
10Base-F	2 Fiber	2000	Manchester

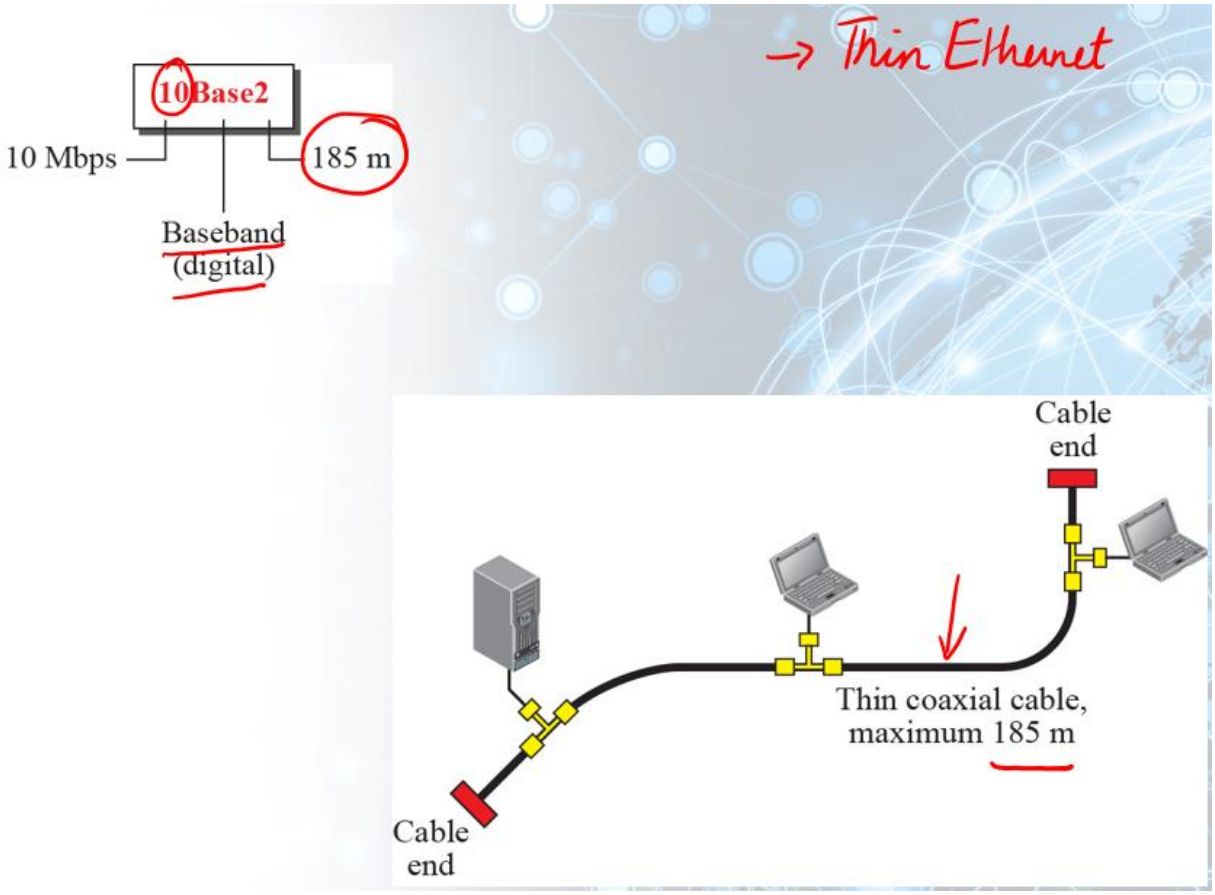
Encoding in Standard Ethernet:



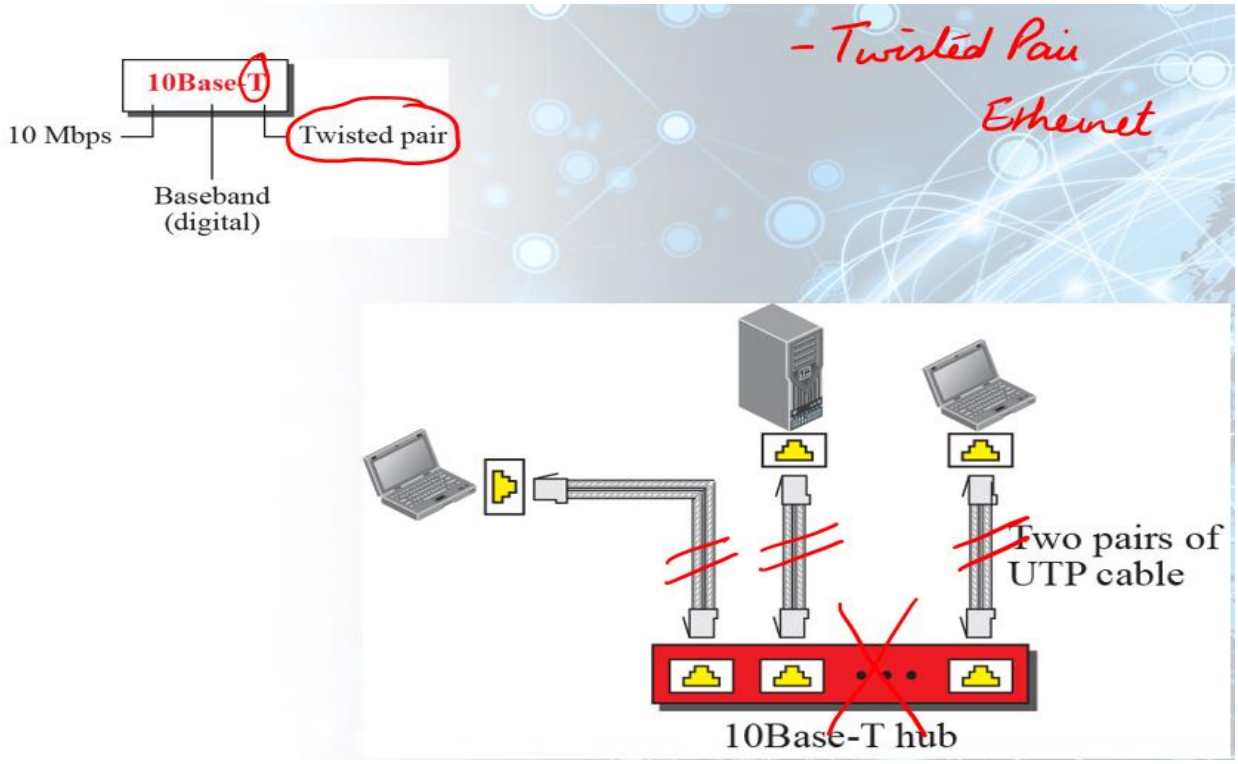
10Base5 implementation:



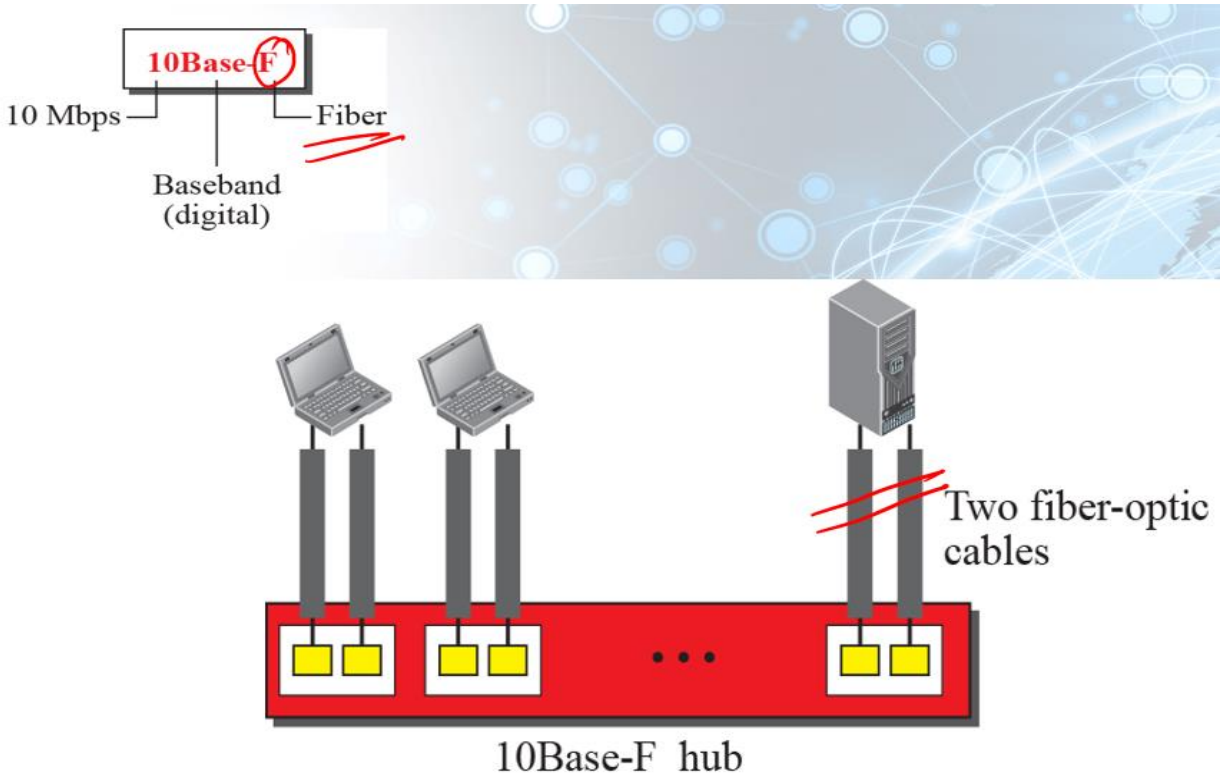
10Base2 implementation:



10Base-T implementation:



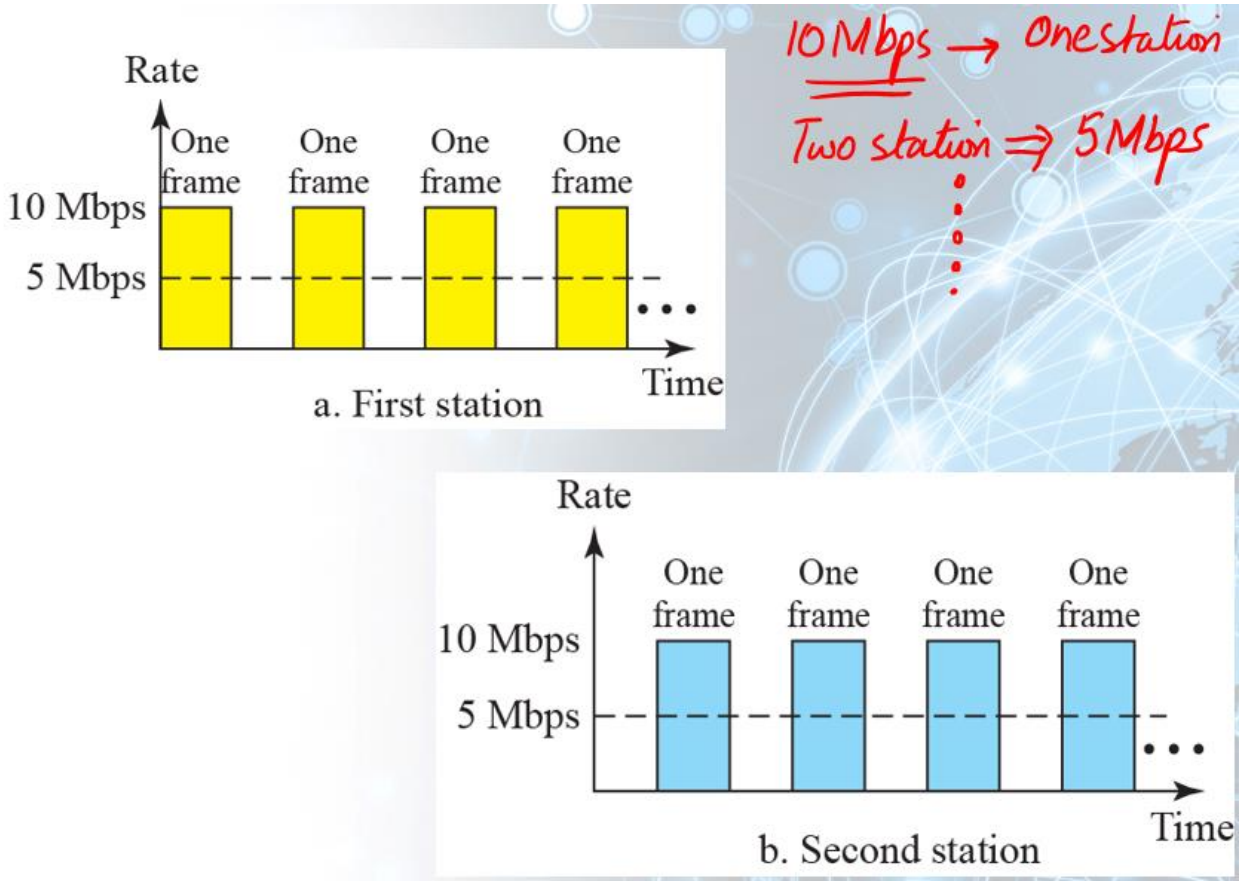
10Base-F implementation:



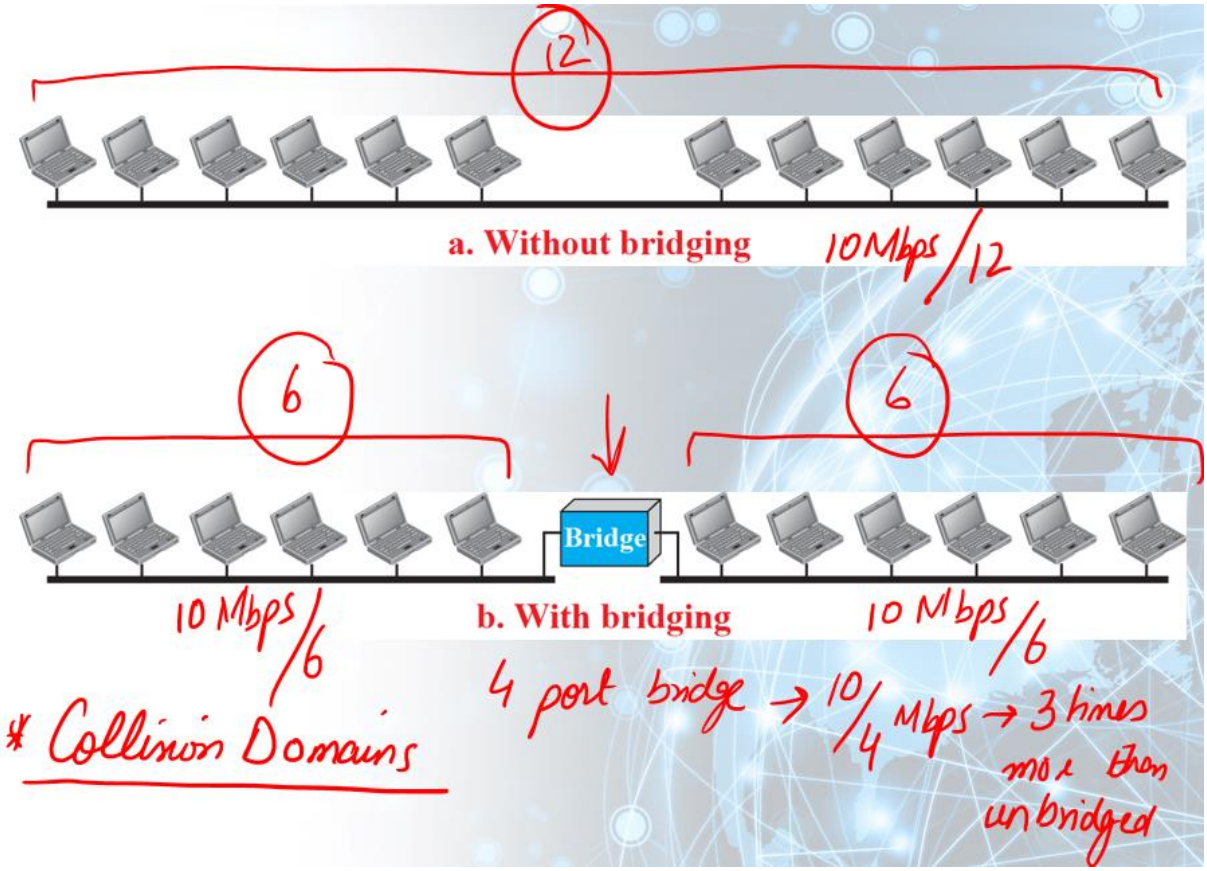
Changes in the Standard

- The changes that occurred to the 10-Mbps Standard Ethernet opened the road to the evolution of the Ethernet to become compatible with other high-data-rate LANs
  - ✓ Bridged Ethernet
  - ✓ Switched Ethernet
  - ✓ Full-Duplex Ethernet

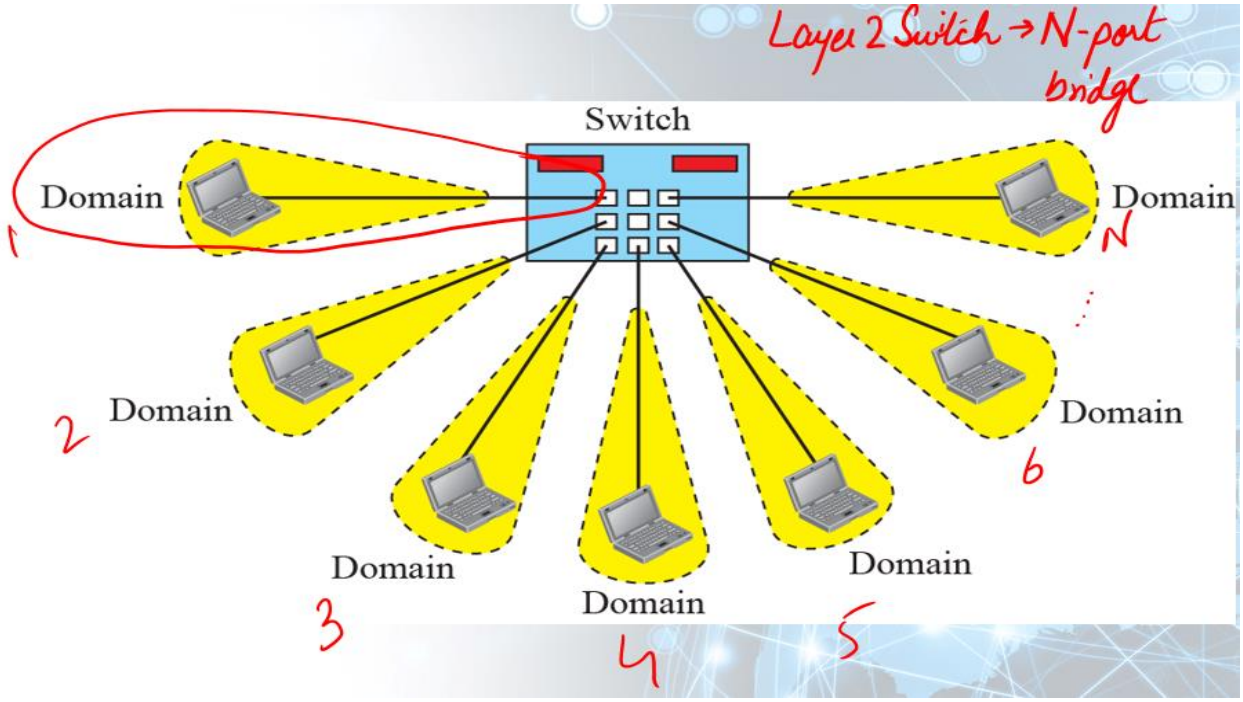
Bridged Ethernet- Sharing Bandwidth:



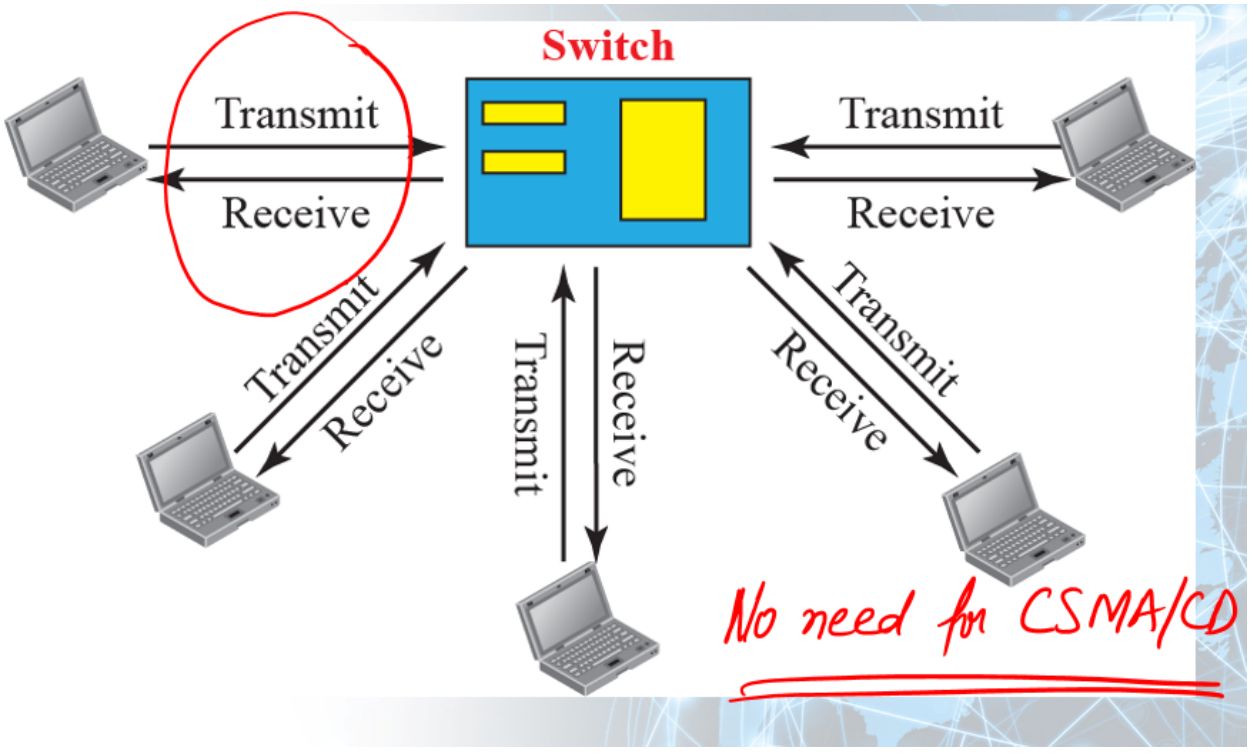
A Network with and without Bridging:



Switched Ethernet:



Full – Duplex Switched Ethernet:



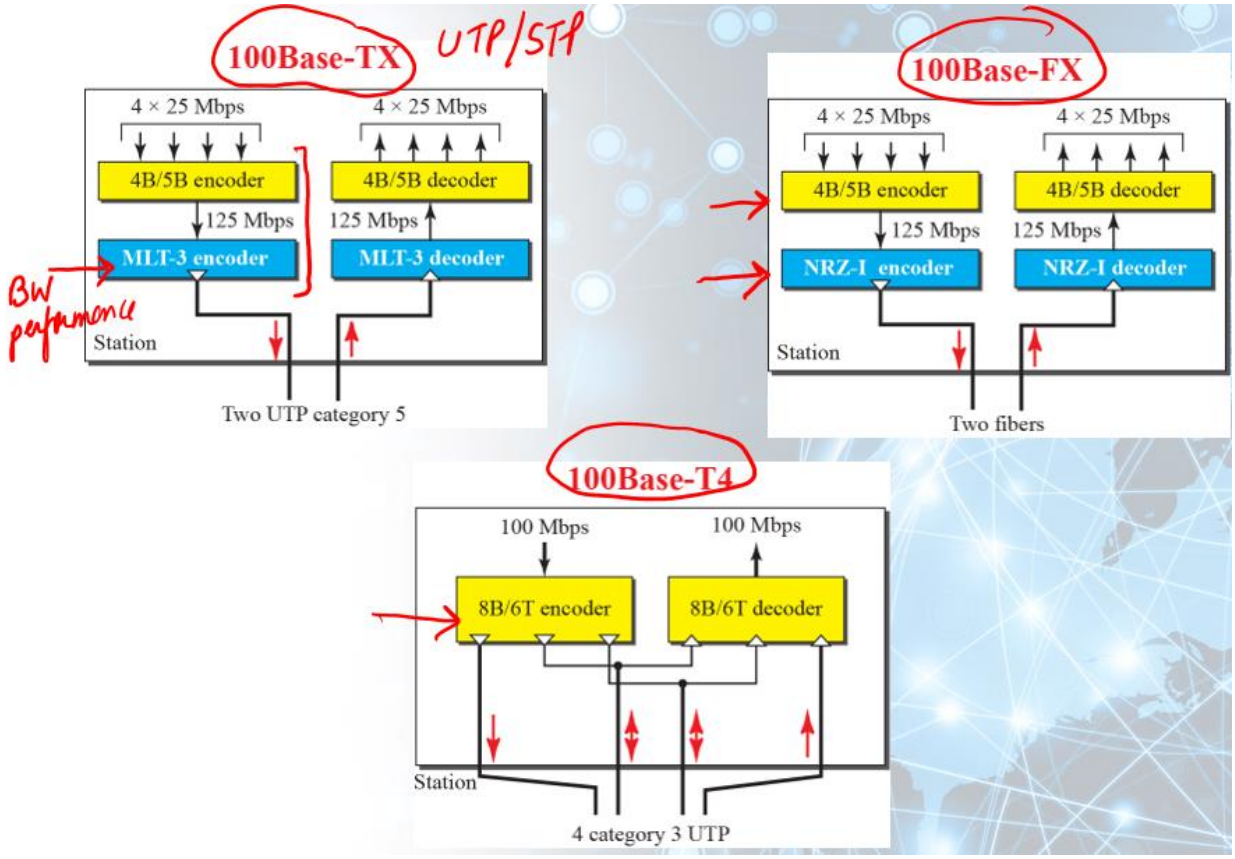
Fast Ethernet

- In the 1990s, Ethernet made a big jump by increasing the transmission rate to 100 Mbps, and the new generation was called the Fast Ethernet
- To make it compatible with the Standard Ethernet, the MAC sublayer was left unchanged
- But the features of the Standard Ethernet that depend on the transmission rate, had to be changed
- Goals of Fast Ethernet:
- Upgrade data rate to 100Mbps
- Make it compatible with Standard Ethernet
- Keep same 48-bit address
- Keep same frame format

Physical Layer:

- To be able to handle a 100 Mbps data rate, several changes need to be made at the physical layer

Encoding for Fast Ethernet:



Implementation of Fast Ethernet implementations:

Implementation	Medium	Medium Length	Wires	Encoding
100Base-TX	STP	100 m	2	4B5B + MLT-3
100Base-FX	Fiber	185 m	2	4B5B + NRZ-I
100Base-T4	UTP	100 m	4	Two 8B/6T

Gigabit Ethernet:

- The goals of the Gigabit Ethernet were:
  - Upgrade the data rate to 1 Gbps
  - Make it compatible with standard or Fast Ethernet
  - Use same 48 bit address
  - Use the same frame format
  - Keep same minimum and maximum frame lengths

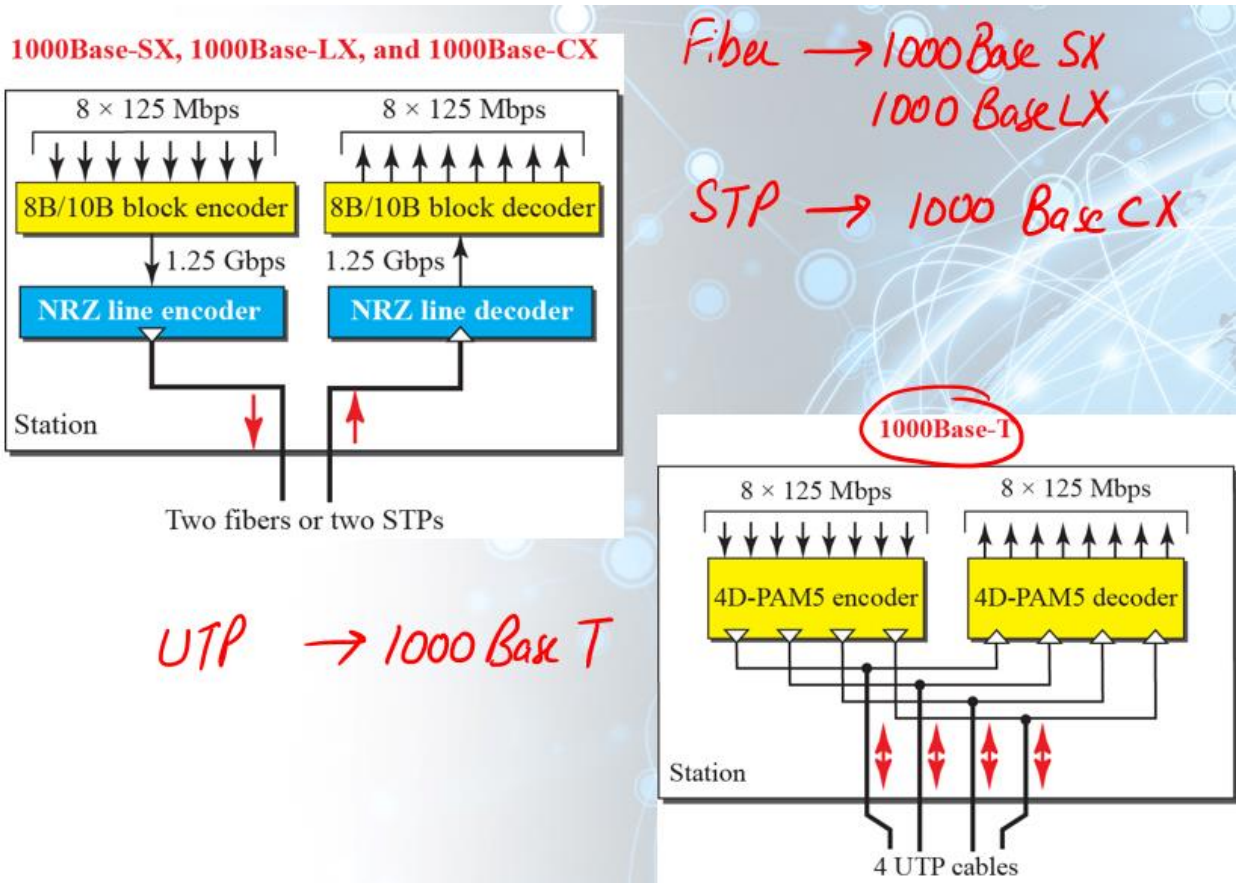
MAC Sub-layer:

- A main consideration in the evolution of Ethernet was to keep the MAC sublayer untouched
- To achieve a data rate of 1 Gbps, this was no longer possible
- Gigabit Ethernet has two distinctive approaches for medium access:
  - Half-duplex
  - Full-duplex

Physical Layer:

- The physical layer in Gigabit Ethernet is more complicated than that in Standard or Fast Ethernet
- We briefly discuss some features of this layer:

Encoding in Gigabit Ethernet:



Summary of Gigabit Ethernet Implementations:

Implementation	*Medium*	*Medium Length*	*Wires*	*Encoding*
1000Base-SX	Fiber S-W	550 m	2	8B/10B + NRZ
1000Base-LX	Fiber L-W	5000 m	2	8B/10B + NRZ
1000Base-CX	STP	25 m	2	8B/10B + NRZ
1000Base-T4	UTP	100 m	4	4D-PAM5

Topic 191 – 196

Access Networks

- Networks that connect a small LAN to an ISP
- Wired networks used to transfer data over long distances

Telephone Network

Lets' see the brief details of telephone network

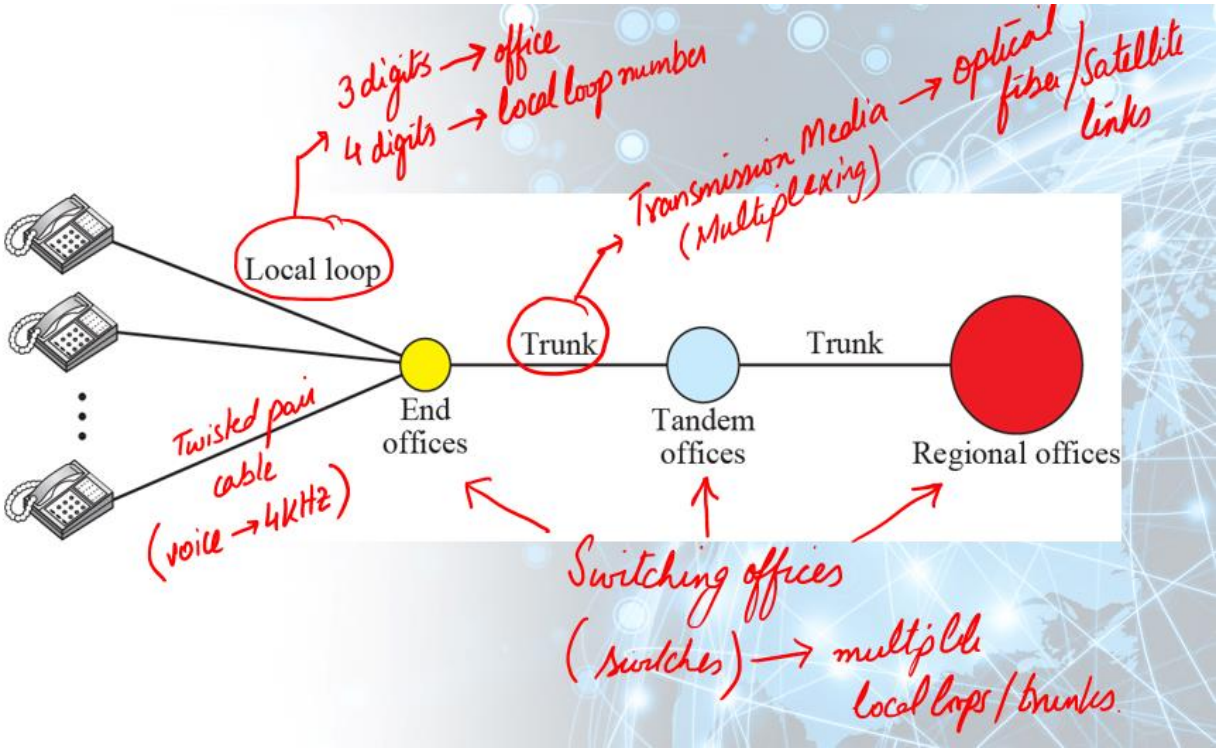
- The telephone network had its beginnings in the late 1800s. Plain Old Telephone System (POTS) was originally an analog system using analog signals to transmit voice. With the advent of the computer era, the network, in the 1980s, began to carry data in addition to voice.
- During the last decade, the telephone network has undergone many technical changes, and the network is now Digital as well as Analog.

Let's see the **Major Components** of the telephone network.

- The telephone network is made of three major components:
  - Local Loops
  - Trunks
  - Switching offices

Moreover, telephone network has several levels of switching offices:

- End offices
- Tandem offices
- Regional offices



Local-Access Transport Areas (LATAs)

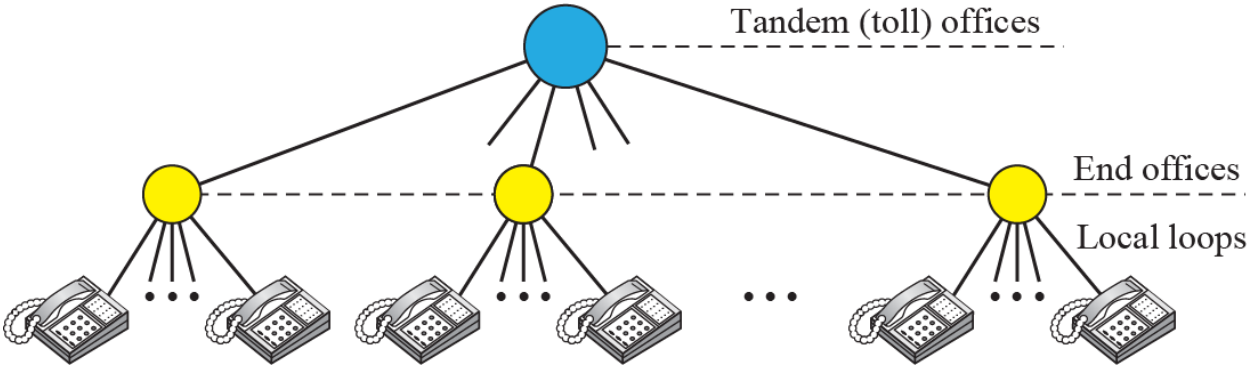
Local Access Transport Area (LATA) is a term used for a geographic region where one or more telephone companies or Local Exchange Companies (LECs) provide telecommunications services. This term LATA is basically used in U.S. telecommunication regulation.

- A LATA can be a small or large metropolitan area. A small state may have a single LATA and a large state may have several LATAs.
- A LATA boundary may overlap with state boundary; part of a LATA can be in one state, part in another state.

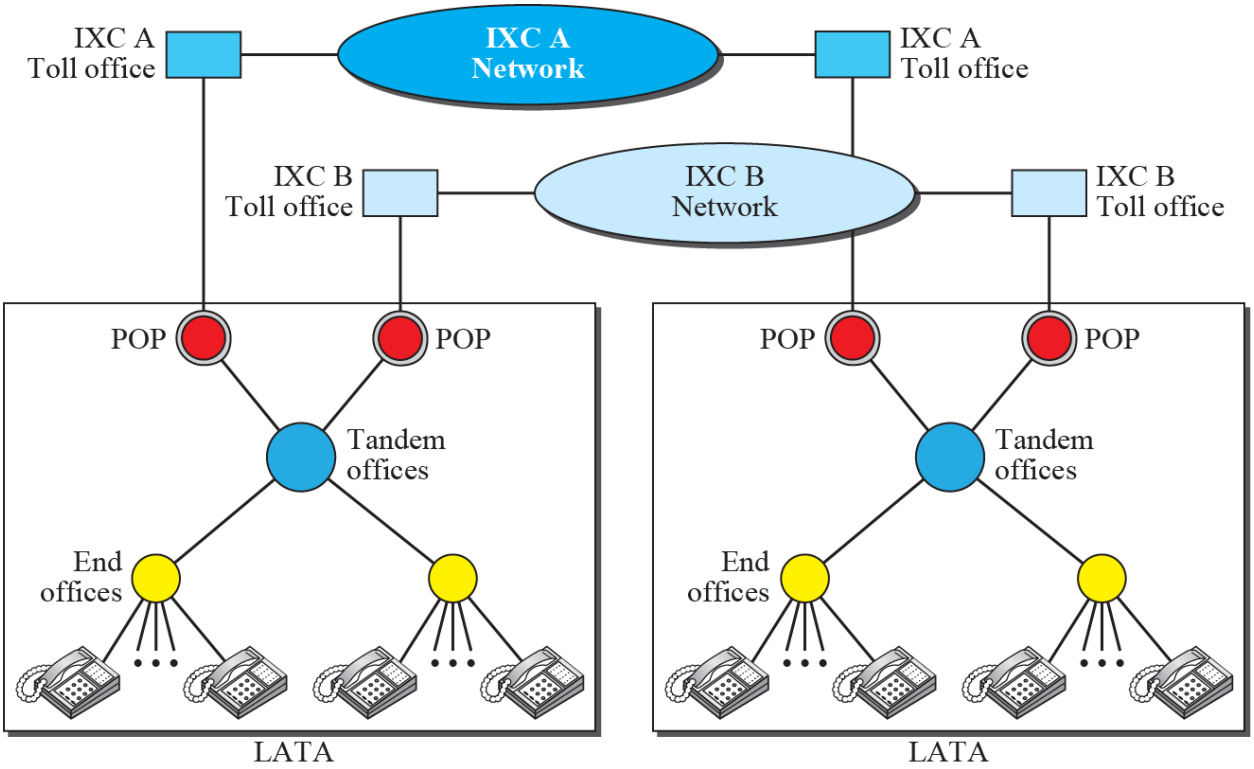
Intra-LATA and Inter-LATA Services

- Services offered by Telephone companies inside a LATA are called Intra-LATA services and between LATAs are called Inter-LATA services.
- Carrier that handles Intra-LATA are called a Local Exchange Carrier (LEC) and the ones that handle Inter-LATA are called Interexchange Carriers (IXCs)

Switching Offices in a LATA



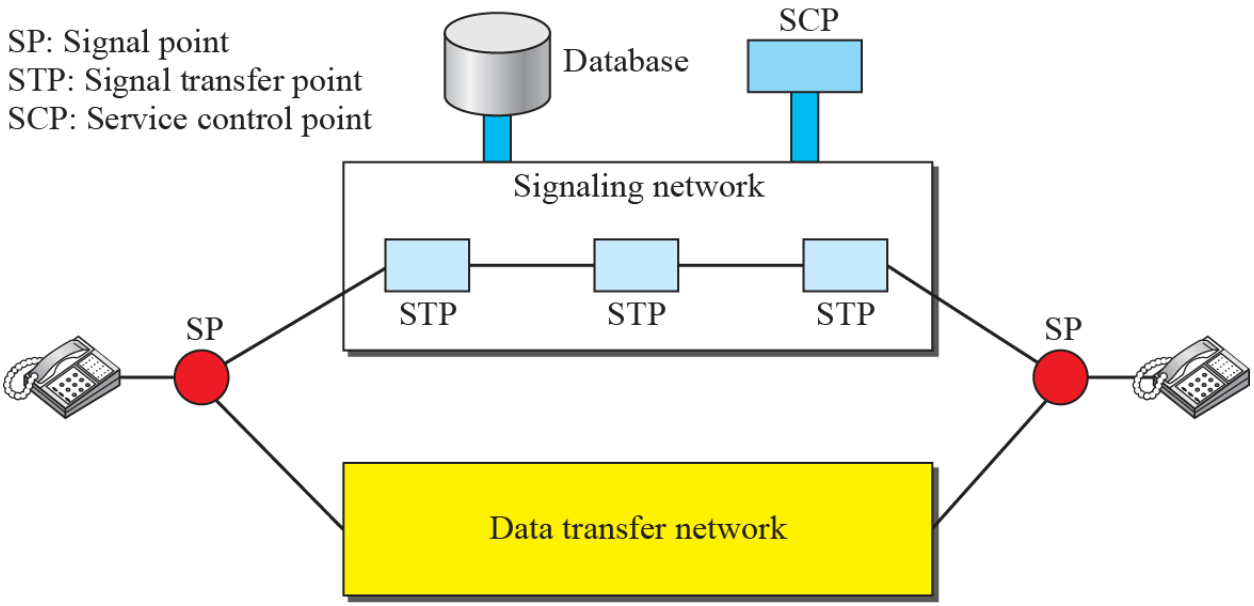
Points of Presence (POPs)



Signaling

- The telephone network in the beginning, used a circuit-switched network with dedicated links to transfer voice communication. The operator connected the two parties by using a wire with two plugs inserted into the corresponding two jacks. Later, the signaling system became automatic.
- Rotary telephones were invented that sent a digital signal defining each digit in a multi-digit telephone number. As telephone networks evolved into a complex network, the functionality of the signaling system increased.

Data Transfer and Signaling Network



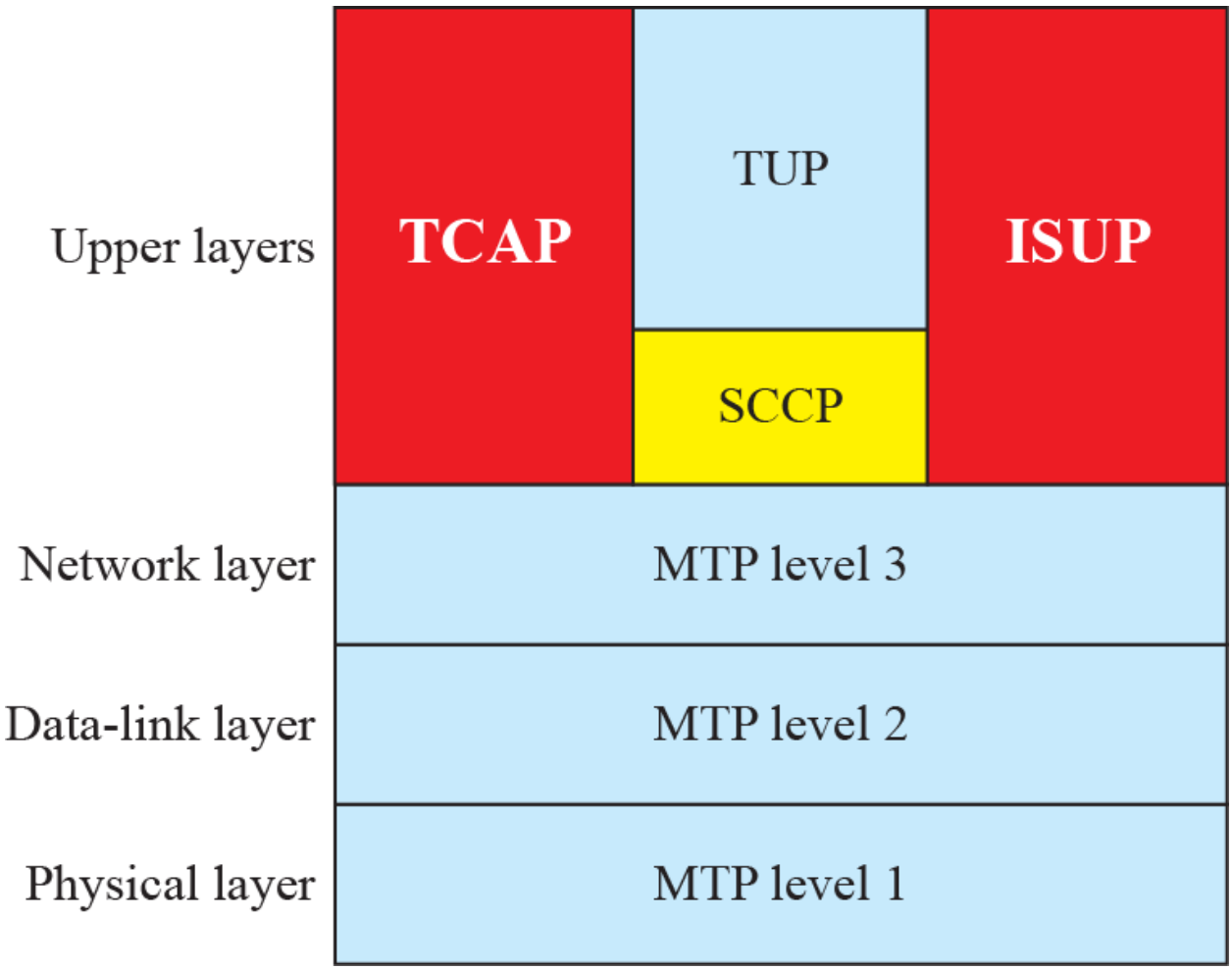
MTP: Message transfer part

SCCP: Signaling connection control point

TCAP: Transaction capabilities application port

TUP: Telephone user port

ISUP: ISDN user port



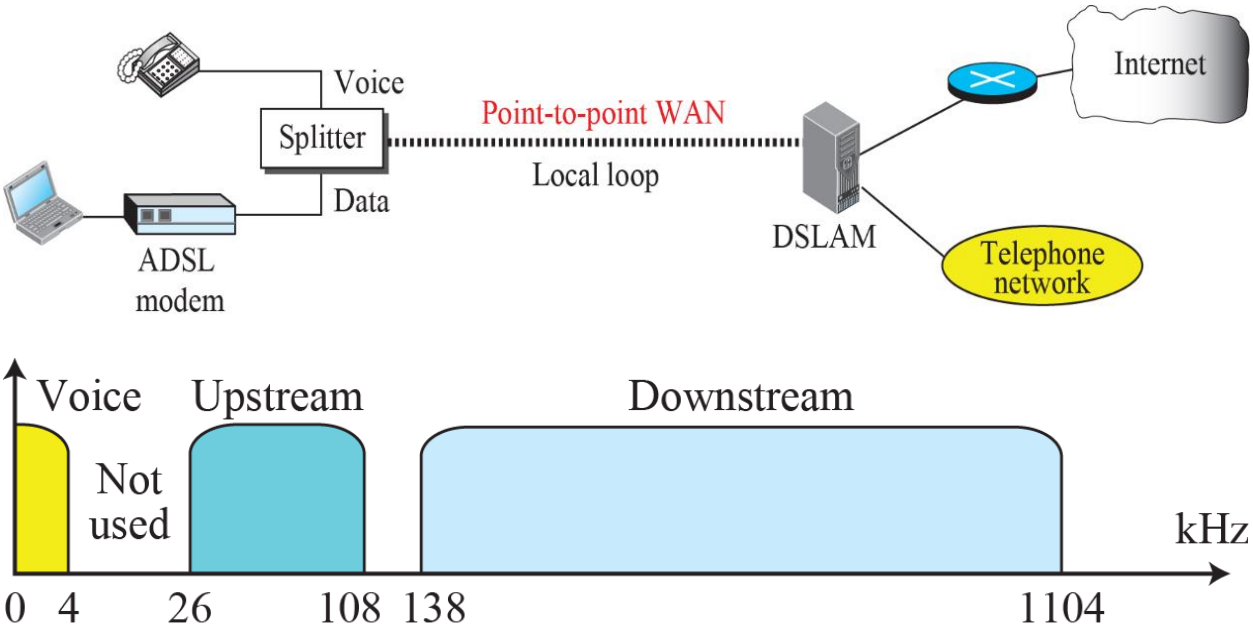
Services

- **Telephone companies provide two types of services:**
  - **Analog Services**
    - Analog Switched Services
    - Analog Leased Services
  - **Digital Services**
    - Switched /56 Service
    - Digital Data Service

Digital Subscriber Line (DSL)

- After traditional dial-up modems reached their peak data rate, telephone companies developed another technology, DSL, to provide higher-speed access to the Internet. DSL supports high-speed digital communication over the existing telephone.
- DSL technology is a set of technologies, each differing in the first letter (ADSL, VDSL, HDSL, and SDSL)

ADSL Point-to-Point Network

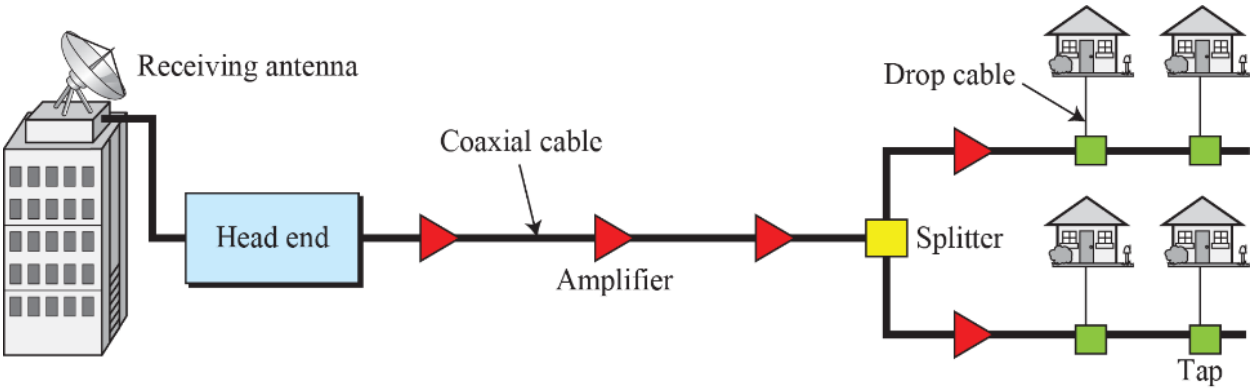


Cable Network

- Cable TV networks were initially created to provide remote subscribers access to TV programs. Cable networks enabled access to remote broadcasting stations via microwave connections. Cable TV also found a good ISP market by using some of the channels originally designed for video.

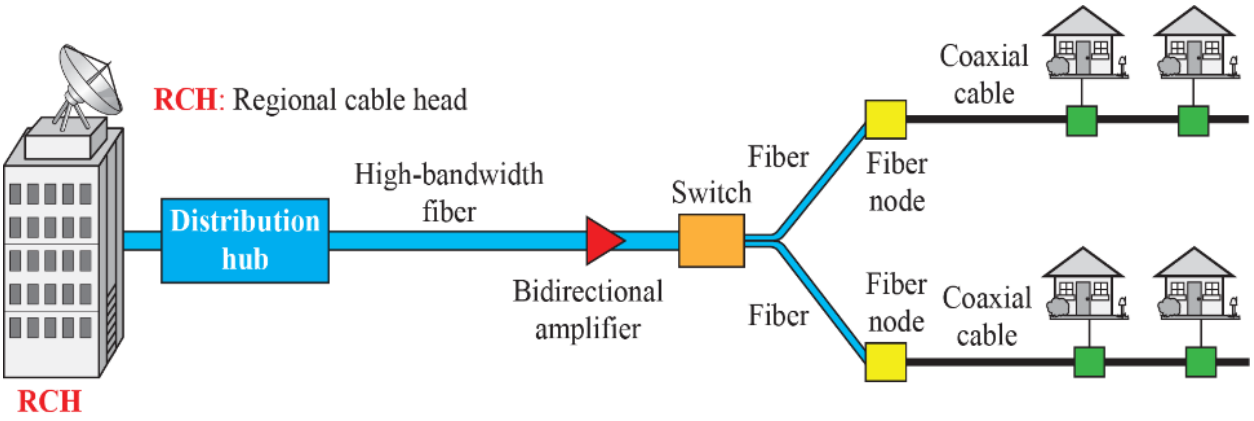
Traditional Cable Networks

- Cable TV started to distribute broadcast video signals to locations with poor or no reception in the late 1940s. It was called community antenna television (CATV) because an antenna at the top of a tall hill or building received the signals from the TV stations.



Hybrid Fiber Coaxial (HFC) Network

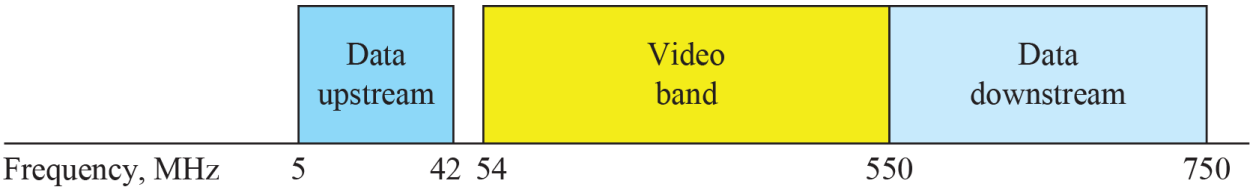
- Second generation of cable network is called a Hybrid Fiber-Coaxial (HFC) network. The network uses a combination of fiber-optic and coaxial cable



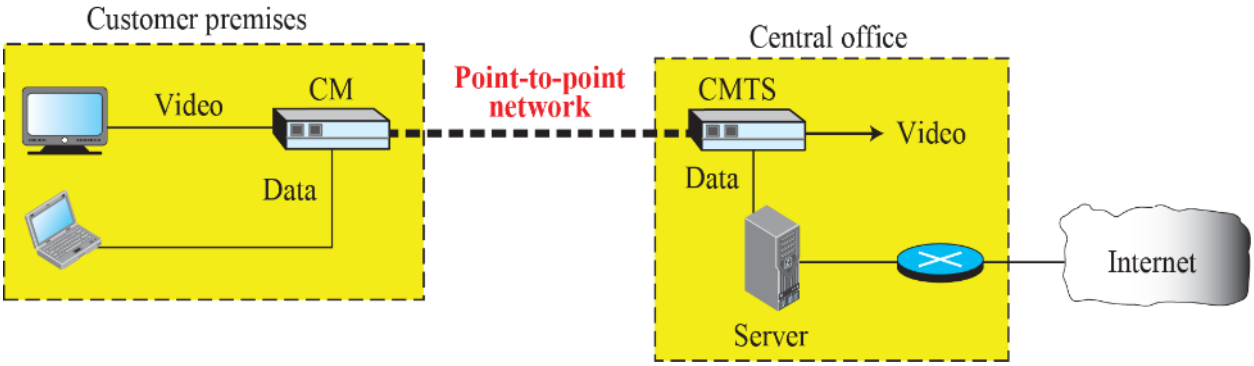
Cable TV for Data Transfer

- Cable companies are now competing with telephone companies for the residential customer who wants high-speed data transfer. DSL technology provides high-data-rate connections for residential subscribers over the local loop BUT UTP is susceptible to Interfence.
- This imposes an upper limit on the data rate. A solution is the use of the cable TV network.

Division of Coaxial Cable Band by CATV



Cable Modem Transmission System (CMTS)



Topic 197 – 204

Synchronous Optical Network (SONET)

- A wide area network (WAN) that is used as a transport network to carry loads from other WANs.
- ITU–T standard called Synchronous Digital Hierarchy (SDH)
- Architecture of a SONET system consists of signals, devices, and connections

SONET Architecture

Signals

-
  - Synchronous Transport Signals (STS)
  - Optical Carriers (OCs)
  - Synchronous Transport Module (STM)

SONET Devcies

-
  - STS Mux/Demux
  - Regenerators
  - Add-Drop Multiplexer and Terminals

Connections

-
  - Section
  - Line
  - Path

SONET Signals

STS	*OC*	*Rate (Mbps)*	*STM*
STS-1	OC-1	51.840	
STS-3	OC-3	155.520	**STM-1**
STS-9	OC-9	466.560	**STM-3**
STS-12	OC-12	622.080	**STM-4**
STS-18	OC-18	933.120	**STM-6**
STS-24	OC-24	1244.160	**STM-8**
STS-36	OC-36	1866.230	**STM-12**
STS-48	OC-48	2488.320	**STM-16**
STS-96	OC-96	4976.640	**STM-32**
STS-192	OC-192	9953.280	**STM-64**

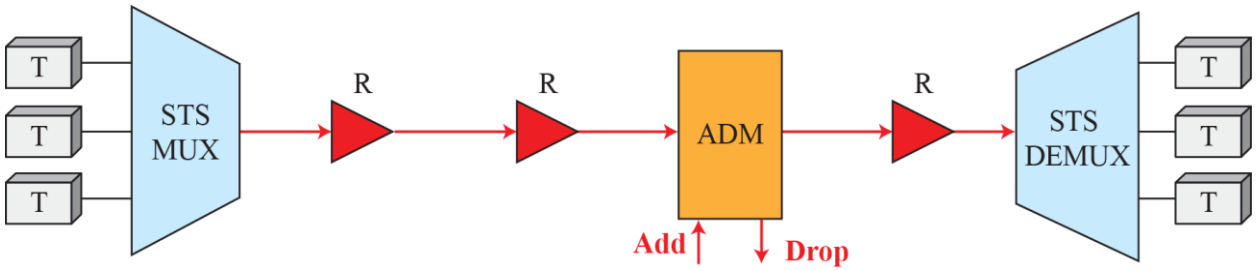
ADM: Add/drop multiplexer

R: Regenerator

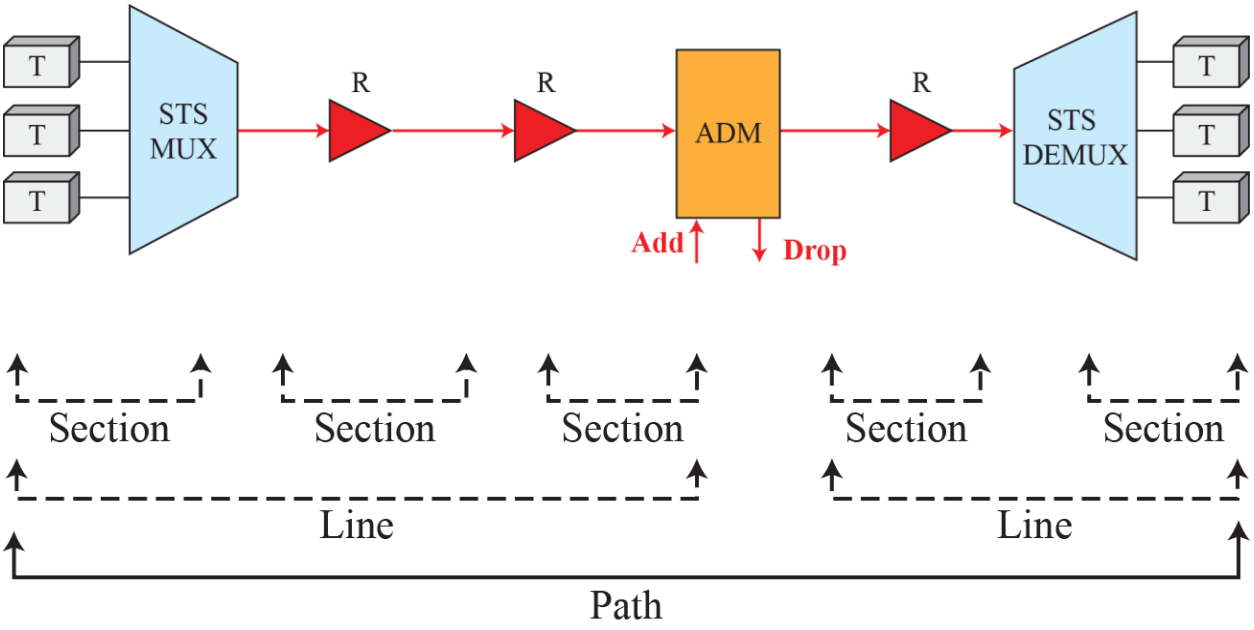
STS MUX: Synchronous transport signal multiplexer

T: Terminal

STS DEMUX: Synchronous transport signal demultiplexer



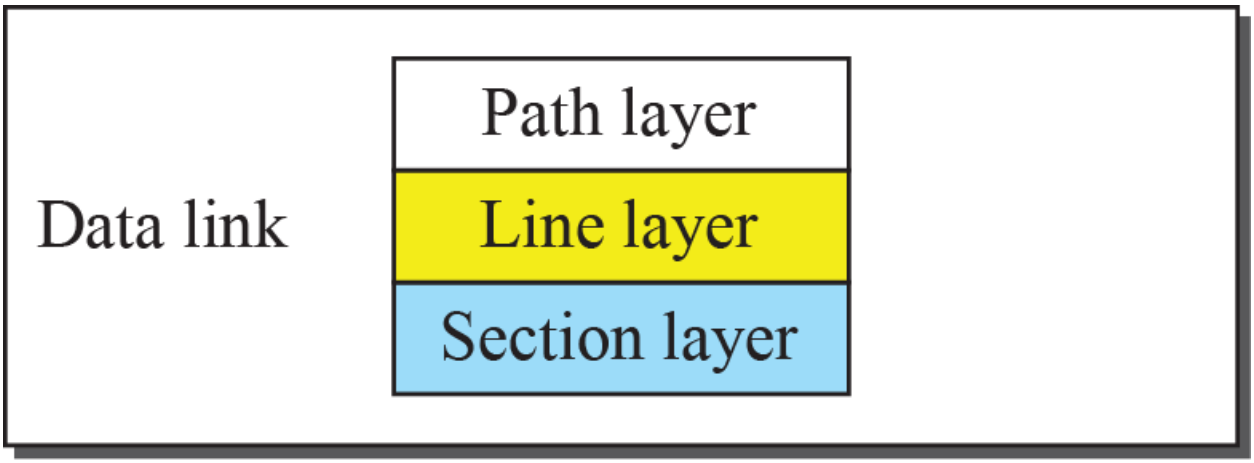
SONET Connections



SONET Layers

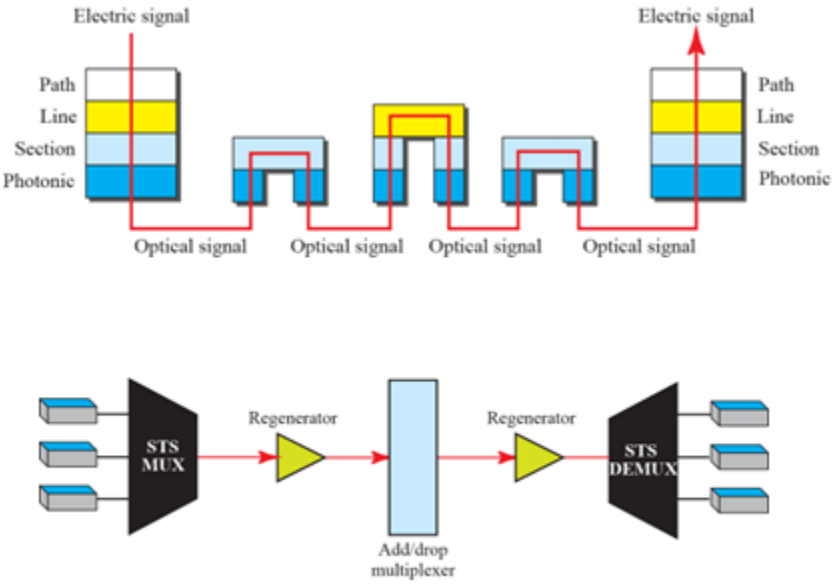
- **The SONET standard includes four functional layers:**
  - **The Path Layer**
  - **The Line Layer**
  - **The Section Layer**
  - **The Photonic Layer**

- **The layers correspond to both the physical and the data-link layers**



Physical **Photonic layer**

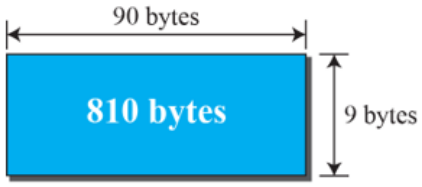
Device-Layer Relationship in SONET



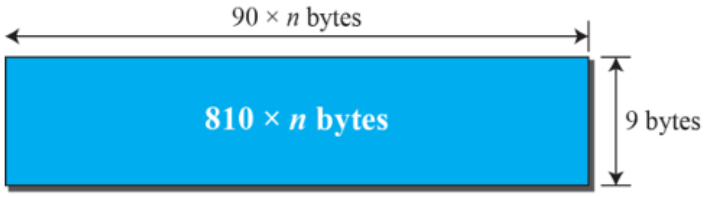
SONET Frames

- Each synchronous transport signal STS-n is composed of 8000 frames.
- Each frame is a two-dimensional matrix of bytes with 9 rows by 90×n columns.
- STS-1 frame is 9 rows by 90 columns (810 bytes), and an STS-3 is 9 rows by 270 columns (2430 bytes).

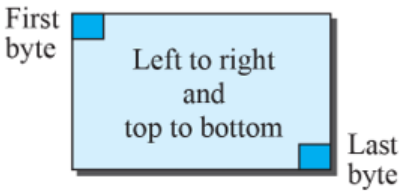
An STS-1 and an STS-n Frame



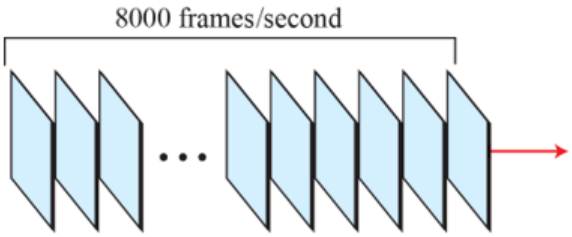
a. STS-1 frame



b. STS-*n* frame



a. Byte transmission



b. Frame transmission

Example

8000 frames per second

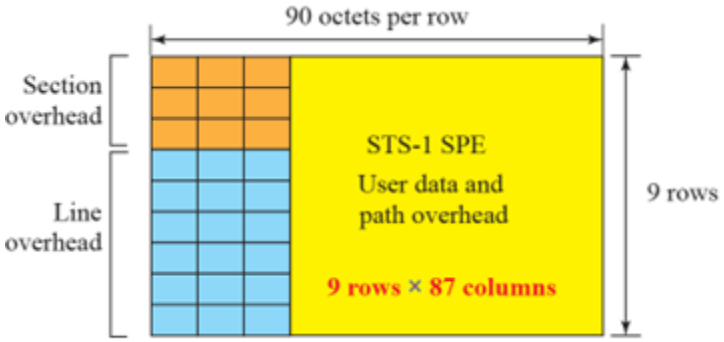
9 bytes × (1 × 90) bytes

Data Rate = 8000 × 9(1×90)

= 51.84 Mbps

Find the data rate of an STS-1 signal

STS-1 Frame Format



Example

Find the data rate of an STS-3 signal

$n = 3$

Data Rate = $8000 \times 9 (3 \times 90)$

$= 155.52$ Mbps

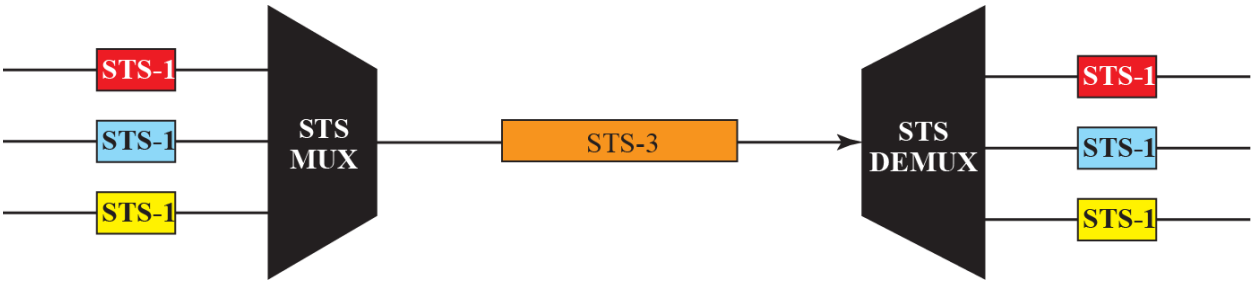
STS-1 Data Rate $\times 3$

$=$ STS-3 Data Rate

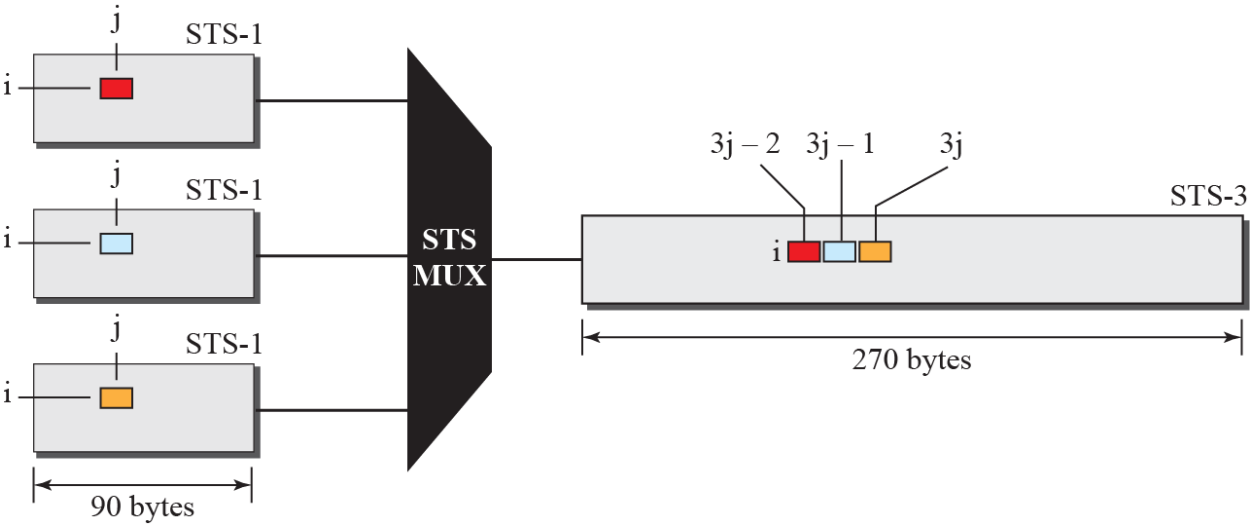
STS Multiplexing

- In SONET, frames of lower rate can be synchronously time-division multiplexed into a higher-rate frame.
- For example, three STS-1 signals (channels) can be combined into one STS-3 signal (channel), four STS-3s can be multiplexed into one STS-12, and so on

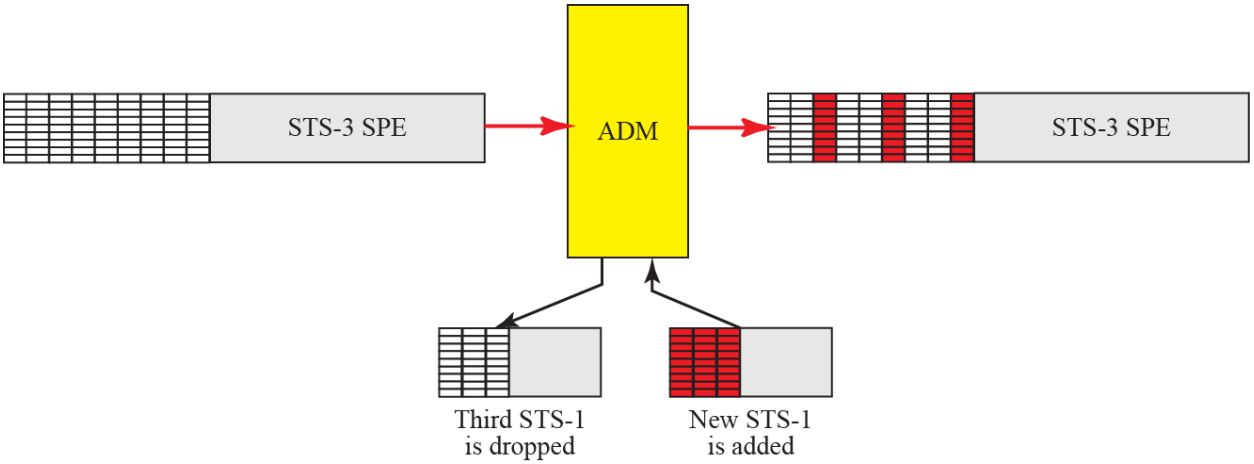
STS Multiplexing/Demultiplexing



Byte Interleaving



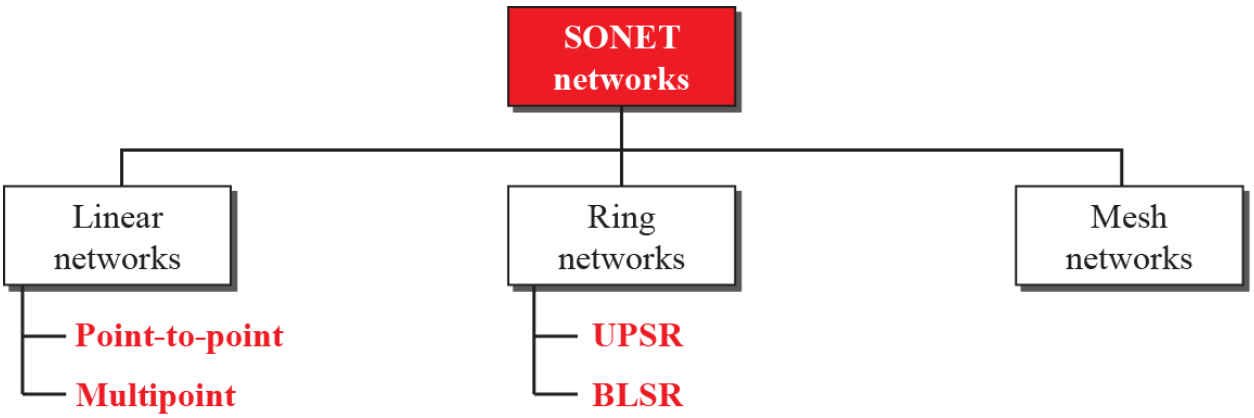
Add/Drop Multiplexer



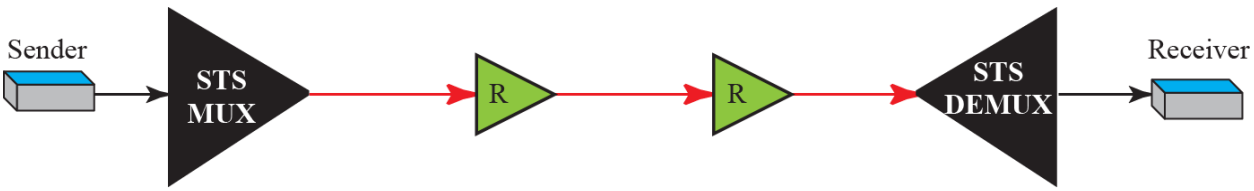
SONET Networks

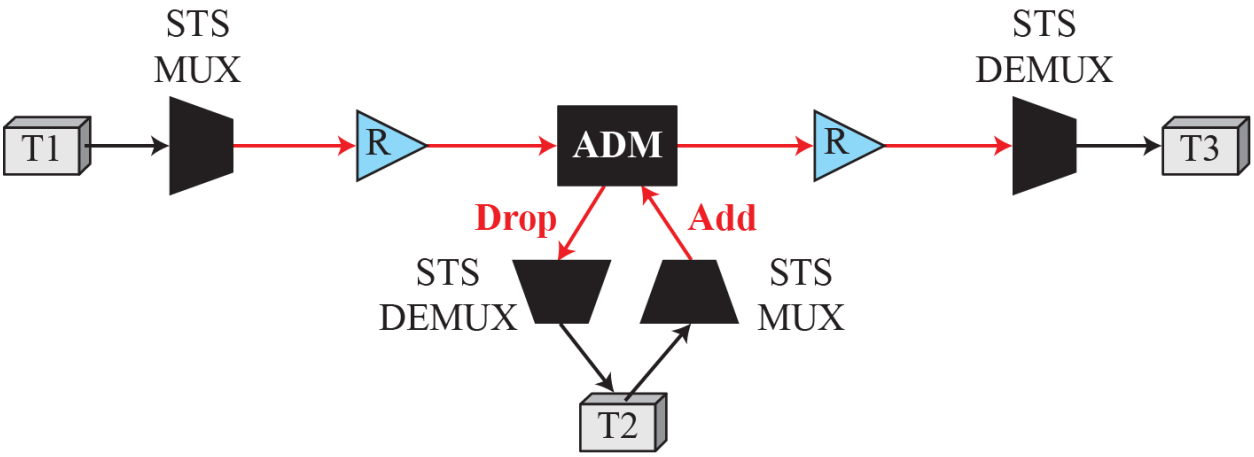
- SONET network can be used as a high-speed backbone carrying loads from other networks such as ATM or IP
- We can roughly divide SONET networks into three categories:
  - Linear Networks
  - Ring Networks
  - Mesh networks

Taxonomy of SONET Networks

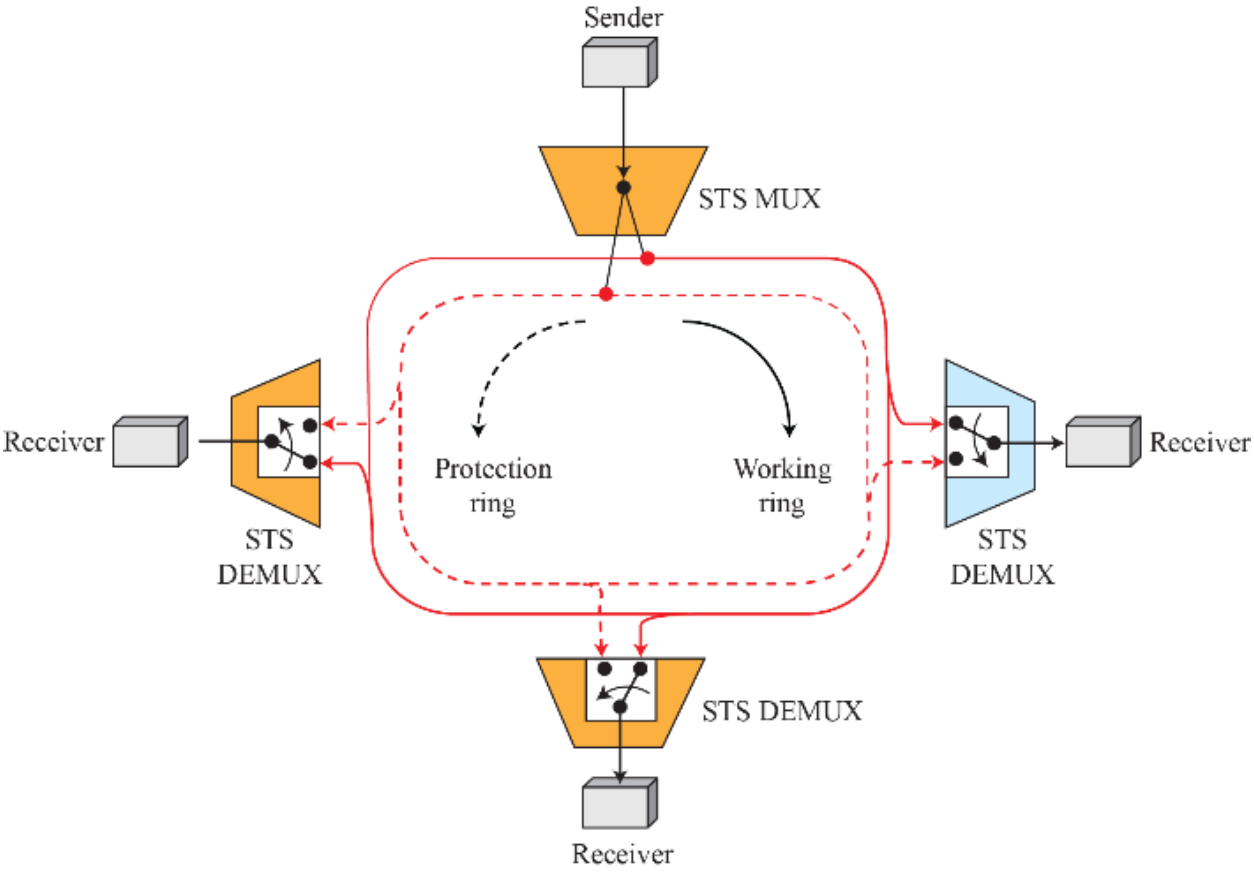


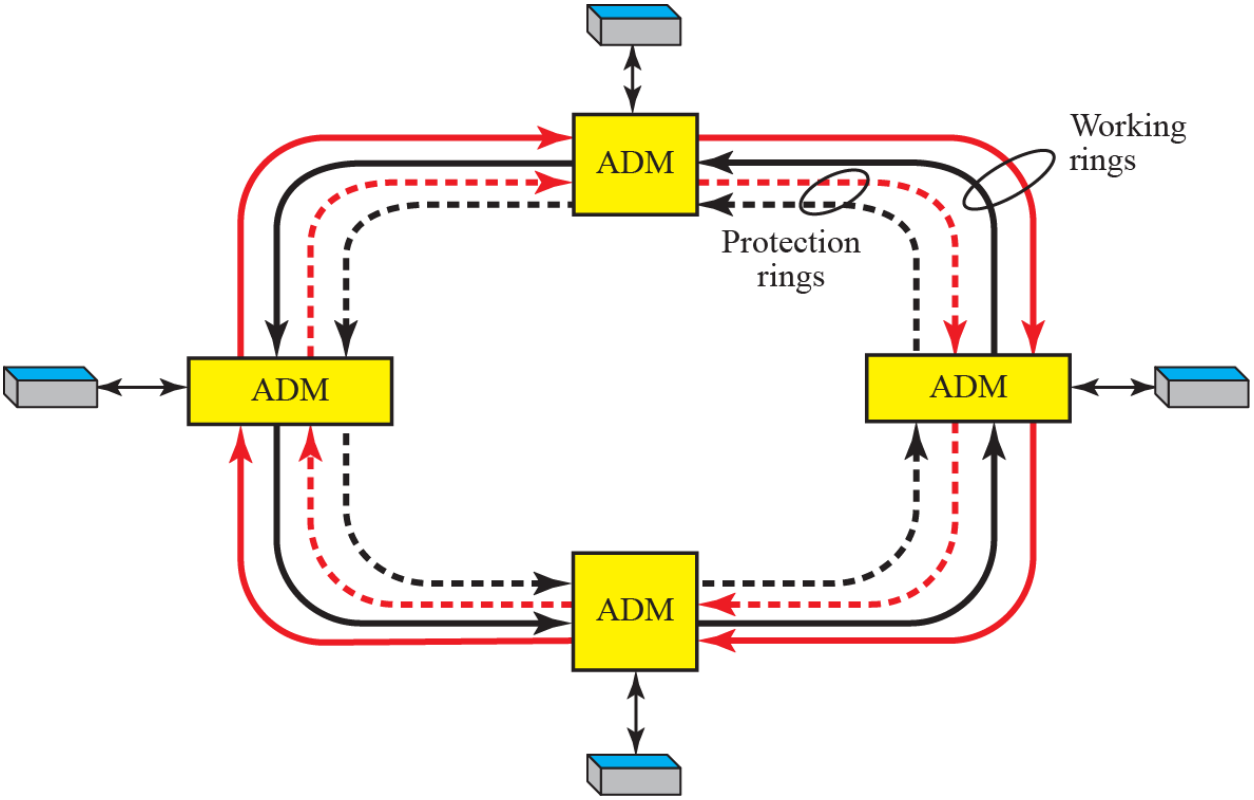
SONET Networks – Linear Networks



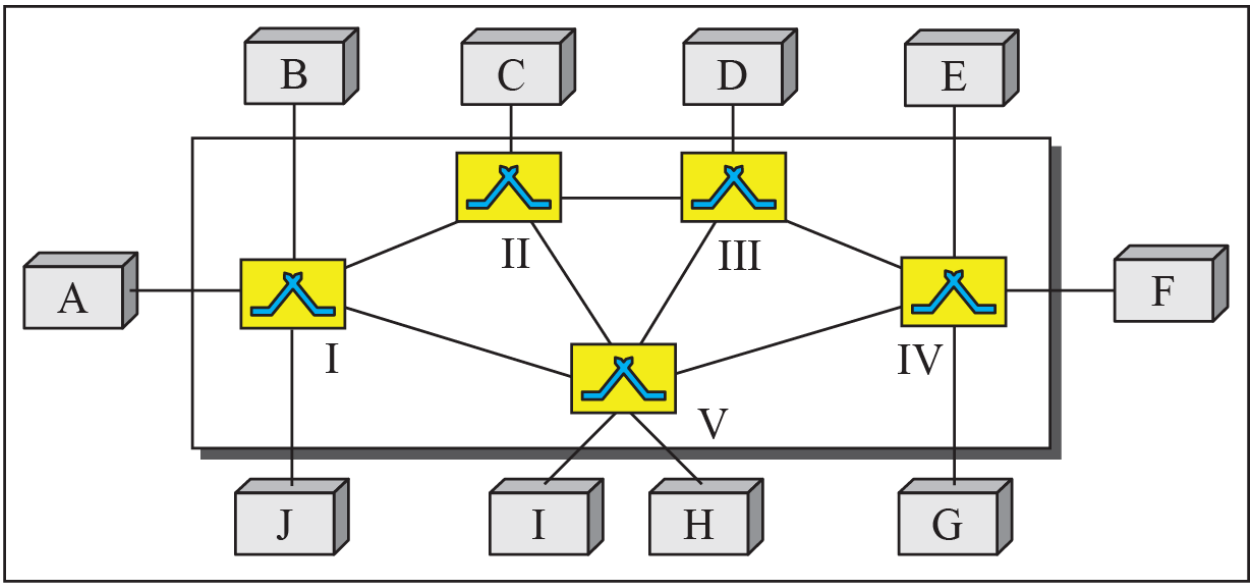


SONET Networks – Ring Networks

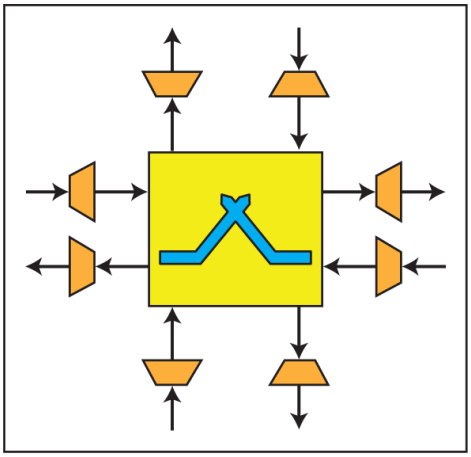




SONET Networks – Mesh Networks



a. SONET mesh network



b. Cross-connect switch

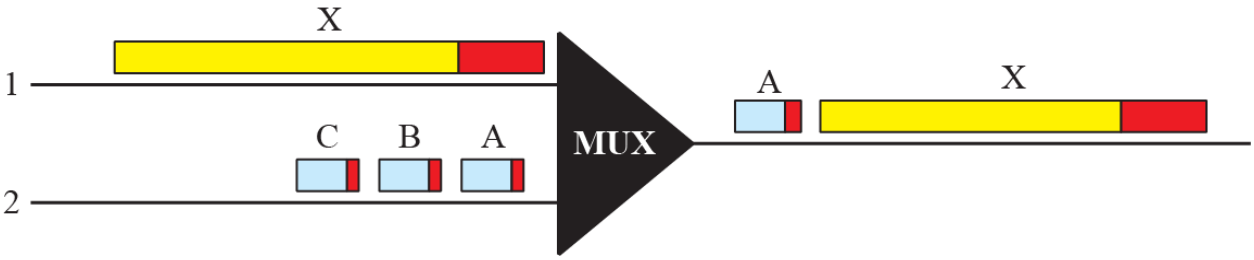
ATM

Asynchronous Transfer Mode (ATM) is a switched wide area network based on the cell relay protocol designed by the ATM forum. The combination of ATM and SONET will allow high-speed interconnection of networks.

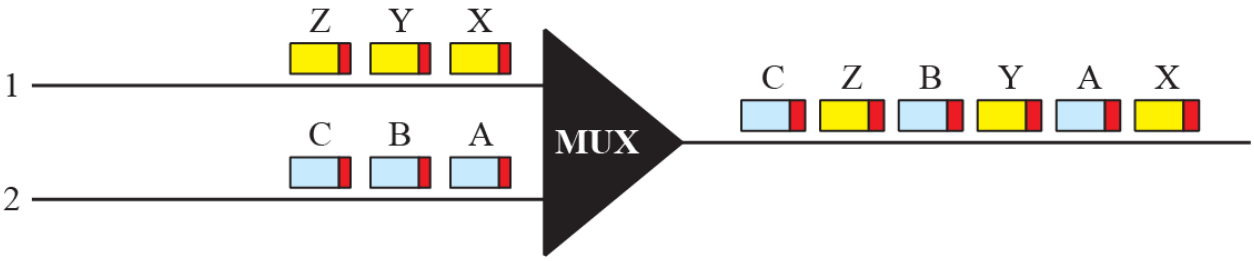
Problems

- Some of the problems associated with existing systems are:
  - Frame Networks
  - Mixed Network Traffic
- Solution
  - Cell Networks
  - Asynchronous TDM

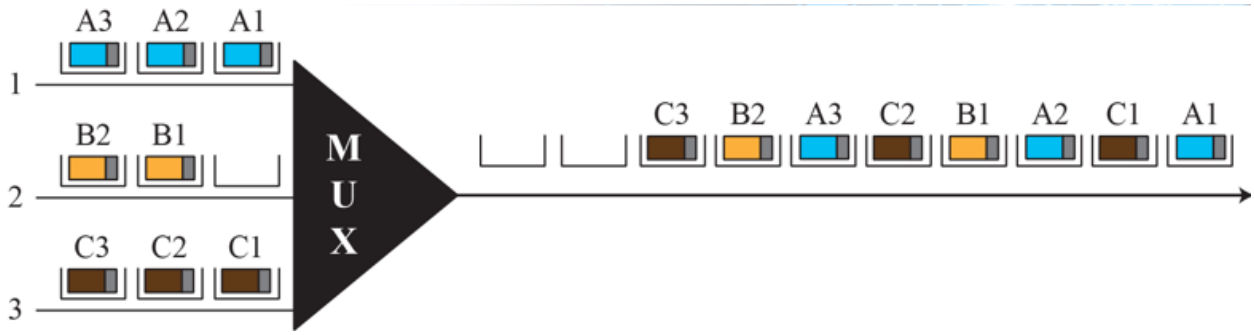
Multiplexing using Different Frame Size



Multiplexing using Cells



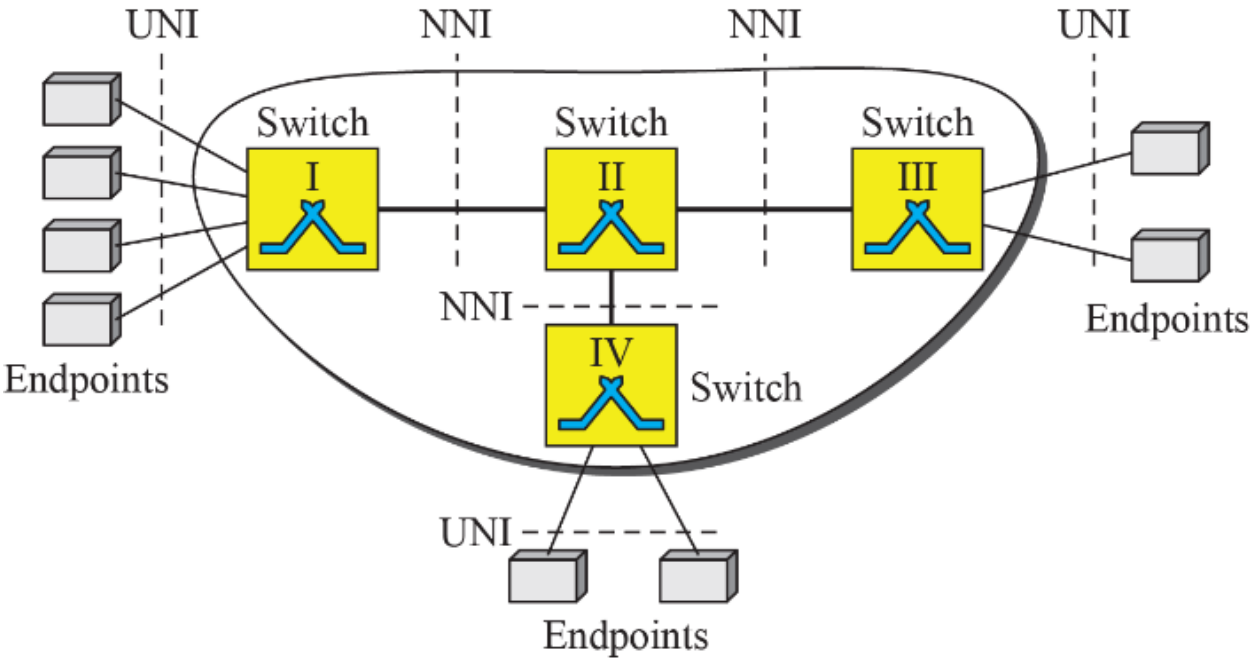
ATM Multiplexing

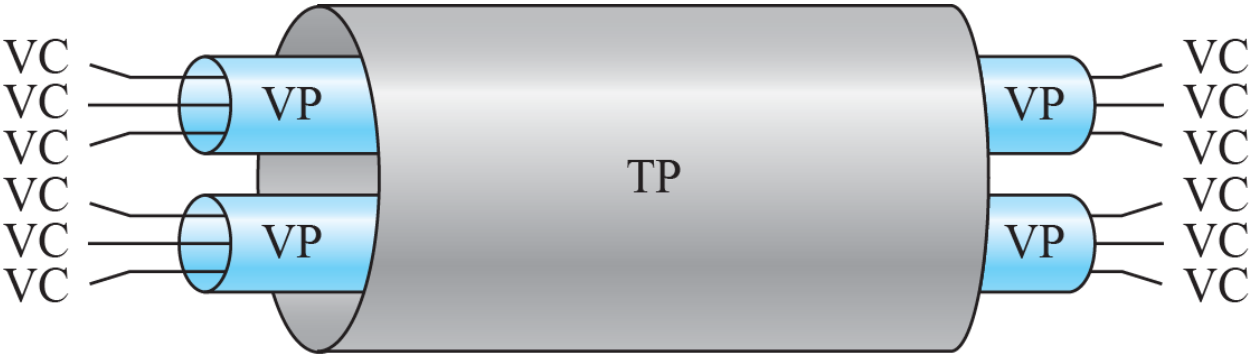


Architecture

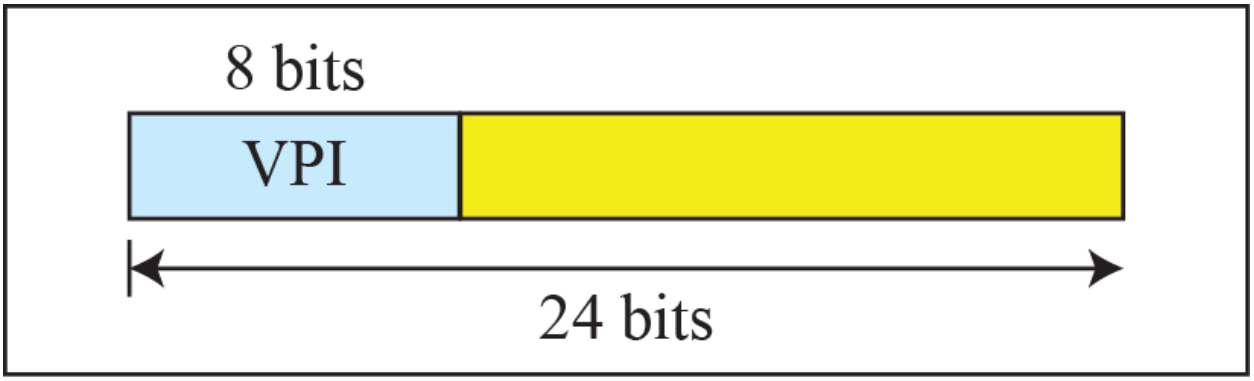
- ATM is a cell-switched network.
- The user access devices, called the endpoints, are connected through a user-to-network interface (UNI) to the switches inside the network.
- The switches are connected through network-to-network interfaces (NNIs).

Architecture of an ATM Network

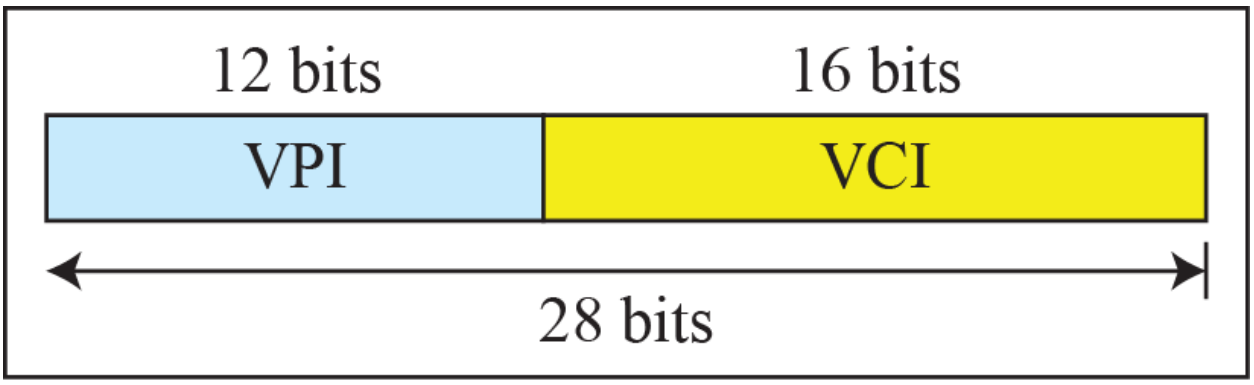




Virtual connection identifiers in UNIs & NNIs

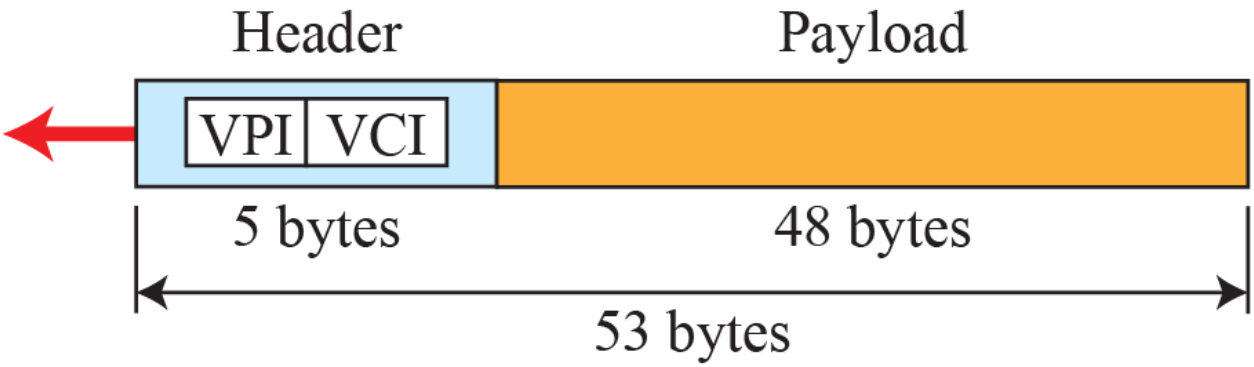


a. VPI and VCI in a UNI

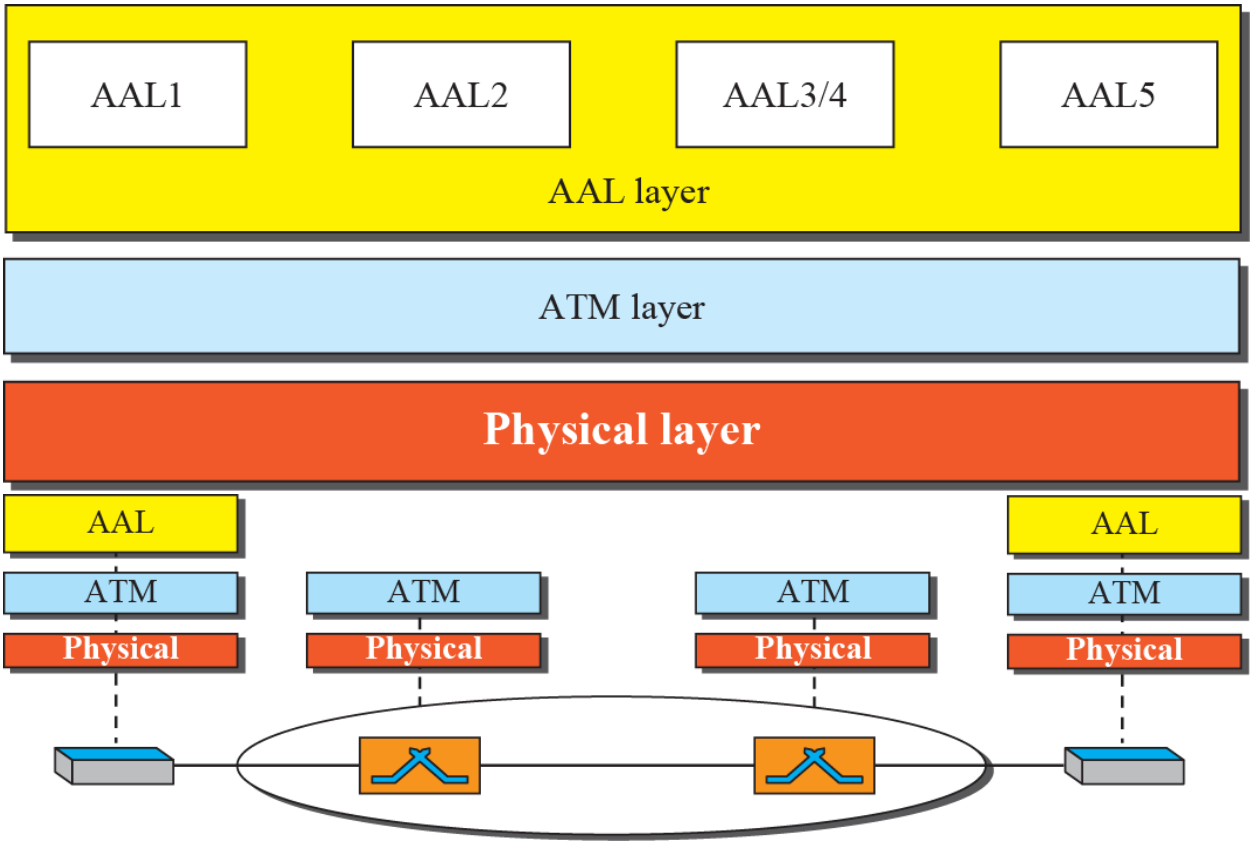


b. VPI and VCI in an NNI

An ATM Cell



ATM Layers


AAL1
AAL2
AAL3/4
AAL5
AAL layer
ATM layer
Physical layer
AAL
ATM
Physical
AAL
ATM
Physical
ATM
Physical
ATM
Physical

Topic 205 – 220

Wireless communication

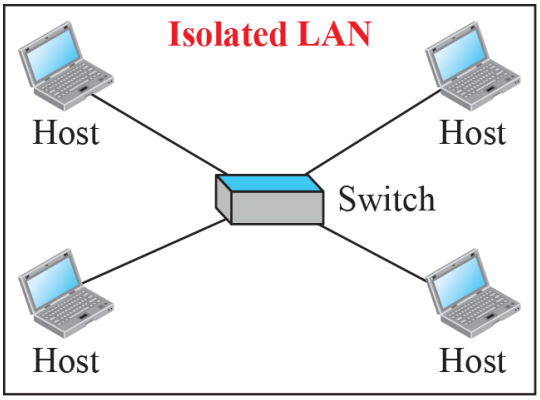
- Wireless communication is one of the fastest-growing technologies as the demand for connecting devices without the use of cables is increasing everywhere.
- Wireless LANs can be found on college campuses, in office buildings, and in many public areas.

Architectural Comparison

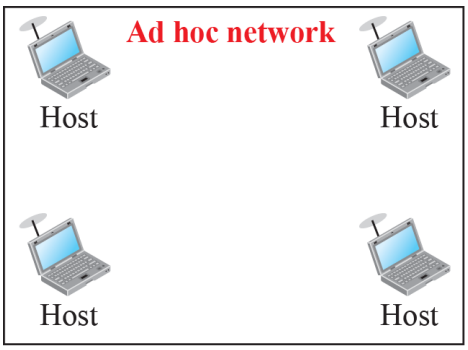
Architecture comparison of wired and wireless LANs

-
  - Medium
  - Hosts
  - Isolated LANs
  - Connection to other Networks
  - Moving between Environments

Isolated LANs: Wired versus Wireless

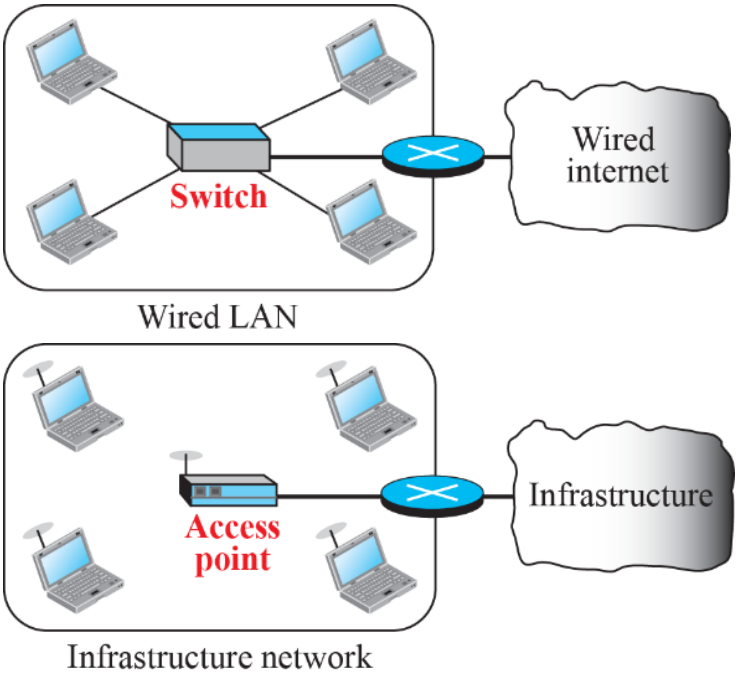


Wired



Wireless

Connection of a Wired/Wireless LAN to other networks



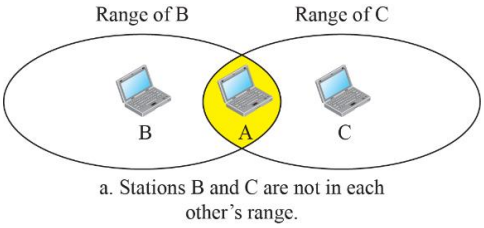
Characteristics of a Wireless LAN

- Several characteristics of wireless LANs either do not apply to wired LANs or the existence of these is negligible and can be ignored
    - Attenuation
    - Interference
    - Multipath Propagation
    - Error

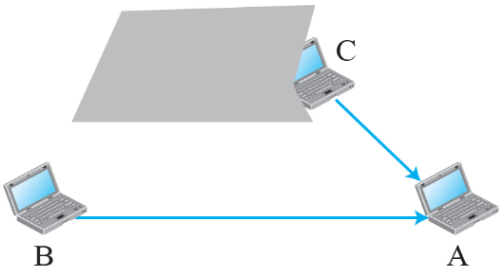
Access Control

- Most important issue in a wireless LAN is how a wireless host can get access to the shared medium (air)
- CSMA/CD does not work in wireless LANs for three reasons:
    1. Wireless hosts don't have power to send and receive at the same time
- The hidden station problem prevents collision detection
- The distance between stations can be large

Hidden Station Problem



a. Stations B and C are not in each other's range.



b. Stations B and C are hidden from each other.

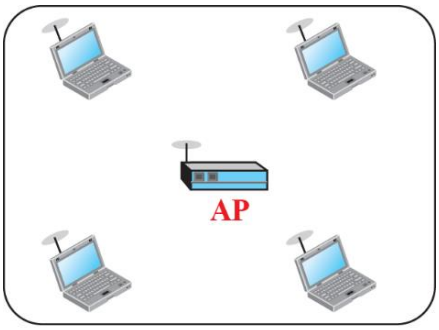
IEEE 802.11 PROJECT

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data-link layers. It is sometimes called Wireless Ethernet. The term WiFi (short for wireless fidelity) as a synonym for wireless LAN (certified by WiFi alliance).

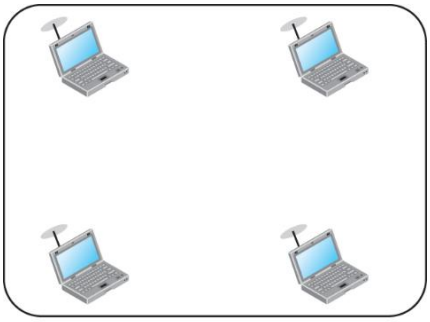
Architecture

- The standard defines two kinds of services:
  - The basic service set (BSS); and
  - The Extended service set (ESS)

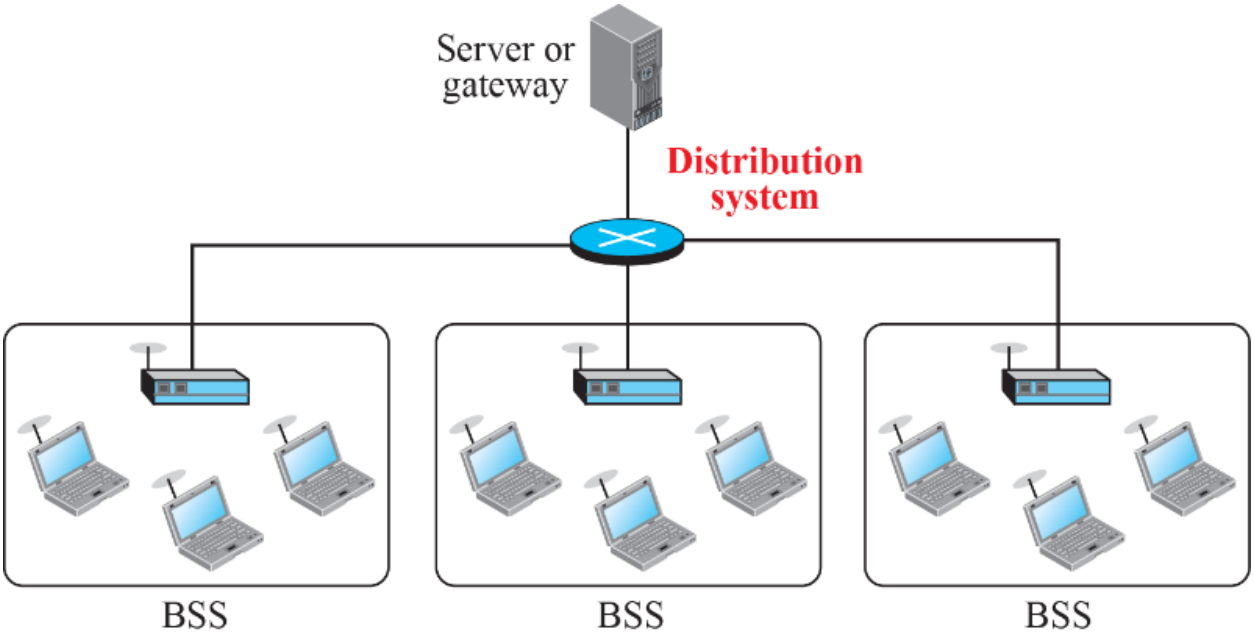
Basic Service Sets (BSSs)



Infrastructure BSS


Ad hoc BSS

Extended Service Set (ESS)



Types of Stations

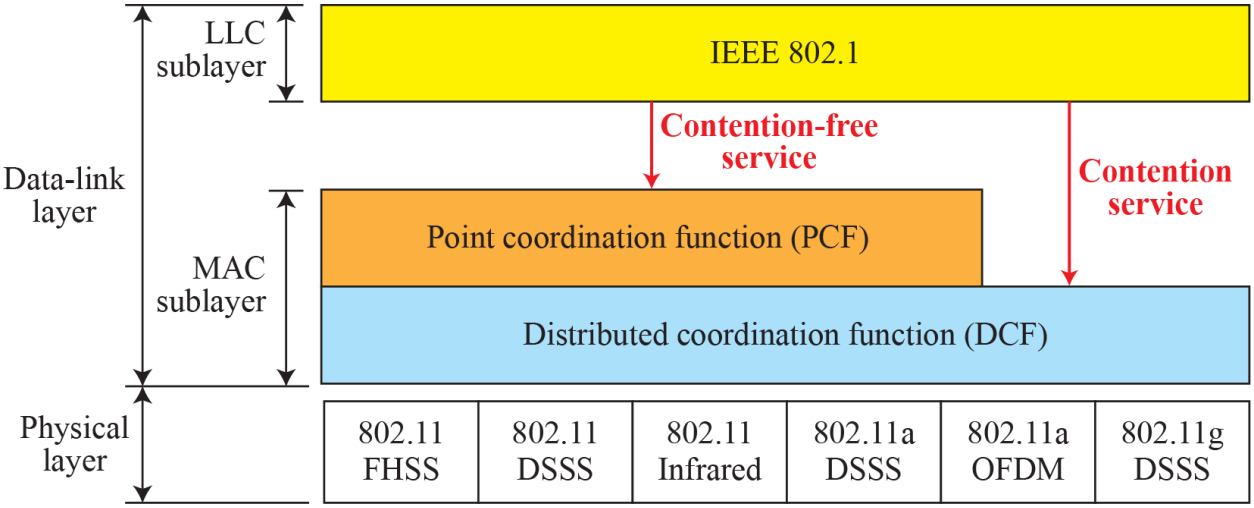
- No-Transition Mobility

- BSS-Transition Mobility
- ESS-Transition Mobility

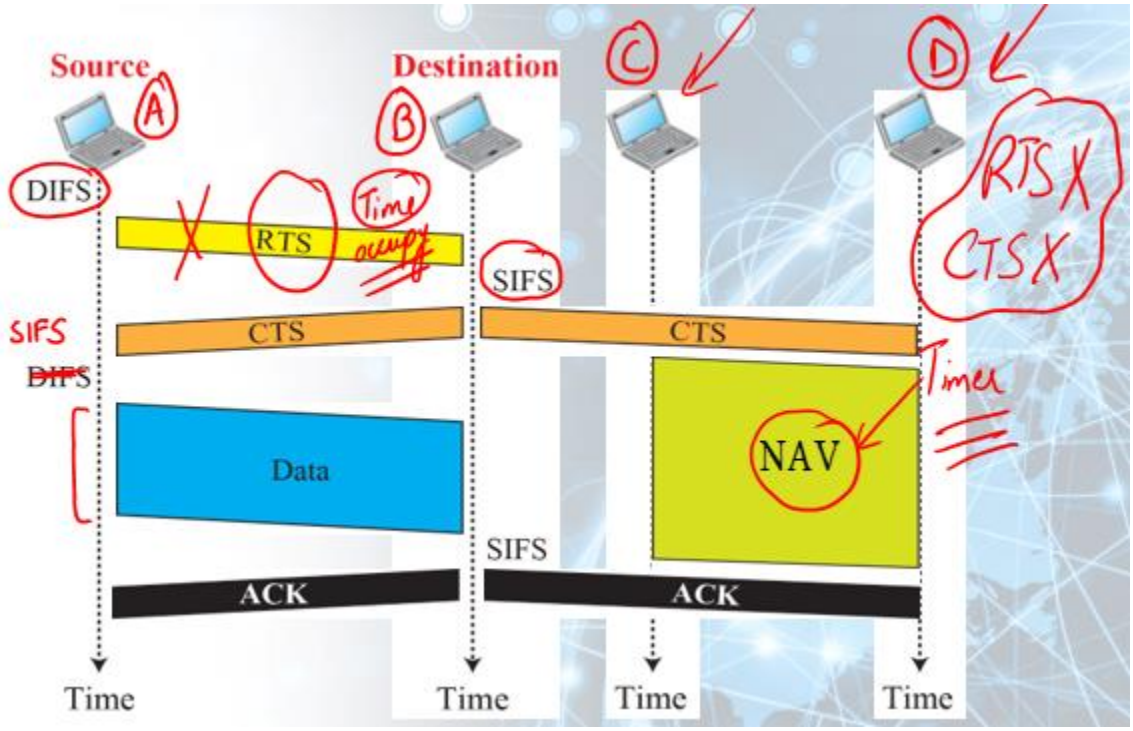
MAC Sub-layer

- IEEE 802.11 defines two MAC sub-layers:
    - The Distributed Coordination Function (DCF) ; and
    - The Point Coordination Function (PCF)

MAC Layers in IEEE 802.11 Standard



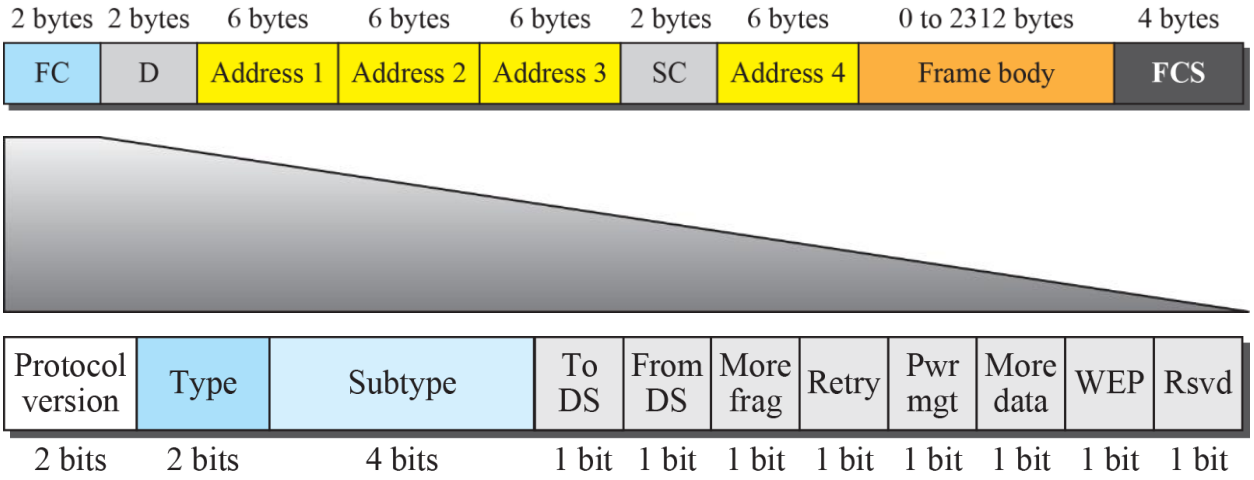
CSMA/CA and NAV



MAC Sub-layer

- IEEE 802.11 defines two MAC sub-layers:
  - The Distributed Coordination Function (DCF) ; and
  - The Point Coordination Function (PCF)

Frame Format



Subfields in FC field

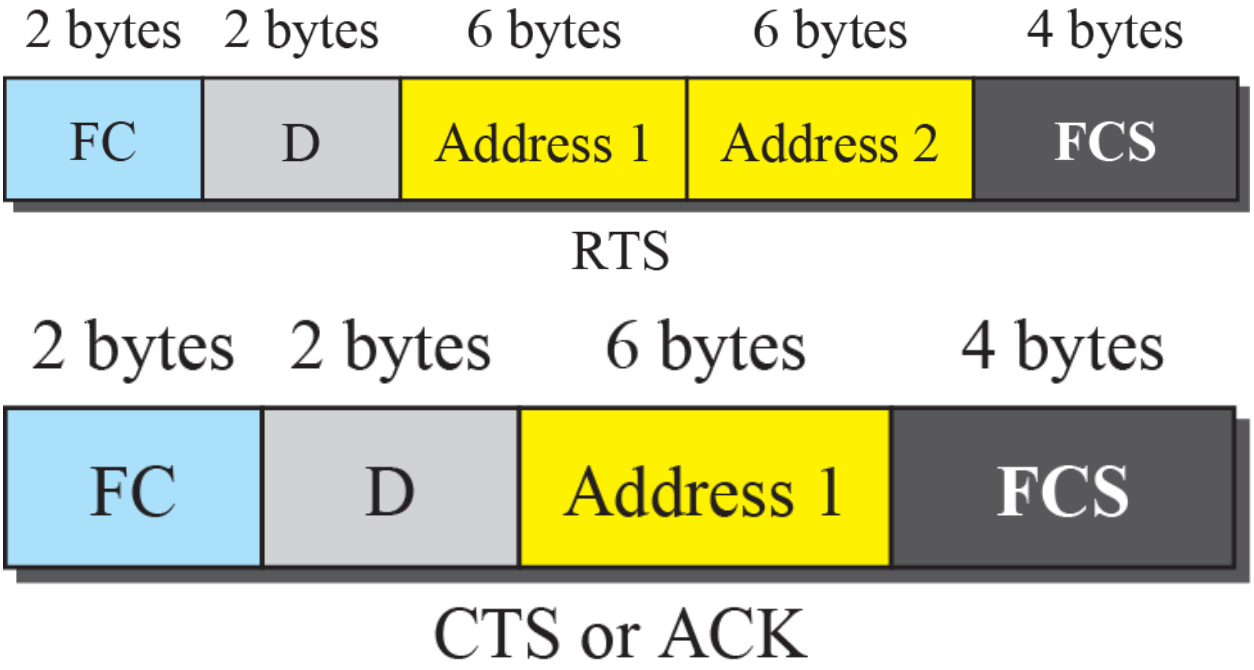
Field	*Explanation*
Version	Current version is 0
Type	Type of information: management (00), control (01), or data (10)
Subtype	Subtype of each type (see Table 6.2)
To DS	Defined later
From DS	Defined later
More flag	When set to 1, means more fragments
Retry	When set to 1, means retransmitted frame
Pwr mgt	When set to 1, means station is in power management mode
More data	When set to 1, means station has more data to send
WEP	Wired equivalent privacy (encryption implemented)
Rsvd	Reserved

Frame Types

Following are the frame types:

- Management Frames
- Control Frames
- Data Frames

Control Frames



Values of Subfields in Control Frames

Subtype	*Meaning*
1011	Request to send (RTS)
1100	Clear to send (CTS)
1101	Acknowledgment (ACK)

Physical Layer

- All physical implementations, except the infrared, operate in the industrial, scientific, and medical (ISM) band, which defines 3 unlicensed bands in 3 ranges:
  - 902–928 MHz
  - 2.400–4.835 GHz
  - 5.725–5.850 GHz

Specifications

IEEE	*Technique*	*Band*	*Modulation*	*Rate (Mbps)*
802.11	FHSS	2.400–4.835 GHz	FSK	1 and 2
	DSSS	2.400–4.835 GHz	PSK	1 and 2
	None	Infrared	PPM	1 and 2
802.11a	OFDM	5.725–5.850 GHz	PSK or QAM	6 to 54
802.11b	DSSS	2.400–4.835 GHz	PSK	5.5 and 11
802.11g	OFDM	2.400–4.835 GHz	Different	22 and 54
802.11n	OFDM	5.725–5.850 GHz	Different	600

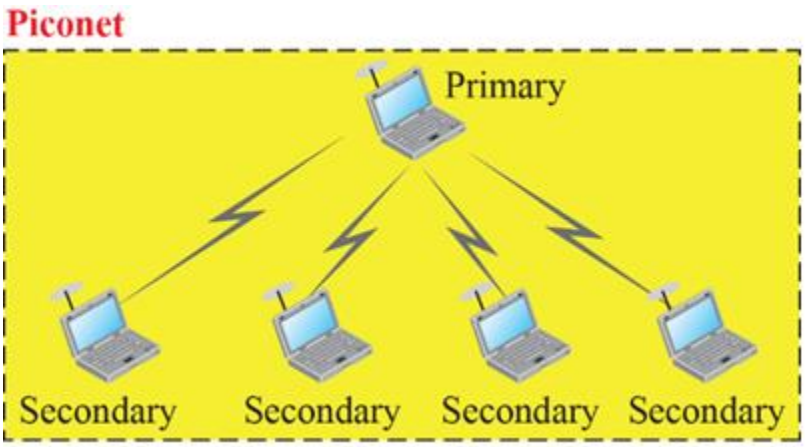
BLUETOOTH

- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other. A Bluetooth LAN is an ad hoc network.
- The devices, sometimes called gadgets, find each other and make a network called a Piconet.

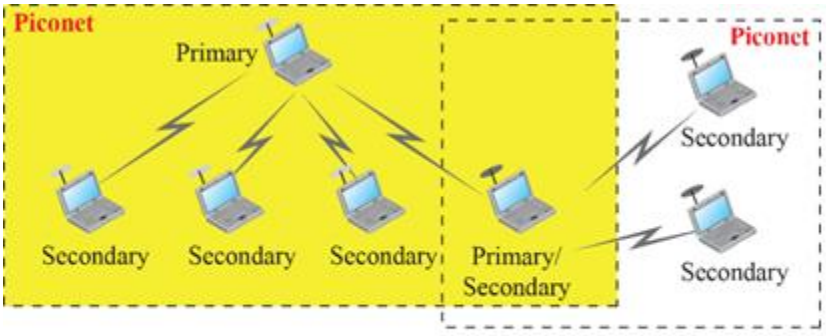
Architecture

- **Bluetooth defines two types of networks:**
  - Piconet
  - Scatternet

Piconet

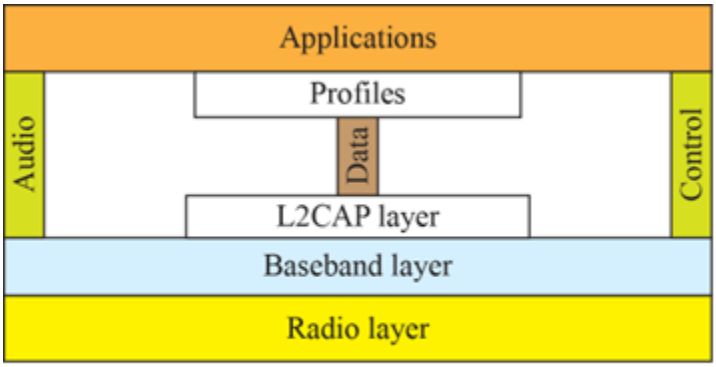


Scatternet



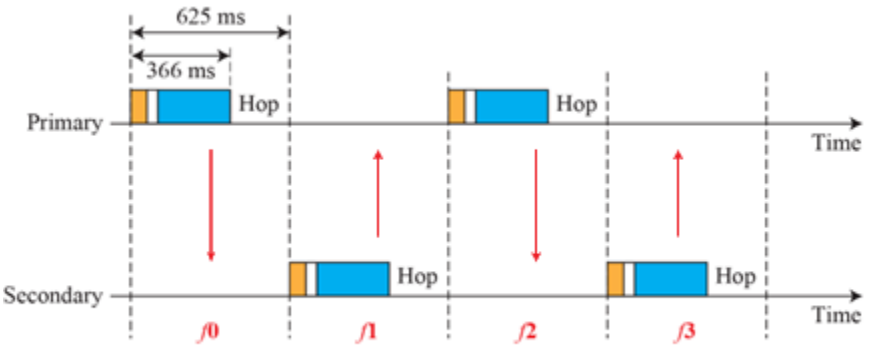
Bluetooth Layers

- Bluetooth uses several layers that do not exactly match those of the Internet model we have defined in this book

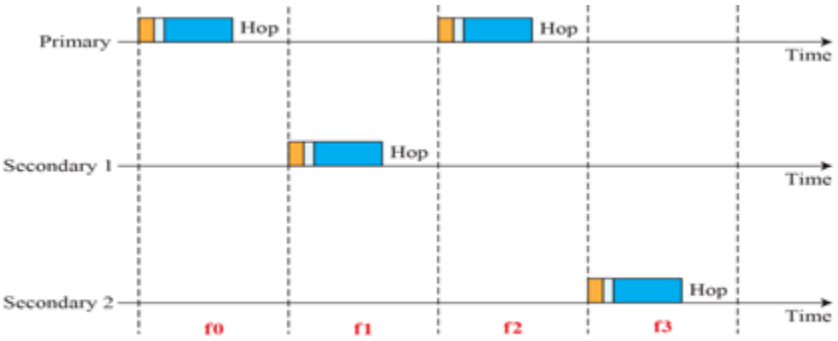


- **L2CAP Data Packet Format**

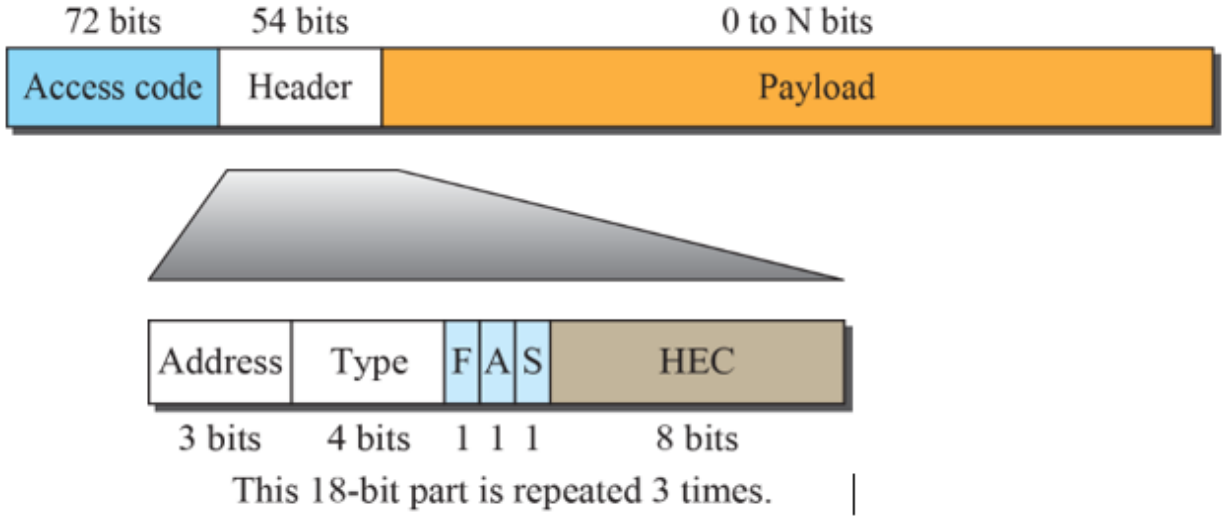
Single-Secondary Communication



Multiple-Secondary Communication



- **Frame Format Types**



Bluetooth

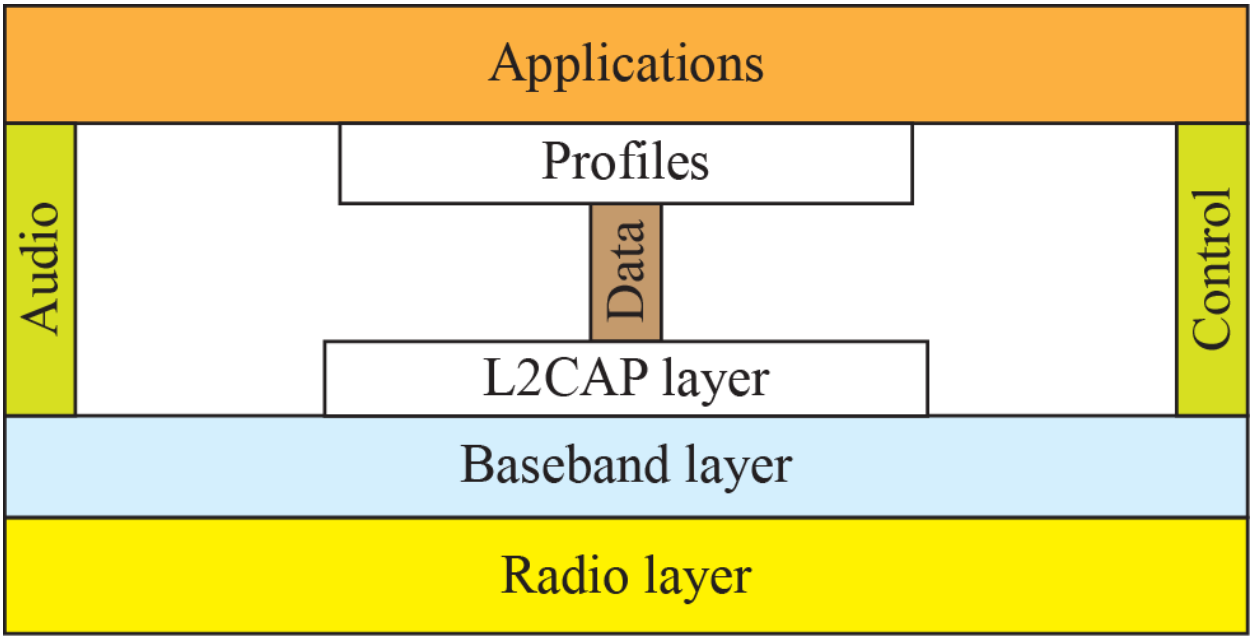
- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other. A Bluetooth LAN is an ad hoc network. The devices, sometimes called gadgets, find each other and make a network called a Piconet.
- Bluetooth technology is the implementation of a protocol defined by the IEEE 802.15 standard. The standard defines a wireless Personal-Area Network (PAN) operable in an area the size of a room or a hall.

Bluetooth Devices

- A Bluetooth device has a built-in short-range radio transmitter. The current data rate is 1 Mbps with a 2.4-GHz bandwidth. This means that there is a possibility of interference between the IEEE 802.11b wireless LANs and Bluetooth LANs

Bluetooth Layers

- Bluetooth uses several layers that do not exactly match those of the Internet model we have defined already.

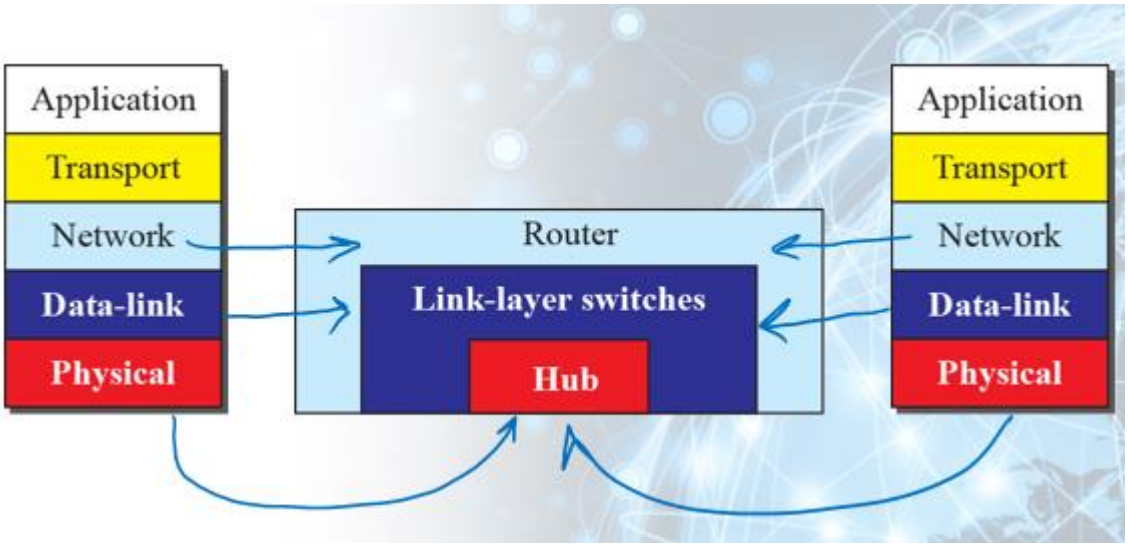


Connecting Devices

- Hosts and networks do not normally operate in isolation
- Connecting devices connect hosts together to make a network or connect networks together to make an internet
- Connecting devices can operate in different layers of the Internet model
- **Three kinds of connecting devices:**

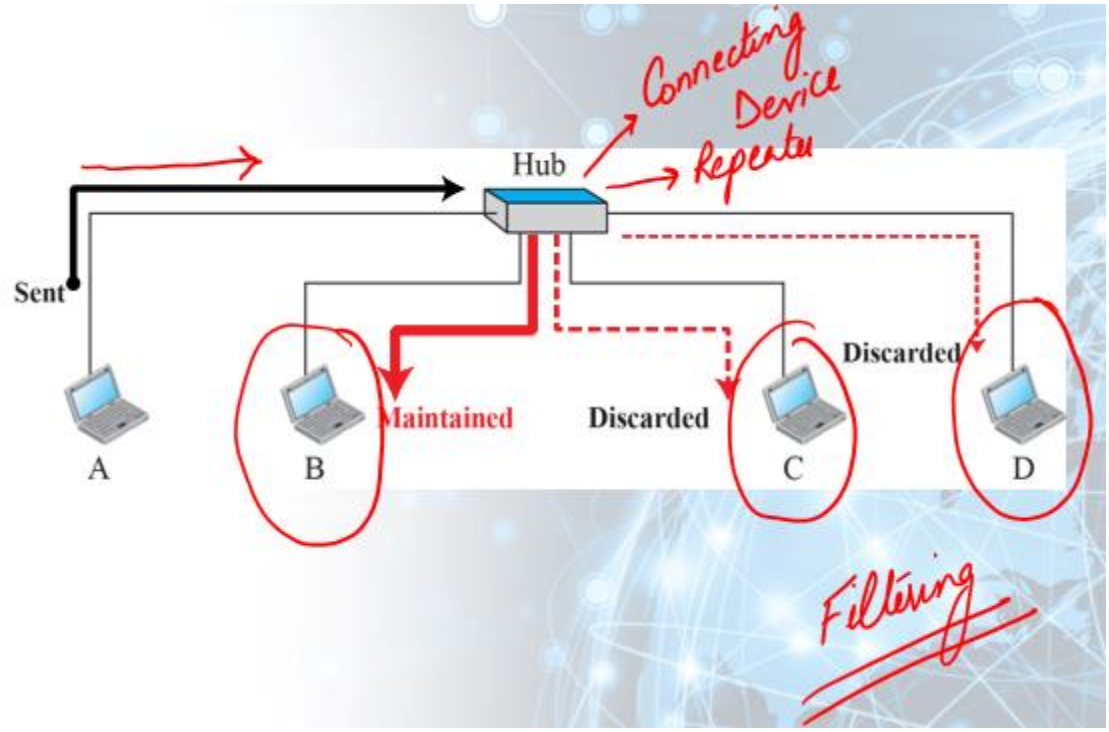
- Hubs
- Link-layer switches
- Routers

Three Categories of Connecting Devices



Hubs

- Hub is a device that operates only in the physical layer
- Signals that carry information within a network can travel a fixed distance before attenuation impacts the data
- A hub (repeater) receives a signal and, before it becomes too weak or corrupted, regenerates it
- Hub is a device that operates only in the physical layer

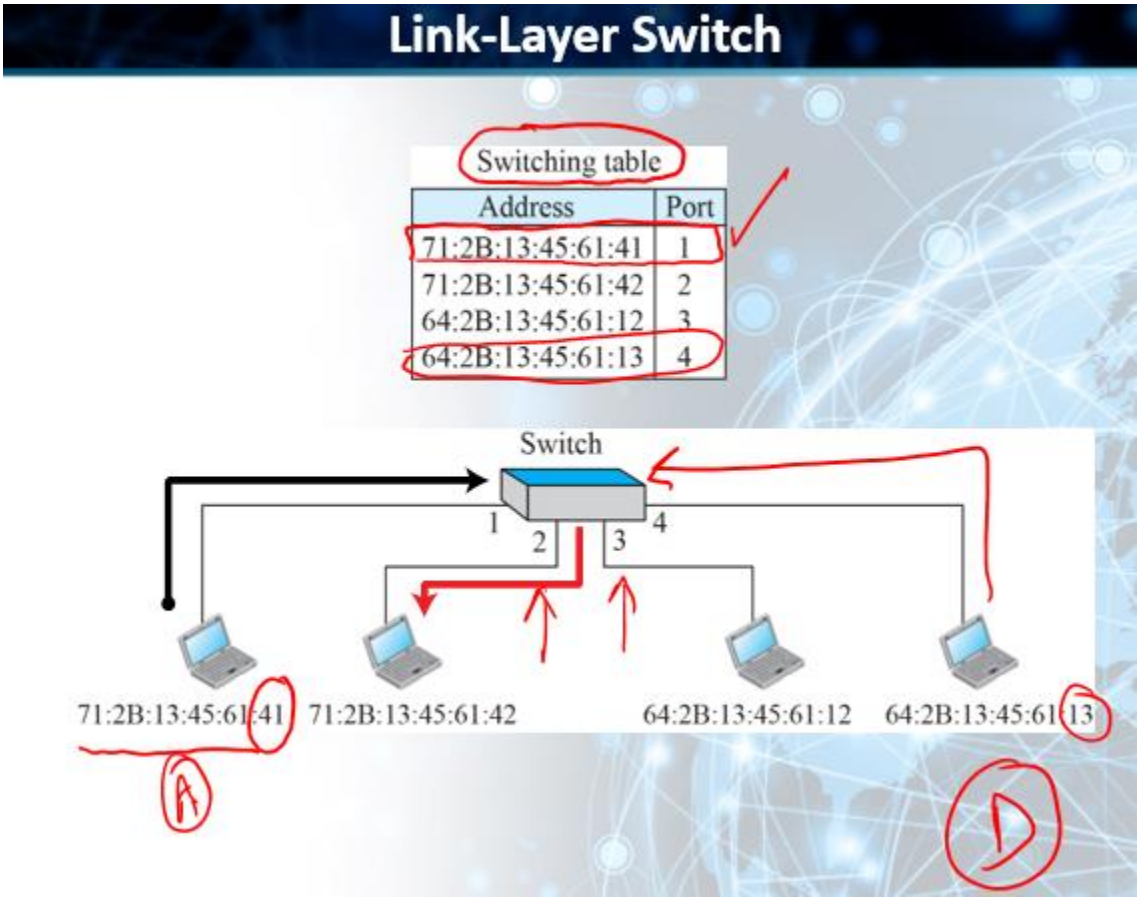


Link-Layer Switches

- A link-layer switch (or switch) operates in both the physical and the data-link layers
- As a physical-layer device, it regenerates the signal it receives
- As a link-layer device, the link-layer switch can check the MAC addresses (source and destination) contained in the frame

Switch versus Hub

- Switch has the ‘Filtering’ capability
- Unlike hub, a switch can check the destination address of a frame and decide on outgoing port
- Switch eliminates collisions and does not require carrier sensing
- Switches connect heterogeneous devices

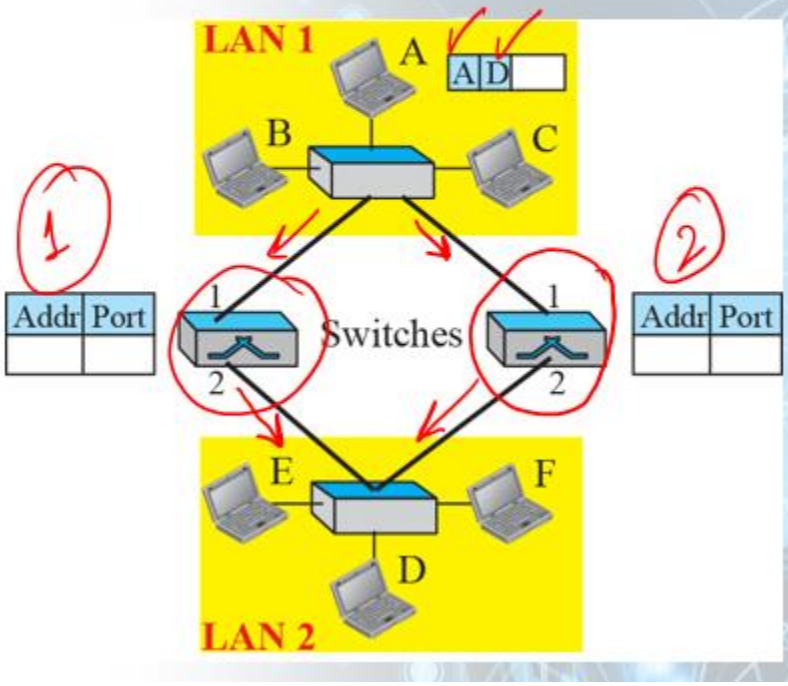


Loop Problem in a Switch

- Redundant switches create Loops in the system
- Created when two or more broadcasting LANs are connected by more than one switch

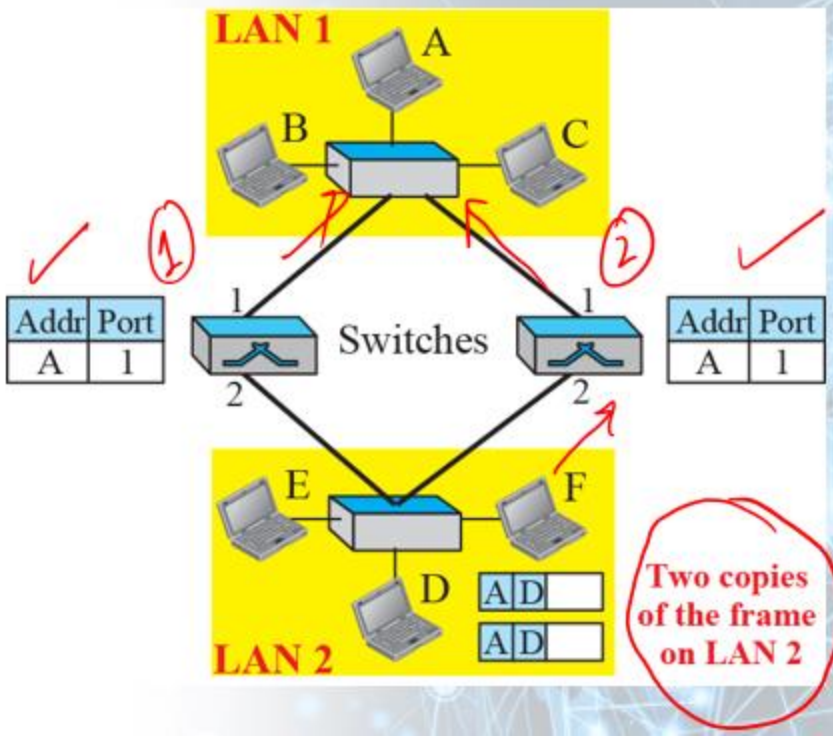
Loop Problem in a Learning Switch (Part a)

a. Station A sends a frame to station D



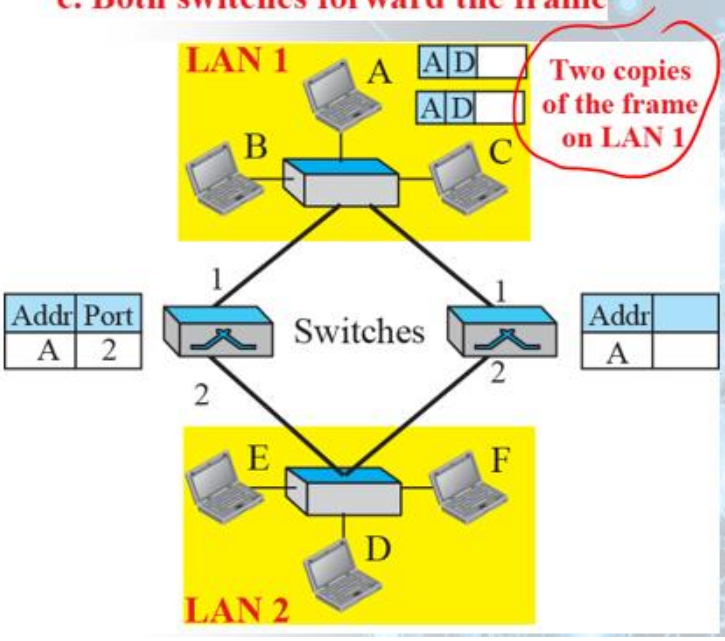
Loop Problem in a Learning Switch (Part b)

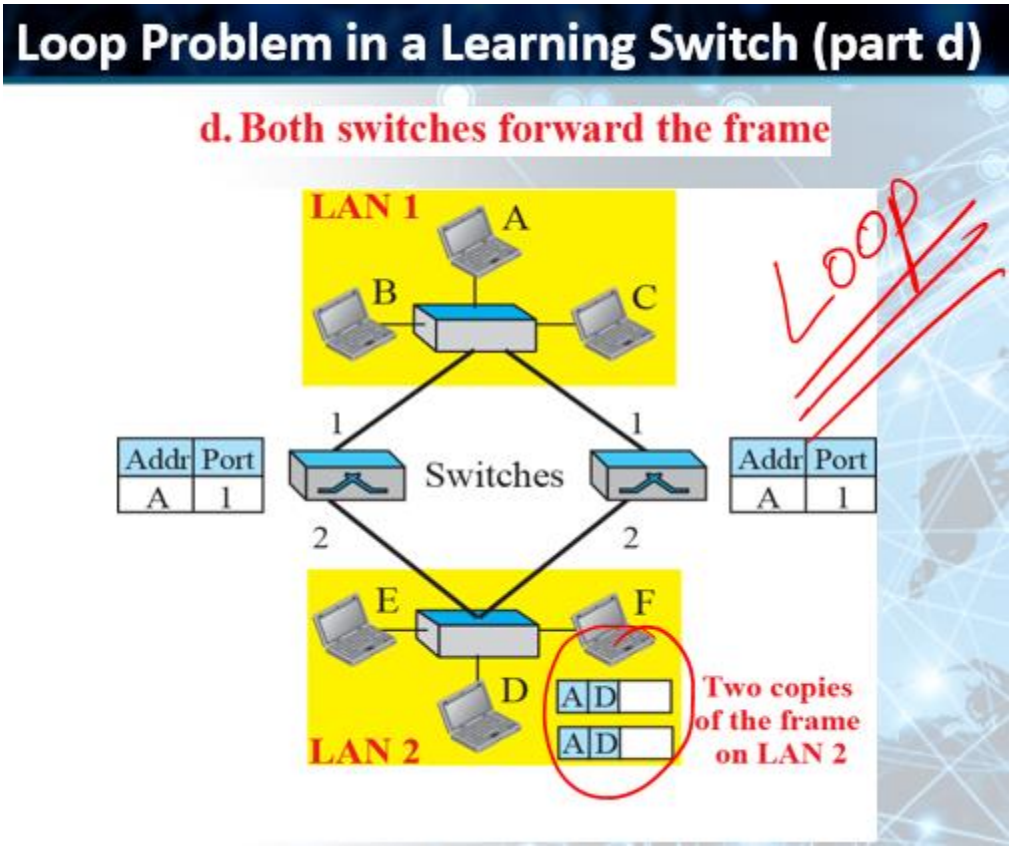
b. Both switches forward the frame



Loop Problem in a Learning Switch (Part c)

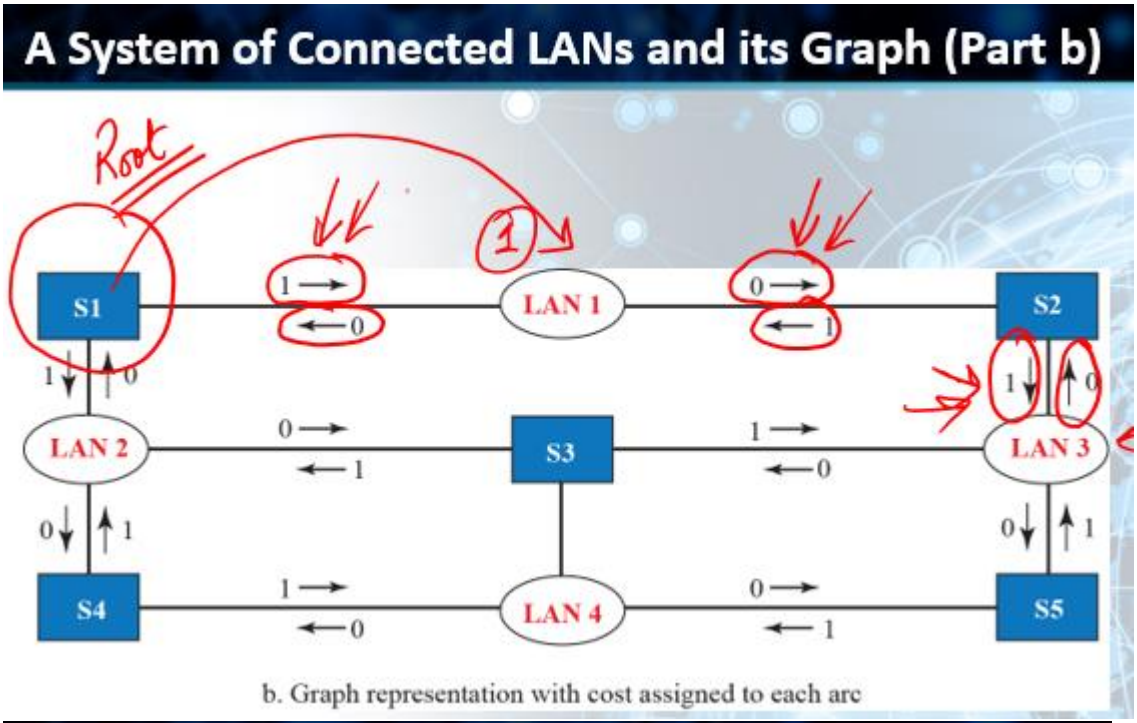
c. Both switches forward the frame

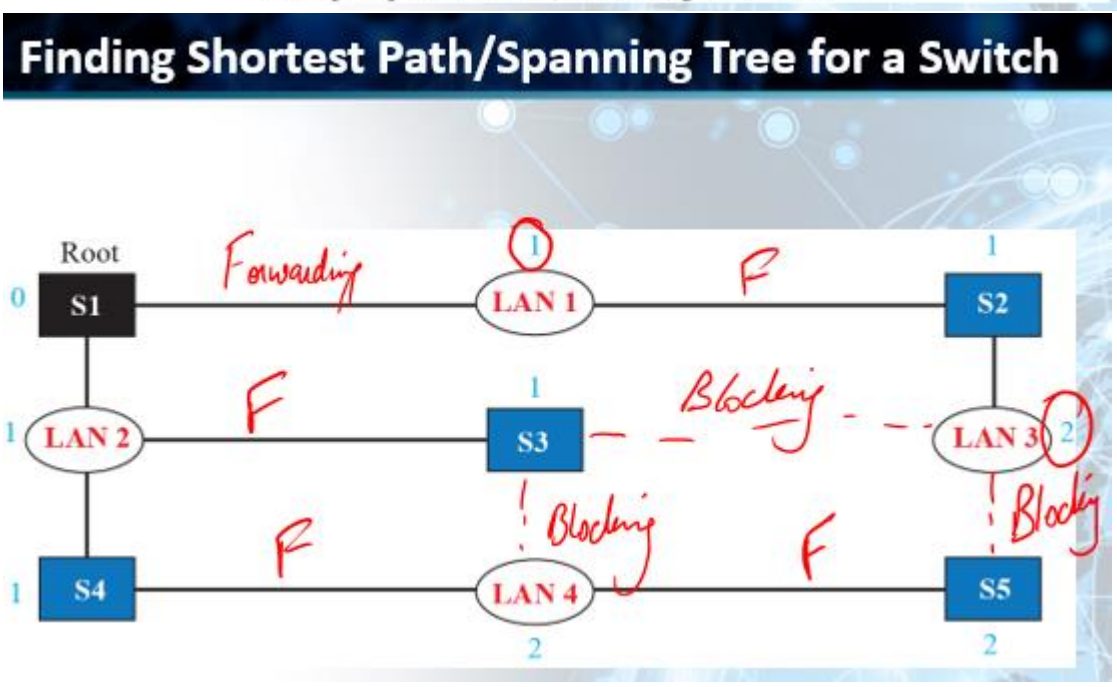




Spanning Tree Algorithm

- In graph theory, Spanning Tree is a graph in which there is no loop
- In a switched LAN, this means creating a topology in which each LAN can be reached from any other LAN through one path only (no loop)
- To find the spanning tree, we assign a cost (metric) to each LAN link





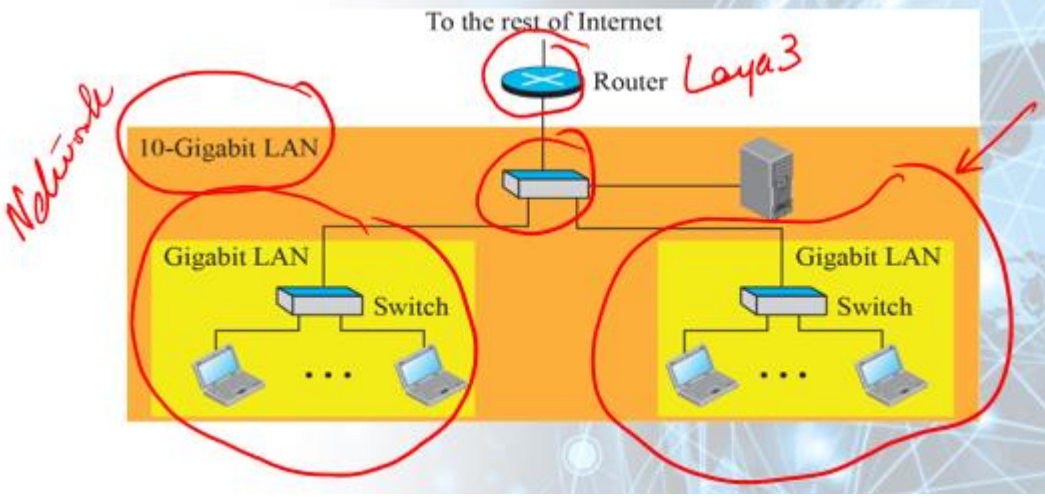
Routers

- We compare routers to two-layer switch and a hub
- A router is a three-layer device; it operates in the physical, data-link, and network layers

Router vs. Switch

Three differences between a router and a repeater or a switch:

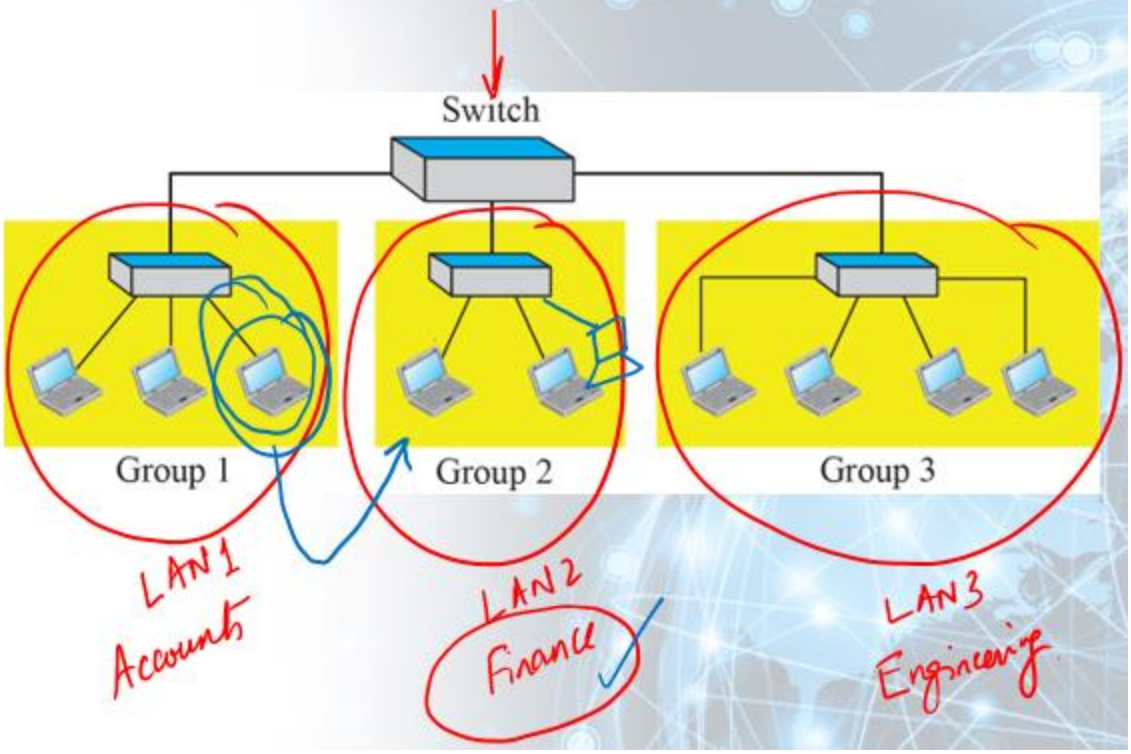
1. A router has a physical and logical (IP) address for each of its interfaces.
2. A router acts only on those packets in which the link-layer destination address matches the address of the interface at which the packet arrives.
3. A router changes the link-layer address of the packet (both source and destination) when it forwards the packet.



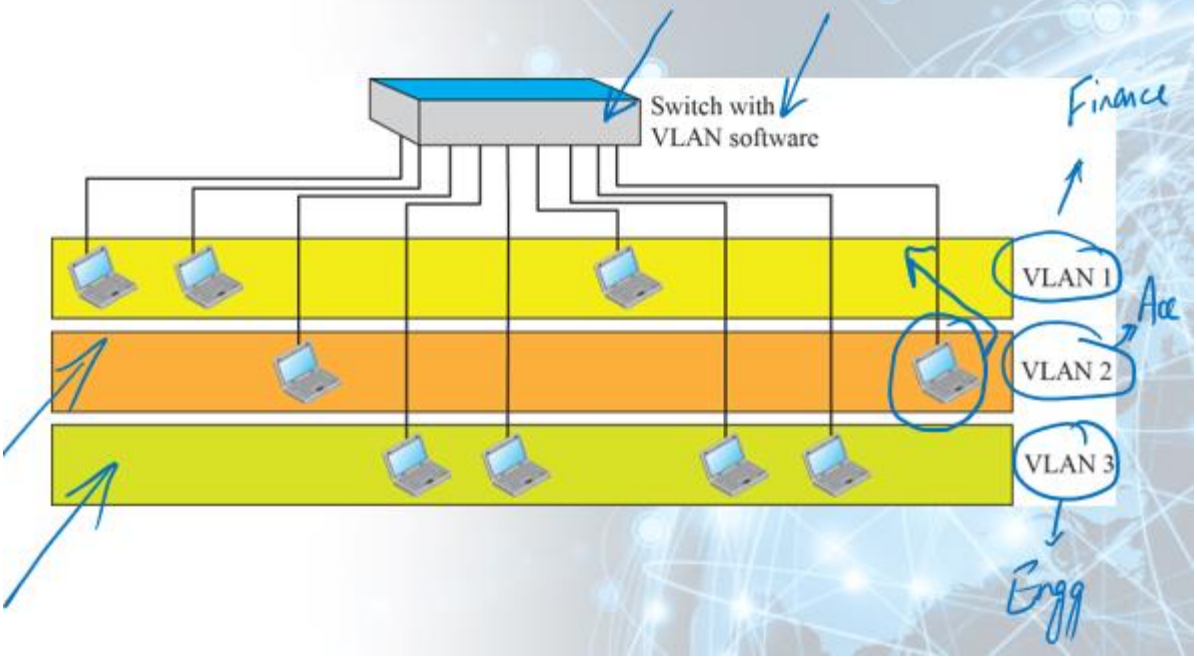
VIRTUAL LANS (VLAN)

- A VLAN is a LAN configured by software, not by physical wiring
- A station is considered part of a LAN if it physically belongs to that LAN i.e. The criterion of membership is geographic
- Provides a virtual connection between two stations belonging to two different physical LANs

A Switch Connecting three LANs



A Switch using VLAN Software



Membership of a VLAN

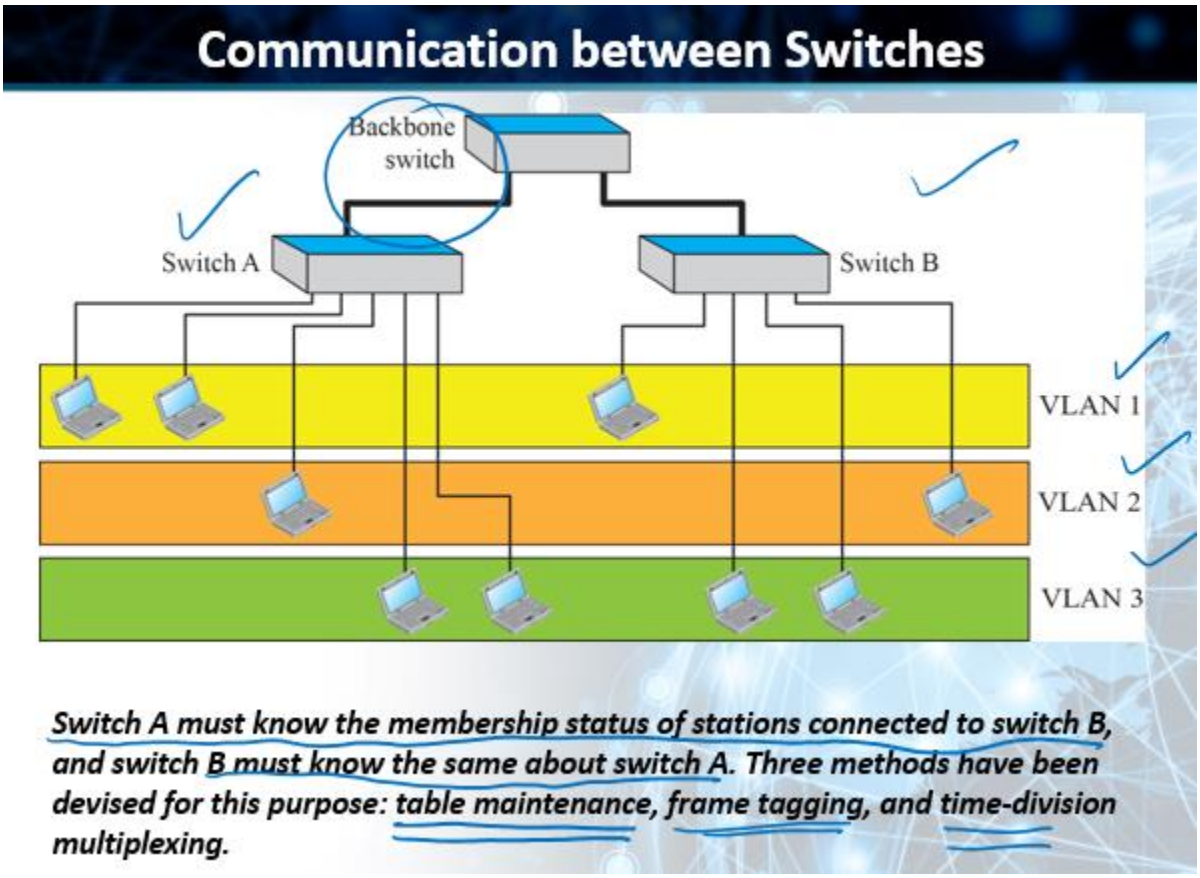
- What characteristic can be used to group stations in a VLAN?
- Vendors use different characteristics such as interface numbers, port numbers, MAC addresses, IP addresses, IP multicast addresses, or a combination of two or more of these

Configuration of a VLAN

- How are the stations grouped into different VLANs?
- Stations are configured in one of three ways:
  - ✓ Manually
  - ✓ Semi-Automatically
  - ✓ Automatically

Communication between Switches

- In a multi-switched backbone, each switch must know:
  - ✓ Which station belongs to which VLAN; and
  - ✓ The membership of stations connected to other switches

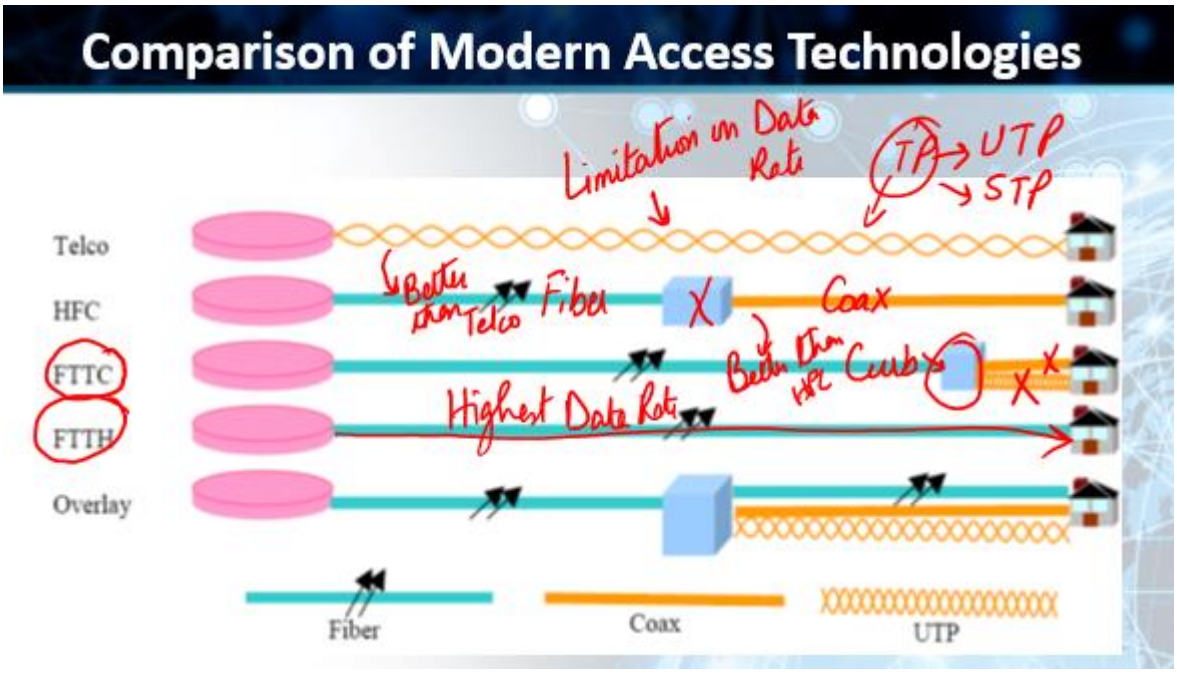


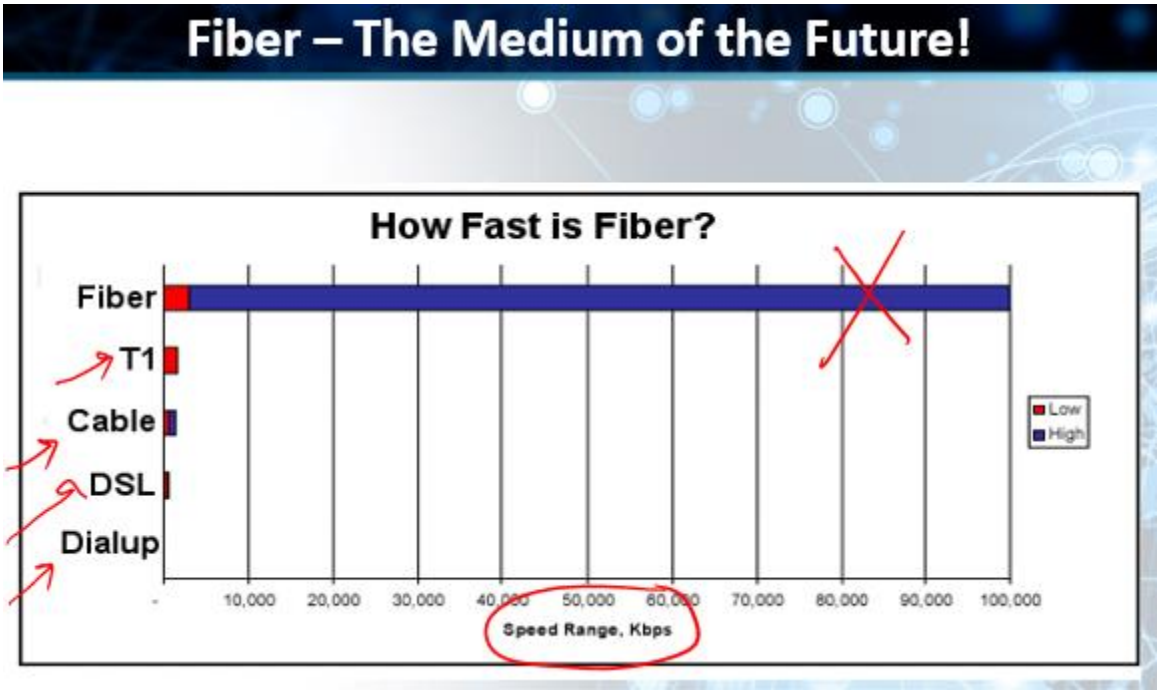
Advantages of using VLANs

- Cost and Time Reduction
- Creating virtual Workgroups
- Security

Comparison of Modern Access Technologies

- **Telco**
- **HFC**
- **FTTx**





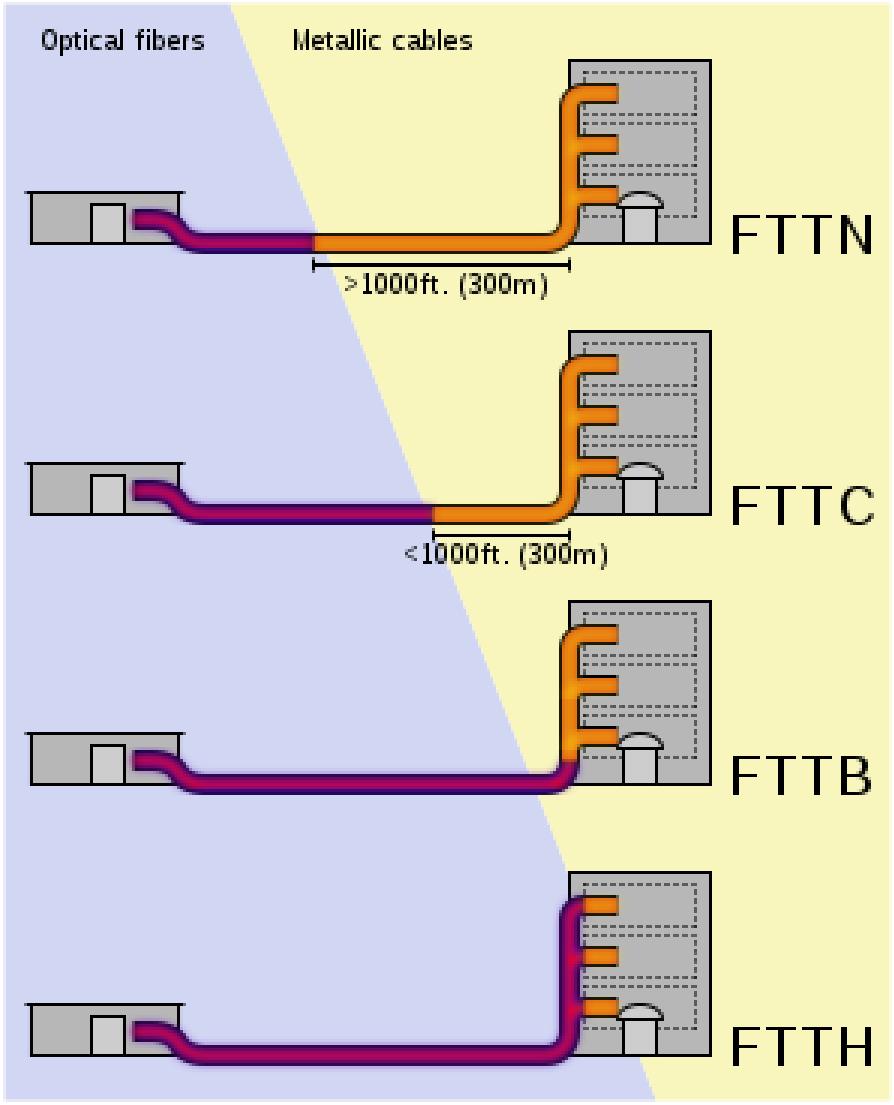
Fiber To The Curb (FTTC)

- An access network in which fiber is used for part, but not the entire link from the provider to the end-user
- An optical to electrical (O/E) conversion takes place somewhere near the end-user
- The terminal network segment of a FTTC network is usually twisted pair or coaxial cable
- The final optical receiver in a FTTC network typically serves several customers

Fiber To The Home (FTTH)

- Need: High-speed data, reliable voice and high-quality video
- Problems:
  - ✓ How to get high speed lines out to each customer?
  - ✓ How to future-proof the architecture?
- Solution: FTTH
- Fiber-to-the-home (FTTH) is the installation of optical fiber from a telephone switch directly into the subscriber's home
- It is one of the latest access technologies
- FTTH is also referred to as Fiber-to-the-Building (FTTB)

Fiber To The Home (FTTH)



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